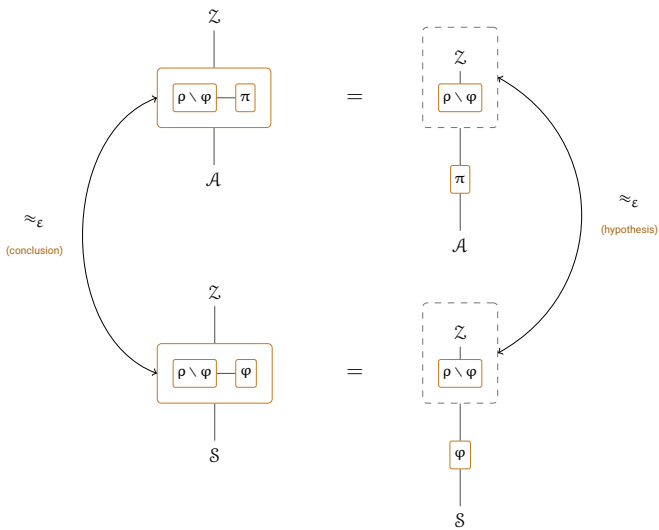


Gamers for UC



Claude Opus 5 (1M context)

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هوش مصنوعی:

پرامپت ها:

بازبینان:

تاریخ:

code, and prose of mixture a in written usually is composability Universal
 verify or test to still harder and falsify, to hard research UC makes which
 we security, property-based of style code-based the Following formally.
 identi- functionalities: UC programming for language rigorous a out set
 a is corruption them, enforce wrappers data, are control access and ties
 passing handler a is execution an and reads, interface every that register
 classes machine explicit over defined then is Emulation token, single a
 and – asymptotics no parameter, security no – bounds concrete with
 a and composition parallel substitution, follows: toolkit composition the
 corollaries three all – adversary dummy the for theorem completeness
 into adversary an or protocol a moves which lemma, absorption one of
 rather execution single a of regrouping exact an is and environment the
 metric the from free coming transitivity with – two of comparison a than
 they and too, functionalities be to out turn properties Game-based itself.
 a and clock global a functionality, signature A emulation, along transfer
 apply helps this hope We work, at style the show network bounded-delay
 proof. UC formal a give ultimately and UC, to methods formal

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Introduction

a in written, is it everywhere almost written, is composability Universal who code, are interfaces functionality's A code, and prose of mixture a what paragraph, a is model corruption The sentence, a is them call may is gap Each reader, the to left is returns actually interface party's corrupt cannot, machine A noticing, without it closes referee human a and small, formally: check to still harder and falsify to hard is proof UC a why is which down, written never was checked be to have would what of much access and Identities code, as down parts those writes paper This Wrappers fields, named in carries functionality a that data become control decides silencer a core, a reach calls which decides guard a them: enforce the to call party's corrupt a hands mediator a and claim, may core a what at twice, reads, interface every that register a is Corruption adversary, single a passes that handler a is execution An names, code the points its of several with ,[7] Canetti's is it — model new a is here Nothing token, is prose than rather code in them stating but — explicit made conventions argued, than rather proved be paper the of rest the lets what

with classes, machine explicit over defined then is Emulation **lemma. One** is that bound a asymptotics, no parameter, security no bounds: concrete is it of shape the and toolkit, composition a is follows What number, a and composition parallel Substitution, claim, structural main paper's the single a of corollaries all are adversary dummy the of completeness the envi- the into adversary an or protocol a moves which lemma, absorption exact an but executions two of comparison a not is Absorption ronment, advantage the so differently, bracketed run, same the one: of regrouping shared a with numbers two than rather number same the is side either on on resting kind, different a of is and exception the is Transitivity bound, it regrouping, any on than rather metric the over inequality triangle the other the machinery the before reason, that for metric the with stated is

need. three

Bellare of style game-based the in property A **functionalities. are Games** out turns — authentication liveness, unforgeability, — [5] Rogaway and as challenger game's the with too. functionality a as expressible be to Properties interface. ordinary an as verdict its and machine ordinary an ideal an emulates protocol a if emulation: along transfer way this written paper the hypotheses under it. to descend properties object's the object. assuming. than rather use each at checks and states

sug- count theorem the than larger is apparatus The **first. read to What** the of most carry definitions Five load-bearing. is it of all not and gests. means it what and execution an fields: five its and functionality a paper: oc- one as machines several runs which bundle. a identity: an occupy to re- the and itself: absorption from: built is absorption what is and cupant machine. by machine executions two pairs that correspondence grouping follow results composition the and spine. the has those has who reader A a scheme. signature a — functionalities concrete The order. in them from Tackmann Maurer. Katz. after network bounded-delay a and clock global an and .[3] Zikas and Tschudi Maurer. Badertscher. and [11] Zikas and show to exist and last come — them over realized channel authenticated them. before anything by needed be to than rather work at style the

the here conventions the find should researcher UC A **for. is this What** the one the is audience second The down. written use. already they ones for type-checker a — UC over tools building Anyone for. is precision extra a needs — harness test a encoding. proof-assistant a code. functionality leave can presentation informal an questions the settles that specification which does: paper this what of most is them settling and reader. its to corrup- a when claim, may core silenced a what admits, guard a caller machine-checked a carry to enough is that Whether read. is register tion the why is It it. claim not do we and here. established not is proof UC way. this written is framework

Systems and Functionalities

several [7] Canetti of framework composability universal the in work We below. explicit make we conventions whose of parts: three has it and instance, protocol running a names id process A

$$\text{pid} = (F, s, i),$$

The .i number instance an and .s id session a ,F name functionality a in- the protocol a of run which session the runs, code what says name one of instances several distinguishes number the and to, belongs stance of parts three the for id.i and id.s ,id.F write We session. a within name id.pid = $\mathcal{F}$.pid bare a $\mathcal{F}$.pid of those for $\mathcal{F}$.i , $\mathcal{F}$.s , $\mathcal{F}$.F likewise and ,id.pid agree. three all means

$\{0, 1\}^* \cup$ of ids party and $\{0, 1\}^* \cup \{A, Z, C\}$ of elements are Names environment the adversary. the for reserved are C and Z ,A where ,{A, Z} el- an is three the of none reserved. Being register. corruption the and of none contains $\{0, 1\}^*$ contain to declared field a so , $\{0, 1\}^*$ of ement parties that machine a being register the party, no names C and them: in- single a as runs three the of Each runs. party any one than rather call their so ,0 instance and 0 session be to fixed session. single a in stance the to these abbreviate We .(C, 0, 0) and (Z, 0, 0) ,(A, 0, 0) are ids process carrying components two other the throughout. C and Z ,A names bare information. no

id.F = A write we Where alone. name the on always almost are Tests for same the and triple, whole the never component. name the mean we test. membership a in named functionality any for and ,C for ,Z collect they and kind, this of names collect below fields the of Two $\{0, 1\}^* \cup \{A, Z\}$ of elements ids, party holds set party A things. different question a is call may who because ids. process whole holds set caller a

written therefore is set caller A code. about only not and instances about abbreviations the with

$$\mathbf{A} := \{\text{pid} : \text{pid}.F = A\}, \quad \mathbf{Z} := \{\text{pid} : \text{pid}.F = Z\},$$

$$\mathbf{Std} := \{\text{pid} : \text{pid}.F \in \{0, 1\}^*\}$$

everything of and environment. the of adversary. the of ids process the for now it "Z and A" admitted have would notation older an Where standard. execution the occur. ever (Z, 0, 0) and (A, 0, 0) only which of. $\mathbf{A} \cup \mathbf{Z}$ admits and $\mathbf{Std}$ admits now it " $\{0, 1\}^*$ " admitted it where each: of one supplying keep sets Party occur. does genuinely instance and session every there declaration a so ids. process not and ids party being those $\{A, Z\}$ bare the split. be to has $\mathbf{P} := \mathbf{N}$ read once that

written component name its by way. same the named is functionality A and session its $\mathcal{F}_{\text{Sig}}.F = \text{Sig}$ with functionality a is $\mathcal{F}_{\text{Sig}}$ subscript: a as (Sig, s, i) id process a for stands notation the so implicit. left instance are they meant is instance particular a Where unspecified. i and s with at written not are they matters. code the only where separately: supplied all.

code own functionality's a because them. suppressing by lost is Little it party the for written is core a global. or Local them. reads ever hardly in- which and to. belongs it session which keeps: it state the and serves own. its than rather business execution's the are is. it name its of stance wrappers the **Guard** of 3 line – only places three in occurs s paper this In read never is i and – 1.3 Definition of check static the and it. invoke that lets what is it way: another place its earns number instance The all. at ids process distinct the carry session one in name one of instances two state- a into observation the makes 7 Chapter for. asks 1.2 Definition that executions. of isomorphism an is instances and sessions renaming ment: a under other. every at holds id process one at proved emulation an and func- every that sessions across called be may who on condition hygiene meets. here tionality

id = pair a is identity An .functionalities) and (Identities 1.1 تعريف
set any is set identity an and id. party a and id process a of (pid, P)
of pairs of set a longer no triple. a being id process a – these of J
fields five the carries $\mathcal{F}$ functionality A names.

id process a	$\mathcal{F}.\text{pid}$
parties served of set	$\mathcal{F}.\mathbf{P}$
admits it ids process caller	$\mathcal{F}.\mathbf{N}$
uses it functionalities	$\mathcal{F}.\mathbf{U}$
parameters further any	$\mathcal{F}.\text{par}$

set identity the occupies and

$$\text{IDs}(\mathcal{F}) := \{(\mathcal{F}.\text{pid}, P) : P \in \mathcal{F}.\mathbf{P}\}.$$

function- of collection a identities which things: two say sets Identity in recur fields five The claim. may machine a which and occupies, alities name owner's the without written below, box every of line parameter the process the :IDs($\mathcal{F}$) fix two first the Only it. supplies heading box the since second the supply parties served the and identities its all by shared is id component.

:($\mathcal{F}'$, R) pair a is $\mathcal{F}.\mathbf{U}$ of entry An dependence. records field fourth The a is R and run. to $\mathcal{F}$ for present be must that functionality a names $\mathcal{F}'$ a Naming there. found is whatever of requires $\mathcal{F}$ what saying predicate of none served that subroutine a — own its on enough rarely is subroutine so — useless be would callers. its of none admitted or parties. caller's its write we

$$\text{serves}(\mathcal{F}, \mathcal{G}) : \Longleftrightarrow \mathcal{G}.\mathbf{P} \supseteq \mathcal{F}.\mathbf{P} \wedge \mathcal{F}.\text{pid} \in \mathcal{G}.\mathbf{N}$$

serves. $\mathcal{F}$ party every serves $\mathcal{G}$ most: up comes that requirement the for would Guard conjunct second the Without callers. its among $\mathcal{F}$ admits and $\mathcal{G}$ to $\mathcal{F}$ from call every refuse

wrapper its calls The names. $\mathcal{F}$ of code the what is records $\mathbf{U}$ What through $\mathcal{A}$ to and .3.5 Section of 6 and 2 lines on $\mathcal{C}$ to — behalf its on makes out. left are and functionality standard every to common are — Mediate so .Std admit and $\{0, 1\}^* \cup \{A, Z\}$ serve targets both nothing: costs which own its whatever $\mathcal{F}$ standard every of hold serves($\mathcal{F}, \mathcal{A}$) and serves($\mathcal{F}, \mathcal{C}$) fail. never could either for entry an and fields.

$\mathcal{F}.\mathbf{F} \notin \{A, Z, C\}$ if standard is functionality A .(Systems) 1.2 تعريف
which .Leak save — 3.5 Section in as wrapped is cores its of each and and procedure a is which .Initialize and .3.2 Section in as wrapped is standard of set finite a is π (world (or system A all. at wrapped not is $\mathcal{F} \neq \mathcal{F}'$ and $\mathcal{F}, \mathcal{F}' \in \pi$ if ids: process distinct carrying functionalities

do. members its whatever occupies It $\mathcal{F}.pid \neq \mathcal{F}'.pid$ then

$$IDs(\pi) := \bigcup_{\mathcal{F} \in \pi} IDs(\mathcal{F}) = \{(\mathcal{F}.pid, P) : \mathcal{F} \in \pi, P \in \mathcal{F}.P\}.$$

protocol use we – down writes designer protocol a what is system A one writing of styles naming real and ideal interchangeably. system and envi- the it: of part no are machines Three classes. disjoint not (below).
 .3.1 Section of $\mathcal{C}$ register corruption the and $\mathcal{A}$ adversary the $\mathcal{Z}$ ronment them of any contains system no 1.2 Definition by so standard. is None of execution the system: every outside lie $IDs(\mathcal{C})$ and $IDs(\mathcal{A})$. $IDs(\mathcal{Z})$ and each lets what is that $\mathcal{A}$ and $\mathcal{Z}$ For instead. three all supplies 3.6 Section it $\mathcal{C}$ For test. under system the on silenced while identity own its claim :6 Chapter and 4.1 Section of operations set the of clear register the keeps comparison every and it. moves replacement no it. for built is blocker no also names three the Reserving register. same the and one over runs admits functionality no so .Std as declared $\mathcal{F}.N$ every of out them keeps callee a being register the right. is which – callers its among register the only.

determined is system A work. real does ids process of Distinctness id the by settled is membership so uses. it id each at places it what by distinct so component. first its in id that records identity every And alone. no and disjoint. is above union the : $IDs(\mathcal{F}) \cap IDs(\mathcal{F}') = \emptyset$ have members twice. occupied is system a of identity

sys- a for $\pi^+ := \pi \cup \{\mathcal{Z}, \mathcal{A}, \mathcal{C}\}$ Write .systems) (Well-formed 1.3 تعريف
 The supplies. execution every machines three the with together tem
 if .wf(π) written well-formed. is π system

$$\begin{aligned} \forall \mathcal{F} \in \pi^+ \quad \forall (\mathcal{F}', R) \in \mathcal{F}.U \quad \exists \mathcal{G} \in \pi^+ : \\ \mathcal{G}.F = \mathcal{F}'.F \quad \wedge \quad (\mathcal{G}.s = \mathcal{F}.s \quad \vee \quad \text{global}(\mathcal{G})) \\ \wedge \quad R(\mathcal{F}, \mathcal{G}). \end{aligned}$$

Guard those exactly are $\mathcal{G}$ on conditions The this. about things Three un- matching session admitted. name – placed is call the when imposes the that precisely says wf(π) so – fitting fields shared. is callee the less by is Matching succeed. can make to built are π of members the calls sur- dependence a lets what is which itself. functionality the by not name.

sits whatever by met is φ_q naming entry an replacement: subsystem vive
 require- the And rest. the meets it as long so real. or ideal name. that at
 almost say would wf together: callee and caller of fields the ties half ment
 it. without nothing

place: its earns enlargement each and π^+ over range quantifiers Both
 machines supplied three the of one naming entry an makes it right the on
 left the on and system. any of member a being none all. at satisfiable
 and serves($\mathcal{Z}, \mathcal{A}$) since here. immediately – entries own their checks it
 is own its on do not does wf What empty. is $\mathcal{C}.\mathbf{U}$ and hold serves($\mathcal{A}, \mathcal{Z}$)
 introduce could members adding follow: that constructions the survive
 requirement every re-evaluates subsystem a replacing and obligations.
 and first the settles (6 (Chapter 6.4 Proposition occupant. new the at
 second. the 4.4 Theorem

condi- static several of one is Well-formedness **collected. Conditions.**
 de- are they and machines. and cores systems. on puts paper this tions
 see to way no reader a leaves which – needed first is each where fined
 is them of None once. them collects table The which. reads result which
 Proposi- as conditions. not fields. reads **Exec** execution: an during read
 particular. in $\mathbf{U}$ for records 7.7 tion

by read	constrains	condition
5.1 Def. ,6.4 Prop. ,4.4 Thm.	entry U every system: a the – witness a has are members its calls can make to built succeed	1.3 Def. ,wf(π)
,4.16 Prop. ,4.4 Thm. ,4.3 Prop. 5.10 Prop. ,4.24 Thm. ,4.20 Thm.	system incoming an no stays: what against common in id process shared a systems: two and serves id process alike admits	well-formed replacement. 4.2 Def. compatible. 6.1 Def.
claim second ,6.3 Thm.	or system a can who environment: does member a reach its on depend not session	session-uniform. 7.2 Def.
,7.12 Cor. ,7.10 Prop. ,7.9 Lem. 7.13 Cor.	members system: a by callers pin I outside id process	,I at tight 6.5 Rem.
(ceding 4.18 Lem. ,4.29 Thm. 5.11 Cor. ,5.9 Prop. half),	system a of cores the	no places call, responsive 9.2 Def.
four same the	the in machine a slot adversary	refusal-oblivious, 4.28 Def.
7.15 Prop. ,7.9 Lem.	standing a code: all than rather convention check per-system a	identity-atomic, 7.4 Def.
used not recorded, nothing:	realization a	leakage-well-formed, 3.2 Section

name a share rows two first The visible. makes table the things Two well-formed own. its on system a about is wf conditions: different are and session- and wf And lands. system incoming an where about replacement who against make, to built is member a calls the – duals are uniformity are They condition. one into them folding invites which – it reach may could definition one because and both, needs result no because apart kept well- be to required is 5.1 Definition of game the anyway: both carry not explains. 5.6 Remark as session-uniform, not deliberately is and formed

par- designated a for $\mathcal{F}.\mathbf{N} = \{\text{pid} : \text{pid}.F = F^\dagger\}$ if local is $\mathcal{F}$ say We **A** whether say to functionality the to it leave (We $.F^\dagger$ name (caller) ent reachable setup a possible: are notions Intermediate it.) call may **Z** and may **N** an and $\{\text{pid} : \text{pid}.F \in \{F_1, F_2\}\}$ admits protocols named two from in- occupied callers' its listing names. than rather ids process whole pin realiza- a of members private the why says 6.5 Remark — outright stances general- of sense the in global. is $\mathcal{F}$ say we end other the At should. tion if [8] setup shared UC's ized

$$\text{global}(\mathcal{F}) : \iff \mathbf{Std} \subseteq \mathcal{F}.\mathbf{N} \vee \mathcal{F}.F \in \{A, Z, C\},$$

stan- any case. ordinary the is disjunct first the two: not and notion one machines three the covers second the and it. call may functionality dard for global — slots their in stands whatever and — supplies execution the tested. is global wherever so treated and reason different a function- local A for. is component session the what is distinction The from only it reach may call a so protocol. one of run one of part is ality number instance The $.id'.s = id.s$ asks Guard of 3 line session: own its name a of instances many as hold may session one so checked. not is shared is functionality global A another. one call may they and likes it as that so — session every from reachable it. of session one — runs across in- shared one call all may $\mathcal{F}$ of sessions the and it. for waived is check global a and G leading a with functionality ideal global a name We stance. has it what stays $\mathcal{G}$ calligraphic the $\mathcal{F}_{\text{GSig}}$ in as γ leading a with one real execu- the machines three The $\mathcal{F}$ alongside functionality second a been. the by $\mathcal{Z}$ and clause. first the by already $\mathcal{C}$ and $\mathcal{A}$ global: are supplies tion and narrow deliberately is $\mathcal{Z}.\mathbf{N} = \mathbf{A} \cup \mathbf{Z}$ because there is which second. session. every from reachable be must environment the yet

dif- protocols real and functionalities Ideal **specifications. ideal vs. Real** corruption the know functionalities ideal First. respects. several in fer execution the pass can they Second. sets. party their in parties of status e.g., (see. interfaces simulator-specific offer may and simulator the to the is sharing and — parties across state share may they Finally. $(\mathcal{F}_{\text{Diffuse}}$ com- parties so party. one to call every ties 2 Chapter of guard the point: a and them. across state share that functionalities through only municate other no in and subroutines ideal through peers its to talks protocol real give: not do we functionalities name $\mathcal{F}_{\text{IO}}$ and $\mathcal{F}_{\text{Diffuse}}$ below. and Here way. de- paper this in Nothing relay. input–output an and channel diffusion a $\mathcal{F}_{\text{Store}}$ store the and $\mathcal{F}_{\text{Rand}}$ source randomness The code. their on pends

the in examples smallest the are they ;14 and 13 Chapters in given. are start. should style the to new reader a where are and paper these of none have protocols real well-specified of interfaces core The access- and blocks building idealized in implicit those beyond features. com- is leakage how specify not do realizations real and wrappers; control ¹tapes. work of contents the being it corruption; upon puted

once; runs it interface: an not and procedure a is Initialize **Initialization.** rest the variables the up sets and activated; first is functionality the when initial- 3.6 Section of execution the — calls no places It uses. code the of be would there so installed; is dispatcher its before machine every izes caller- claimed no takes it it; reach can caller no — one serve to nothing protocols Real one. empty the gets none declaring functionality a and id. a within party a to specific ones only kind; this of initializer global no carry private. are they notation: abusing and (pid, P).Initialize() written process. address. may anyone interfaces not

se- offering functionality ideal an as storage model would approach disciplined more A¹
notation. ease to layer extra the avoid we erasures; cure

Mediators and Silencers, Guards, and Enforcers

con- **Guard** predicate The machinery. of pieces three fixes chapter This
the constrains **Silence** wrapper the machine: a reach that calls the strains
that call a adversary the to hands **Mediate** wrapper the and makes: it calls
3.5 Section of interface full The itself. for answer cannot party corrupt a
composition. their is

cores: the only specify We $\mathcal{F}$ of interface core generic a for Op Write
reaches caller a and interface. full a into each turns below wrapper the
predi- the in collected are checks control access The .Op never .**FullOp**
tar- the takes it algorithm: an but functionality a not is which .**Guard** cate
and $\mathcal{F}$ functionality the and id' identity caller claimed the . id identity get
two in come Boxes presentation. of conventions Two boolean. a returns
and identities. occupies which functionality, a marks one filled a weights:
none. occupies which procedure. or wrapper predicate, a one unfilled an
in claimed identity the is id' where $id' \text{ as } id.Op(in)$ written is call a And
it identity the is id' where $id' \text{ from } id.Op(in)$ as received and it. placing
srequire 1 line where wrapper, the in sits Enforcement under. claimed was
inter- the that checks **1** Line core. a reaches never rejects it call a so it.
for acts it that and admitted is name caller's the that .**2** line exists: face
the unless session, one in lie callee and caller that .**3** line party: right the
sessions. across shared is callee

Guard Predicate

```

:1 return  $id.pid = \mathcal{F}.pid \wedge id.P \in \mathcal{F}.P$  / interfaces that exist
:2       $\wedge id'.pid \in \mathcal{F}.N \wedge id'.P = id.P$  / callers admitted
:3       $\wedge (id'.s = id.s \vee global(\mathcal{F}))$  / one session, unless shared

```

Silence wrapper the machine: a reach that calls the constrains **Guard** pro- a and $\mathcal{I}$ set identity an takes It makes. it calls the constrains below an claiming call A issues. it call every intercepts and it. runs gramme. is other any placed: being ever without **rej** with refused is $\mathcal{I}$ in identity wrapped the where exactly is $\mathcal{I}$ So back. handed answer its and placed outside. anything as speak may it and silenced. is programme it call A further. nothing and programme wrapped the is scope The says machine that what and machine. another to token the passes places func- a otherwise it Were one. this by not wrapper. own its by governed is speak could none and transitively. subroutines its silence would tionality wrap- interface full a shape: this of is below **Silence** of use Every itself. as serves. it core the ping algebraic of sense the in handler. a as this read to convenient is It ambi- the of operation an is issues programme the call A .[4,12] effects wrapper The answer. its for waiting continuation the is k and effect. ent answer the claim. silenced a for **rej** with: resume to what decides **Silence** pro- the passes clause **return** The otherwise. placed actually call the of the what changes silencing so untouched. through result own gramme's shape the fixes only analogy The else. nothing and say may programme wrapper. the of

Silence Wrapper	
id' from id.Op(in) programme $\mathcal{I}$ set identity	
:Silence($\mathcal{I}$, id.Op(in) from id')	
:1 handle id.Op(in) from id' with	
:2 return out	$\mapsto$ out
:3 id".FullOp(in") as id''; k	$\mapsto$ k(rej) if id'' $\in \mathcal{I}$
:4	$\mapsto$ k(id".FullOp(in") as id'') otherwise

corrupt a that call a to happens what say can we hand in **Silence** With a such hands below interface full Every itself. for answer cannot party both and out. way the on and in way the on twice. adversary the to call set corruption a takes It .**Mediate** wrapper the shape: same the in times the and over. hand to x value the served. being id.Op operation the .**C** comes answer the Where in. stands it operation the of id' caller claimed answers party's corrupt a operation. the whatever it: inside fixed is from both argument: ordinary an not is it about thing One .(A, id.P) from come code whose in operation the from return **Mediate** of **return** the and **require** it fails test 's**Mediate** When itself. line the from merely not stand. they on. carries operation that and nothing does

iden- the claiming interface, full 'sA on call ordinary an is hook The
 it which, Guard_A meets therefore It served. being interface the of id tity
 the and target: the of party the is id.P party claimed the because passes
 only admits which silencer, own 'sA by confined is adversary answering
 itself as speak may adversary the id.P for answering while So .(A, id.P)
 own. its of silencing no needs Mediate and else. nothing as and id.P at

Mediate wrapper Corruption			
id' caller claimed	x value	id.Op operation	C set corruption
:Mediate(C, id.Op, x, id')			
:1 if (id'.F ≠ A ∧ id.P ∈ C) then			
:2 return (A, id.P).FullA(id.Op, x) as id			

Model Execution The

Corruption 3.1

a would (what realize protocols real that interface an not is Corruption for exist must it but corrupted?), get to “instructed” when do protocol real but corruptions. reject always protocols real assume can We emulation. corrupted. as registered never are they then

machines Several .C functionality global a as corruptions model We – cores from obtained than rather outright given interfaces full their have

3.2 Section of operation leakage the and adversary, the environment, the sta- the and .Mediate nor Silence neither uses that one only the is C but – call, every on status corruption the reads wrapper that why: is report tus circular. be would it wrapping so implements, interface this what is which needs, it checks the only hand by carries therefore below interface Each first without set corrupted the returns always particular in FullStatus and corrupt. is party the whether asking

3.2 Section that test corruption the so constant, is leakage own 'sC protect: to nothing have would operation leakage other every on imposes the at register the reads adversary the so same. the all kept is guard the other. no at and for acting is it party

holds machine other no so state. own 'sC is C register. the Reading C than other functionality a of code the in appears C Wherever copy. a by first, fetched was it that is understanding the itself.

$C \leftarrow (C, id.P).FullStatus()$ as id,

surrounding the what repeat only would it when line that suppress we and written and read is C fetch: to nothing is there C Inside plain. makes code place. in it updates Corrupt and directly.

C register Corruption

$$\text{par} := \perp \quad .U := \emptyset \quad .N := \text{Std} \cup A \cup Z \quad .P := \{0, 1\}^* \cup \{A, Z\} \quad .pid := C$$

:Initialize()

:1 $C \leftarrow \emptyset$ id' from $id.FullCorrupt()$:2 **require** $\text{Guard}_e(id, id')$:3 **require** $id'.F \in \{A, Z\}$

/ the outside only, never a subroutine

:4 **require** $id.P \in \{0, 1\}^*$

/ reserved names stay honest

:5 $C \leftarrow C \cup \{id.P\}$:6 **return** ok id' from $id.FullStatus()$:7 **require** $\text{Guard}_e(id, id')$:8 **return** C id' from $id.FullLeak()$:9 **require** $\text{Guard}_e(id, id')$:10 **require** $id'.F = A$:11 **return** $\perp$

/ constant: the register keeps no secrets

oper- other the like Wrapped .mediated) not is **FullStatus** (Why 3.1 نکته)
 Suppose matters. it when exactly useless be would **FullStatus** ations.
 $id'.P$ If $id'.P$ of status corruption the ask to id' as out reaches Z
 the to it hands **Mediate** and through call the lets gate the corrupt. is
 ad- the whatever is "corrupts" $id'.P$ "is to answer the so – adversary
 register the reads caller every Since no. say may it and says. versary
 answer adversary-controlled an party. corrupt a treat to how decide to
 will. at off machinery mediation the switch would
 guard the past nothing: silences and mediates therefore **FullStatus**
 whose operation one the is It state. own register's the .C returns it
 the gives it read-only being and register. the from come must answer
 earlier of Status core The infer. already not could it nothing adversary
 of. out keep to left wrapper no being there it. with goes drafts

so $id'.F \in \{A, Z\}$ admits 3 Line do. to outside's the is Corruption
 no and C into party a put each may environment the and adversary the
 it above guard The all. at there anything put may functionality standard
 $id'.P = id.P$ asking corrupts. it party the to interface corrupting the binds
 must environment the and (A, P) at run must adversary the P corrupt to
 as read may Both party. any for do root its lets 3 line which (Z, P) claim
 at ask may either so .C of all returns **FullStatus** and $A \cup Z \subseteq C.N$ well:

be can parties genuine only And set. full the learns always and point any
 stay names reserved the so $\text{id}.P \in \{0, 1\}^*$ asks **FullCorrupt** corrupted:
 construction. by honest
 secures that reading the is it and right. out comes Commensurateness
 in $\mathcal{A}$ is (A, P) at machine The write. may who of division any than rather it
 nothing and other. the in $\mathcal{S}$ and experiment emulation an of execution one
 on machine same the is $\mathbb{Z}$ But parties. same the corrupt to two the forces
 a corrupting slot the of occupant an so both. on $\mathbf{C}$ reads and sides both
 Commensu- asks. simply that environment an by caught is set different
 envi- admissible no assumption: an not consequence. a thus is rateness
 what beyond advantage with apart sets corruption two the tells ronment
 for corrupt may environment the That simulator. the allows 4.11 Definition
 both on own its being makes it corruptions the this. sharpens only itself
 initiative. own occupant's the is catch to left is what sides:
 at corrupt be cannot P so alone. name party by indexed is Corruption
¹another. at honest and id process one

Leakage 3.2

operation leakage $\mathcal{F}$ Functionality	
	par .U .N .P .pid
	id' from <u>$\text{id}.\text{FullLeak}()$</u>
:1 require $\text{id}.pid = \mathcal{F}.pid \wedge \text{id}.P \in \mathcal{F}.P$	/ the interface exists
:2 require $\text{id}'.F = A \wedge \text{id}'.P = \text{id}.P$	/ $\mathcal{A}$. for this party
:3 $\mathbf{C} \leftarrow (C, \text{id}.P).\text{FullStatus}()$ as id	
:4 require $\text{id}.P \in \mathbf{C}$	/ only a corrupt party leaks
:5 return $\text{Silence}(\{\text{id}'' : \text{id}'' \neq \text{id}\}, \text{id}.\text{Leak}() \text{ from } \text{id}')$	

3.5 Section of wrapper general the those are requirements three The
 the in core a from obtained not is leakage because restated imposes.
 deliber- is half caller the **Guard** of half existence the is 1 Line way. usual
 does that functionality a from leak to able be must $\mathcal{A}$ since weaker. ately
 the for act adversary the that only asks 2 line so caller. a as it admit not
 honest an it Without gate. corruption the is 4 Line reading. is it party
 in- full the of 3 line what is which will. at readable be would state party's

$\text{id}.P \in \mathbf{C}$ test every identities. collecting $\mathbf{C}$ — identity an corrupt instead could One¹
 party-wide The subprotocols. to corruptions “spreading” of cost the at — $\text{id} \in \mathbf{C}$ becoming
 once. at family whole a infecting cleaner. is convention

core leakage a so 's.5 line is silencer The else. everywhere refuses terface
 core. other any like identity own its claims calls places that
 mes- the specification: the of part is **Leak** functionality, ideal an For
 For typical. are lists internal of contents the and far so received sages
 at leakage no models which nothing, leak to set be can it protocols real
 fix. the are leak state the of parts which say that functionalities ideal all:
 be- pure is body machine every if leakage-well-formed is protocol A
 all and $\mathcal{F}_{\text{Rand}}$ through drawn is randomness sampled all invocations, tween
 machine a on than rather $\mathcal{F}_{\text{Store}}$ in held invocations across persisting state
 outputs and inputs and — **Leak** own its through it exposing each — tape
 func- input-output an through them routes protocol the where only leak
 leaked. never then are tapes Work default. the not is which, $\mathcal{F}_{\text{IO}}$ tionality
 as- below result No them. on nothing leaving invocations between purity
Leak its for meet must realization a condition the is it this: of any sumes
 it. use than rather it record we and anything, mean to
 indistin- for realized be not need it so interface, this call cannot $\mathcal{Z}$
 it. call can adversary the since simulable, be still must It guishability.

Environment The 3.3

id.**FullZ** interface idper- single a with, $\mathcal{Z}$ functionality special a define We
 deserves $\mathcal{Z}.\mathbf{N} = \mathbf{A} \cup \mathbf{Z}$ field The .id.Z core the silences and guards that
 nothing and itself environment the and adversary the admits it emphasis:
 a let would wider Anything $\mathcal{Z}$ call may functionality standard no so else.
 “open an — subroutines own its of one as environment the call protocol
 rule things distinct Two undefined. left are subroutines whose protocol,”
 caller’s standard a itself: call the refuses $\mathcal{Z}.\mathbf{N}$ of narrowness The out. that
Guard of 2 line so nowhere, $\mathbf{A} \cup \mathbf{Z}$ meets which, **Std** in lies id process
 half other the refuses 1.3 Definition declares. caller the whatever it rejects
 is It supplies. member no dependence a naming member a openness. of
 $\mathcal{Z} \in \pi^+$ since .wf trip not would $\mathcal{Z}$ naming here: work the does that first the
 and well-formed be could system a so satisfiable. entry an such makes
 wider. any field the were open be still
 $\mathbf{Z}$ The itself. call $\mathcal{Z}$ lets it reason: different a for there is $\mathcal{Z}.\mathbf{N}$ in $\mathbf{Z}$ The
 admit- and separate. are parties different at environment the of interfaces
 another. one reach them lets what is $\mathbf{Z}$ ting
 and set. silenced the by settled is which reach may them of Which
 machine other Every systems. the off $\mathcal{Z}$ keep than more does it there
 own its only claim may functionality standard a runs: it where to tied is

its on $\mathcal{Z}$ tie would Nothing $\{id'' \neq id\}$ on silenced is $\mathcal{A}$ and identity, target, the only **Guard** and id' claimed the only tests **Silence** since own, of component party the and interchangeable be would (Z, P') and (Z, P) so loosely as tie, the supplies **3** Line information. no carry would identity $\mathcal{Z}$ a $id'.P = id.P$ when id' claim may id at interface an allow: applications as or party, own your for act words. In $id.P = Z$ when or $id'.P = Z$ when anyone. for act may root the and root: the for and stranded, is root the $id.P = Z$ Without needed. are escapes Both others. the reach can that interface only the and begins **Exec** where is it into back route only adversary's the are: leaves the $id'.P = Z$ Without $(A, id.P)$ at $\mathcal{A}$ wakes **Mediate** and $(A, P) \rightarrow (Z, P)$ is environment the what compare to needing environment an so for. answers it party the for the bring never could P' at said it what against P at said adversary the through it pool cannot and separate, are interfaces these — together two genuinely would **4.11** Definition of class the and — either state shared (Z, Z) at rooted star two-way a is connectivity escapes, both With shrink, star the that do could environment unrestricted an nothing of know we and and (A, P) claiming $\mathcal{Z}$ stops it spirit: same the in restricts **2** Line cannot. honest an driving and **3** line of gate corruption the passing it stops so impersonating be would it adversary the capability a — interface party's on silenced simulator a of records **4.8** Remark as either, have not does **.FullCorrupt** from away $\mathcal{Z}$ keep is do to has longer no line that What less. name. own its under to it admits **3** line which is **A** Admitting more. no and need applications what is **A $\cup$ Z** Together, execu- the into back way other no has environment the because necessary calls something when only again runs it on token the passes $\mathcal{Z}$ once tion: what- hears $\mathcal{Z}$ how is channel same that may: adversary the only and it, because necessary is **Z** Admitting it. tell to chooses adversary the ever and machine, one than rather party per interface one is environment the is: two these beyond Nothing them. of none reach could root the it without not does it subroutine a as it treat would $\mathcal{Z}$ call to had that functionality a what knows adversary the — one no to corruptions reports $\mathcal{C}$ and define. corruption the for ask must $\mathcal{Z}$ why is which — it done having corrupts, it outside machines two the from only reachable is environment the So set. system. every

$\mathcal{Z}$ Environment	
$\text{par} := \mathcal{T}$	$\mathcal{U} := \{(\mathcal{A}, \text{serves})\}$
$\mathcal{N} := \mathcal{A} \cup \mathcal{Z}$	$\mathcal{P} := \{0, 1\}^* \cup \{\mathcal{A}, \mathcal{Z}\}$
$\text{id}' \text{ from } \underline{\text{id.FullZ(in)}}$	
:1 require $\text{Guard}_{\mathcal{Z}}(\text{id}, \text{id}')$	
:2 $\bar{\mathcal{T}} \leftarrow \mathcal{T} \cup \{\text{id}'' : \text{id}'' . \mathcal{F} = \mathcal{A}\}$	/ $\mathcal{A}$ out of reach
:3 $\cup \{\text{id}'' : \text{id} . \mathcal{P} \neq \mathcal{Z} \wedge \text{id}'' . \mathcal{P} \notin \{\text{id} . \mathcal{P}, \mathcal{Z}\}\}$	/ own party, or the root
:4 $\text{out} \leftarrow \text{Silence}(\bar{\mathcal{T}}, \text{id} . \mathcal{Z}(\text{in}) \text{ from } \text{id}')$	
:5 return out	

Adversary The 3.4

id.FullA interface id.per - single a shape: same the has $\mathcal{A}$ adversary The
 envi- the from it separates What id.A core the silencing and guarding
 $\mathcal{A.N} = \text{field the } \mathcal{A} \cup \mathcal{Z}$ only is $\mathcal{Z.N}$ Where it. reach may who is ronment
 is 'sc Only call. a places ever that id process every holds $\text{Std} \cup \mathcal{A} \cup \mathcal{Z}$
 a than rather requirement a is width That none. places $\mathcal{C}$ and missing.
 func- ideal calls the and 2 Chapter of hooks mediation the convenience:
 on placed calls both are $\mathcal{A}(\cdot)$ abbreviation the through place tionalities
 party. some for acting machine a by adversary the
 The notation. this in like looks backdoor a what are interfaces These
 every so call. a begin cannot and own its of token no holds adversary
 party's a control: it handing machine another from starts mounts it attack
 an how is which directly. environment the or Mediate through interface
 is return in say may it What all. at adversary an instructs environment
 may $\mathcal{A}$ of interface an so $\{\text{id}'' : \text{id}'' \neq \text{id}\}$ is set silenced the narrow:
 at. runs it identity the only claim
 call the for $\mathcal{A}(\text{in})$ write we id identity with interface an Inside
 id be must claim The likewise. $\mathcal{Z}(\text{in})$ and id as $(\mathcal{A}, \text{id.P}).\text{FullA}(\text{in})$
 own its but everything on core a silences 5 line since other. no and
 and id. claimed the as caller the learns therefore adversary the identity:
 id' from $\text{id.A}(\text{in})$ as entered is core Its it. for field separate no needs
 in. entered is core other every shape the and declares Silence shape the
 whose on records also call the so id.P is component party target's The
 made. is it behalf

Interface Full The 3.5

and Mediate and Guard place: in now is needs wrapper the Everything
 $\mathcal{C}$ supplies that 3.1 Section of $\mathcal{C}$ register the and 2 Chapter from Silence
 composition. their is functionality standard a of interface full The

$\mathcal{A}$ Adversary				
$\text{par} := \perp$	$\mathcal{U} := \{\mathcal{Z}, \text{ serves}\}$	$\mathcal{N} := \text{Std} \cup \mathcal{A} \cup \mathcal{Z}$	$\mathcal{P} := \{0, 1\}^* \cup \{\mathcal{A}, \mathcal{Z}\}$	$\text{pid} := \mathcal{A}$
id' from <u>$\text{id}.\text{FullA}(\text{in})$</u>				
:1 require $\text{Guard}_{\mathcal{A}}(\text{id}, \text{id}')$				
:2 $\text{out} \leftarrow \text{Silence}(\{\text{id}'' : \text{id}'' \neq \text{id}\}, \text{id}.\mathcal{A}(\text{in}) \text{ from } \text{id}')$				
:3 return out				

$\mathcal{F}$ Functionality				
par	$\mathcal{U}$	$\mathcal{N}$	$\mathcal{P}$	pid
id' from <u>$\text{id}.\text{FullOp}(\text{in})$</u>				
:1 require $\text{Guard}_{\mathcal{F}}(\text{id}, \text{id}')$				
:2 $\mathbf{C} \leftarrow (\mathbf{C}, \text{id}.\mathcal{P}).\text{FullStatus}()$ as id				
:3 require $\neg(\text{id}'.\mathcal{F} = \mathcal{A} \wedge \text{id}.\mathcal{P} \notin \mathcal{C})$ // $\mathcal{A}$ only at a corrupt party				
:4 $\text{Mediate}(\mathbf{C}, \text{id}.\text{Op}, \text{in}, \text{id}')$				
:5 $\text{out} \leftarrow \text{Silence}(\{\text{id}'' : \text{id}'' \neq \text{id}\}, \text{id}.\text{Op}(\text{in}) \text{ from } \text{id}')$				
:6 $\mathbf{C} \leftarrow (\mathbf{C}, \text{id}.\mathcal{P}).\text{FullStatus}()$ as id				
:7 $\text{Mediate}(\mathbf{C}, \text{id}.\text{Op}, \text{out}, \text{id}')$				
:8 return out				

full a place only the are and corruption. handle 7–6 and 4–2 Lines the as changes $\mathbf{C}$ because twice read is It register. the consults interface 6 line before tests the snapshot: a only is read a and proceeds execution monotone. is Membership second. the 7 line on test the first. the use corrupt be may read first the at honest party a so adding. ever only **Corrupt** reverse. the never but second the at party. honest an at adversary the combination: one refuses 3 Line settles **Mediate** and passes. else Everything there. business no has which its runs and $\mathcal{A}$ but everyone by addressable is party honest an So rest. the reaches $\mathcal{A}$ only but everyone. by addressable is party corrupt a core: own The place. party's the in $\mathcal{A}$ by answered being caller other every core. its identity own its claims it whether own: its of case no needs environment either $\mathcal{A}$ through it answers party corrupt a one. external admitted an or way.

call whose $\mathcal{A}$ but anyone by addressed party corrupt a leaves That 7 and 4 Lines place. party's the in answers $\mathcal{A}$ so run. be simply cannot already party a catches first The moments. two at wrapper same the are second The runs. never core the so id over hands and arrival on corrupt over hands and .5 line on ran core its while corrupt turned that one catches the cover two the makes Monotonicity return. to party's the longer no out and returned. already has first the at corrupt party a once: exactly cases

address hooks Both second. the at honest and first the at corrupt is none is adversary answering the so interface. full its through $(A, id.P)$ at A own core's the returns 8 Line callee. other any like silenced and guarded for and throughout. honest party a for cases: two in happens which value. which and admits gate the which itself. A by addressed party corrupt a $.id'.F \neq A$ testing both catches. hook neither

ا interrupts Nothing .preemptive) not is mid-call (Corruption 3.2 نکته
adver- the calls 5 line on core the Suppose running. already is that core
the and — mediates that subroutine a through or $A(\cdot)$ through — sary
records register The value. a returning before $id.P$ corrupts adversary
nor told neither is core the but state. own its being C once. at that
to runs and returned adversary the whatever with resumes it stopped:
it. in $id.P$ find again. C read 6 line does return its on Only completion.
replaced. be to 7 line on Mediate to out hand and
or- this in call. a such over twice token the holds adversary the So
after once and placed. core the call the for core. the inside once der:
in moment No output. its of mediation the for returned. has core the
cor- turns that party a since effect. take corruption the lets between
call the is Nor done. already has it what un-run cannot mid-call rupt
admitted call a and read. first the on decided was 3 line re-gated:
guarantees wrapper the What admitted. stays honest was $id.P$ while
cor- newly a nothing sufficient: and aborting. than weaker therefore is
after arriving Calls unmediated. caller its reaches computed party rupt
.4 line by instead. in way the on caught are corruption the

Executions 3.6

call A token. activation the holds machine one exactly execution an In
continuation a with caller the suspending abbreviates $out \leftarrow id.Op(in)$
be- call Every $.id$ identity at $\mathcal{F}$ of Op interface to token the passing and
dis- Exec what is FullOp and interface. full a addresses machines tween
ma- the inside it. serves that wrapper the by only invoked is core a patches:
addressed call A transfer. token a across never and it. carries that chine
may instance An $.rej$ with answered is occupies machine no identity an to
sus- while re-entered be may and continuations outstanding several hold
 $.A$ by particular in pended.

تعريف 3.3 (Machines) and (occupancy) machine A is anything an execution the whatever and system, the of members the runs: cution identi- of set finite a occupies machine A them. alongside supplies dis- **Exec** and — 1.1 Definition of IDs($\mathcal{F}$) the functionality, a for — ties The occupies. it identities to addressed calls the exactly it to patches and sets. identity disjoint pairwise carry execution one of occupants slot adversary the of occupant An family. other no for defined is **Exec** others: no and A id process of identities the occupying machine a is Chap- of simulator every is so and one. is 3.4 Section of adversary the machine one which during interval maximal a is activation An .4 ter token. the holds

claim may functionalities Standard .id' id claimed a carries call Every oth- claim may $\mathcal{A}$ and $\mathcal{Z}$ call: the placing interface the of identity the only which guard. the are checks Those .2 Chapter of checks the to subject ers. functionality's the against and id identity interface's the against id' tests guard failed A statements. **require** with them specify we and parameters. rather refusal a is rej and sender. the to rej value distinguished the returns en- is rule The output. own its as it return may machine no value: a than and . rej return would code whose core a — assumed than rather forced $\perp$ as delivered are slot. adversary's the from rej of answer mediated a and call. its refused guard a that learns rej receiving caller a so — instead . $\text{rej} \neq \perp$ particular In one. produce can else nothing

tape. coin own its carries machine each machine: per is Randomness only read machine's. other every of independent and uniform. infinite. these over are execution an over Probabilities samples. code its when and 4 Chapter as — occupant one into machines Bundling jointly. tapes what are coins copy's hosted a tapes: merge not does — do 4.4 Section absorption. before were they

envi- the machines: three carries execution the . π system the Besides **C** state whose . $\mathcal{C}$ register corruption the and . $\mathcal{A}$ adversary the . $\mathcal{Z}$ ronment makes which . π of member a is none 1 Chapter By reads. wrapper every a not argument. an as once. initialized is register the — unambiguous 1 line The wrote. designer a object the π keeps and — member a as time second as (Z, Z) at environment the activates then three. the initializes procedure inter- full 's $\mathcal{Z}$ through goes activation first That itself. called had it though and outset. the from silenced is environment the so other. any like face same. the being identities target and claimed the trivially. passes **Guard** $\mathcal{Z}$

outcome. the is returns environment the Whatever

Exec Execution

$\mathcal{C}$ register corruption $\mathcal{A}$ adversary $\mathcal{Z}$ environment π system

$:\text{Exec}(\pi, \mathcal{Z}, \mathcal{A}, \mathcal{C})$

```

:1  $\mathcal{C}.\text{Initialize}(); \mathcal{F}.\text{Initialize}()$  for every  $\mathcal{F} \in \pi$ 
:2 handle  $(Z, Z).\text{FullZ}()$  from  $(Z, Z)$  with    /  $\mathcal{Z}$  is activated first and holds the token
:3 return out                                     $\mapsto$  out
:4  $\text{id}.\text{FullOp}(\text{in})$  from  $\text{id}'$ ;  $k \mapsto k(\text{id}.\text{FullOp}(\text{in})$  from  $\text{id}'$ )
    
```

operation: one exactly has execution the $[4, 12]$ handler a as Read
 not activation only the environment, the activates 2 Line call. a placing
 contin- the and thereafter, call every handles 4 Line machine. a by placed
 passing caller. the suspending back: comes token the where is k uation
 precisely are answer the with caller the resuming and ,id to token the
 the holds machine one Exactly means. answer that on k invoking what
 environment's the when reached is clause **return** The throughout. token
 execution's. the is output its and halts, program

Composition and Emulation UC

Replacement Subsystem 4.1

an- by supplied later subroutine a against written normally is protocol A systems. on operation set a is other the by one Replacing protocol. other

تعريف 4.1 (Subsystem replacement) Let ρ be a system and $\varphi \subseteq \rho$ a subsystem. For it, of subsystem π by φ of replacement the $\rho[\pi/\varphi]$ system is defined as

$$\rho[\pi/\varphi] := (\rho \setminus \varphi) \cup \pi.$$

the id process a on landing from π incoming the stops far so Nothing ambiguously call a leave would which uses, already protocol surrounding it. repair than rather out this rule We addressed.

تعريف 4.2 (Well-formed replacement) The $\rho[\pi/\varphi]$ system is well-formed if ρ is well-formed and π is well-formed and $\rho \setminus \varphi$ in both occurs id process no if well-formed is ρ if

$$\{\mathcal{F}.pid : \mathcal{F} \in \rho \setminus \varphi\} \cap \{\mathcal{F}.pid : \mathcal{F} \in \pi\} = \emptyset.$$

گزاره 4.3 (Replacement) Let ρ and π be systems, and $\varphi \subseteq \rho$ let a system. a is $\rho[\pi/\varphi]$ Then well-formed. be ρ in π by φ of replacement. $IDs(\rho[\pi/\varphi]) = IDs(\rho)$ then, $IDs(\pi) = IDs(\varphi)$ moreover If

اثبات. Those with together $\rho \setminus \varphi$ of those are $\rho[\pi/\varphi]$ of ids process The well-formedness and internally ids distinct carries system Each π of

are members the repeats; id no so disjoint, collections two the makes a is $\rho[\pi/\varphi]$ so, π and ρ from inherited both many, finitely and standard replacement the of well-formedness the on rests claim first That system, separate no need Identities it, in nowhere entering $IDs(\pi) = IDs(\varphi)$ alone, so component, first the in id process its carries identity an hypothesis: identities the For sets, identity disjoint have ids distinct with members themselves.

$$\begin{aligned} IDs(\rho[\pi/\varphi]) &= IDs(\rho \setminus \varphi) \cup IDs(\pi) \\ &= IDs(\rho \setminus \varphi) \cup IDs(\varphi) = IDs(\rho), \end{aligned}$$

□ last, the in $\varphi \subseteq \rho$ and step middle the in $IDs(\pi) = IDs(\varphi)$ using

That below, differently used are proposition the of halves two The well- on rests and all, at it of speak 4 Chapter lets system a is $\rho[\pi/\varphi]$ $IDs(\pi) =$ needs ρ of those are identities its That alone, formedness stated is there admissibility it: assumes 4 Chapter in result no and, $IDs(\varphi)$ Proposi- and sets, identity their of union the through systems of pair a for two the between relation no that outright say 4.20 Theorem and 4.16 tion it, supplies blocking wanted, is equality the Where used, is Nei- identities, 4.3 Proposition and ids, process settles 4.2 Definition can replacement where is dependence and dependence, addresses ther way gives φ after occupied still are on relies $\rho \setminus \varphi$ names the wrong; go Two occupant, new the at re-evaluated is requirement every but, π to gap, the close conditions replacement well-formed a that says below 4.4 Theorem Informally, the in discharged be can π incoming the of entries the when wf preserves of entry an wherever φ for in stands π of member some and world, new be to π for means it what is (ii) Hypothesis, φ inside witnessed was $\rho \setminus \varphi$ it, supply not does emulation and, φ for drop-in a

be π and ρ Let, well-formedness) preserves (Replacement 4.4 **قضيه** well-formed be ρ in π by φ of replacement the let, $\varphi \subseteq \rho$ let systems,

Suppose, wf(ρ) let and, 4.2 Definition of sense the in the in, $(\rho[\pi/\varphi])^+$ in witness a has $\mathcal{F} \in \pi$ every of entry every (i) and 1.3 Definition of sense

wit- some which for $(\mathcal{F}', R) \in \mathcal{F}.U$ entry an has $\mathcal{F} \in \rho \setminus \varphi$ whenever (ii) satisfies $\mathcal{G} \in \pi$ some, φ in lies 1.3 Definition of sense the in ness $.R(\mathcal{F}, \mathcal{G})$ and, $\mathcal{G}.s = \mathcal{F}.s \vee \text{global}(\mathcal{G}), \mathcal{G}.F = \mathcal{F}'.F$

.wf($\rho[\pi/\varphi]$) Then

of claim first the by system a, $\rho' := \rho[\pi/\varphi] = (\rho \setminus \varphi) \cup \pi$ Write اثبات. being $\text{IDs}(\pi) = \text{IDs}(\varphi)$ here, available not is second the – 4.3 Proposition $(\mathcal{F}', R) \in \mathcal{F}.$ entry an and $\mathcal{F} \in \rho'^+$ Take theorem. this of hypothesis no comes $\mathcal{F}$ where by cases, three are There ρ'^+ in witness a place must we from.

claim. the exactly is (i) hypothesis, $\mathcal{F} \in \pi$ If indepen- all, for and once settled are entries its, $\mathcal{C}, \mathcal{A}, \mathcal{Z}$ of one is $\mathcal{F}$ If each name $\mathcal{A}$ and $\mathcal{Z}$ of entries the and empty, is $\mathcal{C}.$ system: the of dently 1 Chapter as them between holding serves and $\mathcal{A}, \mathcal{Z} \in \rho'^+$ with other, waived. is conjunct session the so global, are three All records. this for $\mathcal{G} \in \rho^+$ witness a supplies wf(ρ) so, $\mathcal{F} \in \rho^+$ then, $\mathcal{F} \in \rho \setminus \varphi$ If and $(\rho \setminus \varphi) \cup \{\mathcal{Z}, \mathcal{A}, \mathcal{C}\} \subseteq \rho'^+$ in lies $\mathcal{G}$ case which in, $\mathcal{G} \notin \varphi$ Either entry, and $\mathcal{G}.s, \mathcal{G}.F$ so fields, same the with machine same the is it again: serves and $\mathcal{G} \in \varphi$ Or too, unchanged is $\mathcal{F}$ and unchanged, are reads R whatever conditions three same the meeting π of member a supplies (ii) hypothesis $\square$ ρ'^+ in witness a has entry the way Either place, its in

emula- and φ for drop-in-a be to π for means it what is (ii) Hypothesis bound- the at behaviour compares 4.11 Definition it, supply not does tion that behind machines the of fields declared the about nothing says it ary: caller a admit to fail still may perfectly φ emulates that π a and boundary, ranges it – cheap is check The served, φ party a serve or admitted, φ is it but – else nothing and φ into pointed that $\rho \setminus \varphi$ of entries the over rather replacement the of well-formedness beside belongs and check, a theorem. composition the inside than

Absorption and Emulation 4.2

sys- one when say can we so fixed, now is needs execution an Everything and one under placed are two The another, as secure as least at is tem is that first: settled be must claim may it what so environment, same the below, $\mathcal{Z}$.par of role the

We returns, environment the whatever is $\text{Exec}(\pi, \mathcal{Z}, \mathcal{A}, \mathcal{C})$ of value The which to as guess its – root the at bit a return to environment an take exactly 1 returning as execution the read and – in placed was it system constrains framework the in else Nothing does, environment the when not over, quantified environments the on condition a is this so value, that

not number, a then is executions two between gap The consequence. a
resemblance. informal an

environ- an For systems. be φ and π Let .advantage) (UC 4.5 تعريف
. $\mathcal{S}$ simulator a and $\mathcal{A}$ adversary an , $\mathcal{Z}$ ment

$$\text{Adv}_{\pi, \varphi}^{\text{UC}}(\mathcal{Z}, \mathcal{A}, \mathcal{S}) := \left| \Pr[\text{Exec}(\pi, \mathcal{Z}, \mathcal{A}, \mathcal{C}) = 1] - \Pr[\text{Exec}(\varphi, \mathcal{Z}, \mathcal{S}, \mathcal{C}) = 1] \right|,$$

exe- two the in machine every of coins the over being probabilities the
halts execution the that event the abbreviating $\text{Exec} = 1$ and cutions.
.1 returning environment the with

oc- that machine the in and system the in differ executions two The
are Both other. the on $\mathcal{S}$ side, one on $\mathcal{A}$ identities: adversary's the cupies
definitions. of pair same the by typed

occupies that machine a is adversary An .(Adversaries) 4.6 تعريف
– interfaces adversary's the carries and identities other no and IDs($\mathcal{A}$)
inter- further any and silencer, and guard ,3.4 Section of wrapper the
around – one) adds 9 (Chapter given is machine section's that face
one: is 3.4 Section of machine The them. of each at core arbitrary an
another. is 4.4 Section of dummy the

at admits that adversary an is simulator A .(Simulators) 4.7 تعريف
together functionality standard every of ids process the – $\text{Std}_{\mathcal{U}} \cup \mathcal{Z}$ least
environment's. the with

in stand one lets what is which adversary, an thus is simulator Every
simulators read 4.24 Theorem and 4.12 Proposition class: adversary an
the compare 4.5 Section of containments budget the and adversaries. as
stand- what are callers admitted the and Occupancy directly. classes two
about settles 3.4 Section everything so means, slot adversary's the in ing
of call no begins it unchanged: simulator a of holds ($\mathcal{A}, \mathcal{P}$) at machine a
Proposi- at. runs it identity the only claim it lets silencer its and own, its
uses 4.29 Theorem given: is it whatever of occupancy only asks 4.19 tion
why explains 4.8 Remark and too. callers admitted the and wrapper the

restriction. a is neither
 can emulation so arguments, as machines three the takes metric The
 result. the bounding and them of classes over quantifying by stated be
 com- being pair the by settled is all at used be may environments Which
 pared.

Defini- .slot) adversary the about fixes emulation (What 4.8 نکته
 machines the about nothing says and $\mathbb{Z}$.par constrains below 4.10 tion
 needed, everything say 4.7 and 4.6 Definitions slot: adversary the in
 with free come $\mathbf{P}$ and pid fields The fields. five the than less is it and
 pins IDs($\mathcal{A}$) occupying so $\text{IDs}(\mathcal{F}) = \{(\mathcal{F}.\text{pid}, P) : P \in \mathcal{F}.\mathbf{P}\}$ occupancy:
 pin- and 3 line at $\text{global}(\mathcal{S})$ and 1 line at $\mathcal{S}.\text{pid} = (A, 0, 0)$ giving both,
 leaving parties fewer serving occupant slot a wanted, is what is $\mathbf{P}$ ning
 the where rej answered alone, side one on unoccupied -identity $\mathcal{A}$ an
 the is set silenced the — read never is par field The answers, other
 wrap- required the by code as carried, 2 line of $\{\text{id}'' : \text{id}'' \neq \text{id}\}$ literal
 claim could less on silenced simulator a formality: no is it and per,
 and, 3 line of $\text{id}'.F = A$ gate corruption the pass identity, party's a
 stands it adversary the capability a interface, party's honest an drive
 the names π^+ where, wf by only read inert, is $\mathbf{U}$ field The lacks, for in
 occupant, slot's the than rather $\mathcal{A}$ reserved
 Widen- outright, it requires 4.7 Definition why is which, $\mathbf{N}$ leaves That
 a places that id process every holding already $\mathcal{A}.\mathbf{N}$ impossible, is ing
 be- theorems both costs and nothing simulator a gains narrowing call:
 into party corrupt a at hook mediation every turns **Std** Dropping low,
 ev- over quantifies 4.20 Theorem and — 3.6 Section of $\perp$ deemed the
 in others meets ids process 's φ only admitting simulator a so, ρ ery
 environment the shuts $\mathbf{Z}$ Dropping them, has that world outer first the
 instruc- very the refusing slot the at 2 line alone, side ideal the on out
 mentions this of any in nothing And answers, adversary real the tions
 the hand 4.20 Theorem lets what is which for, simulated being pair the
 unchanged, on simulator hypothesis's

execu- the of machine a is environment $\mathbf{An}$. (Environments) 4.9 تعريف
 many finitely of identities the with together IDs($\mathbb{Z}$) occupies that tion
 3.3 Section of machine the for none — hosts it functionalities standard
 of environments absorbed the for protocol surrounding the of copies

silencer, 3.3 Section of interface the carries It below. 4.15 Definition
 func- hosted each and -identity.Z each at included, par parameter and
 dis- Exec identities: functionality's that at interface full own tionality's
 Z its what is output its and occupies, it identity every at it to patches
 returns. part

$\langle \pi \rangle :=$ write π system a For .environments) (Admissible 4.10 تعريف
 the :closure name its for $\{id : id.pid.F = \mathcal{F}.F \text{ some for } \mathcal{F} \in \pi\}$
 or occupied uses. π name a carrying id process every of identities
 the , φ and π systems For instance. every and session every in not.
 are pair the for admissible environments

$$\mathbb{Z}_{\pi, \varphi} := \{ \mathcal{Z} : \mathcal{Z}.par \supseteq \langle \pi \rangle \cup \langle \varphi \rangle \text{ and } \mathcal{Z} \text{ of identity no at hosts } IDs(\pi) \cup IDs(\varphi) \}.$$

defined is execution an hygiene: occupancy is condition second The
 so , (3.3 (Definition sets identity disjoint pairwise with occupants for only
 could occupies systems the of one identity an at hosting environment an
 at worlds both for out it rules admissibility and it, beside placed be not
 once.

then emulation of statement A below. justified is containment The
 of $\mathbb{S}$ class a , (4.6 (Definition adversaries of $\mathbb{A}$ class a classes: three fixes
 (Defi- environments of $\mathbb{Z} \subseteq \mathbb{Z}_{\pi, \varphi}$ class a and , (4.7 (Definition simulators
 three. the of statement strongest the gives $\mathbb{Z} = \mathbb{Z}_{\pi, \varphi}$ Taking . (4.9 nition
 reading. default the is and

let and , $\varepsilon \geq 0$ let systems, be φ and π Let .emulation) (UC 4.11 تعريف
 if $(\mathbb{A}, \mathbb{S}, \mathbb{Z})$ against ε within φ UC-emulates π that say We . $\mathbb{Z} \subseteq \mathbb{Z}_{\pi, \varphi}$

$$\forall \mathcal{A} \in \mathbb{A} \exists \mathcal{S} \in \mathbb{S} \forall \mathcal{Z} \in \mathbb{Z} : \quad Adv_{\pi, \varphi}^{uc}(\mathcal{Z}, \mathcal{A}, \mathcal{S}) \leq \varepsilon.$$

sim- the definition: the of substance the is quantifiers the of order The
 one so environment, the on not but adversary the on depend may ulator
 executions two the Otherwise once. at $\mathcal{Z} \in \mathbb{Z}$ every against work must $\mathcal{S}$
 the literally being $\mathcal{Z}$ swapped. system single a with experiment one are
 simulator a – $\forall \mathcal{A} \forall \mathcal{Z} \exists \mathcal{S}$ to it Weakening sides. both on machine same
 implies 4.11 Definition notion a gives – faces it environment the for built

$\max_{\mathcal{A}} \min_{\mathcal{S}} \max_{\mathcal{Z}}$ as reading two the prove. not do we converse whose and
 .4.5 Section of form concrete the in quantity one of $\max_{\mathcal{A}} \max_{\mathcal{Z}} \min_{\mathcal{S}}$ and
 instan- below proof each weakening. the survive results composition The
 what conclusion's: the by fixed environment an at hypothesis the tiating
 only since composition. than rather meaning therefore is buys order the
 — watching is whoever $(\pi, \mathcal{A})$ like behaving system one $(\varphi, \mathcal{S})$ is it under
 on. hand and name can one world ideal an

$\mathcal{Z}$.par Since meaningful. comparison the makes what is 4.10 Definition
 kept be must kinds three and it. inside identity no claim may it $\mathcal{Z}$ silences
 and world one in occupied is $\text{IDs}(\pi) \triangle \text{IDs}(\varphi)$ in identity An reach. of out
 which told is one as speak to allowed environment an so other. the in free
 An that. repair can simulator no and happens. anything before in is it world
 claiming environment an but nothing. separates both to internal identity
 which outside. from than rather system the of component a as speaks it
 is name used a of identity unoccupied an And for. not is environment an
 a name: by callers admits **Guard** because one. occupied an as good as
 no asking 2 line name. parent's its of instance every admits callee local
 in- drives name that of instance fresh a claiming environment an so more.
 pair blocked a against and — do callers their as exactly subroutines ternal
 call a refusing blocker the outright. apart worlds the tells it (6 (Chapter
 at three all disallows closure the Silencing admits. occupant genuine the
 ad- what only reach those and names. fresh is claimable stays What once.
 worlds two the between shared functionalities. global the **:Std** of all mits
 issue. at themselves not are they where comparison any in

channel second a is calling and claim. may $\mathcal{Z}$ what governs Silencing
 either in called be may $\text{IDs}(\pi) \triangle \text{IDs}(\varphi)$ of identity An touch. not does it
 where and .(3.6 (Section **rej** is answer the it occupies nothing where world:
 refuses something that provided — again **rej** is answer the does something
 compared being systems the on condition a is proviso The .2 line at **Z**
 not simply is it breaking pair a and class. environment the on than rather
 has world ideal the answer an invent to able being simulator no emulable.
 $\mathcal{Z}$ keep 20 Chapter of subroutines local the why is It give. to machine no
 .N their of out

en- admissible restricted least the restrictive. less being sets Smaller
 blocking the and — exactly closure the is par whose those are vironments
 pair: compatible a for them delivers 6 Chapter of

$$\text{IDs}(\overline{\pi}) = \text{IDs}(\overline{\varphi}) = \text{IDs}(\pi) \cup \text{IDs}(\varphi)$$

from name its copies blocker a and ,6.3 Theorem of claim second the by so for, in stands it member the

$$\langle \overline{\pi} \rangle = \langle \overline{\varphi} \rangle = \langle \pi \rangle \cup \langle \varphi \rangle.$$

share. can systems blocked two the value a merely not is $\mathcal{Z}.\text{par} := \langle \overline{\pi} \rangle$ So¹ admits. pair the restriction weakest the but differ- a of is it and available. already is relation the of property One – below results composition The follows. that everything from kind ent adver- dummy the of completeness the composition, parallel substitution, one regrouping each lemma, absorption single a of corollaries are – sary regroups Transitivity twice. advantage same the reading and execution together 4.5 Definition over inequality triangle the is proof Its nothing, pair, outer the for admissible is class conclusion's the that check the with of none needs and machinery the to than rather metric the to belongs it so reason. that for here stated is It section. this of rest the of apparatus the

گزاره 4.12 (Transitivity). Let $\mathcal{Z} \subseteq \text{let}$ and systems be ψ and φ , π . Let φ UC-emulates π within ε_1 against $(\mathbb{A}, \mathbb{S}, \mathbb{Z})$ and ψ UC-emulates π within ε_2 against $(\mathbb{A}, \mathbb{S}', \mathbb{Z})$ then $\mathcal{Z} \subseteq \text{let}$ UC-emulates π within $\varepsilon_1 + \varepsilon_2$ against $(\mathbb{A}, \mathbb{S}, \mathbb{Z})$.

both in $\mathcal{Z}$. A emulation. of statement legal a is conclusion the First, اثبات. $\mathcal{Z}.\text{par} \supseteq \langle \pi \rangle \cup \langle \varphi \rangle$ hence $\mathcal{Z}.\text{par} \supseteq \langle \varphi \rangle \cup \langle \psi \rangle$ and $\mathcal{Z}.\text{par} \supseteq \langle \pi \rangle \cup \langle \varphi \rangle$ has classes $\text{IDs}(\varphi) \cup \text{IDs}(\psi)$ of nor $\text{IDs}(\pi) \cup \text{IDs}(\varphi)$ of identity no at hosts it and $\langle \psi \rangle$ 4.10 Definition of clauses both so – $\text{IDs}(\pi) \cup \text{IDs}(\psi)$ of none at hence Now this. for related be not need systems three the $\mathcal{Z} \subseteq \mathcal{Z}_{\pi, \psi}$ and hold second the since and $\mathcal{S} \in \mathbb{S}$ supplies emulation first The $\mathcal{A} \in \mathbb{A}$ let φ against adversary an as $\mathcal{S}$ reading, $\mathcal{S}$ over adversaries its quantifies gives inequality triangle the $\mathcal{Z} \in \mathcal{Z}$ every For $\mathcal{S}' \in \mathbb{S}'$ yields

$$\begin{aligned} \text{Adv}_{\pi, \psi}^{\text{uc}}(\mathcal{Z}, \mathcal{A}, \mathcal{S}') &\leq \text{Adv}_{\pi, \varphi}^{\text{uc}}(\mathcal{Z}, \mathcal{A}, \mathcal{S}) + \text{Adv}_{\varphi, \psi}^{\text{uc}}(\mathcal{Z}, \mathcal{S}, \mathcal{S}') \\ &\leq \varepsilon_1 + \varepsilon_2, \end{aligned}$$

on depends $\mathcal{S}'$ As cancels. $\Pr[\text{Exec}(\varphi, \mathcal{Z}, \mathcal{S}, \mathcal{C}) = 1]$ term middle the since $\square$ required. simulator the is it alone. $\mathcal{A}$

be to strong too but coherent is par of regardless environments all over Quantifying¹ the Letting call. by call traffic internal φ reproduce to have would simulator the useful: leaving this, relax would occupy not does φ identities at functionalities spawn simulator in costs spawning what pairing: usual the for empty – withheld be to $\text{IDs}(\varphi) \setminus \text{IDs}(\pi)$ only work. future to leave we composition

the and here. environment absorbed the – below constructions Three
 exe- the of occupant one into machines several fold – 4.4 Section of two
 once. name we which shape. a share They cution.

تعريف 4.13 (Bundle). a as together machines several runs bundle A. as them among token the passing execution. the of occupant single declared a occupies It would. Exec as each initialising and does Exec and cover. must identities own members' the which identities. of set call A member. designated a from parameters and output its takes own bundle's the by inside. served is machines hosted two between from arriving call a :execution the on placed being without and routing through identity. target the carrying member the by served is outside machine hosted a call a included: guard interface. full member's that a in lies target its when execution the on placed is outward places may instance An otherwise. rej answered and set. outward declared in- – hand by members its between calls particular route addition in every outside lies target the when even another. before one terposing it calls the governs routing explicit an such and – identities member's above. rule target the to rest the only leaving names. occupancy. on turns claimed is call outward machine's hosted a How further no reaches and interface one of core the wraps silencer a since leaves occupies bundle the identities whose member A. (2 (Chapter mem- a alone: silencer own its meeting identity. own its under calls its so claim. to none has occupy not does bundle the identities whose ber identity an under leave to calls outward its of each rewrites bundle the declares. instance an what of part being identity which – hold does it inter- that meets claim rewritten the and – set outward its alongside bundle a keeps what is in is member a case Which silencer. face's for. stands it execution the to faithful

bundle a lets what is placed than rather served is traffic internal That responsively. it call cores whose system a of slot adversary the occupy wrapper The quibble. a not is distinction the why saying worth is it and bun- a so places. responder the call every refuses (9.1 (Definition Respond members own its of none reach could calls internal its place to obliged die three drives which .20 Chapter of simulator the and – one answering while at run not could answers. responsive two of each inside hybrids hosted whole the is which was. it where token the leaves inside them Serving all. re- a inside do not may still bundle a What asks. responsiveness what of

the and register the host: not does it anything reach is answer sponsive refused. and placed. are calls their so it. outside are functionalities shared kinds. two in comes machine hosted a that says already 4.13 Definition difference the and identities. its occupies bundle the whether to according keeps bundle the identities whose machine A follows. what of whole the is whose one inside: served is and dispatch execution's the from disappears calls its and somebody. to them leave must cedes bundle the identities — up come moves Both behalf. its on somebody that by placed be must them define we so — adversary an second the protocol. a absorbs first the once. right them prove and together

of set finite a H and environment an be $\mathcal{Z}$ Let .(Absorption) 4.14 تعريف
 of members the , $H = H_{\text{keep}} \uplus H_{\text{cede}}$ as split execution. an of machines
 into H of absorption The slot. adversary the of occupants being H_{cede}
 member each of copy one with $\mathcal{Z}$ of $\mathcal{Z}^H$ (4.13 (Definition bundle the is $\mathcal{Z}$
 , $\mathcal{Z}$ from par and output its takes , $\text{IDs}(\mathcal{Z}) \cup \text{IDs}(H_{\text{keep}})$ occupies it :H of
 still execution surrounding the identities the set outward as has and
 under leave calls its so identities. its keeps H_{keep} of member A serves.
 bundle. the inside served are it to addressed calls and claim own its
 outward each rewrites bundle the so none. keeps H_{cede} of member A
 instruction the as (A, P) at running while places it id'' . $\text{FullOp}(x)$ call

$$(A, P). \text{FullA}(\text{id}''.\text{Op}, x) \text{ as } (Z, P),$$

relay a by execution new the in occupied are cedes it identities the and
 4.4 Section of $\mathcal{D}$ dummy the is throughout which it. for

one only the is It alone. slot adversary the for case ceding the state We
 shape instruction an has interface whose one only the and below. used
 code own its what serves functionality standard a — into calls rewrite to
 calls. machine's another for in stand not could and names

to system a be π let , $\varphi \subseteq \rho$ Let .environment) (Absorbed 4.15 تعريف
 en- absorbed The environment. an be $\mathcal{Z}$ let and , φ of place in put be
 with $\mathcal{Z}$ into $\rho \setminus \varphi$ of (4.14 (Definition absorption the is $\mathcal{Z}^{\rho \setminus \varphi}$ vironment
 one with $\mathcal{Z}$ of bundle the is it so ceded. none and kept member every
 occupies it identities: its keeping copy every , $\mathcal{F} \in \rho \setminus \varphi$ each of copy
 outward has and , $\mathcal{Z}$ from par and output its takes , $\text{IDs}(\mathcal{Z}) \cup \text{IDs}(\rho \setminus \varphi)$

(Defi- environment an is It . $IDs(\pi) \cup IDs(\varphi) \cup \{id : id.F \in \{A, C\}\}$ set
 The ids. process several spans it since functionality, a not , (4.9 nition
 π reads set outward the — only shell hosted the records superscript
 context. by given replacement a to relative fixed is $\mathbb{Z}^{\rho \setminus \varphi}$ so — well as
 different name π in differing but $\rho \setminus \varphi$ sharing replacements two and
 machines.

exe- absorbed the what by forced are set outward the of halves Both
 or π to go identities whose slot. the is first The reproduce. to has cution
 take We in. sits environment absorbed the world which to according φ to
 both, in serves machine same the so alone $IDs(\varphi)$ than rather union the
 The identities. same the occupy not need two the that now matters which
 ma- as than rather ids as named ,C and A ids reserved two the is second
 Propo- while side ideal the on $\mathcal{S}$ holds slot adversary the because chines
 are They both. on machine absorbed same the and one needs 4.19 sition
 bare than rather interfaces full are machines hosted the because there
 each and ,3.5 Section of 6 and 2 lines on register the reads each cores:
 which ,Mediate through adversary the to party corrupt a at call a hands
 half this Without system. any to belongs id Neither ,(A, id.P) to hooks
 a and line. first its at refused be would functionality hosted a to call a
 adversary: the reaching of instead refused be would hook party's corrupt
 exactly regrouped one outer the being stop would execution absorbed the
 it serving nothing ,rej with answered is else Everything . $\rho \setminus \varphi$ met C when
 either. execution outer the in

the is It be. must it and untouched. over carried is parameter The
 calls which changes it to change any so interface. own 's $\mathbb{Z}$ of set silenced
 identically behave to has that machine one the is $\mathbb{Z}$ and — place may $\mathbb{Z}$
 $IDs(\rho \setminus \varphi)$ need functionalities hosted the think might One sides. both on
 full a is each not: do they but themselves. as speak to it from removed
 's $\mathbb{Z}$ claim. its licenses what is wrapper own its of 5 line and interface.
 those Removing ,(4.13 (Definition core own 's $\mathbb{Z}$ than further no reaching
 on $\mathbb{Z}$ un-silence would and them. for new nothing licence would identities
 absorption. before on silenced was it identities the exactly
 the for one into pair outer the for environment an turns thus Absorbing
 treat- special no needs register The admissibility. preserving pair. inner
 the so , $\rho \setminus \varphi$ nor φ neither in lies it system no of member a being ment:
 for blocker no builds blocking and it host not does environment absorbed
 both on machine single same the ,Exec by supplied is it A and $\mathbb{Z}$ Like it.

by so identities. their keep copies hosted Its comparison. every of sides their meet and claims own their under calls their leave they 4.13 Definition the needs: below regrouping the what is which – 's $\mathcal{Z}$ not – silencers own adversary an hosting case. opposite the are 4.4 Section of constructions occupy. not does bundle the identities whose

sys- be π and ρ Let .admissibility) preserves (Absorption 4.16 گزاره
well-formed. be ρ in π by φ of replacement the let and , $\varphi \subseteq \rho$ let tems.

Then

$$\mathcal{Z} \in \mathbb{Z}_{\rho[\pi/\varphi], \rho} \implies \mathcal{Z}^{\rho \setminus \varphi} \in \mathbb{Z}_{\pi, \varphi}.$$

it so untouched, over field the carries absorption half. par the For اثبات. that enough is

$$\langle \rho[\pi/\varphi] \rangle \cup \langle \rho \rangle \supseteq \langle \pi \rangle \cup \langle \varphi \rangle.$$

π name every and , $\varphi \subseteq \rho$ since , ρ by used is uses φ name Every half. hosting the For . $\pi \subseteq \rho[\pi/\varphi]$ since , $\rho[\pi/\varphi]$ by used is uses avoids latter the hosted: $\mathcal{Z}$ whatever beyond $\text{IDs}(\rho \setminus \varphi)$ at hosts $\mathcal{Z}^{\rho \setminus \varphi}$ meets $\text{IDs}(\rho \setminus \varphi)$ and ! $\text{IDs}(\pi) \cup \text{IDs}(\varphi)$ contains which , $\text{IDs}(\rho[\pi/\varphi]) \cup \text{IDs}(\rho)$ $\text{IDs}(\pi)$ meets and , ρ of subsystems disjoint being two the nowhere. $\text{IDs}(\varphi)$ hy- the place one the – replacement the of well-formedness by nowhere between relation No alone. typing for used is it and used, is pothesis ab- the say: not does statement the thing One enters. $\text{IDs}(\varphi)$ and $\text{IDs}(\pi)$ copy hosted each , par own its inside identities claims machine sorbed hosted a 4.13 Definition by – contradiction no is That own. its claiming ad- and environment's. the never and silencer own its meet claims copy's $\square$ applies. field the where read field. the on condition a is missibility

ex- an regroups absorption that is work theorem the makes What
Sec- serves correspondence same the Since it. changing without cution
lemma. the before down it set we , 4.4 tion

state its mean we execution an of configuration a By **Configurations.**
every of part unread the machine. every of state the activations: between the and code. its in where and token the holds machine which tape. coin an lets 3.6 Section and nest. Calls answers. awaiting suspended callers a is these of last the so continuations. outstanding several hold instance in answered. yet not and placed call per entry one set: a than rather chain on. waits it call the and caller the recording placed. order the ab- what is that – machines same the run not do executions two The the as counted is $\mathcal{Z}^H$ once ones same the run they but – does sorbing

ceded a relays the for but and one, as than rather bundles it machines
Exec as those among token the passes 4.13 Definition needs. identity
 the so. Counted side, either on sense same the in it holds each so does.
 the from apart one to one correspond executions two the of machines
 member each of copy hosted the bijection: the for ι write we and relays.
 still execution the everything and $\mathbb{Z}$ to $\mathbb{Z}$ hosted the member, that to H of
 itself. to included, $\mathcal{C}$ runs.

left the of c configuration A . correspondence) (Regrouping 4.17 تعريف
 $c \approx \hat{c}$ written correspond. right the of $\hat{c}$ configuration a and execution
 when

state same the in are ι under image its and c of machine every (C1)
 $\mathcal{C}$ on agree two the so included, $\mathcal{C}$ –

tape: coin unread same the have image its and machine every (C2)

same the at ι under both. in token the holds machine same the (C3)

and values: local same the with and operation same the of point

equal : ι under agree callers suspended of chains two the (C4)

identi- ι machines name entries -thk the k each for and length.

operation. identity. target same – call same the on waiting files.

identity. claimed and input

between bijection a is ι relay. a carries family right-hand the Where
 relay. the than other right the of those and left the of machines the
 con- call relayed a elided: frames own relay's the with read is (C4) and
 with once exactly resumed have. not does left the frame one tributes
 may reads left the rej a that save – returned call left's the answer the
 refusal-obliviousness place one the is which right. the on $\perp$ as arrive
 spent. is

is suspension every left the On bracketing. the from risk at is (C4) Only
 bundle. the and **Exec** between divide they right the on **Exec** by recorded
 the that asks clause The it. to internal calls the of those records which
 were they where counted are own bundle's the once agree chains two
 by kept suspensions. of chain one that: exactly is Regrouping placed.
 dispatchers. two

and correspondence. in begin executions two The **activation. first The**
 case lemma's the activation one the is this since why. saying worth is it
 and $\mathcal{C}$ initialises 1 Line machine. no by placed is it reach: not does analysis
 initialises 4.14 Definition sides. both on runs still execution the everything

every so state, no carries relay a and would. **Exec** as copy hosted each tape No (C1). is which state, same the in start image its and machine so calls, no places Initialize operation The (C2). giving read, been has with and – (C4) giving empty, are chains both and suspended is nothing be cannot copies its initialises bundle the which in order the reason the it 2 Line configuration, the in nothing off read be can order that observed: $\mathcal{Z}$ to goes token the left the On (Z, Z) from (Z, Z) . **FullZ()** activates then which $\mathcal{Z}$ hosted the by carried is (Z, Z) where $\mathcal{Z}^H$ to goes it right the on same the at **FullZ** enter Both occupy, bundle the makes 4.14 Definition (C3). is which input, same the on identity the and returned, value a or placed call a is activation every Thereafter on them of one survives correspondence that showing to comes lemma the with together halt configurations corresponding that and side, either case, by case activations those take paragraphs Its output, same

$\{\mathcal{Z}\} \cup H \cup R$ occupants the run execution an Let (Absorption) 4.18 لم
be $\mathcal{Z}^H$ let and $H = H_{\text{keep}} \uplus H_{\text{cede}}$ let environment, an $\mathcal{Z}$ and $\mathcal{C} \in R$ with
and H_{cede} of member each of Ask 4.14 Definition of absorption the
that H_{keep} in nothing of
:(4.28 (Definition refusal-oblivious be it (I)
(Def- cedes it identities the on call responsive a place R of core no (ب)
and ;(9.2 inition
-identity. $\mathcal{Z}$ no contain $\mathcal{Z}$.par (ج)
of sense the in families occupant legal being families two the Then,
3.3 Definition

Exec $(\{\mathcal{Z}\} \cup H \cup R)$ and

Exec $(\{\mathcal{Z}^H\} \cup R \cup \{\text{identity ceded each at relay a}\})$

close, merely not distributions, as equal – distributed identically are

para- five the and activations, on induction one is argument The اثبات.
fami- occupant two the pairs Machines step, its discharge below graphs
asks Routing alike, start image its and machine every that checks and lies
leaving group, the to internal – cases four finds and lands call each where
the only which of – one ceded a at entering identity, kept a at entering it,
mem- of kind which on splits Claims families, two the between differs last
on silencers same the meets identities its keeping one call: the placed ber

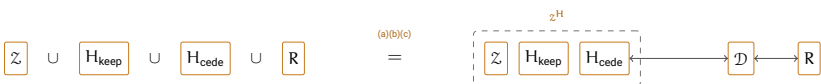
calls outward its so claim. to none has member ceded a while sides. both
 bun- the where — tests three survive must rewriting the and rewritten. are
 passes it whether and silenced. is claim the whether from. speak may die
 the **rej** a differ. can that value one the isolates Refusals guard. relay's the
 hy- it: on (a) hypothesis spends and right. the on $\perp$ as arriving reads left
 slot. relayed a on call responsive a problem. mirror the blocks (b) pothesis
 bun- the once agree suspensions of chains two the that checks Token
 spent is (c) Hypothesis placed. were they where counted are own die's
 conventions Two (Z, P) claim to right member's ceded the on Claims. in
 Dis- restating. worth are and throughout used are 3.6 and 1 Chapters of
 into machines bundling and coins. independent from draw machines tinct
 it what is randomness copy's hosted a so tapes. their merge not does one
 which in order the so calls. no places Initialize And absorption. before was
 an is proof The unobservable. is copies hosted its initialises bundle the
 shown as correspond. configurations initial the activations: on induction
 next same the produce configurations corresponding lemma: the before
 each configurations. corresponding to step and sides both on activation
 which order. same the in coins same the reading image its and machine
 con- halting corresponding and preserves: 4.17 Definition of (C2) what is
 .4.14 Definition by 's $\mathcal{Z}$ hosted the being output the alike. output figurations
 ex- machines which step: inductive the discharge below paragraphs The
 and (C4): claims. it what and goes call a where (C1): alike. start and ist
 both to common are paragraphs two first The (C3). moves. token the how
 costs. member ceded a what are them after two the member: of kinds
 together $\{\mathcal{Z}^H\} \cup R$ runs right the $\{\mathcal{Z}\} \cup H \cup R$ runs left The Machines.
 each of copy one with $\mathcal{Z}$ is $\mathcal{Z}^H$ and identity. ceded each at relay a with
 collections two the bundles. it machines the as Counted $\cdot H$ of member
 copy. its with H of member each pairs ι so relays. the for but coincide
 is which over. relays the leaves and itself. with $\mathcal{C}$ and R of member each
 sides. both on R initialises 1 Line them. gives 4.17 Definition reading the
 relay a and would. **Exec** as copies hosted the initialises 4.14 Definition
 is Occupancy alike. start image its and machine every so stateless. is
 on target: unique a has call each so hypothesis. by sides both on disjoint
 what hold relays the and $IDs(\mathcal{Z}) \cup IDs(H_{keep})$ holds bundle the right the
 go. let H_{cede}
 in endpoints both With lands. it where ask and call a Take Routing.
 the on $\mathcal{Z}^H$ inside served and left the on **Exec** by dispatched is it $\{\mathcal{Z}\} \cup H$
 dispatched is it group that Leaving way. either machine same the by right.
 the in lying target its right. the on outward placed and left the on **Exec** by

still execution the identities the makes 4.14 Definition which set, outward **Exec** by dispatched is it, H_{keep} of identity an at group the Entering serves. identity, that carrying copy hosted the by right the on served and left the on dispatched is it, H_{cede} of identity an at Entering occupies. bundle the which the is which right, the on relay the to and left the on member ceded the to paragraphs two the of subject the and differ families two the place one **Exec** side: either on machine no has it else anywhere Addressed below. callee the at evaluated are Guards .rej answers $\mathbb{Z}^H$ and target no finds do. claims the once agree they so id, claimed the from car- call such Each identities. its keeps that member a for Claims, could that silencers two the and sides, both on id claimed same the ries gov- is interface hosted A both, on silencers same the are one change :4.13 Definition by environment's, the by not and silencer own its by erved inter- full a Being it, reach not does $\mathbb{Z}$ at **Silence** so identities, its keeps it core its silencing 5 line side, either on identity own its exactly claims it face **.Mediate** through and $\mathcal{C}$ on places, wrapper its calls the and $\{\text{id}'' \neq \text{id}\}$ on in- own environment's The sides, both on silencer that even outside stand so unchanged, over carries absorption which, .par by governed is terface exactly admits and identity same the at set same the from computed is $\bar{\mathcal{T}}$ claim, a loses nor gains machine Neither claims, same the identity no has member a Such them, cedes that member a for Claims, out- its rewrites 4.14 Definition so bundle, the inside claim to own its of the where First, tests, three survive to has rewriting the and calls, ward of behalf on runs member hosted the Whenever from, speak may bundle is bundle the — (A, P) at running it has left the where is, that — P party calling $\mathbb{Z}$ by either entered is it : $P = Z$ else or $\text{.}(Z, Z)$ at or (Z, P) at running to party claimed the forcing **Guard** $_{\mathcal{A}}$ with internally served -identity, A an only P party claim (Z, Q) at interface an letting 3 line and target's, the be (Z, P) targets call inward whose relay, the by or : $P = Z$ or $Q \in \{P, Z\}$ when party-tie the by not silenced: is (Z, P) claim the whether Second, exactly, hy- which, .par by not and $\text{.}Z$ is name its since 2 line by not settled, just at **Guard** passes it whether Third, -identities, Z of empties (c) pothesis and $\text{.}2$ line at equal parties the with $Z \subseteq \mathcal{A}.\mathbf{N}$, 1 line at $P \in \mathcal{A}.\mathbf{P}$ relay: the under placed, left the call very the places then relay The $\text{.}3$ line at global at conjuncts same the meet sides both so claimed, left the claim very the way other the travels identity ceded the at arriving call A callee, same the on passed it guard the under relay the reaches it tests: three same the by admitted — payload the in caller its with (Z, P) at $\mathbb{Z}$ to handed is left, the and — $\text{global}(\mathbb{Z})$ and $\mathbf{A} \subseteq \mathbb{Z}.\mathbf{N}$, **Guard** $_{\mathbb{Z}}$ by and silencer own relay's the by

interface's that through (A, P) at interface member's hosted the reaches left. the on passed that guard the guard. places member ceded the call a Where differ. can value One Refusals. relay the right the on while call. own its from rej reads left the refused. is $\perp$ as member hosted the to it delivers 3.6 Section and rej that returns machine refusal-oblivious a this: closes what exactly is (a) Hypothesis (b) Hypothesis unchanged. is distribution its so two. the to alike answers on call responsive a problem. same the of direction other the stops what is the and places. responder the call every refuses **Respond** identity: ceded a slot relayed a on call responsive a so all. at answer to one place must relay which — substance with answered left the where $\perp$ answered be would visible: is does relay the else Nothing one. place may R of core no why is claim. relay's the by met id'. $F = A$ gate its detour. the survives Corrupt the and -identity A an under place must rewriting the call one the is which cannot. bundle

is continuation its and call. a placing operation. one has **Exec** Token. longer no $\mathcal{Z}^H$ to internal calls the right the On back. comes token the where hosted the among token the passes 4.13 Definition but handler. that reach own its on suspended is caller internal each so does. **Exec** as machines call relayed each and answer: same the with resumed and continuation elision the is which once. exactly resumed right. the on frame one adds 3.6 Section and used. are **Exec** of properties Two allows. 4.17 Definition outstanding several hold may machine a instance: every to both grants is second The suspended. while re-entered be may and continuations. outward an on waiting itself is $\mathcal{Z}^H$ while needs R from arriving call a what side. either on moment every at token the holds machine one Exactly call. and order same the in inputs same the sees therefore machine Every every at preserved is correspondence so tape. same the from answers particular in — distributions as agree executions two the and activation $\square$.1 returns $\mathcal{Z}$ that probability the on

side cases two its with equality one is lemma the Diagrammatically. one ceded a outright. bundle the into disappears member kept a side: by relay. a through out back bridged is



above figure's 4.20 Theorem in $\rho \setminus \varphi$ of all like — H_{keep} of member A as only it reaches R so identity, its occupies $\mathcal{Z}^H$ outright: absorbed is —

same the on running too, hosted is H_{cede} of member A bundle. the of part dummy (the $\mathcal{D}$ relay the identity: its occupy not does bundle the but tape, their with tagged $\mathcal{Z}$ to calls 'sR handing place, its in stands (4.4 Section of origi- its under calls outward member's hosted the re-emitting and caller Hypotheses them. between sits relay a tell can side neither so claim, nal is (a) indistinguishable: that keep what are H_{cede} of only asked (a)–(c). $\perp$ a into relays it rej every turns $\mathcal{D}$ since refusal-obliviousness, exactly the claim. own 's $\mathcal{Z}$ on clash a and call responsive a out rule (c) and (b) is figure 's4.20 Theorem show. could hop extra relay's a ways other two the for standing occupant ceded one case, mirror the $H_{\text{cede}} = \emptyset$ case the dummy-adversary 's4.4 Section what is, $(H_{\text{keep}} = \emptyset)$ slot adversary whole on. runs construction protocol a Absorbing apart. used are lemma the of halves two The pure a is left is what and away fall hypotheses three all so nothing, cedes stating worth is it and on. runs 4.20 Theorem case the is that rebracketing: all. at nothing costs it that emphasis the for separately

the let $\varphi \subseteq \rho$ let systems. be π and ρ Let .(Regrouping) 4.19 گزاره
environment an be $\mathcal{Z}$ let and well-formed, be ρ in π by φ of replacement
and φ or π either σ for Then $\text{IDs}(\rho) \cup \text{IDs}(\pi)$ of identity no at hosting
, (3.3 (Definition slot adversary the of $\mathcal{X}$ occupant any for

$$\text{Exec}(\rho[\sigma/\varphi], \mathcal{Z}, \mathcal{X}, \mathcal{C}) \quad \text{and} \quad \text{Exec}(\sigma, \mathcal{Z}^{\rho \setminus \varphi}, \mathcal{X}, \mathcal{C})$$

close. merely not distributions, as equal – distributed identically are

absorbed the is $\mathcal{Z}^H$ that so $H_{\text{cede}} = \emptyset$ with $H = \rho \setminus \varphi$ at 4.18 Lemma اثبات.
applies: lemma the of hypothesis no and 4.15 Definition of environment
Note none. are there and members ceded of asked are (c) and (b) (a).
occupant two The side. ideal the is $\sigma = \varphi$ case the so $\rho[\varphi/\varphi] = \rho$
so cases, both in system a is collection left-hand the legal: are families
says well-formedness because $\sigma = \pi$ for – twice occupied is identity no
is collection that because $\sigma = \varphi$ for and id. process no share π and $\rho \setminus \varphi$
bundle the keeps $\mathcal{Z}$ on hypothesis hosting the right the on while – itself ρ
 $\square$ $\text{IDs}(\sigma)$ of clear

another for subsystem one substituting that says now theorem The
advantage. in nothing costs protocol a inside
then ε within φ UC-emulates π if says: 4.20 Theorem Informally,
simulator and adversary The ε same the within ρ UC-emulates $\rho[\pi/\varphi]$

are conclusion the of environments the unchanged; over carry classes lies $\mathcal{Z}^{\rho \setminus \varphi}$ form absorbed whose $(\rho[\pi/\varphi], \rho)$ pair the for admissible those hypothesis. the of class the in

let and $\varphi \subseteq \rho$ let systems, be π and ρ Let .(Composition) **قضيه 4.20**
 Defini- of sense the in well-formed be ρ in π by φ of replacement the
 Suppose .4.2 tion

π UC-emulates φ within ε against $(\mathbb{A}, \mathbb{S}, \mathbb{Z})$.

Then

$\rho[\pi/\varphi]$ UC-emulates ρ within ε against $(\mathbb{A}', \mathbb{S}', \mathbb{Z}')$,

where

$$\mathbb{A}' = \mathbb{A}, \quad \mathbb{S}' = \mathbb{S}, \quad \mathbb{Z}' = \{ \mathcal{Z} \in \mathbb{Z}_{\rho[\pi/\varphi], \rho} : \mathcal{Z}^{\rho \setminus \varphi} \in \mathbb{Z} \}.$$

Explicitly,

$$\forall \mathcal{A} \in \mathbb{A}' \exists \mathcal{S} \in \mathbb{S}' \forall \mathcal{Z} \in \mathbb{Z}' :$$

$$\text{Adv}_{\rho[\pi/\varphi], \rho}^{\text{uc}}(\mathcal{Z}, \mathcal{A}, \mathcal{S}) \leq \text{Adv}_{\pi, \varphi}^{\text{uc}}(\mathcal{Z}^{\rho \setminus \varphi}, \mathcal{A}, \mathcal{S}) \leq \varepsilon.$$

simulator and adversary The kinds. three of relations. Three اثبات.
 machine same the is adversary an copying: by over carry classes
 the mentions simulator a typing nothing and attacks. it system whichever
 computed thing one the is class environment The for. simulates it pair
 under class hypothesis's the of preimage the being copied, than rather
 once applied ,4.19 Proposition steps. two then is bound The absorption.
 the — exactly one inner the into advantage outer the turns side, each on
 outward its what is which sides. both serving machine absorbed same
 in- the bounds hypothesis emulation the and — allow to chosen was set
 an as shown equality, an is display the of relation first the So one. ner
 of are relations three The second. the with uniformity for only inequality
 reduction the because unchanged over carry adversaries The kinds. three
 as read is $\rho[\pi/\varphi]$ attacking $\mathcal{A} \in \mathbb{A}'$ an machine: same very the on hands
 largest the is equality and serve, would $\mathbb{A}' \subseteq \mathbb{A}$ Any . π attacking $\mathcal{A} \in \mathbb{A}$ an
 $\mathcal{S} \in \mathbb{S}$ the reason: mirror the for over carry simulators The choice. such
 conclusion. the for simulator the as returned is supplies hypothesis the
 be may it That smallest. the is equality and serve, would $\mathbb{S}' \supseteq \mathbb{S}$ any so

pair the mentions simulator a typing nothing :4.8 Remark is all at returned . $(\rho[\pi/\varphi], \rho)$ for slot the in stands already (π, φ) for one so for, simulates it rather computed is something place one the are environments The systems, of pairs different two for classes are $\mathbb{Z}'$ and $\mathbb{Z}$ Here copied. than absorption, under $\mathbb{Z}$ of preimage the is $\mathbb{Z}'$ and

$$\mathbb{Z}' = ((\cdot)^{\rho \setminus \varphi})^{-1}(\mathbb{Z}) \cap \mathbb{Z}_{\rho[\pi/\varphi], \rho},$$

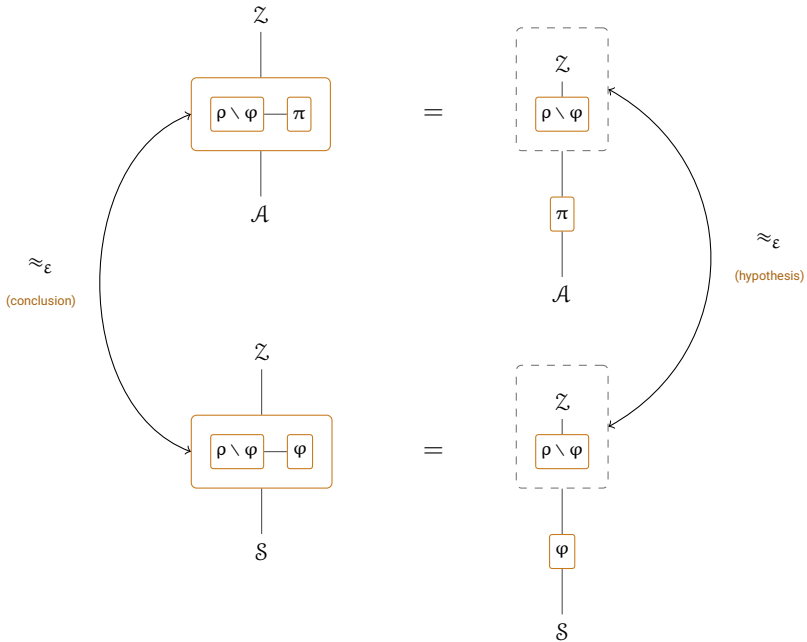
absorp- on becomes it what when exactly counts environment outer an so $\mathbb{Z}' \subseteq \mathbb{Z}_{\rho[\pi/\varphi], \rho}$ since class, genuine a is It covers, hypothesis the one is tion Proposi- , $\mathbb{Z}_{\pi, \varphi}$ class admissible largest the is $\mathbb{Z}$ When construction, by all is preimage the and redundant condition second the makes 4.16 tion class largest the gives class largest the against emulation : $\mathbb{Z}_{\rho[\pi/\varphi], \rho}$ of again.

need- defined, advantage right-hand the makes also 4.16 Proposition between relation no and environments the on hypothesis closure no ing twice, applied 4.19 Proposition is inequality first The .IDs(φ) and IDs(π) ideal the on $\mathcal{X} = \mathcal{S}$ and $\sigma = \varphi$ and side real the on $\mathcal{X} = \mathcal{A}$ and $\sigma = \pi$ with Subtracting, one.

$$\text{Adv}_{\rho[\pi/\varphi], \rho}^{\text{uc}}(\mathbb{Z}, \mathcal{A}, \mathcal{S}) = \text{Adv}_{\pi, \varphi}^{\text{uc}}(\mathbb{Z}^{\rho \setminus \varphi}, \mathcal{A}, \mathcal{S}),$$

brack- experiment one are experiments two the equality: an fact in is it so ideal the because sides both on serves $\mathcal{S}$ same the and differently, eted unequal- second The absorption, same very the by regrouped is execution $\mathbb{Z}$ because $\mathbb{Z}$ in lies which , $\mathbb{Z}^{\rho \setminus \varphi}$ to applied hypothesis emulation the is ity for only inequality an as relation first the displays theorem The . $\mathbb{Z}'$ in lies $\square$ second, the with uniformity

, $\mathbb{Z} \in \mathbb{Z}'$ one and $\mathcal{S} \in \mathcal{S}'$ serving a , $\mathcal{A} \in \mathcal{A}'$ one fixing Diagrammatically, string-diagram the in drawn below, square commuting the is proof the box the and interface, an is wire a system, a is box solid a :[9] of style occu- currently machine whichever is system a of bottom the off hanging slot, adversarial its pies



شكل 1: one is column each down Reading square. commuting a as Substitution. edges horizontal Both exchanged. being pair the marks box dotted the execution: the is edge right the and bounds. than rather equalities are both so absorption. are with one merely not quantity. same the then is left the on conclusion The hypothesis. (.4.20 (Theorem bound. same the

and (π, A) to applied .4.19 Proposition are edges bottom and top The mere not equalities. are both so free. is regrouping – respectively (φ, S) to environ- absorbed (the right the on box dashed the why is which bounds. a between changes that thing only the is (4.19 Proposition of $Z^{\rho \setminus \varphi}$ ment hypothesis. theorem's the is edge right The neighbour. its and column con- theorem's the – edge left The environment. absorbed that to applied not are 's $\approx_\epsilon$ two the shows square the chasing forced: then is – clusion the are equalities. two the by but. ϵ same the by bounded both merely at hybrid one chain. a in squares such n Composing quantity. same very below. 4.24 Theorem exactly is time. a

com- which in form the into combine now transitivity and Substitution metric the with stated was 4.12 Proposition used. actually is position does it where is here machinery: this of none needing proof its above. work. its

نتیجه 4.21 (Composition) Let ρ be a specification with ψ and ρ well-formed. Let φ be a replacement of ρ in π by φ of replacement the let $\varphi \subseteq \rho$ let terms. class the for Z' Write (A, S, Z) against ε_1 within φ UC-emulate π let then (S, S', Z') against ε_2 within ψ UC-emulates ρ If 4.20 Theorem of (A, S', Z') against $\varepsilon_1 + \varepsilon_2$ within ψ UC-emulates $\rho[\pi/\varphi]$

اثبات. 4.20 Theorem By. against ε_1 within ρ UC-emulates $\rho[\pi/\varphi]$. statement a is hypothesis second The $Z' \subseteq Z_{\rho[\pi/\varphi], \rho}$ and (A, S, Z') con- Both well. as $Z' \subseteq Z_{\rho, \psi}$ so class. same that against emulation of the to applies it and hold. therefore 4.12 Proposition by needed tainments $\square$ middle. the in ρ with statements two

with ρ analyzes One used. is composition which in form the is This spec- whatever emulates it proving place. in φ subroutine idealized the whose names. corollary the classes the against – wanted is ψ ification con- that transfers corollary the simulators: step's first the are adversaries emula- miss: to easy is hypothesis One runs. that system the to clusion $Z' \subseteq Z_{\rho, \psi}$ that condition. side 's4.11 Definition as asserts. Z' against tion class the environment every of par the under lie must names 's ψ so – not. do they where vacuous is corollary the and retains.

Composition Parallel 4.3

not is once at several Replacing subsystem. one replaces 4.20 Theorem hybrids: of chain a along one that of applications n but argument new a chain. the down classes three the of bookkeeping the is care needs what satisfi- are and literally. meant are below hypotheses disjointness The machine a is register the :1 Chapter of convention the of because only able one ever only is there and system. no of member a and execution the of once. at π_k every and φ_i every in lie would it member. a Counted it. of carry would π_k two no and disjoint be would ρ of subsystems two no so and $n \geq 2$ for unsatisfiable be would hypotheses the ids: process disjoint systems the of out register the Keeping .C modulo read be to have would written. as read be them lets what is

تعريف 4.22 (Simultaneous replacement) Let ρ be a system and let $\varphi_1, \dots, \varphi_n$ be pairwise disjoint subsystems. For $\varphi_1, \dots, \varphi_n \subseteq \rho$

$\langle \pi_1, \dots, \pi_n \rangle$

$$\rho[\pi_1/\varphi_1, \dots, \pi_n/\varphi_n] := \left(\rho \setminus \bigcup_{i=1}^n \varphi_i \right) \cup \bigcup_{i=1}^n \pi_i.$$

and $\rho_0 = \rho$ that so, $0 \leq k \leq n$ for $\rho_k := \rho[\pi_1/\varphi_1, \dots, \pi_k/\varphi_k]$ Write first the by untouched is φ_k disjoint, pairwise are φ_i the Since $\rho_n = \rho'$ and $\varphi_k \subseteq \rho_{k-1}$ so replacements, $k-1$

$$\rho_k = \rho_{k-1}[\pi_k/\varphi_k].$$

lets which replacement, single a by differ therefore hybrids Consecutive n as chain the walks below proof the step; each on act 4.20 Theorem needs also together steps the Fitting replacement, per one hops, game are. they and adjustable, be to classes the

against ε within φ UC-emulates π Suppose (Monotonicity) 4.23 لم
any $\mathbb{A}^- \subseteq \mathbb{A}$ any for $(\mathbb{A}^-, \mathbb{S}^+, \mathbb{Z}^-)$ against so does it Then $(\mathbb{A}, \mathbb{S}, \mathbb{Z})$
 $\mathbb{Z}^- \subseteq \mathbb{Z}$ any and $\mathbb{S}^+ \supseteq \mathbb{S}$ simulators of class

Narrow- classes. three the over statement $\forall \exists \forall$ a is 4.11 Definition اثبات.
exis- the of that widening and quantifier universal either of range the ing
legal a result the keeps $\mathbb{Z}^- \subseteq \mathbb{Z} \subseteq \mathbb{Z}_{\pi, \varphi}$ and it, preserve each one tential
□ emulation. of statement

sub- disjoint pairwise n replacing that says 4.24 Theorem Informally,
4.20 Theorem is it errors: individual the of sum the costs once at systems
of those being adversaries the hybrids, of chain a along times n applied
environ- the and first, the of those simulators the replacement, last the
once. at step every for admissible ments

$\varphi_1, \dots, \varphi_n \subseteq \rho$ let system, a be ρ Let (composition) (Parallel 4.24 قضيه
 π_k the that such systems be $\pi_1, \dots, \pi_n$ let and disjoint, pairwise be
pro- no shares π_k , k each for and, ids process disjoint pairwise carry
 k each for that Suppose $\rho \setminus \varphi_k$ with id cess

π_k UC-emulates φ_k within ε_k against $(\mathbb{A}_k, \mathbb{S}_k, \mathbb{Z}_k)$,

step, -thk the to attaches 4.20 Theorem class the for $\mathbb{Z}'_k$ write and

$$\mathbb{Z}'_k = \{ \mathcal{Z} \in \mathbb{Z}_{\rho_k, \rho_{k-1}} : \mathcal{Z}^{\rho_{k-1} \setminus \varphi_k} \in \mathbb{Z}_k \}.$$

that finally Assume

$$S_{k+1} \subseteq A_k \quad (1 \leq k < n), \quad Z^\star \subseteq \bigcap_{k=1}^n Z'_k.$$

and $Z^\star \subseteq Z_{\rho', \rho}$ Then

$$\rho' := \rho[\pi_1/\varphi_1, \dots, \pi_n/\varphi_n] \text{ UC-emulates } \rho$$

$$\text{within } \sum_{k=1}^n \varepsilon_k \text{ against } (A_n, S_1, Z^\star).$$

اثبات. that checking in is work the and steps. n over argument hybrid A . taken. is them of any before 4.20 Theorem of instance legal a is step each well- is replacement each shows first The that. do preliminaries Three composition the that so k on induction by system. a ρ_k each and formed ad- is $Z^\star$ class the shows second The step. every at applies theorem over union the taking – and once. at pair intermediate every for missible state- legal a itself is conclusion the that so too. pair outer the for – k downwards. simulators of chain the builds third The emulation. of ment S_{k+1} adversary the for step -thk the by supplied S_k each and $S_{n+1} := \mathcal{A}$ taken: hybrid the is then Only alone. $\mathcal{A}$ on depend S_1 makes what is which adversary. one and subsystem replaced one in differ games consecutive n the summing and 4.20 Theorem of application one is difference that members The well-formed. is step each First. $\sum_k \varepsilon_k$ to telescopes hops place. in leave replacements $k-1$ first the $\rho \setminus \varphi_k$ of those are $\rho_{k-1} \setminus \varphi_k$ of nei- with id process a shares π_k hypothesis by $(\pi_1, \dots, \pi_{k-1})$ with together the in well-formed is ρ_{k-1} in π_k by φ_k of replacement the So group. ther $\rho_0 = \rho$:k on induction by system. a is ρ_k Each 4.2 Definition of sense well- that from 4.3 Proposition by one is $\rho_k = \rho_{k-1}[\pi_k/\varphi_k]$ and one. is ask which 4.16 Proposition and 4.20 Theorem So replacement. formed step. -thk the to apply systems. be to π_k and ρ_{k-1} $Z^\star \subseteq Z'_k \subseteq Z_{\rho_k, \rho_{k-1}}$ From .k fix and $Z \in Z^\star$ Let admissibility. Next, get we

$$Z.\text{par} \supseteq \langle \rho_k \rangle \cup \langle \rho_{k-1} \rangle,$$

$\text{Adv}_{\rho_k, \rho_{k-1}}^{\text{uc}}$ advantage the and hybrids of pair -thk the for admissible is Z so by ρ_k of replacement the to applied 4.16 Proposition defined. is below parameter its Z_{π_k, φ_k} in environment absorbed the places then ρ_{k-1} in π_k in- the so φ_k of and π_k of identities the contains already which 's. Z is

displayed the of union the Taking too. defined is $\text{Adv}_{\pi_k, \varphi_k}^{\text{uc}}$ advantage ner
 .k all over containments

$$\mathcal{Z}.\text{par} \supseteq \bigcup_{k=0}^n \langle \rho_k \rangle \supseteq \langle \rho' \rangle \cup \langle \rho \rangle,$$

the $\bigcup_k \text{IDs}(\rho_k) \supseteq \text{IDs}(\rho') \cup \text{IDs}(\rho)$ of identity no at hosts $\mathcal{Z}$ likewise and
 of clauses both so $k = 1$ and $k = n$ at $\mathcal{Z}_{\rho_k, \rho_{k-1}}$ each of clause hosting
 state- legal a itself is conclusion the and $\mathcal{Z}^\star \subseteq \mathcal{Z}_{\rho', \rho}$ give 4.10 Definition
 emulation. of ment

$\mathcal{S}_k \in \mathbb{S}_k$ let $k = n, n-1, \dots, 1$ For $\mathcal{S}_{n+1} := \mathcal{A}$ put and $\mathcal{A} \in \mathbb{A}_n$ Fix
 is This $\mathcal{S}_{k+1}$ adversary the for supplies step -thk the simulator the be
 $k < n$ at and $\mathcal{S}_{n+1} = \mathcal{A} \in \mathbb{A}_n$ because $k = n$ at index: every at legitimate
 downwards. built is simulators of chain The $\mathcal{S}_{k+1} \in \mathbb{S}_{k+1} \subseteq \mathbb{A}_k$ because
 alone. $\mathcal{A}$ on depends $\mathcal{S}_1$ and it. before one the from each
 games $n+1$ the consider and $\mathcal{Z} \in \mathcal{Z}^\star$ fix Now

$$G_k := \text{Exec}(\rho_k, \mathcal{Z}, \mathcal{S}_{k+1}, \mathcal{C}), \quad p_k := \Pr[G_k = 1], \quad 0 \leq k \leq n.$$

the so $\mathcal{A}$ under ρ' of that G_n and $\mathcal{S}_1$ under ρ of execution the is G_0 Here
 one in differ games Consecutive ρ_n at ends and ρ_0 at starts sequence
 that and adversary, the playing machine the in and subsystem replaced
 $1 \leq k \leq n$ for :4.20 Theorem of application one exactly is difference

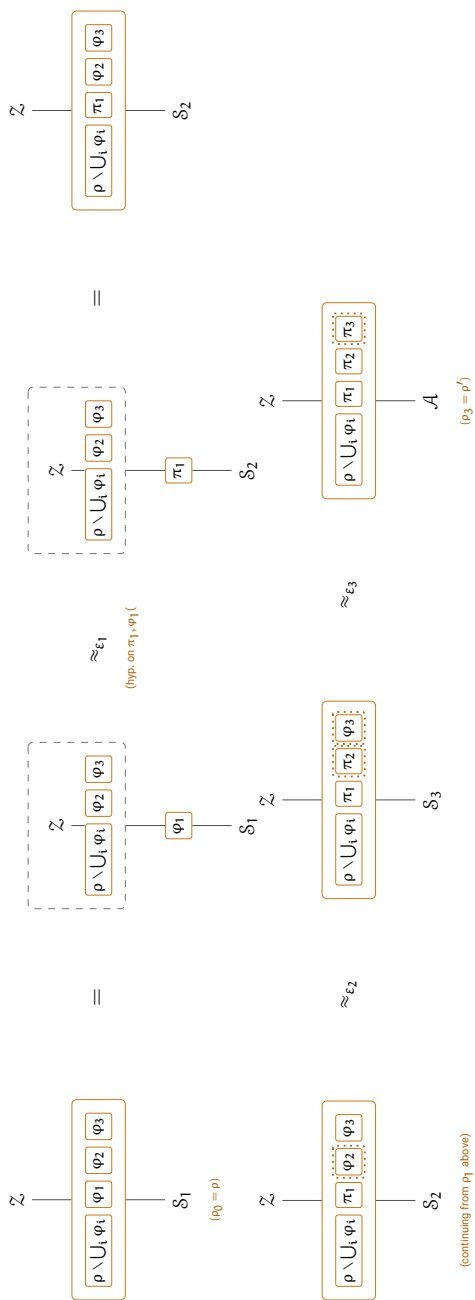
$$\begin{aligned} |p_k - p_{k-1}| &= \text{Adv}_{\rho_k, \rho_{k-1}}^{\text{uc}}(\mathcal{Z}, \mathcal{S}_{k+1}, \mathcal{S}_k) \\ &\leq \text{Adv}_{\pi_k, \varphi_k}^{\text{uc}}(\mathcal{Z}^{\rho_{k-1} \setminus \varphi_k}, \mathcal{S}_{k+1}, \mathcal{S}_k) \leq \varepsilon_k. \end{aligned}$$

the are G_{k-1} and G_k since backwards. read 4.5 Definition is equality The
 inequal- first The compares. $\text{Adv}_{\rho_k, \rho_{k-1}}^{\text{uc}}$ advantage the that executions two
 by $\mathcal{Z} \in \mathcal{Z}^\star \subseteq \mathcal{Z}'_k$ because available, 4.20 Theorem of reduction the is ity
 $\mathcal{Z}^\star$ against statement a step -thk the makes 4.23 Lemma so hypothesis.
 $\mathcal{S}_{k+1}$ adversary the to applied π_k by φ_k of emulation the is second The
 environment. absorbed the and
 telescoping. and hops n the Summing

$$\text{Adv}_{\rho', \rho}^{\text{uc}}(\mathcal{Z}, \mathcal{A}, \mathcal{S}_1) = |p_n - p_0| \leq \sum_{k=1}^n |p_k - p_{k-1}| \leq \sum_{k=1}^n \varepsilon_k.$$

is this arbitrary. was $\mathcal{Z} \in \mathcal{Z}^\star$ and alone $\mathcal{A}$ from chosen was $\mathcal{S}_1 \in \mathbb{S}_1$ As
 $\square$ theorem. the of statement the

figure (the square own 's4.20 Theorem is k hop Diagrammatically,
the and $\mathcal{S} \mapsto \mathcal{S}_k$, $\mathcal{A} \mapsto \mathcal{S}_{k+1}$, $\pi \mapsto \pi_k$, $\varphi \mapsto \varphi_k$ with proof) its above
One ride. the for along slot untouched every carry to widened $\rho \setminus \varphi$ shell
where from compactly continuing 2, 3 hops and full in drawn 1 hop picture.
 $:\rho_3 = \rho'$ to $\rho_0 = \rho$ ends, it



of remainder true the – absorbed is 's) φ (all ρ_0 right: to left Reading φ_1 only leaving it, with along $\mathbb{Z}$ into folding φ_2, φ_3 untouched the and ρ handing ε_1 within π_1 for it swaps hypothesis 's4.20 Theorem exposed. match. to S_2 to S_1 from advanced S , ρ_1 reassembles out back fold the ends flat their to compressed only square, identical the are 3 and 2 Hops dotted the with – add than rather repeat would more twice it redrawing – and flip) to (about φ_k is slot which exactly hop, each at marking, boxes last the in as telescoping, and hops three the Summing flipped). (just π_k appear which of none 's, ε_k separate three turns above, proof the of lines $\cdot \rho \approx_{\varepsilon_1 + \varepsilon_2 + \varepsilon_3} \rho'$ bound one the into statement, own theorem's the in

it as chain the along behaves Corruption **chain. the along Corruption** under- machine one is register the game a Within step, single a at does corrupt cannot hop a so party, by keyed alike, φ_k and π_k , $\rho \setminus \varphi_k$ neath and instances separate are registers the games across halfway; party a parties same the corrupts environment The hops, between nothing carry :3 line under calls same the placing machine same the being hop, every at the – hop per change that machines two the so to, have not does slot the corrupt not need – $S_k \rightarrow S_{k+1}$ occupant slot the and subsystem replaced, (3.1 (Remark unmediated **C** reads environment the but parties: same the commensu- and, ε_k same the by caught is differently corrupting S_k a so parties The assumption, an than rather consequence a again is rateness absorbed the leave hooks **Mediate** their alike: handled are serves $\rho_{k-1} \setminus \varphi_k$ 4.15 Definition of set outward the why is which slot, the for environment environment no that only claims theorem the So .C and A ids the carries $\cdot \sum_k \varepsilon_k$ than more by apart patterns corruption the tells $\mathbb{Z}^\star$ in

has closures hybrid the of union The .classes) largest (The 4.25 نکته together ρ of part untouched the of consists ρ_k Each form, closed a so $\pi_1, \dots, \pi_k$ with

$$\bigcup_{k=0}^n \langle \rho_k \rangle = \langle \rho \rangle \cup \bigcup_{i=1}^n \langle \pi_i \rangle = \langle \rho' \rangle \cup \langle \rho \rangle.$$

$\mathbb{Z}_{\pi_k, \varphi_k}$ class admissible largest the is $\mathbb{Z}_k$ each that now Suppose redun- $\mathbb{Z}'_k$ defining condition second the makes then 4.16 Proposition is $\mathbb{Z}^\star$ of choice permissible largest the and, $\mathbb{Z}'_k = \mathbb{Z}_{\rho_k, \rho_{k-1}}$ so dant,

$$\bigcap_{k=1}^n \mathbb{Z}_{\rho_k, \rho_{k-1}} = \{ \mathbb{Z} : \mathbb{Z}.\text{par} \supseteq \bigcup_{k=0}^n \langle \rho_k \rangle \} = \mathbb{Z}_{\rho', \rho}.$$

the delivers therefore classes largest the against composition Parallel
does. theorem single-step the as again. class largest

are adversaries The suggests. chain the as behave classes three The
adversary an from starting argument the replacement. last the of those
ending chain the first. the of those are simulators the $\rho_n = \rho'$ against
serves class intermediate Each $\rho_0 = \rho$ against simulator a producing by
which next. the of adversaries the and step one of simulators the as twice.
for admissible be must environments The arranges. $S_{k+1} \subseteq A_k$ what is
as additive. is advantage The intersection. the hence once. at step every
it. makes always steps n over argument hybrid a

Adversary Dummy the of Completeness 4.4

re- and it tells environment the whatever out carries adversary dummy The
real a Anything own. its of strategy no has it told: is it whatever back ports
dummy. the driving by achieve can environment an achieves. adversary
quantifying by replaced be can adversaries over quantifying why is which
sys- the for that. proves and it defines section This environments. over
the records 9.4 Remark — precise makes theorem the classes and tems
reach. not does claim the that calls. responsive placing cores regime. one

ma- of kind new a not is dummy The **fixes. already framework the What**
carries it so core. particular a with 3.4 Section of adversary the is It chine.
consequences Three wrapper. 'sA through pass calls its and fields 'sA
written. is it of line a before code the of shape the settle

a silences box A the of 2 Line only. identity one claim may it First.
identity the id claims places dummy the call every so $\{id'' \neq id\}$ on core
cannot therefore " id'' " as id'' .Op "call form the of instruction An at. runs it
not or $(A, id.P)$ as speaks dummy the id'' arbitrary an for out carried be
party. the but id claimed the not is chooses environment the What all. at
may $\mathbb{Z}$ of root the (A, Q) at than rather (A, P) at dummy the instructing by
either. pick may so and party any for act

asks **Guard** predicate The forced. is target the of party the Second.
at dummy the so $(A, id.P)$ is claim the and callee. the of $id'.P = id.P$
does that target A other. no and P party of interfaces reaches (A, P)
the 3 line or 2 line by call the refuses party. honest an or A admit not
to it delivers 3.6 Section of rule the and back. refusal the passes dummy
. $\perp$ as caller own dummy's the

which, $(A, P) \rightarrow (Z, P)$ too: fixed is environment the to route its Third.
agree. parties the and $A \subseteq \mathcal{Z}.N$ because admits **Guard_Z**

environ- the from call A it. called who on dispatches core The **code. The**
mediation a – else anything from call a out: carry to instruction an is ment
In on. hand to message a is – core some inside placed $\mathcal{A}(\cdot)$ an or hook.
oper- an uses. already **Mediate** shape the has payload the directions both
value. a with together ation

D adversary Dummy	
$\text{par} := \perp$	$\text{.U} := \{\{\mathcal{Z}, \text{ serves}\}\}$
$\text{.N} := \text{Std} \cup A \cup \mathcal{Z}$	$\text{.P} := \{0, 1\}^* \cup \{A, \mathcal{Z}\}$
$\text{.pid} := A$	
	$\text{id}' \text{ from } \text{id.A(in)}$
:1 if $\text{id}'.F = \mathcal{Z}$ then	
:2 $\text{parse in as } (\text{id}'.\text{Op}, x)$	/ unparseable: return $\perp$
:3 $\text{return id}''.\text{FullOp}(x)$ as id	/ carry out the instruction
:4 else	
:5 $\text{return } (\mathcal{Z}, \text{id}.P).\text{FullZ}(\text{id}', \text{in})$ as id	/ hand it to $\mathcal{Z}$, caller and all

in- target a names environment the direction: outward the is 3 Line
returns and call that exactly places dummy the and input. an and terface
the – remembered nothing and inspected is Nothing answer. its exactly
one: inward the is 5 Line .Initialize no has it so state. no keeps dummy
together environment the to handed is adversary the reaches whatever
is answers environment the whatever and from. came it identity the with
read relay same the are lines two The unchanged. caller that to returned
transparent. dummy the makes what is which directions. opposite in
an- guards the : $\perp$ with answered is parse cannot 2 line instruction An
names that payload a but .rej with place cannot dummy the anything swer
noth- with machine a what is $\perp$ and own. its of value a needs all at call no
returns. say to ing

5 line on $\mathcal{Z}$ to passed is id' caller The on. pausing worth is detail One
which tell not could environment the it without payload: the merely not and
transparent. be cannot that loses that relay a and asking. is interface

moving about statement a is Completeness **adversary. an Interposing**
needs and it. around machines the into execution the from adversary an
sides: two from seen operation same the are Both move. each for name a
what so interface. dummy the and machine some between placed is $\mathcal{A}$
now place to used $\mathcal{A}$ what and inside. it reaches now $\mathcal{A}$ reach to used
.D to instruction an as leaves

ان be $\mathcal{Z}$ Let .environment) the into absorbed (Adversary 4.26 **تعريف**
 bundle the is $\mathcal{Z}^A$ environment The adversary. an $\mathcal{A}$ and environment
 its keeping not copy the $\mathcal{A}$ of copy one with $\mathcal{Z}$ of (4.13 (Definition
 from par and output its takes and alone IDs($\mathcal{Z}$) occupies it identities:
 interface 's $\mathcal{A}$ by internally served is -identity $\mathcal{A}$ an on places $\mathcal{Z}$ call $\mathcal{A}$. $\mathcal{Z}$
 the call other Any 's. $\mathcal{Z}$ by -identity $\mathcal{Z}$ a on places $\mathcal{A}$ one and there.
 rewritten is $(\mathcal{A}, P)$ at running while id'' . **FullOp**(x) on places. $\mathcal{A}$ hosted
 instruction the to

$$(\mathcal{A}, P). \text{FullA}(\text{id}'', \text{Op}, x) \text{ as } (\mathcal{Z}, P),$$

un- the call a so target's. its never and at runs $\mathcal{A}$ party the to keyed
 party same that at relayed is $(\mathcal{A}, P)$ as only place could $\mathcal{A}$ absorbed
 an from arriving call $\mathcal{A}$ was. it where exactly refused or admitted and
 $(\mathcal{A}, P)$ at interface 's $\mathcal{A}$ to handed is (id', in) payload with -identity $\mathcal{A}$
 id' from call a as included. guard

sim- a be $\mathcal{S}$ Let .simulator) a before interposed (Adversary 4.27 **تعريف**
 (Defini- bundle the is $\mathcal{S}[\mathcal{A}]$ simulator The adversary. an $\mathcal{A}$ and ulator
 's $\mathcal{S}$ carries and IDs($\mathcal{A}$) occupies it $\mathcal{A}$ of copy one with $\mathcal{S}$ of (4.13 tion
 $\mathcal{S}[\mathcal{A}]$ so for. in stands it machine the being $\mathcal{A}$ not $\mathcal{S}$ — fields remaining
 the by served is -identity $\mathcal{Z}$ a from call $\mathcal{A}$ is. $\mathcal{S}$ whenever simulator a is
 partic- in hook mediation a id. process other any from one $\mathcal{A}$ hosted
 to back route its — -identity $\mathcal{Z}$ a on calls 's $\mathcal{A}$ hosted The . $\mathcal{S}$ by ular.
 its $\mathcal{Z}$ external the to $(\mathcal{A}, P)$ under bundle the leave — environment the
 served .4.26 Definition of instruction the become calls outward other
 -identity $\mathcal{Z}$ a on places $\mathcal{S}$ call a and $(\mathcal{Z}, P)$ from call a as $(\mathcal{A}, P)$ at $\mathcal{S}$ by
 $(\mathcal{A}, P)$ at $\mathcal{A}$ to .**Guard** $\mathcal{Z}$ past handed. is

occu- $\mathcal{Z}^A$ bundle The machine. of kind new a is construction Neither
 is it so claimed. $\mathcal{Z}$ what claims and occupies environment an what pies
 the being those -identity. $\mathcal{A}$ no occupies It is. $\mathcal{Z}$ when exactly admissible
 has $\mathcal{A}$ hosted the 4.13 Definition by so execution. right-hand the in relay's
 and above. rewriting the hence under: call a place to own its of identity no
 rewit- the check to — below Siting by discharged — obligation the hence
 of case other the is 4 Chapter in Absorption silencer. 's $\mathcal{Z}$ against claim ten
 claim its keeps there functionality hosted a why is which .4.13 Definition
 the in standing what is which .IDs($\mathcal{A}$) occupies $\mathcal{S}[\mathcal{A}]$ Meanwhile instead.

means. slot adversary

Defini- is alone dummy the to respect with Emulation **theorem. The**
 The $\mathcal{D}$ machine single the to down cut class adversary the with 4.11 tion
 lost. is nothing that is claim
 of price the is it and needed. is adversary the on condition mild One
 relay. a being dummy the

تعريف 4.28 (Refusal-oblivious). An adversary $\mathcal{A}$ is refusal-oblivious if its behaviour is unchanged when it receives rej as to answer the call call is replaced by $\perp$.

of **return** own its 3.6 Section by and answer. an such relays dummy The
 placing adversary unabsorbed an where so $\perp$ as caller the reaches rej a
 dummy the driving adversary same the directly. rej reads call refused a
 is which apart. two the tell cannot adversary refusal-oblivious $\mathcal{A}$. $\perp$ reads
 refusal-oblivious itself is dummy The needs. below regrouping the what
 costs hypothesis the so – it on branching without answer its relays it –
 framework- a reads that adversary real a and nothing. class dummy the
 an than rather artefact modelling a case any in is signal refusal internal
 attack.

the to respect with emulation that says 4.29 Theorem Informally,
 en- an achieves. adversary real a anything nothing: loses alone dummy
 is $\mathcal{A}$ for simulator the and dummy. the driving by achieve can vironment
 be- interposed $\mathcal{A}$ with supplies. hypothesis the simulator one the – $\mathcal{S}_{\mathcal{D}}[\mathcal{A}]$
 $\mathcal{Z}$ on not and alone $\mathcal{A}$ on depends which – it fore

قضيه 4.29 (Completeness) Let φ and π be dummy the of (Completeness 4.29
 $\varepsilon \geq \text{let}$, (9.2 Definition calls responsive no place cores whose systems
 let, (4.28 Definition adversaries refusal-oblivious of class a be $\mathbb{A}$ let. 0
 Suppose $\mathcal{Z} \subseteq \mathcal{Z}_{\pi, \varphi}$ let and simulators. of class a be $\mathbb{S}$

π UC-emulates φ within ε against $(\{\mathcal{D}\}, \mathbb{S}, \mathcal{Z})$.

Then

π UC-emulates φ within ε against $(\mathbb{A}, \mathbb{S}', \mathcal{Z}')$,

where

$$\begin{aligned} \mathbb{S}' &= \{S[A] : S \in \mathbb{S}, A \in \mathbb{A}\}, \\ \mathbb{Z}' &= \{Z \in \mathbb{Z}_{\pi, \varphi} : Z.\text{par} \cap \{\text{id} : \text{id}.F = Z\} = \emptyset \\ &\quad \wedge Z^A \in \mathbb{Z} \text{ every for } A \in \mathbb{A}\}. \end{aligned}$$

. $\mathcal{D}$ for supplies hypothesis the simulator the be $S_{\mathcal{D}} \in \mathbb{S}$ let Explicitly, Then

$$\forall A \in \mathbb{A} \quad \forall Z \in \mathbb{Z}' :$$

$$\text{Adv}_{\pi, \varphi}^{\text{uc}}(Z, A, S_{\mathcal{D}}[A]) \leq \text{Adv}_{\pi, \varphi}^{\text{uc}}(Z^A, \mathcal{D}, S_{\mathcal{D}}) \leq \varepsilon,$$

on depends which $S_{\mathcal{D}}[A]$ is for asks 4.11 Definition simulator the so Z on not and alone A

la- world, per one regroupings, two is argument the of whole The اثبات. the of advantages two the them, Granting below. (R2) and (R1) belled second, the bounds Z^A at hypothesis the and term by term agree display ex- right-hand the — rebracketing pure a is Neither theorem. the is which at standing relay the not, do ones left-hand the machine a carry ecutions rather 4.18 Lemma of half ceding the are both so — identities ceded the hy- three its directly, applied lemma that is (R1) 's. 4.19 Proposition than again run argument same the is (R2) three, theorem's the being potheses because own its of argument an needs it and, $\mathcal{D}$ for $S_{\mathcal{D}}$ and π for φ with of out than rather another to bundle one from moves adversary the there slot the in machine the of used proof lemma's the what execution: the paragraph closing The well, as $S_{\mathcal{D}}[A]$ of hold to shown and recounted is hypothesis the alone, A on depends $S_{\mathcal{D}}[A]$ — order quantifier the checks the be $S_{\mathcal{D}} \in \mathbb{S}$ let and $Z \in \mathbb{Z}'$ and $A \in \mathbb{A}$ Fix Z of nothing knowing $Z^A \in \mathbb{Z}$ gives $\mathbb{Z}'$ in Membership $\mathcal{D}$ for supplies hypothesis the simulator so $Z^A.\text{par} = Z.\text{par} \supseteq \langle \pi \rangle \cup \langle \varphi \rangle$ and it, to applies hypothesis the so of conventions The defined, is display the of right the on advantage the Initialize call-free tapes, unmerged coins, independent — 4.19 Proposition world: per one regroupings, two is work The force, in are —

$$\text{Exec}(\pi, Z, A, \mathcal{C}) \equiv \text{Exec}(\pi, Z^A, \mathcal{D}, \mathcal{C}), \quad (\text{R1})$$

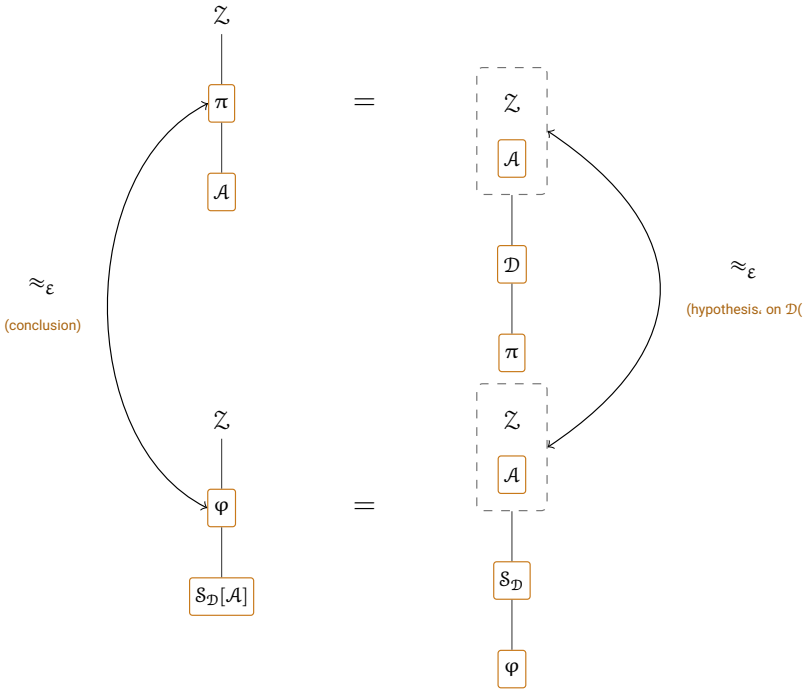
$$\text{Exec}(\varphi, Z, S_{\mathcal{D}}[A], \mathcal{C}) \equiv \text{Exec}(\varphi, Z^A, S_{\mathcal{D}}, \mathcal{C}), \quad (\text{R2})$$

agree advantages two the them, granting distributions: of equality an each is which, ε by second the bounds Z^A at hypothesis the and term by term

executions right-hand the – rebracketing pure a is Neither display, the slot adversary the in relay the not, do ones left-hand the machine a carry 's. **4.19** Proposition than rather **4.18** Lemma of half ceding the are both so – $\mathcal{Z}^H$ that so, $H_{\text{keep}} = \emptyset$ with $H = \{A\}$ at **4.18** Lemma (R1). of Proof Its $\mathcal{D}$ is $\text{IDs}(A)$ ceded the at relay the and **4.26** Definition of $\mathcal{Z}^A$ the is is: A because refusal-oblivious is A theorem's: the are hypotheses three sys- the on hypothesis the is which call, responsive a places π of core no legal, are families occupant The -identity. Z no silences $\mathcal{Z} \in \mathbb{Z}'$ and tems: (R2). of Proof go. let A identities the $\mathcal{D}$ and alone $\text{IDs}(\mathcal{Z})$ occupying $\mathcal{Z}^A$ execu- the of out than rather another to bundle one from moves A Here its is argument the and face its on apply not does **4.18** Lemma so tion, a is transfers it That $\mathcal{D}$ for $\mathcal{S}_{\mathcal{D}}$ and π for φ with again run half ceding which slot, the in machine the of used proof lemma's the what of matter dispatches it that verdicts, 'sGuard $_A$ returns guard its that than more no is a reads A hosted the where – and $\text{id}.F = Z$ has claim the whether on the against left the on bundle the inside $\mathcal{S}_{\mathcal{D}}$ from **rej** instruction, refused ex- refusal-oblivious, is A that – right the on slot the across $\perp$ deemed left, the on $\mathcal{S}_{\mathcal{D}}[A] - A$ holds machine which is differs What (R1). in as actly the perform two the make **4.26** and **4.27** Definitions and – right the on $\mathcal{Z}^A$ in- guarded machine's receiving the through routed each hand-offs, same across and side one on machine a inside travelling instruction the terface, in same the is $\mathcal{Z} \rightarrow A \rightarrow \mathcal{S}_{\mathcal{D}} \rightarrow \varphi$ chain The other. the on execution the same the in instructions same the meets $\mathcal{S}_{\mathcal{D}}$ point: different a at cut both, is occupant the and verbatim, holds Siting claims, same the under order **N** and occupancy, by **P** and **pid** – way either fields same the by judged fields 's $\mathcal{S}_{\mathcal{D}}$ carrying inherits, $\mathcal{S}_{\mathcal{D}}[A]$ and meets $\mathcal{S}_{\mathcal{D}}$ which, **4.7** Definition by left- the routes **4.27** Definition rest, the reads guard no 's. **4.27** Definition by on serves **Exec** that interface guarded same the through hand-off hand under bundle the leave $\mathcal{Z}$ to back calls own 's. A hosted the and right, the right, the on **Exec** cross they as just (A, P) are display the of advantages two the (R2) and (R1) By Conclusion. sim- The ε by second the bounds $(\mathcal{D}, \mathcal{Z}^A)$ at hypothesis the and equal, knowing hypothesis the alone, A on depends and $\mathcal{S}'$ in lies $\mathcal{S}_{\mathcal{D}}[A]$ ulator was $\mathcal{Z} \in \mathbb{Z}'$ met: is **4.11** Definition of order quantifier the so $\mathcal{Z}$ of nothing $\square$ arbitrary.

Theo- as square commuting same the is proof the Diagrammatically, half ceding the from time this built proof), its above figure (the 's. **4.20** rem

part not is $\mathcal{Z}$ into moves what half: keeping the than rather 4.18 Lemma of stood. it where stands relay a and itself, adversary the but system the of



through- untouched φ and π starts: theorem the where is left Bottom before interposed $\mathcal{A}$ with $\mathcal{S}_{\mathcal{D}} - \mathcal{S}_{\mathcal{D}}[\mathcal{A}]$ and occupant real-world the $\mathcal{A}$ out, and (R1) are edges bottom and top The one, ideal-world derived the – it again is read, is it side whichever on, $\mathcal{Z}$ into $\mathcal{A}$ absorbing proof: the of (R2) this but – was '4.19 Proposition as exactly distributions, of equality an (a)–(c) hypotheses it, for paying 4.18 Lemma of half ceding the is it time re- a and itself adversary the is moves what because nothing, of place in serves $\mathcal{Z}^{\mathcal{A}}$ absorbed same The place, its in stand to has ($\mathcal{S}_{\mathcal{D}}$ then, $\mathcal{D}$) lay the is it all: at them compare can edge right the why is which sides, both exactly square the Chasing $\mathcal{D}$ against only stated hypothesis, theorem's for now conclusion, theorem's the – edge left the, 4.20 Theorem in as the forced: is – machine fixed one than rather $\mathcal{A}$ refusal-oblivious every quantity, same the equalities, two the by are, 's $\approx_\epsilon$ two rea- same the for and, 4.20 Theorem in as move classes three The the and over, quantifies conclusion the those are adversaries The sons.

adversary, per one computed, are simulators the ; $\mathcal{D}$ only had hypothesis the and supplies; hypothesis the simulator the before it interposing by it what when exactly counting $\mathcal{Z}$ outer an preimage, a are environments $\mathcal{Z}$ Taking covers. hypothesis the one is $\mathcal{A}$ each absorbing on becomes less – again one largest the $\mathcal{Z}'$ makes class admissible largest the be to no condition a -identities. $\mathcal{Z}$ own their silence that environments the only the because excluded wants. application no and imposes admissibility en- an such which -claim. $\mathcal{Z}$ a under leaves 4.26 Definition of instruction par leaves absorption since – alone side one on refuse would vironment alone.

cost constructions two the .4.5 Section of sense the in concretely Read where $\mathbf{z} \oplus \mathbf{a}$ needs $\mathcal{Z}^{\mathcal{A}}$ then -bounded $\mathbf{a}$ is $\mathcal{A}$ if costs: adversary the what invocation The . $\mathbf{s}$ needed $\mathcal{S}_{\mathcal{D}}$ where $\mathbf{s} \oplus \mathbf{a}$ needs $\mathcal{S}_{\mathcal{D}}[\mathcal{A}]$ and . $\mathbf{z}$ needed $\mathcal{Z}$ indepen- both maximum, a take times per-invocation the and add counts . φ and π of dent

Security Concrete 4.5

about nothing says and classes arbitrary against stated is 4.11 Definition turns resources bounded of machines with them Filling contain. they what rela- the and one, concrete a into section previous the of statement each budgets. on arithmetic into classes between tions

تعريف 4.30 (Budgets bounded and) $\mathbf{a}$ = pair $\mathbf{a}$ is budget A .classes) bounded and (Budgets 4.30 $\mathbf{a}$ steps t most at takes it if -bounded $\mathbf{a}$ is machine A naturals. of (t, r) at for runs it so all: in calls r most at answers and invocation each in in bounds both meeting machine A execution. any in steps t r most rule halting the not or whether -bounded $\mathbf{a}$ counted is execution every are below machines bounded from built bundles the – it into written is of classes the for $\mathbb{Z}_{\pi, \varphi}(\mathbf{a})$ and $\mathbb{S}(\mathbf{a})$. $\mathcal{A}(\mathbf{a})$ Write sense. this in bounded -bounded $\mathbf{a}$ of and simulators, -bounded $\mathbf{a}$ of adversaries, -bounded $\mathbf{a}$ ma- componentwise: ordered are Budgets . $\mathbb{Z}_{\pi, \varphi}$ in lying environments simulator $\mathbf{a}$ or several. of bundle $\mathbf{a}$ – execution an share that chines by combine – adversary an running

$$(t, r) \oplus (t', r') := (\max(t, t'), r + r'),$$

takes time per-invocation the so activation, per running machine one add. counts invocation the while bound larger the

Bounding it. about promise a not machine. the of part is budget The the — execution an bound to enough yet not is machines slot three the it- on recurses that core a and too. machines are system a of members token the carry would forever. another one calling members two or self. Concrete it. reclaim to nothing with machine budgeted every from away ev- of member every :systems budgeted over read therefore statements def- this of sense the in own. its of budget a carries compared system ery machines. many finitely runs then execution An verbatim. applied inition any at steps machine one exactly and steps. many finitely taking each of $\sum_i t_i r_i$ bound step total the within halts execution the so moment. a given is budget a at Halting outright. defined is **Exec** and machines its on suspended caller each resumes mid-call halts that machine a value: read is return root its from away halts that environment an and , $\perp$ with it $A(a) \subseteq A(a')$ budget. the in monotone are classes The .0 returning as 4.23 Lemma makes That . $\mathbb{Z}_{\pi, \varphi}$ and $\mathbb{S}$ for likewise and . $a \leq a'$ whenever the growing budget. adversary the shrinking numbers: about statement a em- preserve each budget environment the shrinking or budget. simulator ulation.

sys- budgeted be φ and π Let .advantage) UC (Concrete 4.31 **تعريف** and simulator. the and adversary the for budgets be s and a let tems. Set environments. of class a be $\mathbb{Z} \subseteq \mathbb{Z}_{\pi, \varphi}$ let

$$\text{Adv}_{\pi, \varphi}^{\text{uc}}[a, s, \mathbb{Z}] := \max_{A \in A(a)} \min_{S \in \mathbb{S}(s)} \max_{Z \in \mathbb{Z}} \text{Adv}_{\pi, \varphi}^{\text{uc}}(Z, A, S),$$

largest the , $\mathbb{Z} = \mathbb{Z}_{\pi, \varphi}(z)$ case the for $\text{Adv}_{\pi, \varphi}^{\text{uc}}[a, s, z]$ abbreviate and budget. that of class

largest the because budget a than rather class a is argument third The compare below statements the and systems. of pair the on depends class needed is class common a Where pairs. different over taken advantages so. say to able be must notation the

read 4.11 Definition of quantifiers three the are operators three The is simulator the reason: same the for and order same the in value. a as are extrema The environment. the before and adversary the after chosen rather finiteness is it and moment. a takes reason the though attained. all. in steps $t r$ most at takes machine -bounded a An compactness. than up coins: $t r$ most at and input any of symbols $t r$ most at reads it so are there and did. it what to read it what from map a is it behaviour to

probabilities dyadic with them weight coins Its maps. such many finitely
 behaviour many finitely induces class bounded a so , 2^{tr} denominator of
 well. as budgeted systems the of members the With all. in distributions
 so behaviours. of choice each at number well-defined a is $\Pr[\text{Exec} = 1]$
 and values. many finitely of one takes 4.5 Definition of advantage each
 an (Over attained. is values many finitely over minimum or maximum a
 Therefore supremum.) a as read is max outer the $\mathbb{Z}$ class arbitrary

$$\pi \text{ UC-emulates } \varphi \text{ within } \varepsilon \text{ against } (\mathbb{A}(\mathbf{a}), \mathbb{S}(\mathbf{s}), \mathbb{Z}_{\pi, \varphi}(\mathbf{z}))$$

is minimum attained the $-\varepsilon$ most at is quantity that when exactly holds
 se- concrete A boundary. the at exact direction reverse the makes what
 proba- a and budgets three relating inequality single a then is claim curity
 sight. in asymptotics no and parameter security no with bility.

Properties

one: of something asks property A systems. two compares Emulation
 5.9 Proposition which – it against played game a win adversary no that
 envi- the into absorbing slot the winning, not dummy's the to reduce will
 func- a is game A either. for wanted is machinery new Little ronment.
 and it. plays that adversary the is environment the other, any like tionalty
 what call may who settles already sections previous the of plumbing the
 and event, an and class, environment an system, a is property a so –
 on do rules game's a that work the of most do 1.1 Definition of fields the
 hy- one comes: it as named is and small, is added be to has What paper.
 4.4 Section conditions three the and system, game the on condition giene
 moves. it anything of asks already

function- standard a is $\mathcal{F}$ for game A .system) game (Game. 5.1 تعريف
 with $\mathcal{G}_m$ ality

$$\begin{aligned}\mathcal{G}_m.\text{pid} &:= (\mathcal{G}_m.F, 0, 0), & \mathcal{G}_m.\mathbf{P} &:= \mathcal{F}.\mathbf{P} \cup \{Z\}, \\ \mathcal{G}_m.\mathbf{N} &:= \mathbf{Z}, & \mathcal{G}_m.\mathbf{U} &:= \{(\mathcal{F}, \text{serves}_{\mathcal{G}_m})\},\end{aligned}$$

interfaces :finalizations more or one interfaces own its beside carrying
 by borne and them for reserved prefix a ,Fin with begin names whose
 and bit, a returning each root, the at served each interface, other no
 one any once served – otherwise or finalization – $\mathcal{G}_m$ of interface no
 Here been. has them of

$$\text{serves}_{\mathcal{G}_m}(\mathcal{G}_m, \mathcal{F}) : \Longleftrightarrow \mathcal{F}.\mathbf{P} \supseteq \mathcal{G}_m.\mathbf{P} \setminus \{Z\} \wedge \mathcal{G}_m.\text{pid} \in \mathcal{F}.\mathbf{N}$$

$\gamma := \{\mathcal{G}_m\} \cup \sigma$ is system game A exempted. root the with serves is
 containing not and uses $\mathcal{F}$ whatever and $\mathcal{F}$ containing σ system a for
 .6.5 Remark of sense the in $\{\mathcal{G}_m\}$ at tight γ with and $\text{wf}(\gamma)$ with , $\mathcal{G}_m$

over FinUF and FinCor properties: several carry may then, game. One was what record that oracles the sharing say, functionality, signature a en- The close. the at them off read verdict the in only differing and signed finalization which choosing by on judged be to which chooses vironment several with game a So does. it thing last the is choice that and call, to several, holding shell a but properties of conjunction a not is finalizations one maximizes that environment an below, advantage own its with each to. it asks here nothing and another, maximize not need

shell the $\mathcal{F}$ for $\mathcal{G}_m$ Game

$\text{par} := \perp$ $\mathcal{U} := \{(\mathcal{F}, \text{serves}_{\mathcal{G}_m})\}$ $\mathcal{N} := \mathcal{Z}$ $\mathcal{P} := \mathcal{F} \cup \{\mathcal{Z}\}$ $\text{pid} := (\mathcal{G}_m.F, 0, 0)$

:Initialize()

:1 $\text{done} \leftarrow \text{false}$
 :2 ... the game's own variables

id' from id.Op(in)
 / closed once finalized

:3 **require** $\neg \text{done}$
 :4 ... the game's own code, which may call $\mathcal{F}$ at party id.P
 :5 **return** out

id' from id.Leak()
 / a game keeps no party's secrets

:6 **return** $\perp$

shared is $\mathcal{F}$ for shell the property: per one $\mathcal{G}_m$ of Finalizations

game the closing each root, the at served each, Fin ... named each

finalization) every of shape (the id' from id.Fin()

:7 **require** $\neg \text{done}$ / one finalization only
 :8 **require** $\text{id.P} = \mathcal{Z}$ / at the root, which stays honest
 :9 $\text{done} \leftarrow \text{true}$
 :10 $w \leftarrow$ this property's verdict, read off the shell's state
 :11 **return** w

id' from ... id.FinUF() , id.FinCor()

:12 ... the same three lines, then each property's own verdict

what, by saying worth is it and forced, is field Each

and session the so experiment, per game one is There **id. process The**
 con- a not is 0 Session .0 at both fix we and information no carry instance
 its claiming by $\mathcal{G}_m$ reaches environment the requirement: a but vention
 then 3 line and $\mathcal{Z} \subseteq \mathcal{G}_m.\mathcal{N}$ because admits 2 line which -identity, $\mathcal{Z}$ own
 obser- that is 7.17 Remark global, being not $\mathcal{G}_m - \mathcal{G}_m.s = \mathcal{Z}.s = 0$ asks
 session, caller's the shares $\mathcal{F}$ local a 1.3 Definition By general, in vation

machines the where is which ,0 session in sits experiment whole the so
too. live supplies execution the

Guard and (5 (line identity own its only claims core A **parties. served The**
other: no at and P party at $\mathcal{F}$ reaches P party at $\mathcal{G}_m$ so $\text{id}' . P = \text{id} . P$ asks
chooses environment the and time. a at party one $\mathcal{F}$ exercises game the
free being root its address. to $\mathcal{G}_m$ of identity which choosing by party the
finalizations. the for added is root The $\mathcal{F} . P$ Hence .3 line by any for act to
why. says paragraph next the and

$\mathcal{G}_m$ make would which .Std Not else. nothing and **Z callers. admitted The**
which .A not game: the finalize functionality standard any let and global
caller every refuses 2 line alone. **Z** With adversary. the to it hand would
environment. the but

and game the renaming so: says 7.16 Corollary and Arbitrary. **name. The**
game a against proved property a so probability. no changes together $\mathcal{F} . N$
because matters This other. every under it against holds name one of
the name to has $\mathcal{F} . N$ so identity. own its only claim game the lets 5 line
the protocol: that names protocol a inside ships that $\mathcal{F}$ the while game
arbitrary not is What property. same the two the makes what is corollary
touches. renaming no which code. caller's the is

the is root the because root the exempts serves $\mathcal{G}_m$ **dependence. The**
game the party a serve not need $\mathcal{F}$ identity: bookkeeping own game's
claimed a verifying – $\mathcal{F}$ query must verdict whose game A at. it calls never
during record ones: usual the are they and options. two has – say forgery.
as root the serve to $\mathcal{F}$ ask or needs. verdict the whatever calls oracle the
.serves becomes serves $\mathcal{G}_m$ when well.

gap the and own. their on suffice not do above fields The **Tightness.**
may it so $\langle \gamma \rangle$ on silenced is environment admissible An 's.6.5 Remark is
from only it forbids 4.10 Definition but – uses γ name a of identity no claim
inter- full a is functionality hosted a and occupies. γ identities at hosting
(.4.13 (Definition silencer own its through identity own its claiming face
of id process unoccupied an at machine a host may environment an So
the of instance second invented an say. $(\mathcal{G}_m.F, 0, 1)$ at name: used a
the – name by caller its admits that σ of member A name. own game's
every pass calls its and machine. that admits – 1 Chapter of pattern local
agree. sessions the agree. parties the .N in lies name the :Guard of line
then would environment The party. honest an at fire not does **Mediate** and
 $\Pr[\text{Win}_{\text{Fin}}] = 1$ and of. nothing records game the that $\mathcal{F}$ on oracle an hold

it something produced adversary the that is verdict whose game any for given. not was mem- Every completely. it refuses and that. refuses $\{G_m\}$ at Tightness in- an so occupies. system the ids process whole admits game the but ber every makes condition party 's6.5 Remark and :2 line fails instance vented 4.10 Definition where occupied. from speak could caller pinned a identity of member a to — open leaves tightness route one The hosting. forbids func- hosted a : $G_m.N = Z$ by closed is — game the is here which itself. I excludes. Z which id. process standard a claims and standard is tionality The own. its but hosted not are -identities Z own environment's the while ids. process of class bare a admit may that member one the thus is game name. standard no is Z because exactly so doing in safe is it and

$\{0, 1\}^*$ in ids Party .root) the at served is finalization a (Why 5.2 نکته party's corrupt a hands interface full the of 4 line and corrupted. be can such at served finalization A runs. core the before adversary the to call that whenever adversary the by answered be therefore would party a whatever be would won?" I "have to answer the and corrupt. is party in- check no and exactly. problem 's3.1 Remark — says adversary the above. wrapper the in happening mediation it. repair can core the side FullCorrupt it: supplies already framework the and fix. the is root The $(G_m.pid, Z)$ at Mediate so .C enters never Z so .id.P $\in \{0, 1\}^*$ asks own. game's the always is computes 10 line bit the and fire never can ma- supplied three the keep to introduced were ids party reserved The 8 Line cost. extra no at honest game the keep they honest: chines place a is other no so identity. that to finalization every confines then win. to

finalization one exactly That .last) and once. finalization. (One 5.3 نکته enforced is answers. game the call last the is it that and answered. is a sets 9 Line jobs. both does flag one the and — assumed than rather finalization. later every refuses 7 line bypass. cannot wrapper the flag interface other every refuses 3 line and another's. or name own its environ- the so .3.6 Section by rej returns **require** refused a thereafter: honest an at — else nothing and closed is game the that learns ment en- is core the before intercepts 4 line one corrupt a At least. at party. refused: than rather adversary the by answered is caller the so tered.

state game's the write to unable being slot the nothing, changes that environ- the stops Nothing already. fixed being event winning the and – right is which either, afterwards functionality global a calling ment won. it what change cannot then does it what and over, is game the tables: its re-tests $\mathcal{F}_{\text{Sig}}$ reason the for matters lines the of order The instance suspended a lets 3.6 Section and calls, place may 10 line rather computed is verdict the before set is flag the so re-entered, be could finalizations nested two about, way other the it Were after, than open, game the find each safe finalizations several makes what is all them governs flag one That environ- an own, its of flag a keep to each Were shell, one off hang to run one in properties two on judged be and twice finalize could ment ver- unforgeability its state a off verdict correctness a say, reading, – them, between oracles the with from, computed been already had dict the why is it and property, one to run a confines what is done Sharing than rather flag shell's the with own their of box a in sit finalizations bookkeeping, own its carrying each

$P \in \text{party } a$ At .costs) it what and buys, corruption (What 5.4 نکته
function- other any like wrapped are interfaces own game's the $\{0, 1\}^*$
does core the :P at $\mathcal{G}_m$ to call a intercepts 4 line $P \in \mathbf{C}$ once so ality's,
the by answered is caller the and nothing, records game the run, not
an replaces Mediate – nothing environment an gains That adversary.
writ- never is state game's the so core, a running than rather answer
Since party, that at oracle the it costs it and – adversary the by ten
party a corrupting, (3.1 Section alone party by indexed is corruption
wants that game A too, party that at oracle game's the closes $\mathcal{F}$ of
the somewhere it serve therefore must party corrupt a at oracle an
outside own its of party a at or root, the at reach: not does corruption
. $\mathcal{F}$ on call no needs oracle that whenever do may it which, $\mathcal{F}.P$
the of either not is cores game's the of out adversary the keeps What
party, corrupt a at $\mathcal{A}$ admits 3 Line for, reach would one lines two
does it so, id'. $F \neq A$ tests Mediate and ones: honest at only it refusing
at slot the from call A places, itself adversary the call a intercept not
3.5 Section as exactly core, 's $\mathcal{G}_m$ run therefore would party corrupt a
 $\mathcal{G}_m.N = \mathbf{Z}$ is it refuses What .. $\mathcal{A}$ by addressed party corrupt a of says
not is field that of narrowness The does, else nothing and, 2 line at

own game's the and adversary the between thing only the but tidiness code.

Guard through go not does leakage because $\perp$ returns $\text{Leak}'s\mathcal{G}_m$ And whatever party corrupt a at it read can adversary the : (3.2 Section to willing not is it there nothing keep must game a so says. $\mathcal{G}_m.N$ over. hand

in- many as address may game A. ($\mathcal{F}$ of instances (Several 5.5 نکته
in Instances differ. can it ways two the and likes, it as $\mathcal{F}$ of stances
sessions comparing 3 line extra, nothing need session own game's the
every for reachable is $(\mathcal{F}.F, 0, i)$ so number, instance the ignoring and
waives what is which. $\text{global}(\mathcal{F})$ need sessions other in Instances. i
silencing 5 line identity. own 's $\mathcal{G}_m$ is claim the way Either line. that
same the and one sees instance every so else, everything on cores its
really instances the And once. decided is control access and caller
differ fields their and code same the run they 7 Chapter by copies: are
and state own its each gives 3.6 Section while alone. id process the in
– sessions n keys, n – property multi-instance a So coins. own its
convention extra no with them, count game the having by written is
means. instance second a what about
Proposi- cheaper. transfer make is do not does them counting What
every instances n over game a so subsystem, one replaces 5.10 tion
Theo- by $n\epsilon$ and applications n costs replaced be to is which of one
records 7.18 Remark as exactly – hybrids of chain a along 4.24 rem
the for property the transfers application One side. relocation the on
others. the of nothing says and contains φ instances

$\mathcal{G}_m.N = \mathbf{Z}$. be) not need and session-uniform, not is game (A 5.6 نکته
the and $\mathcal{G}_m$ of fails 7.2 Definition so global, not is $\mathcal{G}_m$ while $\mathbf{Z}$ meets
it as is That system. game a to apply not do 7 Chapter of results
above. paragraph the by 0 session to pinned is game A be. should
something not is it relocating and experiment, per it of one is there
whose test, under $\mathcal{F}$ the is to applies chapter the what wants: anyone
counts. 5.5 Remark instances

already 4 Chapter class the by settled is play may environments Which
system. single a at taken defines.

for admissible environments The .environments) (Property 5.7 **تعريف** $\mathcal{Z}.\text{par} \supseteq \langle \gamma \rangle$ with those :4.10 Definition of $\mathbb{Z}_{\gamma, \gamma}$ are γ system game a that them among those for $\mathbb{Z}_{\gamma}^{\circ}$ Write .IDs(γ) of identity no at host that can 5.9 Proposition of absorption the ones the -identity, $\mathcal{Z}$ no silence move.

the on Silenced say. to them needs game a what say clauses two The so uses. γ name a of identity no claim may environment an closure. name instance further a as or subroutines. 's $\mathcal{F}$ of one as $\mathcal{F}$ as speak cannot it . $\mathcal{G}_m$ reach which -identities, $\mathcal{Z}$ own its is claim may it what them: of any of envi- the So .Std of all admits what only reach which names. fresh and directly $\mathcal{F}$ touches and interfaces 's $\mathcal{G}_m$ through game the plays ronment is which may. anyone as it touches it then and – global is $\mathcal{F}$ when only the in grants already 4.10 Definition what and means global being what against property a therefore is $\mathcal{F}$ global a of property A setting. emulation that. knowing written be must game the and it. shares that adversary an also it and .3.3 Definition of hygiene occupancy the is clause hosting The game. second a installing from environment the stops

game a be $\gamma = \{\mathcal{G}_m\} \cup \sigma$ Let .advantage) property (Win, 5.8 **تعريف** admissible be $\mathcal{Z}$ let finalizations. 's $\mathcal{G}_m$ of one be Fin let system. Write slot. adversary the occupy $\mathcal{X}$ let and .(5.7 (Definition γ for in 1 returns $(\mathcal{G}_m.\text{pid}, \mathcal{Z})$.FullFin interface the that event the for Win_{Fin} of one is σ of Fin property the in $\mathcal{Z}$ of advantage The .Exec($\gamma, \mathcal{Z}, \mathcal{X}, \mathcal{C}$)

$$\text{Adv}_{\mathcal{G}_m, \sigma, \mathcal{Z}, \mathcal{X}}^{\text{Fin}} := 2 \Pr[\text{Win}_{\text{Fin}}] - 1,$$

$$\text{Adv}_{\mathcal{G}_m, \sigma, \mathcal{Z}, \mathcal{X}}^{\text{Fin}} := \Pr[\text{Win}_{\text{Fin}}],$$

over being probability the reading. search the and reading decision the meant is two the of Which execution. the in machine every of coins the this by not and given. is game the where finalization each for fixed is choose. to how says below calibration on paragraph the definition: when σ for $\mathcal{F}$ write and . $\mathcal{D}$ is it when subscript the from $\mathcal{X}$ drop We class a and environments of $\mathcal{Z} \subseteq \mathbb{Z}_{\gamma, \gamma}$ class a . $\varepsilon \geq 0$ For . $\sigma = \{\mathcal{F}\}$ against ε within Fin property the has σ say we occupants. slot of $\mathbb{X}$ within and . $\mathcal{X} \in \mathbb{X}$ every and $\mathcal{Z} \in \mathbb{Z}$ every for $\text{Adv}_{\mathcal{G}_m, \sigma, \mathcal{Z}, \mathcal{X}}^{\text{Fin}} \leq \varepsilon$ if $(\mathcal{Z}, \mathbb{X})$. $\mathbb{X} = \{\mathcal{D}\}$ when $\mathcal{Z}$ against ε

oc- Win_{Fin} one most at .(5.3 Remark game the closes flag one Since game a and disjoint. are finalizations distinct of events the run: any in curs executions several in measured properties several is them of several with one. off read numbers several than rather

Def- of that where signed. is advantage the reading decision the Under cannot environment an point: the is asymmetry the and not. is 4.5 intion rather verdict game's the being event the guess. a flip can it as Win_{Fin} flip to nothing is there and -1 scores win cannot it game a so own. its than of. value absolute an take

that and output. its than rather execution the of event an is Win_{Fin} returns environment the whatever is output execution's an deliberate: is that thing a but reported be to guess a not is property a and .(4 (Chapter makes 5.2 Remark because distinction. the by lost is Nothing happened. the bit the is computes 10 line bit The matters. it where agree two the en- the name may we so unmediated. environment. the hands finalization at return its with $\mathcal{Z}$ for $\mathcal{Z}^\star$ write .Fin fixed a for it: reports that vironment ($\mathcal{G}_m.\text{pid}, \mathcal{Z}$).FullFin on placed I call any if 1 "return by replaced root the place. its earns that of part Every otherwise". 0 and .1 with answered was party another at addressed finalization a because named is identity The adversary the by answered is and honest is party that where 8 line fails game's: the be not need back coming value the so corrupt. is it where be- named is operation The .5.2 Remark by happens. neither root the at the root. the at $\mathcal{G}_m$ of interface every serves 1 line so . $\mathcal{Z} \in \mathcal{G}_m.\mathbf{P}$ cause finalizations several with because and $-$ included oracles own game's per $\mathcal{Z}^\star$ one on. reporting is it property which say must form reporting the be- bit. a as answer the of reading a than rather 1 against is test The .Fin bit no is rej and .5.3 Remark by rej back comes finalization later a cause one first the because first. the than rather call any is it And .(3.6 (Section fails $\mathbf{P} \neq \mathbf{Z}$ for $(\mathbf{Z}, \mathbf{P})$ claiming environment an $-$ refused be may placed noth- served: is one later a while $-$ unset done leaving test. party 's2 line answered ever is call one most at since quantifying. by weakened is ing Exec So rest. the refusing and finalization one serving game the .1 with on call every it. compute can $\mathcal{Z}^\star$ and occurs. Win_{Fin} when exactly 1 returns caller. other no admits $\mathcal{G}_m.\mathbf{N} = \mathbf{Z}$ itself: placed it one being interface that $\mathbf{Z}$ which id. standard a claims hosts environment the functionality a and Then excludes.

$$\Pr[\text{Exec}(\gamma, \mathcal{Z}^\star, \mathcal{X}, \mathcal{C}) = 1] = \Pr[\text{Win}_{\text{Fin}} \text{ in } \text{Exec}(\gamma, \mathcal{Z}, \mathcal{X}, \mathcal{C})],$$

value the in only differing and calls same the placing machines two the

membership whose in is $\mathcal{Z}$ class every in $\mathcal{Z}^\star$ puts also That with. halt they budgeted the and them. among $\mathbb{Z}_{\gamma,\gamma}$ – alone hosting and par on turns back. bit a reading of step further one the to up 4.30 Definition of classes used. is this of all where is 5.10 Proposition

4.4 Section of dummy the is occupant slot default The **dummy. the Why** environment. the is it and adversary one has game a that is reason the and envi- an so strategy. no keeps and directions both in relays dummy The own 's $\mathcal{F}$ where and do: can slot the everything holds it driving ronment ideal an how is which 3.4 Section of $\mathcal{A}(\cdot)$ the – adversary the reach cores an- that environment the then is it – unpinned value a leaves functionality $\mathcal{X}$ Dropping it. give to means game a freedom the exactly is which swers. but generality of loss no thus is case dummy's the for subscript the from 5.10 Proposition because kept is form longer the plays: who of choice a not is nothing costs abbreviation the That slot. that in simulator a needs the 4.4 Section of construction the is reason the and taste. of matter a environment. the into absorbs slot

be $\gamma = \{\mathcal{G}_m\} \cup \sigma$ Let .environment) the into absorbs slot (The 5.9 گزاره
(9.2 (Definition calls responsive no place cores whose system game a
(Defini- slot adversary the of occupant refusal-oblivious a be $\mathcal{X}$ let
in lies again 4.26 Definition of $\mathcal{Z}^\mathcal{X}$ Then . $\mathcal{Z} \in \mathbb{Z}_\gamma^\circ$ let and (4.28 tion
 $\mathcal{G}_m$ of Fin finalization every for and , $\mathbb{Z}_\gamma^\circ$

$$\text{Adv}_{\mathcal{G}_m, \sigma, \mathcal{Z}, \mathcal{X}}^{\text{Fin}} = \text{Adv}_{\mathcal{G}_m, \sigma, \mathcal{Z}^\mathcal{X}}^{\text{Fin}}.$$

ε' within it has it . $\mathbb{Z}_\gamma^\circ$ against ε' within Fin property the has σ if So
Read occupants. refusal-oblivious of $\mathbb{X}$ class every for $(\mathbb{Z}_\gamma^\circ, \mathbb{X})$ against
occupant -bounded $\mathbf{x}$ an and environment -bounded $\mathbf{z}$ an concretely.
-bounded. $\mathbf{z} \oplus \mathbf{x}$ is that environment absorbed an give

gives $H_{\text{keep}} = \emptyset$ with $H = \{\mathcal{X}\}$ at 4.18 Lemma اثبات.

$$\text{Exec}(\gamma, \mathcal{Z}, \mathcal{X}, \mathcal{C}) \equiv \text{Exec}(\gamma, \mathcal{Z}^\mathcal{X}, \mathcal{D}, \mathcal{C}),$$

Now one. for one here. assumed three the being hypotheses three its
and .1 returns $(\mathcal{G}_m.\text{pid}, Z)$ at FullFin interface the that event the is Win_{Fin}
identity that occupies $\mathcal{G}_m$ – alone slot adversary the touches absorption
and alike. Fin each for probability one has event the so – sides both on
Sec- is Admissibility reading. either under number one gives 5.8 Definition
occupies environment an what occupies $\mathcal{Z}^\mathcal{X}$ that observation 's4.4 tion

is budget the is: $\mathbb{Z}$ when exactly admissible is it so ,par 's $\mathbb{Z}$ carries and in $\mathbb{X}$ each at read display the is sentence final The .8 Chapter of $\mathbf{z} \oplus \mathbf{a}$ the $\square$ turn.

plac- system game a more: no and 's4.29 Theorem are hypotheses The environ- an is so and ,9.4 Remark by this. outside is calls responsive ing 4.26 Definition of instruction the -identities. $\mathbb{Z}$ own its silences that ment occupant the what on condition a is Neither -claim. $\mathbb{Z}$ a under leaving in machine the whether care not need game a – point the is which does. moved. be can it that only fairly. plays slot the one same the is it and this. of all with comes condition scope One Theo- of regrouping the on rests above proposition The throughout. respon- no place cores whose systems for only holds which ,4.29 rem same the otherwise: fails relay the where showing 9.4 Remark calls. sive emula- the supplies what is which itself. 4.29 Theorem governs condition and class. adversary wider a from wants 5.10 Proposition hypothesis tion whose system game A once. at sides both on it inherits 5.11 Corollary must and three all outside sits therefore $\mathcal{A}^1$ through slot the reaches $\mathcal{F}$ responsive a What directly. established hypothesis dummy-only its have it. leaves 9.4 Remark where leave we like look would games of theory

takes finalization a readings two 's5.8 Definition of Which **calibration. The** reads $2 \Pr[\text{Win}_{\text{Fin}}] - 1$ factor The this. is rule the and taste. of matter a not is prob- with wins guesses that environment an :decision a as property the right is That .1 scores wins always that one and .0 scores and $\frac{1}{2}$ ability environment the bit a is verdict the where – strategy a is guessing where unforge- – property search a for wrong is It for. coin a flipped have could there because – outright hard be to meant is winning where say. ability. on ε bound a so , $\Pr[\text{Win}_{\text{Fin}}] = (1 + \text{Adv}_{\text{gm}, \sigma, \mathbb{Z}}^{\text{Fin}})/2$ reads quantity same the for which probability. winning the on $(1 + \varepsilon)/2$ bound a is advantage the half succeeding forger a permit would 0 of ε an scale: wrong the is ε small the Declaring itself. $\Pr[\text{Win}_{\text{Fin}}]$ take therefore properties Search time. the single a lets what is globally one fixing than rather finalization per reading second the of is paper this in property Every once. at kinds both carry shell out- impossible be to meant is below games the of each winning – kind search the take them of all so – guess a than better no merely not right. the is guessing where properties for there is reading decision the reading: them. among game a as written indistinguishability an strategy.

Its for. paid is it where is 5.10 Proposition and free. not is choice The decision the for 2ε – scales two the at bound same the are displays two

transfer inherits finalization each so – reading search the for ε reading, at inherits two the mixing game a and fixes. reading own its scale the at both.

the beside than rather way this properties define to reason the Finally, them. carries emulation framework:

be $\gamma = \{\mathcal{G}_m\} \cup \sigma$ Let .emulation) along transfer (Properties 5.10 گزاره well- be γ in π by φ of replacement the let , $\varphi \subseteq \sigma$ let system. game a a system game a be $\gamma[\pi/\varphi]$ let and ,4.2 Definition of sense the in formed Suppose well. as

π UC-emulates φ within ε against $(\{\mathcal{D}\}, \mathbb{S}, \mathbb{Z})$.

every and $\mathcal{G}_m$ of Fin finalization every for that such $\mathcal{S} \in \mathbb{S}$ is there Then condition a – $\mathcal{Z}^{\star \gamma \setminus \varphi} \in \mathbb{Z}$ has form reporting whose $\mathcal{Z} \in \mathbb{Z}_{\gamma[\pi/\varphi], \gamma}$ admissible largest the is $\mathbb{Z}$ whenever vacuous makes 4.16 Proposition absorption budget. a at class largest the is it when not though class. – more step one form reporting the and $\mathbf{c}_{\gamma \setminus \varphi}$ costing

$$\text{Adv}_{\mathcal{G}_m, \sigma[\pi/\varphi], \mathcal{Z}}^{\text{Fin}} \leq \text{Adv}_{\mathcal{G}_m, \sigma, \mathcal{Z}, \mathcal{S}}^{\text{Fin}} + 2\varepsilon \quad \text{reading), (decision}$$

exe- two the of events winning the for Win'_{Fin} and Win_{Fin} writing and. one ideal the and one real the – compares above display the cutions

$$\Pr[\text{Win}_{\text{Fin}}] \leq \Pr[\text{Win}'_{\text{Fin}}] + \varepsilon \quad \text{reading). (search}$$

be to checked is statement the First bound. one and moves Three اثبات. $\gamma[\pi/\varphi]$ on hypothesis the where is which names, it systems the about oc- new the at set party game's the neither supplies emulation spent: is of part is tightness and .N occupant's that in id process its nor cupant the turns 5.8 Definition of form reporting the Then besides. 5.1 Definition and root the at returns $\mathcal{Z}$ what rewriting bit. output an into event winning envi- the inside itself game the moves absorption Then run. no changing distin- one becomes game the playing environment an that so ronment. that at applied 4.20 Theorem is left is What systems. two the guishing deci- the gives: it bound the is reading search The environment. absorbed .Fin reads argument the in nothing and doubled. bound that is reading sion are advantages two the First. separately. finalization each of holds it so $\mathcal{G}_m \notin \sigma$ Since names. statement the systems the about are and defined both in lies $\mathbb{Z}_{\gamma[\pi/\varphi], \gamma}$ and $:\gamma[\pi/\varphi] = \{\mathcal{G}_m\} \cup \sigma[\pi/\varphi]$ so , $\mathcal{G}_m \notin \varphi$ have we

a when weakening 4.10 Definition of clause each $\mathcal{Z}_{\gamma[\pi/\varphi], \gamma[\pi/\varphi]}$ and $\mathcal{Z}_{\gamma, \gamma}$ property a is side left-hand the That one. to shrinks closures two of union not is it and spent. is $\gamma[\pi/\varphi]$ on hypothesis the where is all at advantage occupant new the at $\mathcal{G}_m. \mathbf{P} = \mathcal{F}'. \mathbf{P} \cup \{Z\}$ neither supplies emulation free: 4.4 Theorem of (ii) hypothesis being second the $\mathcal{G}_m. \text{pid} \in \mathcal{F}'. \mathbf{N}$ nor $\mathcal{F}'$ of part is tightness buys: it all that is Nor 4.20 Theorem of part no and pin members ' π ' that addition in asks hypothesis the so 5.1 Definition least that supplies Emulation did. ' φ ' where id process by callers their the of nothing saying and boundary the at behaviour comparing all. of it. behind fields declared

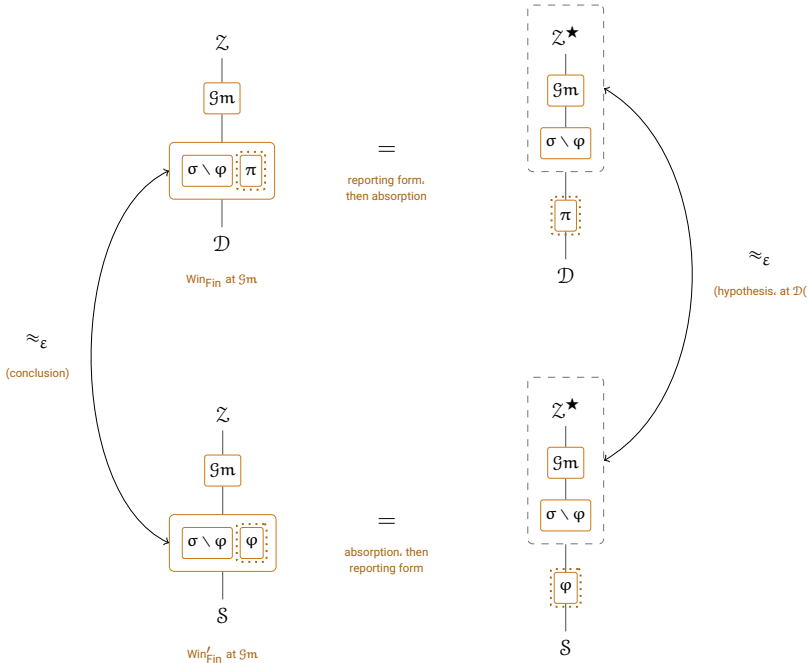
depends it $\mathcal{D}$ for supplies hypothesis the simulator the be $\mathcal{S} \in \mathbb{S}$ Let the take and Fin fix $\mathcal{Z}$ a such Fix $\mathcal{Z}$ every serves $\mathcal{S}$ one so alone. $\mathcal{D}$ on 5.8 Definition after paragraph the from Fin that for $\mathcal{Z}^\star$ form reporting gives which

$$\Pr[\text{Exec}(\delta, \mathcal{Z}^\star, \mathcal{X}, \mathcal{C}) = 1] = \Pr[\text{Win}_{\text{Fin}} \text{ in } \text{Exec}(\delta, \mathcal{Z}, \mathcal{X}, \mathcal{C})]$$

$\mathcal{Z}^\star$ places hypothesis the $\mathcal{X}$ occupant either and δ system game either for $\mathcal{Z}^\star \in$ and replacement. this to attaches 4.20 Theorem that $\mathcal{Z}'$ class the in host. they what on and par on agree two the and is $\mathcal{Z}$ because $\mathcal{Z}_{\gamma[\pi/\varphi], \gamma}$ adversary the at γ in π by φ of replacement the to applied 4.20 Theorem left-hand two the of difference the bounds $\mathcal{Z}^\star$ environment the and $\mathcal{D}$ $\mathcal{X} = \mathcal{S}$ and $\delta = \gamma$ against $\mathcal{X} = \mathcal{D}$ and $\delta = \gamma[\pi/\varphi]$ with – probabilities which two. the between ε most at by differs $\Pr[\text{Win}_{\text{Fin}}]$ Hence ε by – being reading. decision the under advantage the display: second the is order The first. the is which 2ε most at by differs then $2\Pr[\text{Win}_{\text{Fin}}] - 1$ the what is bound probability the proof: for than rather use for matters the Under doubled. it is bound decision-reading the and gives. argument is display second the so probability. the is advantage the reading search Nothing arises. never two of factor the and statement transfer the already $\square$ separately. finalization each of holds it so Fin reads argument the in

and one. textbook the is reduction The proof. that about things Two Defi- of $\mathcal{Z}^{\star \gamma \setminus \varphi}$ environment absorbed the so: it makes what is absorption environ- “an so $\gamma \setminus \varphi$ of rest the with along itself game the hosts 4.15 nition distinguish- environment “an becomes “ π against game the playing ment 4.20 Theorem what is which it. inside game the moving by “ φ from π ing the in $\mathcal{S}$ against stated is right the on property the And about. proved was the keeps 5.8 Definition why is which dummy. the against than rather slot meet need anyone obligation an not is it but – argument an as occupant well. as environment the into absorbs slot the because form. that in

move the and move one is it because drawing, worth is reduction The the in again. 1 Figure of square commuting the is It past. read to easy is being pair the marking again box dotted the with and conventions same moves: that game the is it time this but — exchanged



the of instead moving game the with square same the as transfer, Property 2: شکل in belongs: game a where system, the inside sits G_m column left the In protocol. distinguishing. the doing machine the of part box. dashed the inside sits it right the (.5.10 (Proposition reduction. whole the is boundary that Crossing

system, the inside sits it column left the In G_m watching by it Read verdict the is Win_{Fin} and it, plays environment the belongs: game a where the of part box. dashed the inside sits it column right the In returns. it whole the is boundary that Crossing distinguishing. the doing machine en- an becomes π against game the playing environment an — reduction a is left is what across, is game the once and — φ from π telling vironment about. proved was 4.20 Theorem what is which question. emulation bare rewrites only form reporting The free. are edges horizontal two The it so untouched, places it call every leaving root, the at returns Z what executions of identity an ,4.18 Lemma is absorption and run: no changes

is it and for, paid thing only the is edge right the So bound. a than rather in $\mathcal{S}$ carries row bottom the why is which — $\mathcal{D}$ at applied hypothesis the occupant the keeps 5.8 Definition why so and dummy, the not and slot the :1 Figure in as exactly forced, then is edge left The all. at argument an as two than rather quantity same the 's $_{\approx \varepsilon}$ two the makes square the chasing . ε an share to happen that bounds

double. its is other the proposition's, the is quantity that of reading One left the so , $\Pr[\text{Win}_{\text{Fin}}]$ is advantage property the reading search the Under under appears: two of factor no and statement transfer the already is edge it. doubles which , $2\Pr[\text{Win}_{\text{Fin}}] - 1$ is advantage the reading decision the this of instance an is paper the in proof property per-functionality Every not. do instances the where diagram a earns it why is which square. one

Propo- of setting the In .adversary) every against (Transfer, 5.11 نتیجه place $\gamma[\pi/\varphi]$ of and γ of cores the that addition in suppose ,5.10 sition proposi- that $\mathcal{S}$ simulator the that , (9.2 (Definition calls responsive no (Defini- refusal-oblivious are below $\mathcal{A}$ occupant the and supplies tion finalization a Fix . $\mathcal{Z}_{\gamma}^{\circ}$ in lie named environments the that and , (4.28 tion sense the in $\mathbb{Z}_{\gamma, \gamma}$ against ε' within Fin property the has σ If . $\mathcal{G}_{\text{m}}$ of Fin then ,5.8 Definition of

$$\text{Adv}_{\mathcal{G}_{\text{m}}, \sigma[\pi/\varphi], \mathcal{Z}, \mathcal{A}}^{\text{Fin}} \leq \varepsilon' + 2\varepsilon$$

proposition the $\mathcal{Z}$ every and slot adversary the of $\mathcal{A}$ occupant every for advan- on hypothesis the of place in If them. among $\mathcal{Z}^{\mathcal{A}}$ with admits quantifi- same the under $\Pr[\text{Win}'_{\text{Fin}}] \leq p$ satisfies side ideal the tages real the naming Win'_{Fin} and Win_{Fin} with , $\Pr[\text{Win}_{\text{Fin}}] \leq p + \varepsilon$ then cation. search a form the is which — 5.10 Proposition in as events ideal and reading's. decision the being display first the in. inherits property

Proposi- By absorption. are two outer the which of steps. Three اثبات. the , $\text{Adv}_{\mathcal{G}_{\text{m}}, \sigma[\pi/\varphi], \mathcal{Z}, \mathcal{A}}^{\text{Fin}} = \text{Adv}_{\mathcal{G}_{\text{m}}, \sigma[\pi/\varphi], \mathcal{Z}^{\mathcal{A}}}^{\text{Fin}}$ system, game real the at 5.9 tion bounds $\mathcal{Z}^{\mathcal{A}}$ at 5.10 Proposition case. dummy's the being side right-hand game ideal the at now again. 5.9 Proposition And . $\text{Adv}_{\mathcal{G}_{\text{m}}, \sigma, \mathcal{Z}^{\mathcal{A}}, \mathcal{S}}^{\text{Fin}} + 2\varepsilon$ by it σ on hypothesis the which , $\text{Adv}_{\mathcal{G}_{\text{m}}, \sigma, (\mathcal{Z}^{\mathcal{A}})^{\mathcal{S}}}^{\text{Fin}}$ as term first the rewrites system. environ- admissible an being machine absorbed doubly the , ε' by bounds form probability The proposition. same the of applications two by ment 5.10 Proposition of display second the on read steps three same the is $\square$ all. at probability no changing absorption first. the than rather

prop- the now is side ideal the on hypothesis The buys. this things Two over quantified σ about statement one — alone dummy the against erty asked own its on 5.10 Proposition where — else nothing and environments supplied. emulation the simulator particular the against property the for — leaks or corrupts simulator that how about assumed be need Nothing Sec- which refusal. framework-internal a on branch not does it that only absorb- does. it else whatever moves: it machine every of asks 4.4 tion ideal-side the machine a of part into it turns environment the into it ing real the of property the is conclusion the And covers. already hypothesis same the by dummy. the against only not adversary. every against system are absorptions two the concretely. Read side. other the on applied move innermost the $\mathcal{S}$ -boundeds an and $\mathcal{A}$ -boundeda an for price: whole the cost hosting one $\mathbf{z} \oplus \mathbf{a} \oplus \mathbf{s}$ budget environment at reached is statement else. nothing charges 8 Chapter and moved. machine per

Noth- .side) ideal the on does slot the of occupant the (What 5.12 نکته the whatever absorbs which above. proof the for needed is here ing instead is stake at is What supplies. 4.20 Theorem ε the into does slot propo- the executions two the right: the on hypothesis the of cost the Corruption system. the in as well as slot the in differ compares sition is not is that half The issue. at is it of half only and two. in divides $\mathbf{d}' . F \in \{A, Z\}$ admits 3 line itself: for does environment the what it corruptions the for cooperation nobody's needs environment an so on identically them makes it root its from (Z, P) claiming and wants. initiative: own occupant's the is issue at is that half The sides. both asked is it what decline or to. it asked nobody parties corrupt may $\mathcal{S}$ right-hand the why is That relay. would $\mathcal{D}$ where slot. the through the against not and $\mathcal{S}$ against property the is 5.10 Proposition of side argument. an as occupant the keeps 5.8 Definition why and dummy. every against game the is form that in satisfy to has σ ideal an What the to it reduces 5.11 Corollary produce: may emulation the simulator environ- the into simulator the moving by so does and again. dummy it. of anything asking by than rather ment reads environment The why. is register the and ε the escapes Nothing whole the returns and nothing mediates which **FullStatus** through **C** com- and point any at **C** read may it so. (3.1 Remark set corrupted whatever with together itself corrupted it parties the against it pare ad- class the environment an be $\mathbb{Z}$ Let add. to slot the instructed it

0 of return a and added comparison that with $\mathcal{Z}$ be $\mathcal{Z}'$ let and mits, is it what corrupting $\mathcal{D}$ fails, never it side real the On fails. it where prob- whatever with fails it side ideal the on besides: nothing and told what declining — direction either in with, strays simulator the δ ability advan- with distinguishes $\mathcal{Z}'$ So unbidden, corrupting or asked, is it therefore may One $\delta \leq \varepsilon$ gives hypothesis emulation the and δ tage make, to instructed is it corruptions the echoes $\mathcal{S}$ though as reason commensurateness, 's3.1 Section is which — paid already price a at the discharges what not is it but having, worth is It properties. for read does, slot the absorbing :5.10 Proposition of right the on hypothesis conduct. simulator's the of all at nothing needs it and

the between divides Leakage .side) ideal the on (Leakage 5.13 نکته because own its of check one needs and does, corruption as sides 2 line :Leak a read can slot the Only .Guard through go not does it identity. an such claim cannot environment the and $\text{id}' \cdot F = A$ asks must: else something and there, game the protect not does $\mathcal{G}_m$. N So read- is leakage game's the since party, the with together gate 's4 line other every at and corrupt, never is root the party, corrupt a at only able therefore state game's The .5.4 Remark by $\perp$ returns Leak 's $\mathcal{G}_m$ party differ does What side. neither on route that by adversary the reaches produce: must $\mathcal{S}$ which 's π against leakage 's φ is sides the between a and ε same the by bounded emulation, of part ordinary an is that check. to further nothing with bound the inherits property

Blockers

same the and one on environment an silences 4 Chapter of 4.10 Definition on depend not do it to open claims the so worlds, both in closure name re- no use 4.20 Theorem and 4.16 Proposition and — faces it world which narrower: is buys blocking What all. at sets identity two the between lation delivers so and set, silenced exact an as attainable closure that makes it 4.10 Definition of discussion the as class, admissible restricted least the It else. nothing and identities occupies blocker A records. 4 Chapter in for false is Guard of 2 line on conjunct the so .N := $\emptyset$ caller, no admits en- never therefore is core Block Its .rej receives caller the and call every so, Guard through go not does Leakage formality. a is body its and tered with machine a all, $\perp$ returns it reachable: is core Leak party's corrupt a requires 1.2 Definition as wrapped cores. are Both give. to has state no first the makes that wrapper the is it and functionality. standard any of unreachable.

$\mathcal{B}[\mathcal{F}]$ Blocker	
$\text{par} := \perp$	$\text{.U} := \emptyset$
$\text{.N} := \emptyset$	$\text{.P} := \mathcal{F}.\text{P}$
$\text{.pid} := \mathcal{F}.\text{pid}$	
$\text{:1 return } \perp$	$\text{id}' \text{ from } \text{id}.\text{Block()}$ / unreachable due to Guard
$\text{:2 return } \perp$	$\text{id}' \text{ from } \text{id}.\text{Leak()}$ / reachable, but there is no state

from for. in stands it functionality the by indexed is $\mathcal{B}[\mathcal{F}]$ blocker The By else. nothing and set identity an fix that fields two the copies it which . $\mathcal{F}$ of identities the precisely occupies it : $\text{IDs}(\mathcal{B}[\mathcal{F}]) = \text{IDs}(\mathcal{F})$ construction it unreachable. Being besides. nothing and included. instance and session respect. other no in and identities those of occupant an as $\mathcal{F}$ for in stands By uses. system a ids process the for $I_\pi := \{\mathcal{F}.\text{pid} : \mathcal{F} \in \pi\}$ Write we which member, one exactly by carried is $q \in I_\pi$ each 1.2 Definition

Blockers $.IDs(\pi) = \bigcup_{q \in I_\pi} IDs(\pi_q)$ and $\pi = \{\pi_q : q \in I_\pi\}$ so π_q write a matters: distinction the and members. by not and ids by indexed are case which in code. different carrying worlds both in occur may id process q a such for blocker a and q uses π although π of member a not is φ_q id. same the at machines two put would already is what of clear blocker a keeps instead $I_\varphi \setminus I_\pi$ by Indexing q has $IDs(\pi)$ of identity no hence q a such carries π of member No there. φ_q from built blocker A $.IDs(\varphi_q) \cap IDs(\pi) = \emptyset$ and component first its in exist- an of id the on nor serves already π identity an on neither lands thus them of all and blockers. disjoint pairwise give ids Distinct member. ing throughout. ids distinct has again system enlarged the so π to new are wherever agree addition in must compared being systems of pair The overlap. they

when- if. compatible are φ and π Systems .(Compatibility) **تعريف 6.1** it at sitting functionalities two the both. in occurs id process a ever remaining the admit: they whom on and parties served their on agree may — run they code the and parameters. their use. they what — fields differ.

both to common ids the point: one exactly at used is Compatibility contributes it whatever so id. an such for added is blocker No worlds. serve could id shared a it Without sides. two the on agree already must uncovered. go would difference the and φ in and π in parties different only since needs. argument the than more is well as agree to **N** Requiring nothing. costs it but fixed: is id process the once set identity an enters **P** every in sides both on functionality same the by carried being id shared a use. we case

φ and π systems For .systems) blocked system. (Blocker **تعريف 6.2** is Blockers($\varphi \setminus \pi$) system blocker the

$$\text{Blockers}(\varphi \setminus \pi) := \{ \mathcal{B}[\varphi_q] : q \in I_\varphi \setminus I_\pi \},$$

are pair the of systems blocked the and

$$\bar{\pi} := \pi \cup \text{Blockers}(\varphi \setminus \pi), \quad \bar{\varphi} := \varphi \cup \text{Blockers}(\pi \setminus \varphi).$$

con- the input: an than rather notation is **Blockers** of argument The
 mem- the not there. occupants 's φ and $I_\varphi \setminus I_\pi$ ids the reads struction
 code. different carrying id shared a at — denotes difference the set ber
 by covered thus is $I_\varphi \setminus I_\pi$ in id Each built. is blocker no yet and $\varphi_q \in \varphi \setminus \pi$
 party functionality's that with there. living functionality the of blocker the
 lacks. it and has other the ids process the exactly gains world Each set.
 blocker. no get worlds both to common Ids achieve. to has blocking all
 work. its does 6.1 Definition where is which
 each worlds: two the equalises blocking says 6.3 Theorem Informally,
 oc- two the pair compatible a for and system. a again is system blocked
 having world each — $IDs(\pi) \cup IDs(\varphi)$ set. identity same the and one cupy
 lacks. it and has other the ids process the exactly gained

blocked the Then systems. be φ and π Let .(Equalisation) 6.3 **قضيه**
 moreover are φ and π If systems. are 6.2 Definition of $\bar{\varphi}$ and $\bar{\pi}$ systems
 then compatible.

$$IDs(\bar{\pi}) = IDs(\bar{\varphi}) = IDs(\pi) \cup IDs(\varphi).$$

asks first The compatibility. needs second the only and claims. Two اثبات.
 standard- finiteness. — system a of asks 1.2 Definition things three the
 from them gets and system. blocked each of — ids process distinct ness.
 this and has world other the ids the by indexed are blockers indexing: the
 compares second The member. existing no on land they so lacks. one
 world one only id an At $I_\pi \cup I_\varphi$ over index by index sets identity two the
 fields two the copying blocker a set. same the contribute sides both uses.
 con- two the so built. is blocker no id shared a At set. identity an fix that
 the supplies compatibility and — stand they as $IDs(\varphi_q)$ and $IDs(\pi_q)$ tribute
 is That agree. sets party served the that not. does indexing the thing one
 three asks system a being claim. first the For used. is it place only the
 they that and standard. be members its that finite. be it that $\bar{\pi}$ of things
 members added the immediate. is Finiteness ids. process distinct carry
 car- blocker a inherited: is Standardness $I_\varphi \setminus I_\pi \subseteq I_\varphi$ by indexed being
 Block functionality. standard a of asks 1.2 Definition wrappers the ries
 copied is id process its and 3.2 Section in as Leak and 3.5 Section in as
 distinctness. For C, Z, A of none is name its so φ of member a from
 and $q \in I_\varphi \setminus I_\pi$ with $\mathcal{B}[\varphi_q]$ the are **Blockers** $(\varphi \setminus \pi)$ of members the
 in lies none and distinct pairwise are ids These $\mathcal{B}[\varphi_q].pid = \varphi_q.pid = q$

The $I_{\bar{\pi}} = I_{\pi} \cup I_{\varphi}$ and ids process distinct carry $\bar{\pi}$ of members the so I_{π} $\bar{\varphi}$ to applies argument same blocker a since $\mathcal{F}$ every for holds $IDs(\mathcal{B}[\mathcal{F}]) = IDs(\mathcal{F})$ second. the For Hence set. identity an determine that fields two the copies

$$IDs(\bar{\pi}) = \bigcup_{q \in I_{\pi}} IDs(\pi_q) \cup \bigcup_{q \in I_{\varphi} \setminus I_{\pi}} IDs(\varphi_q),$$

$$IDs(\bar{\varphi}) = \bigcup_{q \in I_{\varphi}} IDs(\varphi_q) \cup \bigcup_{q \in I_{\pi} \setminus I_{\varphi}} IDs(\pi_q),$$

that in q each at them Compare $I_{\pi} \cup I_{\varphi}$ by indexed are unions both and both then $q \in I_{\varphi} \setminus I_{\pi}$ if and $IDs(\pi_q)$ contribute both then $q \in I_{\pi} \setminus I_{\varphi}$ If set. the and $IDs(\pi_q)$ contributes first the both, in lies q If $IDs(\varphi_q)$ contribute $\pi_q \cdot pid = \varphi_q \cdot pid =$ so q index the at carried are Both $IDs(\varphi_q)$ second follow not does that thing one the supplies compatibility and outright, q two those by fixed is set identity An $\pi_q \cdot P = \varphi_q \cdot P$ indexing, the from by term agree therefore unions two The $IDs(\pi_q) = IDs(\varphi_q)$ so fields. equals each and term.

$$\bigcup_{q \in I_{\pi}} IDs(\pi_q) \cup \bigcup_{q \in I_{\varphi}} IDs(\varphi_q) = IDs(\pi) \cup IDs(\varphi).$$

□

places. three of one in falls id process A table. a is claim second The each: at sits what is theorem the and

one column per process id $q \in I_{\pi} \cup I_{\varphi}$

	$I_{\pi} \setminus I_{\varphi}$	$I_{\pi} \cap I_{\varphi}$	$I_{\varphi} \setminus I_{\pi}$	
$\bar{\pi}$	π_q	π_q	$\mathcal{B}[\varphi_q]$	
$\bar{\varphi}$	$\mathcal{B}[\pi_q]$	φ_q	φ_q	shaded: a blocker – occupies the identities. admits no caller
	$IDs(\mathcal{B}[\pi_q]) = IDs(\pi_q)$ a blocker copies pid and P	compatibility: $\pi_q \cdot P = \varphi_q \cdot P$ the one place it is used	$IDs(\mathcal{B}[\varphi_q]) = IDs(\varphi_q)$ a blocker copies pid and P	

so both rows occupy $IDs(\pi) \cup IDs(\varphi)$

three The place. each at does blocking what and fall. can id process a Where **شكل 3:** **6.3** Theorem of claim second the of cases three the are columns

a and blocker, a gains id the lacks that world the columns outer the In for, in stands it functionality the of identities the exactly occupies blocker

rows two the So set. identity an fix that fields two only the copied having working a contributes one although — there identities same the contribute the is which nothing. answers that machine a other the and functionality blocking all is sets identity Equalising differently. them shading of point answered that blocker a achieves: it all be better had it and achieve. to has compar- the and learn. could environment an what change would anything worthless. be would enable to built was it is on built is blocker No lives. hypothesis the where is column middle The they as $IDs(\varphi_q)$ and $IDs(\pi_q)$ contribute rows two the so id. shared a for carried are both — equal those makes indexing the in nothing and stand. not. need sets party served the but outright. agree ids process the so, q at is column this and do. they that assumption the exactly is Compatibility assump- the what also is it way. other the Read used. is it place only the a is worlds two the in parties different serving id shared a costs: tion adding — has it device one the because cover. cannot blocking difference present. already is occupant an where unavailable is — occupant an at systems are $\bar{\varphi}$ and $\bar{\pi}$ That this. of any needs claim second the Only two the states theorem the why is which compatibility. of nothing asks all alone. first the on lean may below 6.4 Proposition why and separately two the of one is system a enlarging and system. a enlarges Blocking undone. come could 1.3 Definition ways

sys- be φ and π Let .well-formedness) preserves (Blocking 6.4 گزاره
 $.wf(\bar{\varphi})$ and $.wf(\bar{\pi})$ Then $.wf(\varphi)$ and $.wf(\pi)$ with tems

com- no asks which 6.3 Theorem of claim first the by system a is $\bar{\pi}$. اثبات.
 $\bar{\varphi}$ for argument the $\bar{\pi}$ for argue We question. legal a is $.wf(\bar{\pi})$ so patibility.
 $(\mathcal{F}', R) \in \mathcal{F}.U$ and $\mathcal{F} \in \bar{\pi}^+$ Take $.wf(\pi)$ of place in $.wf(\varphi)$ with same the is
 blocker a check: to nothing is there blockers added the of one is $\mathcal{F}$ If
 in lies $\mathcal{F}$ Otherwise entry. no contributes it so construction. by $U = \emptyset$ has
 is witness That entry. that for $\mathcal{G} \in \pi^+$ witness a supplies $.wf(\pi)$ and π^+
 so fields. member's no alters blocking and $\pi^+ \subseteq \bar{\pi}^+$ since present. still
 conditions three The were. they what are reads R everything and $\mathcal{G}.s, \mathcal{G}.F$
 $\mathcal{G}$ same the of hold still therefore
 en- U empty their beyond blockers about nothing uses argument The
 whenever well-formedness preserves system another to system a larging
 once witness a because own. their of entries no carry added members the
 Nei- change. never on judged is it fields the and removed never is present
 □ used. is claim second 's 6.3 Theorem nor compatibility ther

own environment's the keeps closure The .realizations) (Tight 6.5 نکته
 ad- An it. around route one the is hosting names: used the off claims
 identity unoccupied an at machine a host may environment missible
 and – say caller, a of instance second invented an – name used a of
 through identity own its claiming interface. full a is machine hosted a
 a Against .(4.13 (Definition par through than rather silencer own its
 an- then can worlds two the name. that admit members whose pair
 con- no and occupies, them of one only identity an at differently swer
 absorp- refusing without intruder the refuse can class the on dition
 of machines hosts 4.15 Definition of environment absorbed the tion:
 – names admitted at occupants outside – shape same the exactly
 admissibil- which runs. it code the in only intruder the from differs and
 realization. the to belongs repair the of Part read. not does rightly ity
 out- member every if members its of I subset a at tight system a Call
 than rather system, the by occupied ids process whole admits I side
 – serves callers admitted its of each parties only serves and names.
 callers', its exactly set party member's private a makes this serves with
 route host the closes Tightness is. subroutine private a what is which
 every outside lies id process unoccupied an :I outside members the to
 identity every makes condition party the and ,2 line fails and N pinned
 for- 4.10 Definition where occupied. from speak could caller pinned a
 I outside member a to addresses intruder an call a so hosting: bids
 Exec by or N empty blocker's the by too, side ideal the on refused. is
 to route the is close not does tightness What occupant. no finding
 used bare a admits that member interface an itself: I of member a
 fake hosted a by reachable is – 1 Chapter of pattern local the – name
 con- tightness and be, would member private a as exactly instance
 $Z \cup A$ only admitting or too, those Pinning .I of member no strains
 do we it: close would directly, them reaches environment the where
 hygiene a is Tightness here. condition sufficient general a pursue not
 necessary protocol. subroutine-respecting a of spirit the in condition
 when of characterization full a leak: to not members private the for
 open. leave we faithful are pairs blocked against comparisons

Renaming and Relocation

Chapter 1 records a functionality's code that reads its own- ses- in the **Guard** of 3 line at occurs s number: instance its never and sion This nowhere. occurs i and **1.3** Definition in and it. invoke that wrappers and sessions Renaming statement. a into observation that turns chapter at proved emulation an so executions. of isomorphism an is instances same the at and ε same the within other. every at holds id process one func- a that — more no and needed is condition hygiene One budgets. every from reachable be session every outside from reachable tionality isolat- and — satisfies specifies paper the machine every which session. single a for written be realization a lets what is That work. the half is it ing rest. the covering as read and is. **15** Chapter of $\mathcal{F}_{\text{Sig}}$ as session.

تعريف 7.1 (Relocation). A is relocation θ permutation the of process that such ids

(i) $\forall q, q'. F \in \text{ids} \implies \theta(q).s = \theta(q').s$

(ii) $\forall q, q'. F \in \text{ids} \implies \theta(q).s = \theta(q').s \iff \{0, 1\}^*$

(iii) $\forall q \in \{(A, 0, 0), (Z, 0, 0), (C, 0, 0)\} \implies \theta(q) = q$

the execution supplies.

Conditions (i) and (ii) exactly say that the standard names

$$\theta(F, s, i) = (F, \hat{\theta}(s), \theta_{F,s}(i))$$

for a session. and name each for and. ids session of $\hat{\theta}$ permutation a for a of components two the write we — numbers instance of $\theta_{F,s}$ mutation spo- being ι and σ own. their of letters with than rather way this relocation proofs the notation whose **4.17** Definition and **4.19** Proposition by for ken name by match wf and **Guard** because kept. are Names import. below

permuted are sessions functionality; different a become may nothing and not and session a share ids two whether asks 3 line because blocks. as component that block. a within freely permuted are instances and which: nowhere. read being

code: in literals as occur that ids process three the pins (iii) Condition of $(C, id.P)$ read register the in $.C$;Mediate of $(A, id.P)$ hook the in $.A$ identity Every .5 line of relay the in and 2 line at $.Z$ and ;6 and 2 lines in party a with paired three these of one is paper the in outright named no and three those pins it and – all them pins three the pinning so scope. the and name. reserved a of id every pinning as read be not must It more. (A, s, i) triple a holds **A** set the 1 Chapter by cosmetic: not is difference of all fixing condition a so occurring. ever $(A, 0, 0)$ only $.i$ and s every for every at $\hat{\theta}(s) = s$ give would unrestricted (ii) with together read and them to (ii) Restricting all. at session a moves that relocation no leave and s determined is $\hat{\theta}$ care: same the of half other the is names standard the move to left are occur never that name reserved a of ids the and there. where is shows 7.5 Proposition which class. own their inside harmlessly stay. they

freedom That it. move to free is $\hat{\theta}$ and $.0$ session pins here Nothing only the is this since where. exactly seeing worth is it and price. a has com- 3 Line another. from session one tells framework the which at point three the and callee. the of tested is global while session caller's the pares to Z or A from call a so ;0 session in sit supplies execution the machines own callee's the when exactly line that passes global not is that callee a session moving relocation a tell would shape that of callee A .0 is session the callee other every for can: else Nothing alone. it leaving one from 0 before 2 line at refused or global by waived either are machines supplied gap. the closes callee the on condition one So at. looked is session the .0 on restriction a than rather condition hygiene a is it and

تعريف 7.2 (Session-uniform) A functionality $\mathcal{F}$ session-uniform is if

$$\mathcal{F}.N \cap (A \cup Z) \neq \emptyset \implies \text{global}(\mathcal{F}),$$

reach- is it if only session every outside from caller a admits it that so its of each if session-uniform is system A session. every from able is. hosts it functionality each if environment an and is. members

do from reached be can functionality session-uniform a sessions The

it. of needs relocation what is which in. sits it session the on depend not being three all outright. condition the meet machines supplied three The and – second the by $\mathbb{Z}$. global of disjunct first the by $\mathcal{C}$ and $\mathcal{A}$ – global sec- same that by global slot. adversary the of occupant every does so 2 line which and empty is $\mathbf{N}$ whose blocker. a does so and disjunct. and de- paper the $\mathbf{N}$ every are Those session. the whatever refuses therefore to line one is condition the where parameter. a as its takes $\mathcal{F}_{\text{Sig}}$ clares: open leaves 1 Chapter shape the is excludes condition the What check. a it: call may $\mathbf{Z}$ and $\mathbf{A}$ whether declare itself may $\mathcal{F}$ local a that saying in in slot the from reachable is adversary the admitting functionality local cannot relocation a thing one the is that and other. no in and 0 session 3 line of widening one-line a alternative. the records 7.17 Remark carry. altogether. condition the with dispense would that

identities on acts θ relocation $\mathbf{A}$. relocation) a of (تعريف 7.3 Action
on and alone. component party the leaving. $\theta(\text{pid}, \mathbf{P}) := (\theta(\text{pid}), \mathbf{P})$ by
with $\theta\mathcal{F}$ to $\mathcal{F}$ functionality a sends It pointwise. sets identity

$$\begin{aligned}\theta\mathcal{F}.\text{pid} &:= \theta(\mathcal{F}.\text{pid}), & \theta\mathcal{F}.\mathbf{P} &:= \mathcal{F}.\mathbf{P}, & \theta\mathcal{F}.\mathbf{N} &:= \theta(\mathcal{F}.\mathbf{N}), \\ \theta\mathcal{F}.\mathbf{U} &:= \theta(\mathcal{F}.\mathbf{U}), & \theta\mathcal{F}.\text{par} &:= \theta(\mathcal{F}.\text{par}),\end{aligned}$$

through par on and $\theta(\mathcal{F}', \mathbf{R}) := (\theta\mathcal{F}', \mathbf{R})$ by entry $\mathbf{U}$ a on acts θ where the is $\theta\mathcal{F}$ of code the contains: it ids process and identities whatever a target. call a in – it in occurring id process every with $\mathcal{F}$ of code image: its by replaced – value stored a or test. a identity. claimed re- is computes than rather declares machine a set identity every and its means (4.13 (Definition bundle a for which too. image its by placed acts It hand. by fixes it routing any and set outward its set. occupied execution the of machine other any on and memberwise. system a on hosts $\theta\mathbb{Z}$ that so included. members hosted bundle's a way. same the to predicates requirement the require We hosts. $\mathbb{Z}$ what of images the $\mathbf{P}$ only reading it is. serves as $\mathcal{R}(\mathcal{F}, \mathcal{G}) \iff \mathcal{R}(\theta\mathcal{F}, \theta\mathcal{G})$ equivariant. be the so pointwise. relabels above clause Every $\mathbf{N}$ of membership a and particular In $(\theta'\theta)\mathcal{F} = \theta'(\theta\mathcal{F})$ and identity the as acts id one: is action what is which ids. process on does it as machines on θ undoes θ^{-1} directions. both in argue below results the lets

relabelling itself by not is names program a ids process the Relabelling machine the on condition a is two the between gap the and does. it what

it. without false is 7.9 Lemma because it, state We model.

handles it if identity-atomic is Code .code) (Identity-atomic 7.4 تعريف
to one project may It tokens. opaque as ids process and identities
name, their of two or them, of two test component: party or name its
mem- for one test equality: for components, instance or session party,
F.N tests 2 line as given, been has code the them of set a in bership
3 and 2 lines as tests, those by them of set a form ;J tests Silence and
destruc- and outright: one name $\{id'' \neq id\}$ forms 5 line and J form
or session a with compute not may It one, carries that value a ture
or session no and ordering, no arithmetic, no number: instance an
pro- or identity an inside from apart own, its on down written instance
whatever step, one as counts these of Each outright, named id cess
answer, one as call a counts 4.30 Definition as involved, identities the

Guard predicate The identity-atomic, is paper this in machine Every
all, at i reads core no else, nothing and equality for sessions compares
We (iii), condition of three the are outright named ids process only the and
7.3 Definition makes what is it and throughout, this be to model the take
merely not does there substitution the code identity-atomic on sound:
op- permitted each behaviour, relabelled the computes but literals rewrite
par- and names leaves $\emptyset$ because projections — equivariant being eration
sets the and injective is $\emptyset$ because membership and equality alone, ties
because identity named a code, the with relabelled are formed and tested
and machine a makes what also is It image, its by it replaces substitution
the in operations same the perform they steps: same the take image its
clause last the and both, on same the being flow control the order, same
have it inside ids the however each for step one charges definition the of
— unchanged over carry 4.30 Definition of budgets the So renamed, been
relocation a grant, not would symbol the by charging model cost a which
session, a of encoding the lengthen to free being
names $id.i \bmod 2$ returning core A fail, both restriction the Without
in- another to carried untouched: it leaves substitution so id, process no
not is machine relabelled the and differently, answers it number stance
in target its misses $(F, id.s, id.i + 1)$ calling core A relabelled, original the
4.6 Definition — write would anyone machine a is Neither way, same the
stated be to has exclusion the why is which core, arbitrary an permits
assumed, than rather

functionality a $\mathcal{F}$ relocation. a be θ Let .invariances) (Basic 7.5 گزاره
 Then system. a π and
 disjoint pairwise carrying occupants and $IDs(\theta\mathcal{F}) = \theta(IDs(\mathcal{F}))$ (i)
 them: keep sets identity
 $\theta(\mathbf{A}) = \mathbf{A}$. $\theta(\mathbf{Std}) = \mathbf{Std}$ so itself, onto class name each carries θ (ii)
 $\{pid : local\ a - names\ by\ declared\ \mathbf{N}\ every\ and\ .\theta(\mathbf{Z}) = \mathbf{Z}\ and$
 $image: own\ its\ is - them\ among\ pid.\mathcal{F} = \mathcal{F}^\uparrow\}$
 $;\mathbf{global}(\theta\mathcal{F}) \iff \mathbf{global}(\mathcal{F})$ (iii)
 $;\langle\theta\pi\rangle = \theta(\langle\pi\rangle) = \langle\pi\rangle$ (iv)
 and $;\mathbf{wf}(\pi) \iff \mathbf{wf}(\theta\pi)$ and system. a is $\theta\pi$ (v)
 relocation. a again is θ^{-1} (vi)

's7.1 Definition of which is see to thing useful the and items. Six اثبات.
 an injectivity: and (i) conditions is (i) spends. one each conditions three
 (ii) untouched. one and relabelled one fields. two by fixed is set identity
 a being and themselves. into map classes name – alone (i) condition is
 three the makes what is which well. as them onto partition. a on bijection
 only reading global (ii). from follows then (iii) invariant. sets id reserved
 closures name again. (i) condition is (iv) . $\mathbf{Std}$ of subset a and name a
 well-formedness one: substantial the is (v) names. by specified being
 waived either is conjunct session the match. names because transports
 preserves (ii) condition where ids standard two between read or global by
 the runs converse the :7.3 Definition by equivariant is R relation the and it.
 and (i) conditions – legitimate is that why is (vi) . θ^{-1} on argument same
 (iii) and inverse. the of hold they so equivalence. an and identity an are (ii)
 pid by fixed is set identity An (i) ids. three those fixes θ because holds
 alone: second the leaving while first the relabels relocation a and . $\mathbf{P}$ and
 identities. on injective is θ because survives disjointness

The itself. into maps class name each 7.1 Definition of (i) By (ii)
 $\theta^{-1}(x)$ point the $\mathcal{F}$ of class the in x for so ids. process the partition classes
 $;\mathcal{F}$ of class the is therefore and itself into maps which class. some in lies
 unions are sets displayed three The well. as class that onto is map the so
 them. of union a is name a by specified ids of set a and classes. name of
 is second The . $\mathcal{F}.\mathcal{F} \in \{A, Z, C\}$ or $\mathbf{Std} \subseteq \mathcal{F}.\mathbf{N}$ reads $\mathbf{global}(\mathcal{F})$ (iii)
 first. the for :7.1 Definition of (i) by untouched

$$\mathbf{Std} \subseteq \theta(\mathcal{F}.\mathbf{N}) \iff \theta^{-1}(\mathbf{Std}) \subseteq \mathcal{F}.\mathbf{N},$$

bijection. a being θ above. (ii) item by $\theta^{-1}(\mathbf{Std}) = \mathbf{Std}$ and

keeps θ uses: system its names the by specified is closure name A (iv)
 union a is $\langle \pi \rangle$ And $\langle \theta\pi \rangle = \langle \pi \rangle$ and does π names the uses $\theta\pi$ so those.
 as $\theta(\langle \pi \rangle) = \langle \pi \rangle$ gives (ii) item so ids. party with crossed classes name of
 well.

injective. being θ ids. process distinct carry $\theta\pi$ of members The (v)
 un- wrappers their and unchanged being names their standard. are and
 an and $\mathcal{F} \in \pi^+$ take .wf For finite. is family the relabelling: by touched
 session the For sides. both on match Names $\mathcal{G}$ witness with $(\mathcal{F}', R)$ entry
 con- the waives above (iii) item and global($\mathcal{G}$) either $\mathcal{G}.s = \mathcal{F}.s$ conjunct
 the of entries the $\mathcal{F}$ is so then and – standard is $\mathcal{G}$ or sides. both on junct
 – empty being $\mathcal{C}.U$ and other each only naming machines supplied three
 test. the preserves 7.1 Definition of (ii) and standard are ids both that so
 requires. 7.3 Definition equivariance the by $R(\mathcal{F}, \mathcal{G}) \iff R(\theta\mathcal{F}, \theta\mathcal{G})$ Finally
 both on identically settled are machines supplied three the of entries The
 and above (ii) items by fixed being them of reads serves field every sides.
 a θ^{-1} on argument same the run converse. the For 7.1 Definition of (iii)
 .7.3 Definition from $\theta^{-1}(\theta\pi) = \pi$ using (vi). item by relocation

of identity an as stated are 7.1 Definition of (ii) and (i) Conditions (vi)
 does (iii) inverse: the of hold so and equivalence. an as and components
 $\square$ ids. three those fixes θ because

declarations N pinned the only (ii). By word. a deserve these of Two
 pin. they members the with move they and all. at move 6.5 Remark of
 Proposi- makes what is which invariant. are closures name (iv) by And
 as exactly constrained class environment an against out come 7.10 tion
 it. to corresponding merely one than rather before

every for $\theta Op = Op$ if pid-blind is core A .core) (Pid-blind 7.6 تعريف
 may It moves. relocation a that id process no names it : θ relocation
 it and does. Guard as scope. in identities two of sessions the compare
 fixes 7.1 Definition of (iii) that ids process three the of one name may
 to free leaves (iii) which name. reserved a of id other some not but –
 move.

read 2 Chapter of wrappers The pid-blind. is paper this in core Every
 to 6 and 2 lines and $(A, id.P)$ to hooks Mediate only: parties and names
 names on tests by par from $\bar{J}$ computes box $\mathcal{Z}$ the reserved: both $(C, id.P)$
 and $id.P$ reads $\mathcal{F}_{Sig}$ and names: payload its call the places $\mathcal{D}$ parties: and
 identity. an of else nothing and P own its

different a and needs. 7.9 Lemma than property stronger a is That
 execu- an of machine every which ,7.4 Definition needs lemma The one.
 slot adversary the of occupant the and environment the meet. must tion
 the name to having each pid-blind. be can those of neither and – included
 con- the of reading the is buys pid-blindness What drives. it ids process
 relabelled fields five with π is $\theta\pi$ cores. pid-blind of system a for clusion:
 protocol. the is protocol relocated a so rewritten. code of line a not and
 elsewhere. run
 answers: definition the question the of half other the lemma. the Before
 all. at reads execution an fields which

agreeing functionalities standard be $\mathcal{F}'$ and $\mathcal{F}$ Let .fields) (Inert 7.7 گزاره
 code. and par , $\mathbf{P}$,pid on
 no then alone. $\mathbf{U}$ in differ they that so well. as $\mathbf{N}$ on agree they If (i)
 putting by got systems two the in them: distinguishes execution
 the produces activation every id. process that at other the or one
 configuration. same
 same the on and state same the from then alone. $\mathbf{N}$ in differ they If (ii)
 1 line at refuse wrappers both either ,id' from id.FullOp(in) call
 point same the at core same the enter and admit both or ,rej with
 state. same the in

execution. an during read are fields two the where of inspection By اثبات.
 results the and 1.3 Definition of check static the by only read is $\mathbf{U}$ field The
 of readers The (i). gives which interface. any by or Exec by never it. about
 global predicate the and ,Guard of 2 line two: are execution an inside $\mathbf{N}$
 interface full the of 1 line which ,Guard _{$\mathcal{F}$} inside sit Both ,3 line hence and
 too execution an outside read is field (The else. anything before **srequire**
 typing the by and ,6.1 Definition by two. those through 1.3 Definition by –
 on conditions are those but – 6.5 Remark and 4.7 Definition of conditions
 read is and verdict that decides field the So one.) of steps not system. a
 the – core the and 1 line between lines the wrapper: the in else nowhere
 set silenced the hook. Mediate first the gate. corruption the read. register
 two so call. the and code the ,par , $\mathbf{P}$,pid of functions are – {id'' $\neq$ id}
 core same the enter and together 5 line reach admit both that wrappers
 □ (ii). is That state. same the in point same the at

execu- every in inert is $\mathbf{U}$ and else. nothing and control access is $\mathbf{N}$ So
 past Inertness slot. adversary the for records already 4.8 Remark as tion.

to strengthened be cannot it and step. one about statement a is guard the admits one call a that is reason obvious The returned. value the about one is one obvious less The apart. states two the sends refuses other the and entered core the because differently. end can admit both call a even that instance an permits 3.6 Section – wrapper very this re-enter may 5 line at the and – identity own its claim core a lets 5 line and re-entered. be to places that core A differs. field the where again. $\text{Guard}_{\mathcal{F}}$ meets call inner $\mathcal{F}$ an separates rej not is answer the if just 1 returns and id as id. $\text{FullOp}'()$ ad- both call a on not. does that $\mathcal{F}'$ an from id process own its admitting verdicts guard in differ two the claim: exact the is survives What mitted. separate to enough is inner. or outer verdict. differing single a and alone. them.

a admits par nor $\mathbf{P}$ Neither .read) are that fields two (The 7.8 نکته fixes $\mathbf{P}$ field The reasons. different for and kind. this of statement second the and all at $\mathcal{F}$ to dispatches Exec calls which hence . $\text{IDs}(\mathcal{F})$ types $\mathcal{F}_{\text{Sig}}$ of Initialize – directly it read cores and :1 line of conjunct over quantifies Verify of $\exists P' \notin \mathbf{C}$ test the and . $\forall \mathbf{K} : \mathcal{F}_{\text{Sig}} . \mathbf{P} \rightarrow \mathcal{K} \cup \{\square\}$ symmetry. a not is set party a shrinking or enlarging So domain. that A fix to extended $\{0, 1\}^*$ of β permutation a :renaming is one is What state. party-indexed the on and $\mathbf{C}$ on . $\mathbf{P}$ on identities. on acting . $\mathcal{Z}$ and .7.9 Lemma of argument the by executions of isomorphism an gives as – indexing and equality through only ids party read cores provided places two The equivariant. being $\exists P' \notin \mathbf{C} : \mathbf{vk} = \mathbf{vk}[P']$ its does. $\mathcal{F}_{\text{Sig}}$. $\mathcal{Z}$ party the names 3 line well: as it survive otherwise enters id party a permutes. β set a .id. $P \in \{0, 1\}^*$ tests FullCorrupt and fixes. β which is $\mathcal{Z}$.par and field. the of definition by code the by read is par field The A in. invariant be to nothing is there besides: 4.10 Definition by read by declared is it whenever itself to it moves and it. moves relocation is. $\mathcal{Z}.\text{par} := \langle \pi \rangle$ which names.

instances and sessions relabelling that says 7.9 Lemma Informally. Proposi- is it observable: nothing changes execution an throughout every and bijection. absorption the of place in θ with induction 's4.19 tion wrapper. one of line one of equivariance the is check

system. session-uniform a π relocation. a be θ Let .(Relocation) 7.9 لم session- a $\mathcal{Z}$ and .(3.3 (Definition slot adversary the of occupant an $\mathcal{X}$

is **Exec** that so ,IDs(π) of identity no at hosting environment uniform
Then four. the at defined

$$\text{Exec}(\theta\pi, \theta\mathbb{Z}, \theta\mathbb{X}, \mathbb{C}) \quad \text{and} \quad \text{Exec}(\pi, \mathbb{Z}, \mathbb{X}, \mathbb{C})$$

distributed. identically are

two : θ to up read , $\iota := \theta$ at 4.17 Definition of correspondence The اثبات. other's the is one when correspond chains caller or points program states. is proof the and symmetric, is relation the bijection a is θ Since image. below paragraphs four the step whose activations, on induction an again so bijectively, relabelled is occupancy that checks Machines discharge. takes Guards correspondence. in start and defined are executions both the and injectivity, are two first the turn: in **Guard** of conjuncts three the place only the is 3 line since work. real does that paragraph one the is third session- where is this — another from session one tells framework the would 0 session moving relocation a it without and spent, is uniformity wrap- the checks register the mediation, Silencers, verdict, the change corruption all, at image no needs register the that observes and pers. step, the closes token claims, Routing, alone, party by indexed being cor- The sides, both on branches same the taking code identity-atomic paired machine each — $\iota := \theta$ with 4.17 Definition of that is response to up read (C4) and (C3) (C1), with and — itself with $\mathbb{C}$ image, its with cor- callers suspended of chains two points, program two states, two : θ on bijection a is θ Since other. the of -image θ the is one when respond induction an is proof the before as and symmetric, is relation the identities step, the discharge below paragraphs The activations, on

(i) by bijectively, relabelled is occupancy and system a is $\theta\pi$ Machines. the at hosts $\theta\mathbb{Z}$ and IDs(π) of clear hosts $\mathbb{Z}$ since :7.5 Proposition of (v) and right the on disjoint pairwise are occupants the hosts, $\mathbb{Z}$ what of images 1 Line defined. are executions both so left, the on are they where exactly state initial member's a and sides, both on member every and $\mathbb{C}$ initialises outset, the at (C1) giving relabelled, image's its is

, $\theta\mathcal{F}$ at image its and $\mathcal{F}$ at id' from id. **FullOp**(in) call a Take Guards. id.P $\in \mathbf{P}$ and injectivity, by $\theta(\text{id}.pid) = \theta(\mathcal{F}.pid)$ iff $\text{id}.pid = \mathcal{F}.pid$:1 Line by again , $\theta(\text{id}'.pid) \in \theta(\mathcal{F}.\mathbf{N})$ iff $\text{id}'.pid \in \mathcal{F}.\mathbf{N}$:2 Line untouched, is 3 Line alone, leaves θ components reads test party the and injectivity, of (iii) by sides two the on agrees disjunct second whose disjunction a is together sides both on fails it — fails global($\mathcal{F}$) suppose so ,7.5 Proposition

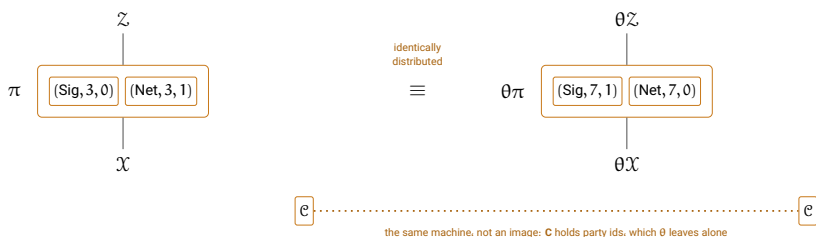
is execution the of machine Every first. the consider and – neither on or and $\mathcal{C}$, $\mathcal{X}$ and hypothesis, by copies hosted 's $\mathcal{Z}$ and π session-uniform: global which, $\mathcal{Z}$ and $\mathcal{C}$, $\mathcal{A}$ are ids process their because interfaces own 's $\mathcal{Z}$ claim a and, $\mathcal{Z}$ nor $\mathcal{A}$ neither meets $\mathcal{F}$. $\mathcal{N}$ So disjunct. second its by admits fixing (iii) – sides both on 2 line at refused is $\mathcal{Z}$ of or $\mathcal{A}$ of id the carrying sets two those does. $\mathcal{F}$. $\mathcal{N}$ as just $\mathcal{Z}$ and $\mathcal{A}$ missing $\theta(\mathcal{F}. \mathcal{N})$ and ids, those cannot id supplied third The .7.5 Proposition of (ii) by -invariant θ being carry therefore 3 line reach that claims The calls. no placing $\mathcal{C}$ all. at arrive once $\{0, 1\}^*$ in lying $\mathcal{F}$ of name the, id does so and ids, process standard holds it when exactly images the of holds $\text{id}' . s = \text{id} . s$ so fails: $\text{global}(\mathcal{F})$ session- paragraph one the is This .7.1 Definition of (ii) by originals, the of change would 0 session moving relocation a it without for: is uniformity callee, every at sides, both on same the is verdict the So here, verdict the requirements three whose .3.2 Section of interface leakage the including read. lines these what only read

$\{\text{id}'' \neq$ is 5 line of set silenced The register. the mediation. Silencers, both on identity own its claims core a so $\{\text{id}'' \neq \theta \text{id}\}$ is image whose, $\text{id}\}$ $\theta(\text{par})$ from computed is box $\mathcal{Z}$ the of $\bar{\mathcal{T}}$ set The else, nothing and sides on computed set the of image the is it so parties, and names on tests by wrapper The admits, that what of image the exactly admits and left, the ad- 6 and 2 lines $\{(A, \text{id}.P) \text{ to hooks and } \text{id}' . F \text{ and } \text{id}.P, \mathcal{C} \text{ reads Mediate (iii) that three the of one is here outright named id Every } .(C, \text{id}.P) \text{ dress collects } \mathcal{C} \text{ image: no needs itself register the And fixes. 7.1 Definition of ma- same the literally is } \mathcal{C} \text{ so touch, not does relocation a which ids, party Corruption configurations, corresponding at set same the holding chine this, buys what is (3.1 (Section alone party by indexed being its of occupant the to call a dispatches Exec token, claims, Routing, by does occupancy while } \theta \text{ under correspond targets and identity, target answered is identity unoccupied an to addressed call a paragraph: first the by correspond Claims right. the on is image its when exactly left the on rej a, (7.4 (Definition identity-atomic being code The paragraph, previous the } -\theta \text{ are that calls place and branches same the take image its and machine values with point program same the reach they so another, one of images is which order, same the in coins same the reads each and } \theta \text{ by related on operation one having Exec unchanged, is discipline token The (C2), and (C3) is which arising, suspensions of chain same the and sides both (C4).$

being output the alike, output configurations halting Corresponding the So fixes. 7.1 Definition of (iii) identity an, $\langle \mathcal{Z}, \mathcal{Z} \rangle$ at return 's $\theta \mathcal{Z}$ and 's $\mathcal{Z}$

of probability the on particular in distributions. as agree executions two
 □ .1 returning

same the and execution. One state. to than see to easier is lemma The
 side: by side renamed. id process every with execution



$\theta : (F, s, i) \mapsto (F, \theta(s), \theta_{F,s}(i))$ — the name kept, the session block moved, the instance free within it:
 $(Z, 0, 0)$, $(A, 0, 0)$ and $(C, 0, 0)$ pinned, so Z , X and C keep their identities and only the process ids inside their code move

renamed. id process every with execution same the beside execution One :4 شكل
 distributions. as agree two the that is 7.9 Lemma

name The systems. two the in legible is do may relocation a Everything
 quietly may nothing and name by match wf and Guard because kept. is
 both — block a as moves session The functionality. different a become
 and caller whether asks 3 line because — together 7 to 3 from go members
 instance the And in. are they one which never and session a share callee
 because places. changing here 1 and 0 block. that within move to free is
 pinned. are literals as occur that ids three The it. reads code any of line no
 process the only and sides both on identities same the at sit C and X, Z so
 rewritten. are code their inside ids

be- set. same the holding machine same the is it that: even not is C
 Index- party. a touch not does relocation a and ids party collects C cause
 worth is it and this. buys what is (3.1) (Section alone party by corruption ing
 have would id process by indexed register a — buys it much how noticing
 pattern corruption the and else. everything like θ by across carried be to
 disturb. could relocation a thing a be then would

looks lemma the way this Drawn give. cannot picture the warning One
 mediation every silencer. every conjunct. guard every is: nearly it and free.
 either which of all ids. pinned three the and parties names. only reads hook
 and test. session single the is exception The it. by fixed are or θ survive
 — remark a than rather hypothesis a is session-uniformity reason the is it
 them of one in sitting but session every outside from reachable callee a

that θ a after and before differently machines supplied the answer would one that at apart come would executions two the and .0 session moves is rest the work: the of whole the is proof the of paragraph That line. shows. already picture the bookkeeping

sys- φ , π and relocation a be θ Let .transports) (Emulation 7.10 **گزاره**
 the of environment every and them of two the let (iii) and (ii) for tems:
 Then need. not does (i) which session-uniform, be named classes
 and adversaries to adversaries carries θ and $\mathbb{Z}_{\theta\pi, \theta\varphi} = \theta(\mathbb{Z}_{\pi, \varphi})$ (i)
 4.4 Section of dummy the fixing simulators, to simulators
 UC- $\theta\pi$ then $(\mathbb{A}, \mathbb{S}, \mathbb{Z})$ against ε within φ UC-emulates π if (ii)
 and $(\theta\mathbb{A}, \theta\mathbb{S}, \theta\mathbb{Z})$ against ε within $\theta\varphi$ emulates
 $\text{Adv}_{\theta\pi, \theta\varphi}^{\text{uc}}[\mathbf{a}, \mathbf{s}, \mathbf{z}] = \mathbf{a}, \mathbf{s}, \mathbf{z}$ every and systems budgeted for (iii)
 $\text{Adv}_{\pi, \varphi}^{\text{uc}}[\mathbf{a}, \mathbf{s}, \mathbf{z}]$

.7.5 Proposition of part different a on rests each and claims. Three اثبات.
 trans- 4.10 Definition of clauses both classes: the about bookkeeping is (i)
 host- the and -invariant θ are closures name because clause par the port.
 because ways both going containment the with injectivity. by clause ing
 to simulators and adversaries to go adversaries too: relocation a is θ^{-1}
 fixes. θ ones the are read definitions those fields the because simulators
 im- then is (ii) pid-blind. being core its outright. fixed is dummy the and
 simulator returned the world. each to applied 7.9 Lemma from mediate
 a is θ form: concrete the is (iii) alone. $\theta\mathbb{A}$ on depends which. $\theta\mathbb{S}$ being
 conclu- the class the onto classes budgeted three the of each of bijection
 changes bijections along function a reindexing and over. quantifies sion
 preserved is Session-uniformity infimum. an nor supremum a neither
 Def- (i). For part. each in than rather once checked is which throughout.
 outright: -invariant θ is clause par The $\mathbb{Z}$ of things two asks 4.10 inition
 $\mathbb{Z}.\text{par}$ when exactly $\theta(\langle\pi\rangle \cup \langle\varphi\rangle) = \langle\theta\pi\rangle \cup \langle\theta\varphi\rangle$ contains $\theta\mathbb{Z}.\text{par} = \theta(\mathbb{Z}.\text{par})$
 .7.5 Proposition of (iv) by invariant being closures name, $\langle\pi\rangle \cup \langle\varphi\rangle$ contains
 of images the at hosting $\theta\mathbb{Z}$ injectivity. by transports clause hosting The
 IDs(π) $\cup$ IDs(φ) of image the being IDs($\theta\pi$) $\cup$ IDs($\theta\varphi$) and hosts $\mathbb{Z}$ what
 a being θ^{-1} ways. both goes containment The proposition. that of (i) by
 iden- whose IDs($\mathbb{A}$) occupies adversary an slot. the For (vi). by relocation
 components party and fixes 7.1 Definition of (iii) id process the carry titles
 relabels θ so pid-blind: is which wrapper. 's $\mathbb{A}$ carries it and alone. leaves θ
 4.7 Definition shape. same the of adversary an is $\theta\mathbb{A}$ and alone core the
 The simulators. to go simulators so (ii), by fixes θ set a $\mathbf{N} \supseteq \mathbf{Std} \cup \mathbf{Z}$ asks

so – names payload its call the places 3 line – pid-blind is core dummy's whose .3.4 Section of $\mathcal{A}$ the of claimed is kind the of Nothing . $\theta\mathcal{D} = \mathcal{D}$ it. needs below nothing and arbitrary. leaves 4.6 Definition core hypothesis the simulator the be $\mathcal{S} \in \mathbb{S}$ let and $\mathcal{A} \in \mathbb{A}$ fix (ii). For so alone. $\theta\mathcal{A}$ on depends it : $\theta\mathcal{S} \in \theta\mathbb{S}$ return . $\theta\mathcal{A} \in \theta\mathbb{A}$ Given it. for supplies 7.9 Lemma . $\theta\mathbb{Z} \in \theta\mathbb{Z}$ any For met. is 4.11 Definition of order quantifier the gives world each to applied

$$\text{Adv}_{\theta\pi, \theta\varphi}^{\text{uc}}(\theta\mathbb{Z}, \theta\mathcal{A}, \theta\mathcal{S}) = \text{Adv}_{\pi, \varphi}^{\text{uc}}(\mathbb{Z}, \mathcal{A}, \mathcal{S}) \leq \varepsilon,$$

right. the on two the of -images θ the being left the on executions two the . $\theta\mathbb{Z} \subseteq \mathbb{Z}_{\theta\pi, \theta\varphi}$ emulation. of statement legal a is conclusion the (i) By of (ii) by $\mathbf{Z}$ and $\mathbf{A}$ fixes θ throughout: preserved is Session-uniformity exactly session-uniform is $\theta\mathcal{F}$ so (iii). its by global and 7.5 Proposition memberwise. environment an of and system a of same the and is. $\mathcal{F}$ when well. as respect this in θ under closed therefore is below class Every the answer and steps same the take image its and machine a (iii). For therefore flow control its and identity-atomic being code the calls. same and system budgeted a of member a : (7.4 (Definition both on same the biject- a is θ So reason. same the for budget same the carry image its quantifies conclusion the class the onto classes three the of each of tion (i). by $\mathbb{Z}_{\theta\pi, \theta\varphi}(\mathbf{z})$ onto $\mathbb{Z}_{\pi, \varphi}(\mathbf{z})$ and themselves. onto $\mathbb{S}(\mathbf{s})$ and $\mathbb{A}(\mathbf{a})$ over: of advantage the makes 7.9 Lemma . θ^{-1} by induced inverse with each extrema nested two the so triples. corresponding at equal 4.5 Definition function one to applied operators three same the are 4.31 Definition of supremum a neither changes reindexing and bijections. along reindexed Def- of finiteness the what here: needed is finiteness No infimum. an nor separate a is which attained. are extrema the that is settles 4.31 inition $\square$ alike. sides both on holds and matter

proof a and relocation. to up system a of property a thus is Emulation family a for so say To outset. the at instance the and session the fix may related are ids process of lists two when know to needs one instances. of conditions. three is answer the and all. at relocation a by

$\vec{q} = (q_1, \dots, q_m)$ Let .another) to relocates tuple one (When 7.11 گزاره name. standard of ids process of tuples be $\vec{q}' = (q'_1, \dots, q'_m)$ and for if. only and if j every for $\theta(q_j) = q'_j$ with θ relocation a is There

,k and j all

$$q_j.F = q'_j.F, \quad (q_j.s = q_k.s \iff q'_j.s = q'_k.s), \\ (q_j = q_k \iff q'_j = q'_k).$$

reloca- a by related are name one of ids process two any particular In
tion.

or- in off read 7.1 Definition of conditions three the is Necessity اثبات. assignment. session the from $\hat{\theta}$ build construction: a is Sufficiency der. the between bijection well-defined a makes condition second the which be- permutation a to it extend and sides. two the on occurring sessions name each at then. side: each on size one of is over left is what cause and third the which assignment. instance the from $\theta_{F,s}$ build session. and the Taking extensible. again hence and injective make conditions fourth (iii) and $\vec{q}'$ to $\vec{q}$ carrying relocation a assembles else everywhere identity con- the is Necessity touched. never were ids reserved the because holds shared sessions (i), by names order: in off read 7.1 Definition of ditions

injective. being θ by equality and (ii), by not or occurring sessions many finitely the be S' and S let sufficiency. For well-defined a $q_j.s \mapsto q'_j.s$ makes condition second The $\vec{q}'$ in and $\vec{q}$ in over left is what so – image an being $q'_j.s$ every onto, – $S \rightarrow S'$ bijection to extends it and alike, infinite or finite size, one of is sides two the on $s \in S$ session a and F name a fix Now ids. session of $\hat{\theta}$ permutation a instance Their $q_j.s = s$ and $q_j.F = F$ with j indices the consider and then $q_j.i = q_k.i$ if injective: and defined well is $q_j.i \mapsto q'_j.i$ assignment and agree: instances the and condition fourth the by $q'_j = q'_k$ so, $q_j = q_k$ sessions and names their while $q'_j \neq q'_k$ so, $q_j \neq q_k$ then $q_j.i \neq q_k.i$ if in- the between bijection a again is Each differ. instances the and agree. permutation a to extends it so sides. two the on occurring numbers stance θ let and session. and name other every at identity the take did: $\hat{\theta}$ as $\theta_{F,s}$ iden- the as and names standard the on $(F, s, i) \mapsto (F, \hat{\theta}(s), \theta_{F,s}(i))$ as act the because (iii) .7.1 Definition meets θ Then ones. reserved the on tity a claim, last the For $\vec{q}'$ to $\vec{q}$ carries it and outright, fixed are ids reserved □ vacuously. conditions three the meets name one of q' , q pair single

نتیجه 7.12 One instance suffices. Call maps of pair a $\vec{q} \mapsto \pi(\vec{q})$ and $\varphi(\theta\vec{q}) = \theta(\varphi(\vec{q}))$ and $\pi(\theta\vec{q}) = \theta(\pi(\vec{q}))$ if family instance an $\vec{q} \mapsto \varphi(\vec{q})$

$\pi(\vec{q})$ by occupied ids process the lists $\vec{q}$ where θ relocation every for speci- the and subroutines private its protocol. a – together $\varphi(\vec{q})$ and with moving and tuple one by indexed against measured is it fication UC- $\pi(\vec{q})$ that and session-uniform are $\varphi(\vec{q})$ and $\pi(\vec{q})$ Suppose it. to down cut $(A(a), S(s), Z_{\pi(\vec{q}), \varphi(\vec{q})}(z))$ against ε within $\varphi(\vec{q})$ emulates $\vec{q}'$ every at Then $\vec{q}$ single a for environments. session-uniform the UC-emulates $\pi(\vec{q}')$. 7.11 Proposition of conditions three the meeting cut-down same the and $S(s)$, $A(a)$ against ε same the within $\varphi(\vec{q}')$ id. process one occupying family a for particular. In pair. that for class name. that of id every at emulation gives id one at emulation

equivari- $\vec{q}'$ to $\vec{q}$ carrying relocation a supplies 7.11 Proposition اثبات. and $\vec{q}'$ at pair the to systems of pair the carrying one into it turns ance re- the of session-uniformity – statement the transports 7.10 Proposition proposi- that of proof the as original. the of that from coming pair located one-id a :7.11 Proposition of clause final the is claim last The records. tion $\square$ name. its of id other any against conditions three the meets tuple

satisfiable. condition the makes what is tuple whole the by Indexing id process principal the by indexing since recording. worth is point the and given a fix may relocation A not. does – writes one thing first the – alone in equivariance so id. another at sitting member private a move still and q whole the by fixed id an have to $\pi(q)$ of member every force would alone q three. reserved the and itself q are ids such only the and q of stabiliser Equivariance. q fixes that relocation some by movable being other every all. at members private no with systems only admits therefore alone q in wants. anyone case the not is which

to has else what asking and instead. systems of pair a for Written be can name a of instance new A two. in splits statement the move. (ii) condition is reason the and cannot. session new a itself: by up taken nothing so blocks. as sessions permutes relocation a :7.1 Definition of it. in behind another leaving while s session of out member one carry can

φ and π Let .session) one moving instance. one (نتیجه 7.13 Moving the and one at member a carrying each systems session-uniform be occu- $q.F = q_0.F$ with id process a be q let and q_0 id process same against ε within φ UC-emulates π Suppose system. neither by pied the that is conclusion the below case each In $(A(a), S(s), Z_{\pi, \varphi}(z))$

the and $\mathbb{S}(\mathbf{s})$, $\mathbb{A}(\mathbf{a})$ against ε same the within UC-emulates pair moved
 $\mathbf{z}$ at pair that of class admissible largest

numbers instance of $\theta := (q_0 \ q)$ transposition the $\langle q.s = q_0.s$ If (i)
 the fixes it and relocation, $\mathbf{a}$ is session that in $q_0.F$ name the at
 $\theta\varphi$ and $\theta\pi$ So system, either of member other every of id process
 else nothing and q to carried q_0 at instance the with φ and π are
 a in instance moved the naming member $\mathbf{a}$ that save – disturbed
 6.5 Remark of manner the in callers its among it pinning by field,
 it, with along carried mention that has otherwise, or
 process a fixing while q to q_0 carries relocation no, $\langle q.s \neq q_0.s$ If (ii)
 system $\mathbf{a}$ of member no so – $q_0.s$ session in name standard of id
 q_0 carrying One entire, moves block the and fixed, is there lying
 ses- in member $\mathbf{a}$ has system neither addition in if and exists; q to
 and π of member each carries that chosen be may one, $\langle q.s$ sion
 process the fixes and $q.s$ session into $q_0.s$ session in lying φ of
 hypoth- further that Without elsewhere, lying member every of id
 same the by – too move $q.s$ session in already members the esis
 one fix can q to q_0 carrying relocation no, $\langle q$ at applied argument
 blocks two with pair $\mathbf{a}$ about then is conclusion the so – them of
 one, than rather moved

exhib- been has relocation $\mathbf{a}$ once 7.10 Proposition are parts Both. اثبات.
 of ids two of transposition the (i) For one, exhibiting is work the so ited,
 else nothing moves it that is check the and it, does session common a
 q_0 carrying relocation $\mathbf{a}$ – first comes half negative the (ii) For occupied,
 session, that of id standard every moves session different $\mathbf{a}$ of id an to
 constructs then half positive the and – 7.1 Definition of (ii) condition by
 The session, third every fixes and $q_0.s$ session moves that relocation the
 (vi) by relocation $\mathbf{a}$ is which, θ^{-1} to half negative the applies sentence last
 is relocation the once 7.10 Proposition are parts Both. 7.5 Proposition of
 has transposition the (i), For classes, the supplying (iii) part its exhibited,
 be- does (iii) and hold, 7.1 Definition of (ii) and (i) conditions so, $\hat{\theta} = \text{id}$
 $\mathbf{C}$, $\mathbf{Z}$, $\mathbf{A}$ of none so and system $\mathbf{a}$ of member $\mathbf{a}$ of name the is $q_0.F$ cause
 system, neither by occupied is q and $\langle q$ and q_0 ids two the only moves It
 $\langle q_0$ naming field $\mathbf{a}$ is fixed not is what fixed; is id occupied other every so
 relabels, 7.3 Definition which

standard of id process any be p let and $\theta(q_0) = q$ suppose (ii), For
 $\theta(p).s = \theta(q_0).s =$, 7.1 Definition of (ii) condition by $\langle p.s = q_0.s$ with name
 the of transposition the $\hat{\theta}$ take existence, For $\langle \theta(p) \neq p$ so, $\langle q.s \neq p.s$

name the at $q.i$ to $q_0.i$ carrying permutations instance and sessions two $q_0.s$ session carries θ That elsewhere. identity the $q_0.s$ session in $q_0.F$ lies member no if session: third a of id every fixes and $q.s$ session into $q_0.s$ session of those exactly are moves it members the $q.s$ session in in lying member a be m let and q to q_0 carry θ let sentence. final the For carries and 7.5 Proposition of (vi) by relocation a is θ^{-1} Then $q.s$ session every moves it to applied claim negative the so $q_0.s \neq q.s$ with q_0 to q give would $\theta(m) = m$ and them: among m $q.s$ session of id standard $\square$ too. m moves θ so $\theta^{-1}(m) = m$

permutation a components only the not are instances and Sessions a raises: 5 Chapter question a settles what is third the and move. can (5 (line name own game's the carrying claim a under $\mathcal{F}$ exercises game protocol a inside ships that $\mathcal{F}$ the while game. the name must $\mathcal{F}.N$ so field the is it and field. one in differ two The instead. protocol that admits reads. Guard

process the of ν permutation a is renaming A .(Renaming) 7.14 تعريف
form the of ids

$$\nu(F, s, i) = (\bar{\nu}(F), s, i)$$

identi- on acts It .C and Z ,A fixing names the of $\bar{\nu}$ permutation a for in and prescribes. 7.3 Definition as sets declared and code fields. ties. image. its by code in outright occurring name every replaces addition

substi- 7.3 Definition before. not was and needed is clause extra The a what being names enough. is relocation a for which ids. process tutes name bare a against test a so them. moves renaming a keeps: relocation dis- 4.4 Section of dummy the $id'.F = Z$ the or ,Mediate in $id'.F \neq A$ — travels it paper this of machines the For too. travel to has — on patches so fixes. $\bar{\nu}$ three the are outright mention they names only the nowhere: fields. only all. at code of line no changes renaming a

Then renaming. a be ν Let .(Renaming) 7.15 گزاره

$\nu(Z) = Z$ and $\nu(A) = A$, $\nu(\mathbf{Std}) = \mathbf{Std}$: $IDS(\nu\mathcal{F}) = \nu(IDs(\mathcal{F}))$ (i)
when exactly session-uniform is $\nu\mathcal{F}$: $global(\nu\mathcal{F}) \iff global(\mathcal{F})$
and : $wf(\pi) \iff wf(\nu\pi)$ with system a is $\nu\pi$: $\langle \nu\pi \rangle = \nu(\langle \pi \rangle)$ is: $\mathcal{F}$
renaming: a is ν^{-1}

and slot adversary the of $\mathcal{X}$ occupant every , π system every for (ii)

no with – IDs(π) of identity no at hosting $\mathcal{Z}$ environment every
– them of any of asked session-uniformity

$\text{Exec}(\nu\pi, \nu\mathcal{Z}, \nu\mathcal{X}, \mathcal{C})$ and $\text{Exec}(\pi, \mathcal{Z}, \mathcal{X}, \mathcal{C})$

and distributed: identically are

($\mathcal{A}, \mathcal{S}, \mathcal{Z}$) against ε within φ UC-emulates π if $;\mathbb{Z}_{\nu\pi, \nu\varphi} = \nu(\mathbb{Z}_{\pi, \varphi})$ (iii)
the and $:(\nu\mathcal{A}, \nu\mathcal{S}, \nu\mathcal{Z})$ against ε within $\nu\varphi$ UC-emulates $\nu\pi$ then
budget. every at agree 4.31 Definition of advantages concrete

اثبات. and re-run. proofs relocation corresponding the are parts three All. car- than rather permuted now are Names changes. what is overview the reserved the definition: the of point the is which themselves. onto ried session-uniformity, and global them with and fixed. stay **Std** and classes hard one the (ii) In fixed. than rather carried is closure name a while re- a trivial, becomes – conjunct session the – 7.9 Lemma of paragraph hypoth- session-uniformity no why is which session. no moving naming result: the not reading, the is lost is what (iii) In all. at here appears esis hypothesis's the to corresponds conclusion the of class environment the of changes two with 7.5 Proposition of proof the is (i) it. being than rather onto carried than rather permuted now are classes Name bookkeeping. the by declared one the becomes names by declared **N** an so themselves. the being .**Std** while – definition the of point whole the is which – images classes, reserved single being .**Z** and **A** and class, standard every of union (iii) of ids reserved the and 7.2 Definition .global them with and fixed, are being $\langle \rangle$ fixed, than rather carried is closure name a And .7.1 Definition of is conjunct session the wf For uses. system a names the by specified side one on holds match name the and session, no moving ν untouched.

other. the on when exactly paragraph, hard one its for but verbatim 7.9 Lemma of proof the is (ii) leav- ν unchanged. literally is disjunct first 's3 line trivial: becomes which needs Nothing (i). by preserved is second its and alone. session every ing other Every here. appears session-uniformity no why is which excluding, travels that set a in membership a or equality for name a reads check Defi- of clause extra the with together 7.4 Definition and code, the with as behaviour relabelled the compute substitution the makes 7.14 nition before.

only closures the of used which .7.10 Proposition of proof the is (iii)
is What put. stay they that never and $\nu(\langle \pi \rangle \cup \langle \varphi \rangle) = \langle \nu\pi \rangle \cup \langle \nu\varphi \rangle$ that

class environment the :7.5 Proposition after recorded reading the is lost than rather hypothesis the of one the to corresponds conclusion the of
 □ images. the name now must par since identically, constrained being

and renaming a be ν Let .immaterial) is name caller's (The 7.16 نتیجه
 $\nu\gamma =$ Then .5.1 (Definition $\mathcal{F}$ for system game a be $\gamma = \{\mathcal{G}_m\} \cup \sigma$ let
 every for and , $\nu\mathcal{F}$ for game a is $\nu\mathcal{G}_m$ system. game a is $\{\nu\mathcal{G}_m\} \cup \nu\sigma$
 $\mathcal{G}_m$ of Fin finalization

$$\text{Adv}_{\nu\mathcal{G}_m, \nu\sigma, \nu\mathcal{Z}, \nu\mathcal{X}}^{\text{Fin}} = \text{Adv}_{\mathcal{G}_m, \sigma, \mathcal{Z}, \mathcal{X}}^{\text{Fin}}$$

Since slot. adversary the of $\mathcal{X}$ occupant every and $\mathcal{Z} \in \mathbb{Z}_{\gamma, \gamma}$ every for
 reads form abbreviated the , $\nu\mathcal{D} = \mathcal{D}$

$$\text{Adv}_{\nu\mathcal{G}_m, \nu\sigma, \nu\mathcal{Z}}^{\text{Fin}} = \text{Adv}_{\mathcal{G}_m, \sigma, \mathcal{Z}}^{\text{Fin}}.$$

environments of class a against ε within Fin property the has σ So
 class. that of image the against ε within it has $\nu\sigma$ when exactly

check game. a being $\nu\mathcal{G}_m$ For (i). is wf($\nu\gamma$) with system a is $\nu\gamma$ That اثبات.
 moving ν .($\tilde{\nu}(\mathcal{G}_m.F), 0, 0$) is id process its :5.1 Definition of fields four the
 is it so , $\nu\mathcal{F}.\mathbf{P} = \mathcal{F}.\mathbf{P}$ and untouched is $\mathbf{P}$ its instance: nor session neither
 ,($\nu\mathcal{F}$, serves $_{\mathcal{G}_m}$) is entry its (i): by $\nu(\mathbf{Z}) = \mathbf{Z}$ is $\mathbf{N}$ its required: as $\nu\mathcal{F}.\mathbf{P} \cup \{\mathbf{Z}\}$
 still 8 Line . $\mathbf{N}$ of membership a and $\mathbf{P}$ reading equivariant, is serves $_{\mathcal{G}_m}$ and
 id, process a of part no being ids party root, the at finalization every serves
 on one same the names Fin so name. interface no moves renaming a and
 FullFin interface the that event the is Win $_{\text{Fin}}$ advantage. the For sides. both
 (ii) by :($\nu\mathcal{G}_m.\text{pid}$, $\mathbf{Z}$) to identity that carries ν and ,1 returns ($\mathcal{G}_m.\text{pid}$, $\mathbf{Z}$) at
 correspondence. the under distributed identically are executions two the
 5.8 Definition and sides both on probability same the has event the so
 the For (iii). is $\nu\gamma$ for admissible is $\nu\mathcal{Z}$ That number. same the gives
 of sets and $\text{pid} := A$ are fields dummy's the : $\nu\mathcal{D} = \mathcal{D}$ form. abbreviated
 call the places and id'. $F = \mathbf{Z}$ on dispatches core its and fixes. (i) that ids
 □ moves. mentions it name no so names. payload its

and use. not does γ name a with $\mathcal{G}_m.F$ of transposition the at Read
 .5.1 Definition of wants one what says this alone. $\mathcal{G}_m$ by carried $\mathcal{G}_m.F$ with
 other no match: to renamed $\mathcal{F}.\mathbf{N}$ and name any given be may game The
 property A either. does code of line no 7.14 Definition by and moves. field

property whose $\mathcal{F}$ the and called, is game its what to tied not therefore is use. will deployment the name caller whatever admit may proves one a carries It programme. caller different a is gives renaming no What protocol the to game a never protocol, a to protocol a and game a to game the to answers 's $\mathcal{F}$ hands that protocol a and — earnest in $\mathcal{F}$ use will that 7.7 Proposition could. nothing so , $\mathcal{F}$ of property no satisfies adversary inside but nowhere read being **N** side, other the from gap the narrows control access the and code 's $\mathcal{F}$ about statement a is property a :Guard $\mathcal{F}$ rest the Crossing carry. could field that else nothing about and it, around Relocations crossing, the is 5.10 Proposition and for, is emulation what is so executions, of isomorphism an being each compose, renamings and once, at instance and session name, in moved be may system a

an not is Session-uniformity .distinguished) is 0 (Session 7.17 نکته
one, real a is excludes it functionality the and proofs, the of artefact
,0 session in sit $\mathcal{Z}$ and $\mathcal{A}$ and session, caller's the compares 3 Line
A admits yet 1 Chapter of sense the in local is that functionality a so
session own its when exactly slot adversary the from addressable is
does global($\mathcal{F}$) and fails $\text{id}.s = \text{id}.s$ conjunct the $\mathcal{F}.s \neq 0$ with :0 is
an out carries $\mathcal{D}$ how is it — observable is difference The it, save not
no pair a such for so — party corrupt a at 4.4 Section of instruction
is 7.2 Definition and isomorphism, an is 0 session moving relocation
exclude: can one least the also is It convenient, than rather necessary
or global by waived are machines supplied the callee other every for
one larger no and do would condition smaller no so ,2 line at refused
needed, is
itself paper the place one and exotic, not is excludes it shape The
of way a as offers, 6.5 Remark plainly, said be should it for reaches
member a such that member, interface an to route hosting the closing
else, one no and adversary the and environment the — $\mathbf{Z} \cup \mathbf{A}$ only admit
not is it so ,**N** its outside lying **Std** global, not is shape that of member A
artefact the but 7.2 Definition of defect no is That session-uniform,
environment the lets 3 line most: costs it place the at through showing
whose protocol a so other, no in and 0 session in member a such reach
,0 session in only drivable is $\mathbf{Z} \cup \mathbf{A}$ exactly admit members interface
admissible an routes two The it, about say to has relocation whatever
a claim may It accordingly, divide protocol a into has environment
Std of all admitting member a at only admits 2 line which name, fresh

one of run one not hence sessions. across shared hence global. –
 a at admits 2 line which -identity. Z own its claim may it Or protocol.
 .0 session to member that pins 3 line then and – Z admitting member
 session a in interface protocol session-local a reaches route Neither
 choosing. own its of
 re- worth is all at -identity Z own its claim can environment the That
 name the available: route second the makes what is it since calling.
 Defi- and $\langle \pi \rangle \cup \langle \varphi \rangle$ outside lie -identities Z so system. no by used is Z
 par- its only is N whose member A claimable. them leaves 4.10 nition
 artefact the there and route. neither by reachable is class name ent's
 alone. slot adversary's the through shows
 as 3 line Reading

$$(id'.s = id.s \vee \text{global}(\mathcal{F}) \vee id'.F \in \{A, Z\})$$

confer then would 0 Session condition. the with dispense would
 reach- be would adversary the admitting functionality local a nothing.
 be could 7.2 Definition and alike. session every in slot the from able
 all of hold would results the above: statement every from dropped
 the in else Nothing ones. session-uniform the than rather systems
 being ids reserved the stays. (iii) condition – change would chapter
 con- tuple the and is. rule session the whatever code in outright named
 widening The .0 session of free already are 7.11 Proposition of ditions
 are names it machines two the rest: the of spirit the in and small is
 them. of tested is global wherever disjunct second 'sglobal by global
 $\mathcal{Z}$ and parties. corrupt to $\mathcal{A}$ confines still 3 line of gate corruption the
 than more is buy would it What .par own its by and N by out stays
 and session its to local be protocol a let would what is it relocation:
 the pairing the is which time. same the at environment the by driven
 as- silently suggestion 's6.5 Remark which and deny above routes two
 record and [7] Canetti's is which rule. narrower the keep We sumes.
 here. price its and option the both

Proposi- .theorem) many-instance a not is (Relocation 7.18 نکته
 and follow. not does It nothing. charges and copy one moves 7.10 tion
 execu- the : n ε than rather ε cost indices n at copies n that true. not is
 other's each not are $\varphi(\vec{q}_1), \dots, \varphi(\vec{q}_n)$ of and $\pi(\vec{q}_1), \dots, \pi(\vec{q}_n)$ of tions

– itself of copies n to system a carries relocation no – relabellings
 .4.24 Theorem of chain hybrid the is them between route only the and
 one statement the is together give results two the What adds. which
 Corol- $\vec{q}$ one at $\varphi(\vec{q})$ emulates $\pi(\vec{q})$ prove proof: single a from wants
 whose .4.24 Theorem and ε within other every at it gives 7.12 lary
 and disjoint pairwise being $\vec{q}_k$ the to come hypotheses disjointness
 at $n\varepsilon$ within once at n all replaces shell, surrounding the of clear
 the in functorial also is Relocation charges. 8 Chapter budgets the
 only reads which of each .6 Chapter and 4.1 Section of constructions
 in- an since $\theta(\rho[\pi/\varphi]) = (\theta\rho)[\theta\pi/\theta\varphi]$ sets: identity and ids names.
 right- the – $\theta\pi = \overline{\theta\pi}$ difference, set the with commutes action jective
 of system blocked the as requires. 6.2 Definition as read. side hand
 relabels θ fields two the copies blocker a since – $(\theta\pi, \theta\varphi)$ pair the
 Defini- since $\theta(\mathcal{Z}^{\rho\setminus\varphi}) = (\theta\mathcal{Z})^{\theta\rho\setminus\theta\varphi}$ and $I_{\theta\varphi} \setminus I_{\theta\pi} = \theta(I_\varphi \setminus I_\pi)$ and
 half reserved whose set, outward declared bundle's a relabels 7.3 tion
 relocated a So .7.5 Proposition of (ii) by fixed is $\{\text{id} : \text{id}.F \in \{A, C\}\}$
 proof. relocated the of composition the is composition of proof

Costs Concrete

collected are and line main the outside sit cost concrete of threads Two the cost the and members, its off read itself, in system a of cost the here: of budgets the on arithmetic charge. 4.5 Section of results composition .4.31 Definition

whole, a as cost a has also system a member, by member Budgeted up. level one read cost functionality's a is it and

call a is π system a of invocation An .system) a of (Cost 8.1 تعريف cas- a by it services π ; π outside from members its of one reaching cost The outside. returns control before activations internal of cade total largest the , T_π time per-invocation the :(T_π , R_π) pair the is π of activations the over summed – cascade such one in steps of number count invocation the and – time a at running machine one triggers, it $T_\pi R_\pi$ most at in runs π So answers. π invocations of number the , R_π steps.

with $\pi = \{\mathcal{F}_1, \dots, \mathcal{F}_n\}$ Let .costs) member from cost (System 8.2 گزاره within $\mathcal{F}_i$ of activations the bound k_i let and ,(t_i , r_i) cost of $\mathcal{F}_i$ each Then . π of invocation one

$$T_\pi \leq \sum_i k_i t_i, \quad R_\pi \leq \sum_i r_i,$$

each activates invocation every If steps. $\sum_i t_i r_i$ most at in runs π and $T_\pi \leq$ then – $k_i \leq 1$ subroutine-respecting, π – once most at member . $R_\pi \leq \min_i r_i$ members, all drives invocation each moreover if ; $\sum_i t_i$

t_i most at for each times. k_i most at $\mathcal{F}_i$ activates invocation An اثبات.

whole the over $\mathcal{F}_i$ of activations The $\sum_i k_i t_i$ most at spends it so steps.
 ac- one least at is invocation every and r_i most at number execution
 are steps total the $\sum_i r_i$ most at number invocations the so tivation.
 mem- all drives invocation each When $\sum_i (\text{of activations } \mathcal{F}_i) t_i \leq \sum_i r_i t_i$
 for $R_\pi \leq r_i$ so times. R_π exactly $\mathcal{F}_i$ each activate invocations R_π bers.
 □ .i every

pro- disjoint with systems be π_2 and π_1 Let .union) a of (Cost 8.3 نتیجه
 invokes one of member no If (T_2, R_2) and (T_1, R_1) costs and ids cess
 then other. the of member a

$$\begin{aligned} (T_{\pi_1 \cup \pi_2}, R_{\pi_1 \cup \pi_2}) &= (T_1, R_1) \oplus (T_2, R_2) \\ &= (\max(T_1, T_2), R_1 + R_2). \end{aligned}$$

the and π_2 into calls k most at makes π_1 of invocation each instead If
 and entry. π_1 a from $T_{\pi_1 \cup \pi_2} \leq T_1 + k T_2$ then acyclic. is graph cross-call
 unbounded cross-calls the with $R_{\pi_1 \cup \pi_2} \leq R_1 + R_2$ with symmetrically.
 union's. the determine longer no costs summary the

and π_2 or π_1 enters $\pi_1 \cup \pi_2$ outside from call a cross-calls no With اثبات.
 that to belonging and T_2 or T_1 most at costing there. stays cascade its
 $\max(T_1, T_2)$ is time per-invocation the so count: invocation subsystem's
 union the to internal is π_2 into π_1 from cross-call A add. counts the and
 at steps: T_2 most at π_2 of invocation one by cascade the extends and
 π_2 to external once call a while $T_1 + k T_2$ give acyclically. them. of k most
 number still invocations the so counted. longer no is π_1 from placed but
 □ $R_1 + R_2$ most at

can call one it without idle: not is proviso subroutine-respecting The
 towards degenerates T_π and end. without system the around driven be
 member's private a keeps that discipline same the $-\sum_i t_i r_i$ of whole the
 Def- of $\oplus$ the to dual is cost system This 6.5 Remark in pinned callers
 activation. per member one bundle. a as system a hosting 4.30 initiation
 the in pays $c_{\rho \setminus \varphi}$ cost the $(\max_i t_i, \sum_i r_i)$ into $\oplus$ by members its combines
 over times the sums subsystem a as it calling below: costs composition
 synchronous the in $(\sum_i t_i, \min_i r_i)$ calls. external counts and cascade a
 when tightly more second the $\sum_i t_i r_i$ total same the bound Both case.
 by itself: level system the at recurs $\oplus$ And pass. full a is invocation every
 at operator one it. by compose subsystems non-interacting 8.3 Corollary
scales. both

turns time per-invocation the another one call π of members the When
clean. is case one and calling, that of shape the on

$\pi = \{\mathcal{F}_1, \dots, \mathcal{F}_n\}$ of graph call the Suppose .systems) (DAG 8.4 گزاره
at $\mathcal{F}_i$ member internal each activates invocation every and acyclic is
the among time per-invocation largest the – times $\max_{j: \mathcal{F}_j \rightarrow \mathcal{F}_i} t_j$ most
Then it. call that members

$$T_\pi \leq \sum_i \left(\max_{j: \mathcal{F}_j \rightarrow \mathcal{F}_i} t_j \right) t_i \leq \left(\max_i t_i \right) \sum_i t_i,$$

graph. the of depth the of independent time per-invocation a

times a_i some $\mathcal{F}_i$ each activates source a entering invocation An اثبات.
at $\mathcal{F}_i$ internal each and once runs source A steps. $\sum_i a_i t_i$ spends and
whence throughout. $a_i \leq \max_i t_i$ so hypothesis, by times $\max_{j \rightarrow i} t_j$ most
□ $T_\pi \leq \sum_i (\max_{j \rightarrow i} t_j) t_i \leq (\max_i t_i) \sum_i t_i$

bound not does alone acyclicity for place. its earns hypothesis The
per- the activation, one in $\mathcal{F}_i$ to makes $\mathcal{F}_j$ calls the for n_{ji} Writing T_π
the counts a_i so, $a_i = \sum_{j \rightarrow i} n_{ji} a_j$ solve counts activation invocation
them: multiplies graph reconvergent a and paths, $\mathcal{F}_i$ source-to- weighted
 $s \rightarrow \{A_k, B_k\} \rightarrow C_k \rightarrow \{A_{k+1}, B_{k+1}\} \rightarrow \dots$ – diamonds N of chain a
per- the and 2^N is foot the at α so level, each at count the doubles –
the by node each Capping depth. the in exponential is time invocation
stops what exactly is sum their by than rather callers its over maximum
and more. once theme -over-summation $\oplus$ the – accumulating paths the
Propo- of $k_i \leq 1$ case subroutine-respecting the of counterpart DAG the
bound caps invocation The . $\max_{j \rightarrow i} t_j$ to relaxed activation one, 8.2 sition
indep- alone, times the in bound a is here point the case: any in $a_i \leq r_i$
depth. the of and caps the of dent
alone caps the statement: general the of whole the is remark last That
structure. call the on condition no with class. cost the close already

mem- whose $\pi = \{\mathcal{F}_1, \dots, \mathcal{F}_n\}$ system A .composition) (Cost 8.5 نتیجه
most at over steps $\sum_i t_i r_i$ most at in runs (t_i, r_i) costs have bers
call its whatever $R_\pi \leq \sum_i r_i$ and $T_\pi \leq \sum_i t_i r_i$ so invocations, $\sum_i r_i$
and λ parameter a in polynomial is r_i and t_i each if Hence structure.
polynomially the polynomial: is (T_π, R_π) bounded. polynomially is n

formation. under closed are systems bounded

invo- one and ,8.2 Proposition are bounds invocation and step The اثبات. A too. $T_\pi \leq \sum_i t_i r_i$ so execution. whole the of part is cascade cation's □ polynomial. a is polynomials many polynomially of sum

depth-free the unlike here. needed is graph call the on condition No reconvergent A it. carry caps the because ,8.4 Proposition of bound time that if only so do can times 2^N member a activate would that cascade the outside already member a — $r_i \geq 2^N$ so calls. 2^N answers member trun- and excess the refuses cap the class the within class: polynomial counterpart efficiency the is This activations. $\sum_i r_i$ at cascade the cates em- $\rho[\pi/\varphi]$ — security preserves composition where :4.20 Theorem of efficiency. preserves composition cost — φ emulates π once ρ ulates itself is functionalities polynomial-time from assembled protocol a so is apparatus concrete the what are two the Together polynomial-time. polynomial-time into compose protocols UC-secure polynomial-time for: be below reading asymptotic the lets what is which protocols. UC-secure own. its of theory a than rather corollary a theorem composition the operation the survives closure same The runs.

and ρ Let .boundedness) polynomial preserves (Replacement 8.6 گزاره many. polynomially and polynomial are members whose systems be π $\rho[\pi/\varphi]$ Then well-formed. be ρ in π by φ of replacement the let and bounded. polynomially is

those are (4.1 (Definition $\rho[\pi/\varphi] = (\rho \setminus \varphi) \cup \pi$ of members The اثبات. those with together , ρ of member a as polynomial each , φ outside ρ of many. polynomially , $|\rho| + |\pi|$ most at number they polynomial: again , π of □ applies. 8.5 Corollary

the whenever efficient is replacement a after runs that system the So cost for read 4.20 Theorem — are π realization the and ρ specification operation. same the on and security. than rather

sim- the and adversaries the 4.20 Theorem In **numerically. Composition.** are environments The do. budgets their so unchanged. over carry ulators the is costs absorption what and computed. is something place one the cost the so member. by member budgeted are Systems hosting. of cost

of budgets the of -combination $\oplus$ the for $\mathbf{c}_{\rho \setminus \varphi}$ write up: looked be can the times. per-invocation their of largest the $-\rho \setminus \varphi$ of members the called- the to dual cost hosted-bundle the counts. invocation their of sum execution. any in them bounds which — 8.2 Proposition of cost subsystem 4.15 Definition of environment absorbed The them. driving is whoever so else. nothing and functionalities those and $\mathcal{Z}$ runs

$$\mathcal{Z} \in \mathbb{Z}_{\rho[\pi/\varphi], \rho}(\mathbf{z}) \implies \mathcal{Z}^{\rho \setminus \varphi} \in \mathbb{Z}_{\pi, \varphi}(\mathbf{z} \oplus \mathbf{c}_{\rho \setminus \varphi}),$$

the- the into this Feeding .4.16 Proposition being $\mathbb{Z}_{\pi, \varphi}$ in containment the is conclusion the orem.

$$\text{Adv}_{\rho[\pi/\varphi], \rho}^{\text{uc}}[\mathbf{a}, \mathbf{s}, \mathbf{z}] \leq \text{Adv}_{\pi, \varphi}^{\text{uc}}[\mathbf{a}, \mathbf{s}, \mathbf{z} \oplus \mathbf{c}_{\rho \setminus \varphi}].$$

the costs and simulator. the in and adversary the in free is Composition side. environment the on once. system surrounding

sim- step's first the that asks 4.12 Proposition **numerically. Transitivity.** in- an becomes which class. adversary step's second the be class ulator A . $\mathbf{s} \leq \mathbf{a}'$ as soon as holds $\mathbb{S}(\mathbf{s}) \subseteq \mathbb{A}(\mathbf{a}')$ budgets: between equality will- is step next the adversary an be must build to willing is one simulator both for admissible class one for asks also proposition The face. to ing $\mathcal{Z} \subseteq \mathbb{Z}_{\pi, \varphi} \cap \mathbb{Z}_{\varphi, \psi}$ for here: carried be must hypothesis that and steps. well, as $\mathbb{Z}_{\pi, \psi}$ in lies 4.12 Proposition of proof the by which

$$\text{Adv}_{\pi, \psi}^{\text{uc}}[\mathbf{a}, \mathbf{s}', \mathcal{Z}] \leq \text{Adv}_{\pi, \varphi}^{\text{uc}}[\mathbf{a}, \mathbf{s}, \mathcal{Z}] + \text{Adv}_{\varphi, \psi}^{\text{uc}}[\mathbf{s}, \mathbf{s}', \mathcal{Z}].$$

brack- the places three all in $\mathbf{z}$ with Written here. do not will alone Budgets con- is first the and $\mathbb{Z}_{\varphi, \psi}(\mathbf{z})$ and $\mathbb{Z}_{\pi, \varphi}(\mathbf{z})$, $\mathbb{Z}_{\pi, \psi}(\mathbf{z})$ over range would ets outer the for admissible environment an — others the of neither in tained mid- the cover would hypothesis no so — IDs(φ) about nothing told is pair dis- composition The cancels. inequality triangle the that probability dle the into class one its maps absorption trouble: such no has above play of chain the Along together. put displays two the is 4.21 Corollary other. hop each add. errors the accumulate: effects two same the 4.24 Theorem at absorbs it what hosting of cost the by budget environment the widens inequalities the become 4.25 Remark of relations class the and step. that $\mathbf{s}_{k+1} \leq \mathbf{a}_k$

emulation an that Say **up. blows simulator the whether and Nesting.** every for $(\mathbb{A}(\mathbf{a}), \mathbb{S}(f(\mathbf{a})), \cdot)$ against holds it if f overhead has statement one simulator the given. is adversary the whatever :a budget adversary

components two the budget per-invocation the In it. of f costs builds differently, behave

$$f(t, r) = (\max(t, \tau), \gamma r + \delta),$$

of call each for places simulator the calls the γ – affine count query the maximum. a time per-invocation the and – own its δ runs. it adversary the per runs machine one since work, per-activation own simulator's the τ each. to work bounded only adds simulator the and activation simulator and adversary its all. at f touch not does 4.20 Theorem the returns replacement single a so hypothesis. the of those are classes the on paid is price whole the and supplied hypothesis the machine very sim- a where only: place one in compose Overheads side. environment is $\mathcal{S}'$ There .4.12 Proposition is which adversary, an as re-read is ulator . $f_2 \circ f_1$ overhead has composite the so „ $\mathcal{A}$ from built itself „ $\mathcal{S}$ from built that , $r_j = \gamma r_{j-1} + \delta$ obeys budget query the times, k results the Applying is

$$r_k = \gamma^k r_0 + \delta \frac{\gamma^k - 1}{\gamma - 1} \quad (\gamma \neq 1), \quad r_k = r_0 + k\delta \quad (\gamma = 1),$$

bounded , $t_k = \max(t_0, \tau_1, \dots, \tau_k)$ saturates. time per-invocation the while depth. in constant and work own level's largest the by dichotomy a count. query the in entirely lives therefore blow-up The for call one than more place must that simulator A alone. γ on turning : γ^k cost applications k and , $\gamma > 1$ has runs it adversary the of call some of factor a costs already 20 of depth a $\gamma = 2$ at and exponential. genuinely call each relays that simulator A count. query the on million. a about , 2^{20} applications k and , $\gamma = 1$ has own its of traffic bounded adds and once check to property the is This level. per once paid δ linear. : $r_0 + k\delta$ cost why also is It it. implies here proved nothing built: is simulator a when being rewinding reasons. usual the beyond matter simulators straight-line in anywhere simulator rewinding one : $\gamma > 1$ acquire to way standard the it. below level every on γ factor a puts tower the

4.24 Theorem depth. as well as breadth governs formula same The adver- its as step next the to simulator each handing by $\mathcal{S}_n, \dots, \mathcal{S}_1$ builds -deepn an as exactly overheads n compose instances parallel n so sary. at $\mathbf{a}_k \geq f_{k+1}(\mathbf{a}_{k+1})$ asks budgets with read 4.25 Remark and does. tower extra $n\delta$ costs protocol $\gamma = 1$ a of instances n Composing link. every mod- even for hopeless is one $\gamma = 2$ a of instances n composing queries: . n erate

per- the in or side, environment the on happens comparable Nothing absorbs, it shell the hosting of cost the adds level Each time. invocation inner- the reaches nesting kdepth- a so disjoint, are shells tower's a and run- of that count invocation its $-z \oplus \bigoplus_{i \leq k} c_i$ budget at statement most shell's, largest the time per-invocation its once, system ambient the ning of applications k likewise, behave errors The tower, the deep however exponen- degrades claim security a Where $\sum_i \varepsilon_i$ giving 4.12 Proposition possi- only the is $\gamma > 1$ overhead query simulator's the nesting, under tially per-invocation the neither so activation, per runs machine one source: ble multiply, can budget environment the nor time

security a mentions above Nothing **reading. asymptotic the Recovering** the let λ by everything index statement, familiar the get To parameter. negligible be advantage the that ask and λ in polynomials be budgets and machines bounded polynomially the become then classes the it: in is form concrete The emulation. UC of definition usual the 4.11 Definition is form asymptotic the and in, proved are results composition the one the practice-oriented ordering the — round way other the not it, of corollary a .[2] for argued long has security provable

Calls Responsive

write to way no had far so has framework the promise a needs 15 Chapter back, one get must it signature a for adversary the asks Sign When down. of rest the drive and off go to free adversary the were :now back it get and mid- suspended left be would party the answering, before execution the asks UC synchronous immediacy same the — lost locality and signature the restricting for settles there text The .[11] environment responsive a of instead. code in thing the says chapter This class. simulator

it — helpfully answer adversary the that Not **forbidden. be to has What** to has What answer. bad a with deals 12 Chapter and anything, return may returning, and call the receiving between leaving: token the is forbidden be a calls the on restriction a is That all. at call no place must responder the that. exactly for wrapper a has already paper the and makes, machine

maxi- its at **Silence** is **Respond** .answering) (Responsive 9.1 **تعريف**
identities, all of set the $\mathcal{J}_{all}$ with mum:

$$\text{Respond}(\text{id.Op(in) from id'})$$

$$:= \text{Silence}(\mathcal{J}_{all}, \text{id.Op(in) from id'}).$$

pro- wrapped the call every 2 Chapter by so , $\mathcal{J}_{all}$ in lies claim Every
pro- The placed. being ever without **rej** with refused is issues gramme
value: a return still must and likes it as long as compute may gramme
on. token the hand is do cannot it what

clause last its with **Silence** is **Respond** out, Written **handler. a as Read**
so permits, it claim whose call a forwards **Silence** wrapper The deleted.
a only re-raise, no has **Respond** re-raise: a partly is clause operation its
ef- call ambient the discharges it 2 Chapter of vocabulary the In constant.

Respond Wrapper

id' from id.Op(in) programme

:Respond(id.Op(in) from id')

```

:1 handle id.Op(in) from id' with
:2   return out                ↦ out
:3   id''.FullOp(in'') as id''; k ↦ k(rej)

```

locally handled is effect the scope its inside it: forwarding than rather fect
 It it. to respect with pure is programme wrapped the so completely, and
 on call a wanted: 15 Chapter property the locality, of definition the also is
 never token the activation, own responder's the within served is slot the
 responsive a – responsiveness of statement crisp the is That it. leaving
 oper- the and – Exec to escape not does effect call whose one is core
 do that calls the exactly dispatches Exec since follows, reading ational
 more. no and responder the of activation one is call responsive A escape.

car- adversary The .call) responsive interface. (Responsive 9.2 تعريف
 and guard same the with FullA¹ interface second a ,FullA beside ries.
 :Silence of place in Respond but core same the

```

id.FullA1(in) from id' :   require GuardA(id, id');
                           return Respond(id.A(in) from id').

```

identity with interface an Inside .FullA¹ on call a is call responsive A
 abbreviates $\mathcal{A}(\text{in})$ as ,id as (A, id.P).FullA¹(in) for $\mathcal{A}^1(\text{in})$ write we id
 one. ordinary the

price. the are they and follow. things Three
 calling by reached is C set The register. the read cannot responder The
 computed is answer responsive a so refused. is call that and ,FullStatus
 subrou- no delegate: cannot It knows. already responder the what from
 being Corrupt corrupt. cannot it And environment. no party. other no tine.
 Sec- because own. its on stating deserves last The other. any like call a
 is reads two the between gap the and twice register the reads 3.5 tion
 in. slip can corruption a where exactly

reg- the execution. any In .corrupt) cannot call responsive (A 9.3 گزاره
 call the when state its is returns call responsive a when C state ister's

placed. was

اثبات. 3.1 Section By **C** only changes 5 line on **FullCorrupt** and only
 re- the and placing the Between call. a by reached is interface that when
 and throughout. token the holds responder the call responsive a of turn
 so placed. being before refused is issues it call every 9.1 Definition by
 in- that in token the holds machine other No reached. not is **FullCorrupt**
 □ either. it reach can none so terval.

serves it party the $\mathcal{A}^1$ through only adversary the reaches core a if So
 second the and 3 line at corrupt was it when exactly 7 line at corrupt is
 describes 3.2 Remark call. that of account on fire cannot hook mediation
 which under condition the is responsiveness does: it when happens what
 not. does it

Sign ,Gen in $\mathcal{A}$ of place in $\mathcal{A}^1$ Writing wanted. 15 Chapter what is This
 condition a than rather code the of property a locality makes Verify and
 every of holds it so it. enforces wrapper the and class. simulator the on
 4.7 Definition included. simulator the – slot adversary the of occupant
 What them. among **FullA**¹ interfaces. adversary's the simulator a giving
 may adversary responsive a good: answer the make is do not does it
 back: token the buys Responsiveness that. repairs **San** and $\perp$ return still
 value. the buys sanitization

نکته 9.4 (Responsiveness) the and (The dummy) pull chapters two
 The where. about explicit being worth is it and other. each against
 relaying by functionality a from call a answers 4.4 Section of dummy
Respond which call. a is that and – 5 line on environment the to it
 refused. is relay the call: responsive a serve cannot $\mathcal{D}$ So refuses.
 ad- real a where $\perp$ as answer its delivers 3.6 Section of rule the and
 would (R1) regrouping The substance. with answer would versary
 respon- first the at $\mathcal{A}^1$ use cores whose system a for fail therefore
 whose systems hypothesises 4.29 Theorem why is which – call sive
 systems responsive for completeness Recovering none. place cores
 relayed the that so side. environment's the on notion matching a needs
 that leave we on: moving token the without answered be can question
 open.

Language Core

protocols functionalities. book's this of code the give 21 to 13 Chapters an as not program, a as read be to meant is code that and games, and convention stated a by kept be to has that promise a — one of illustration for promise this exactly keep [5] will. good reader's the by than rather language terminating typed, strongly small, a with proofs game-based short a and (15 Figure (their grammar context-free a by given, $\mathcal{L}$ call they the takes chapter This it. of top on allow they sugars the of list named here from chapter every what for grammar closed a here: steps two same it for word own book's this already body, operation's an — core a calls on each grammar, that from departures of list short own book's this and — it against checked be can come to still core every that so once named faith. on taken than rather

.Guard so. deliberately and exempt. is chapter this before Everything adver- dummy the .(3.6 (Section Exec .(2 (Chapter Mediate and Silence they core: a inside run not do (9 (Chapter Respond and (4.4 (Section sary guards, evaluating calls, dispatching cores, of system a runs what are exe- of definition A next. the to activation one from token the handing as exactly — executes it language the in program a itself not is cution informal their in sits adversary an meeting game a of account own]'s5] no- one the and — 15 Figure their of grammar the outside .3 Section form continuation the own, core's a for mistaken be could that tation be- .k resumption unbound explicit, an carrying id".FullOp(in") as id""; k operation's an inside occurs once never it layer: outer that to entirely longs — constantly does. it of relative plainer A on. 13 Chapter from code own the under resumed operation, functionality's another to call ordinary an that and — continuation fresh a handed than rather identity own caller's below. right, own its in matter core-language a is one

op- every and statements. of sequence a is core A **grammar. core The** book's this what to Scoped core. one exactly is writes book this eration grammar: full 's $\mathcal{L}$ to than rather use. actually cores

```

core ::=  $\varepsilon$  | stmt ; core
stmt ::= lv  $\leftarrow$  e | lv  $\leftarrow_{\mathcal{S}}$  S
        | if e then core [ else core ]
        | for  $x \in S$  do core
        | return e
lv ::= x | x[e]
e ::= ... | call
call ::= id".Op( $e_1, \dots, e_k$ ) as id

```

شكل 5: The core The grammar 's $\mathcal{L}$ usage. own book's this to scoped grammar. core The (15 Figure their mathematical ordinary to left expressions of sub-grammar the with six the over standard already is writes book this expression every since notation. (below). restriction Adversary-only their past generalized own. 's $\mathcal{L}$ is call below: types (Chapter 10.)

's $\mathcal{L}$ – table set. string. boolean. integer. five: 's $\mathcal{L}$ not types. base Six reading $\mathbb{L} : \mathbb{N} \rightarrow \{0, 1\}^n \cup \{\square\}$ throughout. type function's a as kept array. book's this $\square$ there. does $A : \{0, 1\}^* \rightarrow \{0, 1\}^* \cup \{\text{undefined}\}$ as exactly -for one Only below. addition. own book's this identity. and – undefined pair a binds it of instance one the and ,for $x \in S$ do attested. is form obvious an – (16 and 17 lines ,20 (Chapter for $(m, y) \in \mathbb{N}$ do directly. integer- bounded The it. destructuring and variable one binding for sugar the licensed and far, so unused reserve. in is allows also $\mathcal{L}$ form range it. needs core a moment

id' from id.Op(in) headed is core operation's Every **implicit. is caller The** though core the in anywhere readable is – identity caller's the – id' and it: like nothing carries $\mathcal{L}$ in oracle plain A .Op of argument no names it is who exactly asking by begin many so because it need cores book's this representative being operations 's $\mathcal{G}_{\text{Clock}}$ runs. else anything before calling does core the before corruption 'sid.P and id'. F tests 13 line at guard (the else). anything

the for expression call the reserves $\mathcal{L}$ **core. another call may core A** one. place not may games their of one inside oracle an alone: Adversary of protocol the – does book this in game and protocol functionality. Every neighbours), and 22 (lines mid-operation $\mathcal{F}_{\text{Sig}}$ and $\mathcal{F}_{\text{PKI}}$ reads 20 Chapter (5 (line itself $\mathcal{F}_{\text{Rand}}$ into back calls game unpredictability own 's $\mathcal{F}_{\text{Rand}}$ and

others from built are protocols and functionalities book's this because – and, (1 (Chapter call may core a what exactly naming **U** declaration, by own, 's $\mathcal{L}$ is itself call The other. any like functionality a is shell game's a re- and expression other any like evaluates $\text{id}''.$ Op($e_1, \dots$) **as** id unrelaxed: stays What identity. own its under line, same the at core calling the sumes k continuation unbound an with form. general the is core a outside strictly this of line next "the than rather point resumption arbitrary an for standing call a what for notation own model's execution the is that – core" same defined. is it where is 3.6 Section and core's, a never general. in means

the throughout used line, **require** e own core's A **sugar. is Require** sugar is come. to cores the of plenty inside and chapters execution-model guard failed a though even rej literally not – **not if** e **then return** $\perp$ for no value one the rej makes 3.6 Section that. exactly returns elsewhere be- $\perp$ into rej core-level a laundering output, own its as return may core com- case common the names merely **require** own core's a delivery; fore or- an with hand. by out guard same the spell operations 's $\mathcal{G}_{\text{Clock}}$ pactly. core-language one the are both again): 13 (line **if ... then return** dinary not. or sugared statement.

string- is neither :L[ctr], reg[id.P] **rule. new no need indices Table** – ([5]) enhancement third own its answer. the has already $\mathcal{L}$ and indexed. table Every silently. first, encode($\cdot$) through passed is index non-string a is there way; this reads already string-indexed not is that writes book this it. for them citing beyond here add to nothing

makes 1.1 Definition sugar. not is departure One **type. sixth a is Identity** process a party, a and id process a of $\text{id} = (\text{pid}, P)$ pair a identity an once. at levels both through reads core a and – (F, s, i) triple a itself id reads ever core no field one the $\text{id}.i$ projections, named $\text{id}.P$, $\text{id}.s$, $\text{id}.F$ projection, a carries types five 's $\mathcal{L}$ of None outright). so says 1 (Chapter of account own 's5] by equality-preservation. only gives encode($\cdot$) and identity So serve. to that for often too far apart read is identity an – it with atomic, than rather structured type. base sixth a as list type the joins one the – operations only its as equality componentwise and projection have. already not does it something for $\mathcal{L}$ ask cores book's this place

how compute: may core a what is above grammar The **code. the Reading** kept. mostly already one and promise. separate a is page the on set is it **.return .do .for .else .then .if** default: algorithmic's bold. are Keywords declared is it wherever link a carries name operation's An **.require** and

caps small **accentsoft** in rendered both to, back called is it wherever and ev- from it to back point more two declaration, the anchors macro one – mutable own core's a for reserved typewriter, ∞ in lives State site. call ery the and state, than rather parameters bound italic, stay out in id fields: own core's a what deciding reader future a – together run be not must two Com- other, no and distinction this exactly needs like looks type record That load-bearing, never side, the to off italic, **commenttint** in print ments nota- new not done, is written already chapters the over pass copy-editing already-declared an of mention prose first a sweep: uniformity a but tion constant the and throughout, declaration its to back linked is operation stor- and tag trace every with together **done**, **false**, **true**, **rej**, **ok** – atoms op- an than rather **commenttint** carry – book this in elsewhere key age a is “this separate does now colour bare so colour, own name's eration value.” a is “this from name”

sin- a is together, buy, above departures the and grammar the What own core's a is math book's this of line a on, **13** Chapter from test: gle depar- the plus above table the against type-checks it when exactly code property's a verdict, game's a – specification a is it and here, named tures to gap the Closing not, does it when exactly – pick own 's**San** quantifier, compiling of matter a then is otherwise, or OCaml language, actual an deciding, first of not once, forms statement of handful a and types six to claiming ever was symbols of display given a what chapter, by chapter compute.

Bad Until Identical

same the are that games two :[13, 5] hops by advance proofs Code-based
 probabill- the on bound a and raised, been has flag a where except program
 licence the is playing game of lemma fundamental The raising, the of ity
 probabill- with differ outcomes their ,bad until identical are games the if –
 the on games both run coupling: a is proof its and – $\Pr[\text{Bad}]$ most at ity
 this In up, goes flag the before company part cannot they and coins same
 a environment, an runs execution An program, one not is game a paper
 them: among wanders token the system: a and register a occupant, slot
 rest the drive and mid-operation token the handed be may occupant the
 dis- to exists 9 Chapter freedom the – answering before execution the of
 lemma the replays chapter This it, of all survive must hop a and – cipline
 the nothing: costs wandering the that is finding the and executions, over
 all, at hypothesis no has below lemma

ordinary an of variable state ordinary an is `bad` **lives, flag the Where**
 but value any assigned never ,Initialize in `bad ← false` declared machine:
 machine the by state machine's a of rest the like owned afterwards, `true`
 – it off hand other every keeps already framework The it, declares that
 calls provoke is own not does it state to do can machine a thing one the
 answers, replaces **Mediate** party corrupt a at and it, extend or read that
 or code own owner's its by raised is flag a so – (5.4 (Remark state never
 interface an through immaterial, is it read may anyone Whether all, at not
 two the raised is flag the until :**Guard** bypasses which ,Leak through or
 lemma the raised is it after and alike, read any answer below machines
 members – once at flags carry may machines Several paid, already has
 the is below Bad and – own its owning each included, name a sharing
 flag, some raises machine some that event

تعريف 11.1 (Identical) until (bad) functionalities $\mathcal{F}'$ and $\mathcal{F}$ are $\mathcal{F}'$ and $\mathcal{F}$ functionalities Standard. line agree initialize their agree: fields five their if bad until identical among $\text{bad} \leftarrow \text{false}$ and sides both to common declarations line. for bad-guarded inside except agrees code remaining their and them: $\text{bad} \leftarrow \text{true}$ assignment an of continuation the shapes: two of blocks. the — block enclosing its of end the to assignment the past just from ev- that so in. stands assignment the body operation or loop branch. — assignment the across only reached is continuation the in erything bad- the Erasing. bad is test whose conditional a of body the and operation functionality. one leaves sides both from blocks guarded assign- the particular in included: Leak line. for line and operation for divergent own its heading each code. common are themselves ments Equal. true but value any bad assigns side neither and continuation. occu- Two erase. to nothing with bad. until identical are functionalities initialize their when bad until identical are slot adversary the of pants until identical are π' and π Systems way. same the read are. cores and the of member a with one of member every pairs bijection some if bad it. with bad until identical other

by checked is Membership]'s.5] both shape. the about things Two — behaviour about claim a not code. of comparison a is it — inspection divergent a that so built is it And audit. to cheap hops keeps what is which its are block bad-guarded a into ways only the early: run cannot statement the of test a and crossed. is it as flag the raises that assignment an head. while runs bad-guarded nothing so — monotone is flag the and itself. flag the fields. equal carry members paired Because down. is flag owner's its $\langle \pi \rangle = \text{names}$. same the use and identities same the occupy systems two $\mathbb{Z}_{\pi, \pi'} = \mathbb{Z}_{\pi, \pi} = \mathbb{Z}_{\pi', \pi'}$ both: serves environments of class one and $\langle \pi' \rangle$

الم 11.2 (The fundamental lemma over systems) Let π' and π systems. Let $\mathcal{X}'$ and $\mathcal{X}$ occupants let bad. until identical be some that event the for Bad Write $\mathcal{Z} \in \mathbb{Z}_{\pi, \pi'}$ let and bad. until identical execution whichever in $\text{bad} \leftarrow \text{true}$ assignment an executes machine Then named. is

$$\Pr[\text{Bad in } \text{Exec}(\pi, \mathcal{Z}, \mathcal{X}, \mathcal{C})] = \Pr[\text{Bad in } \text{Exec}(\pi', \mathcal{Z}, \mathcal{X}', \mathcal{C})],$$

value, common the for $\Pr[\text{Bad}]$ writing and,

$$\begin{aligned} & \left| \Pr[\text{Exec}(\pi, \mathcal{Z}, \mathcal{X}, \mathcal{C}) = 1] \right. \\ & \left. - \Pr[\text{Exec}(\pi', \mathcal{Z}, \mathcal{X}', \mathcal{C}) = 1] \right| \leq \Pr[\text{Bad}]. \end{aligned}$$

then pairing the – systems game are systems two the moreover If
of holds same the – preserved being fields game, to game matches
share: games two the which ,Fin finalization every

$$\begin{aligned} & \left| \Pr[\text{Win}_{\text{Fin}} \text{ in } \text{Exec}(\pi, \mathcal{Z}, \mathcal{X}, \mathcal{C})] \right. \\ & \left. - \Pr[\text{Win}_{\text{Fin}} \text{ in } \text{Exec}(\pi', \mathcal{Z}, \mathcal{X}', \mathcal{C})] \right| \\ & \leq \Pr[\text{Bad}]. \end{aligned}$$

execu- both fixes tape every Fixing pointwise. read coupling. One اثبات.
compared and sample one from taken be may two the so outright. tions
corre- two the up is flag no while that shows induction The step. by step
exactly is which – code common on stands token the and strictly spond
reach- being neither buy. shapes bad-guarded two 's11.1 Definition what
test a through or flag the raises that assignment the across except able
step same the therefore is raising first The down. is it while fails that
rather number one is $\Pr[\text{Bad}]$ display: first the is that and sides. both on
equal are runs two the Bad Off bound. a share to happen that two than
each and there. indicator single a has run the of fact every so outright.
indi- where set the of measure the most at is probabilities of difference
the used, are 3.6 and 1 Chapters of conventions Two differ. may cators
own its from draws machine each restates: lemma absorption the ones
its when only read other's. every of independent and uniform tape, coin
their of tapes keeping copies hosted environment's an – samples code
ev- Fixing calls. no places Initialize and – nothing merging bundling own.
the holds machine which outright: execution an fixes therefore tape eryl
next its what and state. what with operation, which of line which at token.
as only entering coins configuration, the of functions are does statement
tape. own holder's the of positions unread next the
machine to machine matches pairing The executions. two the Couple
the and environment the occupant, to occupant member, to member –
iden- same the occupy machines paired and – themselves to each register
separately execution Each tapes. equal machines paired give so titles.
indep- and uniform being side either on tapes distribution, its keeps

point- compared and tapes of sample one off read be may both so dent.
 of (C1)–(C4) if configuration a at strictly correspond two the Say wise.
 tapes. unread equal states. equal — pairing the under hold 4.17 Definition
 sus- values. local same the with line same the at hands paired in token the
 and anywhere relay no with — frame for frame matching callers pended
 elided. nothing

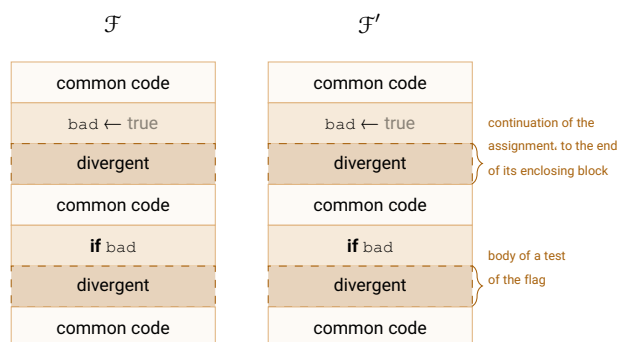
register the initialises 1 Line strictly. correspond they start the At
 places Initialize — order whatever in sides. both on member every and
 trace no leaves order the so owner's. its but state no touches and calls no
 line. for line agree that sInitialize run members paired configuration: the in
 draw. they where drawing. and calls no placing them. among $\text{bad} \leftarrow \text{false}$
 one the to tokens both hands 2 line and tapes: equal of positions same the
 identity. same the at environment

side. either on raised yet flag no strictly. correspond two the Suppose
 the at machines paired in stands token The run. statement one let and
 is statement bad-guarded a code: common is line that and line. same
 raises head the flag. the of test a through or head its through only reached
 flag no and down. is flag the while fails test the crossed. is it as flag the
 state same the over read statement. same the is statement the So up. is
 If alike. tape for tape consumed having histories coins. next same the and
 frame same the resume both returns. it if alike: write sides both writes. it
 tar- — call same the place both call. a places it if value: same the with
 equal over wrappers same the by met — identity claimed and payload get.
 no at or machines. paired at landing state. register equal an and fields
 rej — fixes 3.6 Section delivery every and alike: rej answered and machine
 slot mediated a or rej return would core a where $\perp$. **require** refused a for
 the survives correspondence Strict choice. a not rule. a is — one answers
 code. common — $\text{bad} \leftarrow \text{true}$ assignment an is statement the if And step.
 it. execute sides both — continuation divergent its of head the at standing
 once. at executions both in raising first the is step the and

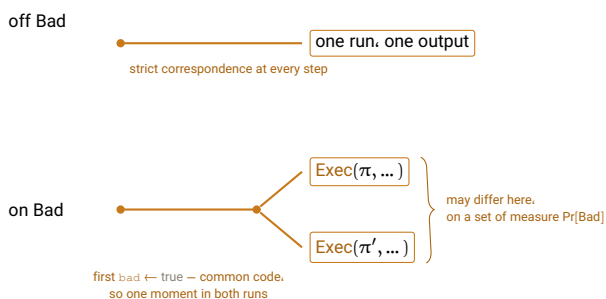
and raised ever is flag no either then: space. coupled the on Pointwise
 exactly halts one so — step every at strictly correspond executions two the
 one being environment the output. same the with does. other the when
 and simultaneous. is it raising. first a is there or — alike resumed machine
 has therefore Bad event The it. through and to up agree executions the
 equality. with display. first the is which space. coupled the on indicator one
 indicator one has run the of fact any so outright. equal are runs the Bad Off
 the that Win_{Fin} event the systems. game for and. output. the too: there
 of difference Each .(5.8 (Definition 1 returns $(\mathcal{G}\text{m}.\text{pid}, Z)$). **FullFin** interface

may indicators its where set the of measure the most at is probabilities
 $\square$.Pr[Bad] is which differ.

worth are they and each, picture a have proof the and definition The
 Above consumes. other the what exactly is one since together, seeing
 then coupling the what is below code: the of asks 11.1 Definition what is
 earlier the of vocabulary box-and-wire the uses panel Neither it. with does
 to machine one from moved is here nothing point: the is that and figures.
 between. wire a draw to nothing is there so another.



erase the two shaded blocks and one functionality remains.
 operation for operation and line for line



شکل 6: Bad Until Identical. panels. two in bad, until Identical
 box-and-wire the uses panel Neither it. with does coupling the what below. code:
 to machine one from moves here nothing because figures. earlier the of vocabulary
 (11.2 Lemma and 11.1 (Definition another.

hop a keeps what is which inspection. by checkable is panel upper The
 behaviour. about claim a not code. of comparison a is it audit: to cheap

they them between and allowed. ones only the are shapes shaded two The in ways only the — down is flag the while run can shaded nothing why are flag. the of test a and crossed. is it as flag the raises which head. the are runs what exactly is above shading the So down. is it while fails which else. nowhere and below. fork the beyond

The argument. the of whole the is that and point. single a is fork The both so code. common itself is block divergent each heading statement which — sample coupled same the of step same the at it execute runs two the in number same the is $\text{Pr}[\text{Bad}]$ that display. first lemma's the gives no is there Bad Off bound. one with numbers two than rather executions indicator single a has it of fact every so run. one are runs two the all: at fork difference each and — Win_{Fin} systems. game for and. output. the — there may indicators where set the of measure the most at is probabilities of $\text{Pr}[\text{Bad}]$ is measure its and branch. lower the is set That differ.

sys- the of nothing asks lemma The .assumed) is (Nothing 11.3 نکته refusal- not tightness. not well-formedness. not slot: the of or tems responsive of absence the not — savouring worth — and obliviousness. the for hypothesis a pays machine a moves that result Every calls. cores. responsive exclude 5.9 Proposition and 4.29 Theorem move: the and once. at sides both on exclusion the inherits 5.11 Corollary for three all outside sit channel the and network the clock. notifying respon- their — indifferent is above lemma The reason. that exactly nothing because — else anything as cheaply as hop notifications sive on resting runs. about identity an is it regrouping the Like moved. is fixed. are tapes its once execution an of determinism the but nothing for appears budget No achieve. runs two what of comparison a not enters 4.30 Definition and none. needs exactness reason: same the $\text{Pr}[\text{Bad}]$ bounding about sets application an when only

route: shorter tempting a is There .absorption) by not (Why 11.4 نکته exists 4.18 Lemma — environment the into execution whole the absorb [5] quote and program. single a is side each until — machines move to why see to instructive is it and works. it where works It pair. the on slot the along: come hypotheses Absorption's lemma. worse the is it re- a place may core no refusal-oblivious. be must moved occupant environment's the and (9.4 Remark identity ceded a on call sponse

only lemma the proves reduction the so — -identityZ no hold must par
 notifying the and covers, already theorem dummy the systems the for
 than longer no is coupling direct The it. outside falls one, for clock,
 4.17 Definition of machinery the nothing: carries and reduction the
 coupled executions two and rebracketed, execution one for built was
 — analysis case easier an with induction same the is tapes equal on
 over. smooth to refusal no claims, of rewriting no relay, no

between hop arguments Emulation concretely, for, is lemma the What
 simula- a variant, lazier a and functionality a objects: ideal neighbouring
 pair a is hop Each removed, check one with simulator same the and tor
 the and company, part two the where raised flags the bad, until identical
 the against execution, same the in — $\Pr[\text{Bad}]$ at hop the prices lemma
 proof a where alongside hopping occupants the with environment, same
 prop- a generally is that and , $\Pr[\text{Bad}]$ bound to is remains What it. needs
 condition raising the that precisely is verdict whose finalization a :erty
 applies, transfer where inherited, and 5.8 Definition by priced occurred,
 func- signature the at sits instance canonical The .5.11 Corollary through
 honest- 'sVerify where bad raise to $\mathcal{F}_{\text{Sig}}$ Instrument .15 Chapter of tionality
 and written being flag the see, can caller no addition an — fires test key
 stand, forgery the lets then that variant the it beside set and — read never
 proba- the costs them between hop the and bad, until identical are two The
 police, to exists FinUF quantity the — submitted ever is forgery a that bility

Sanitization

af- not do attacks so outputs. simulator sanitize to need often will We
for settings. honest-majority In adversely. too functionalities ideal fect
preserve to one $\perp$ non- a to output $\perp$ a sanitize typically we instance.
a is Clean where $\text{San}[\text{Clean}](v; c)$ procedure a such write We liveness.
re- and consults. it context whatever with together value a on predicate
as operates which $v' \leftarrow \text{San}[\text{Clean}](v; c)$ by v output simulator a place
lists semicolon the after c context The otherwise. stated unless below
different and read-only. is it reads: predicate the state functionality the
contexts. different take predicates

San[Clean] Sanitizer		
c context	v value	Clean predicate
:San[Clean](v; c)		
:1	if Clean(v; c) then	
:2	return v	
:3	pick $v' \in \{v' : \text{Clean}(v'; c)\}$	
:4	return v'	

returns it what and fails. Clean(v; c) when only acts sanitization So
access read-only than more no need sanitizers Our ¹.Clean satisfies then
specification the of part is picks 3 line How state. functionality the to
observably are rules different so caller. the reaches value picked the –
value good first lexicographically the fix we and – functionalities different
only surfaces pick the looks: it than less matters choice The throughout.

it everything and infinite is from draws Clean space the whenever exists value good A¹
needed are halves Both .c context the in named value a is non-membership beyond excludes
of conjunct first the is Σ or $\mathcal{K}$ infinite an in membership :15 Chapter of predicates the for
4.30 Definition Under .c in passed table a of entry an is exclude they else everything and each.
many finitely records run bounded a – execution any in supported finitely is context the
space. infinite an inside finite is set excluded the so – entries

some preferring adversary an and .Clean fails answer own slot's the when
it. submit simply may value good particular

Randomness

Functionality 13.1

read to one right the and paper this in functionality smallest the is $\mathcal{F}_{\text{Rand}}$ out, handed it what remembers strings, -bitn uniform out hands It first.

1 Chapter remembers, still it what leaks and entry, an forget party a lets code, the is here code: no owed and it named

It footnote, a than rather chapter own its worth it make things Two is it so – adversary the on not clock, the on not – all at call no places has 9 Chapter and slot, the to left is all at nothing where example one the not is paper, this in other every unlike property, its And price, to nothing bound the probability a to up unpredictable is randomness :0 at attained decorative, being stops 5.8 Definition of ε the where is which name, to has

per instance one parameters, are **N** and **P**, pid pattern, 's $\mathcal{F}_{\text{Sig}}$ **fields. The** .1 Chapter of pattern local the in served protocol the naming **N** session, strings, the of width the ,par := n, make, to call no being there ,**U** := $\emptyset$.5 line at reads **Rnd** which

party's corrupt a so entire, $\perp$ returns Leak **for. is leak the what is Erasure** the but defect a not is That adversary's, the is draws of history whole must it randomness draws that protocol a it: beside **Erase** having of point Read leak, cannot erased been has what and entry, the erases retain not and sampling, of as much as erasure secure of model a is $\mathcal{F}_{\text{Rand}}$ way that One way, same the shaped is 14 Chapter of store the why is pair the wrongly: state to easy is it since plainly, stating worth is consequence times of number the is $\perp$ in entries of number the erased, is nothing while erasing values: the as well as count the reports leak the so called, was Rnd count, that of out entry an takes what exactly is

$\mathcal{F}_{\text{Rand}}$ Functionality	
$\text{par} := n$	$\mathcal{U} := \emptyset$
$\mathcal{N}$	$\mathcal{P}$
pid	
$\text{id}' \text{ from } \text{id.Erase}(i)$ $:8 \text{ } \mathcal{L}[i] \leftarrow \square$ $:9 \text{ return ok}$	$:\text{Initialize}()$ $:1 \text{ ctr} \leftarrow 0$ $:2 \text{ } \mathcal{L} : \mathbb{N} \rightarrow \{0, 1\}^n \cup \{\square\}$ $:3 \text{ } \mathcal{L}[*] \leftarrow \square$
$\text{id}' \text{ from } \text{id.Leak}()$ $:10 \text{ return } \mathcal{L}$	$\text{id}' \text{ from } \text{id.Rnd}()$ $:4 \text{ ctr} \leftarrow \text{ctr} + 1$ $:5 \text{ } r \leftarrow_s \{0, 1\}^n \text{ / the one sampling in the paper}$ $:6 \text{ } \mathcal{L}[\text{ctr}] \leftarrow r$ $:7 \text{ return } (\text{ctr}, r) \text{ / the index, so a caller may erase}$

index the with answers Rnd operation The code. the on remarks Three
 be must draw a erase to means that caller A string. the as well as wrote it
 so caller's, the not and state functionality's the is `ctr` and it, name to able
 to willing caller a by used be only could it beside Erase the 7 line without
 this and draws, it coin every erases 21.2 Section of protocol the – guess
 so rewritten. never and counter a by indexed are Entries it. lets what is
 never-written between distinction 's $\mathcal{F}_{\text{Store}}$ to companion no needs Erase
 index an at $\square$ and reused. never is passed has Rnd index an erased: and
 counter the says it above $\square$ while erased was entry the says `ctr` below
 any as parties, across shared is counter the And there. reached not has
 dis- draw instance one from drawing parties two is: state functionality's
 per-party a than rather source a $\mathcal{F}_{\text{Rand}}$ makes what is which entries, tinct
 generator.

Properties 13.2

advance in names environment no :unpredictability, **FinUnp** property. One
 .5.8 Definition of reading search the takes It returns. draw later a value a
 $\mathcal{F}_{\text{Rand}}.\mathcal{N} := \{\mathcal{Gm}.\text{pid}\}$ with $\gamma := \{\mathcal{Gm}\} \cup \{\mathcal{F}_{\text{Rand}}\}$ over played is game The
 once argument tightness 's15.2 Section, $\mathcal{F}_{\text{Rand}}.\mathcal{P} \subseteq \mathcal{Gm}.\mathcal{P}$ and pinned
 the draw a is see not does game the draw a reason: its for and more.
 on. rule cannot verdict

Guess operation The relay. a not is that interface one carries shell The
 environment the that so exists it all: at not $\mathcal{F}_{\text{Rand}}$ touches and call no places
 verdict the lets what is sequence a being trace the and record, on go can
 order. of use 's17.2 Section is That first. record on went it whether ask
 answers, of order the read steadiness purpose: different a for borrowed
 answer. an against claim a of order the reads unpredictability

follows finalization the $\mathcal{F}_{\text{Rand}}$ over shell Game

$\mathcal{U} := \{(\mathcal{F}_{\text{Rand}}, \text{serve}_{\text{gm}})\}$ $\mathcal{N} := \mathbf{Z}$ $\mathcal{P} := \mathcal{F}_{\text{Rand}} \cdot \mathbf{P} \cup \{\mathbf{Z}\}$ $\text{pid} := (\text{gm.F}, 0, 0)$
 $\text{par} := \perp$

<pre> :13 require $\neg \text{done}$:14 $\text{tr} \leftarrow \text{tr} \cdot (\text{guess}, \text{id.P}, y)$ / no call: the root may guess too :15 return ok :16 return $\perp$ </pre>	<pre> id' from <u>id.Guess(y)</u> :1 $\text{done} \leftarrow \text{false}$:2 $\text{tr} \leftarrow ()$ / the trace. a sequence id' from <u>id.Rnd()</u> :3 require $\neg \text{done}$:4 require $\text{id.P} \neq \mathbf{Z}$ / the root only finalizes :5 $(i, r) \leftarrow \text{id}_{\mathcal{F}_{\text{Rand}}} \cdot \text{FullRnd}()$ as id / the index is not recorded :6 $\text{tr} \leftarrow \text{tr} \cdot (\text{rnd}, \text{id.P}, r)$:7 return r id' from <u>id.Erase(i)</u> :8 require $\neg \text{done}$:9 require $\text{id.P} \neq \mathbf{Z}$:10 $\text{id}_{\mathcal{F}_{\text{Rand}}} \cdot \text{FullErase}(i)$ as id :11 $\text{tr} \leftarrow \text{tr} \cdot (\text{erase}, \text{id.P}, i)$:12 return ok </pre>
--	--

shared is $\mathcal{F}_{\text{Rand}}$ over shell the $\mathcal{G}_{\text{m}}$ of finalization The

(unpredictability) id' from id.FinUnp()

```

:17 require  $\neg \text{done}$ 
:18 require  $\text{id.P} = \mathbf{Z}$ 
:19  $\text{done} \leftarrow \text{true}$ 
:20  $\text{parse tr as } (e_j)_{j=1}^m$ 
:21  $w \leftarrow 1$  if  $\exists j < k \exists y : e_j = (\text{guess}, \cdot, y) \wedge e_k = (\text{rnd}, \cdot, y)$ 
:22 else  $w \leftarrow 0$ 
:23 return w

```

does Guess and root the refuse relays The code. the about things Two would root the at relay a 's:17.2 Section exactly is asymmetry the and not. ap- and ,1 line at rej answered be serve, not does $\mathcal{F}_{\text{Rand}}$ party a address guessed that environment an whereupon – trace the to $(\text{rnd}, \mathbf{Z}, \text{rej})$ pend relay, the at Refused nothing. predicted having outright, win would rej actually $\mathcal{F}_{\text{Rand}}$ string a carries entry rnd every and appended, is nothing refused be to nothing has it so call, no places Guess operation The drew. hon- asks verdict the And included. root the everywhere. served is and for intercepts wrapper own game's the party corrupt a at nobody: of esty core's honest an is tr in entry every so , (5.4 (Remark runs core the before .17.2 Section in as exactly already.

گزاره 13.1 $\mathcal{F}_{\text{Rand}}$ is unpredictable. Let $\mathcal{Z}$ be a place most at q_g on calls q_r most at and Guess slot, sary
adver- the of $\mathcal{X}$ occupant every for Then .Rnd on q_r most at

$$\Pr[\text{Win}_{\text{FinUnp}}] \leq q_g q_r 2^{-n}.$$

any $\mathbb{Z}$ for $(\mathbb{Z}, \mathbb{X})$ against $q_g q_r 2^{-n}$ within unpredictability has $\{\mathcal{F}_{\text{Rand}}\}$ So
occupants. all of class the $\mathbb{X}$ and counts those meeting class

اثبات. Fix $k = (r, \cdot, \text{rnd})$. By tightness no caller reaches $\mathcal{G}_m$. $\mathcal{F}_{\text{Rand}}$ reaches $\mathcal{G}_m$ only through relay (5) so r is $\mathcal{F}_{\text{Rand}}$ indepen- and $\{0, 1\}^n$ on uniform :5 line at drew Rnd own 's $\mathcal{F}_{\text{Rand}}$ value the
and call no placing $\mathcal{F}_{\text{Rand}}$ it, before did execution the everything of dent
it. draw to state no reading

Every draw. that before moment the to up execution the on Condition
most at are there and history. that by determined is $j < k$ with entry guess
most at is values their of any equals r that probability the so them. of q_g
returns 21 line at verdict the and , k of choices q_r most at are There . $q_g 2^{-n}$
. $q_g q_r 2^{-n}$ gives k over bound union a so matches. (j, k) pair some if only 1
ap- not does slot the of occupant The on. rests bound the details Two
sampling. the reaches does $\mathcal{X}$ nothing so call. no place cores 's $\mathcal{F}_{\text{Rand}}$ pear:
absorption an without occupant every against holds bound same the and
construction. by worthless is draw a after made guess a And argument.
it guesses then and r reads that environment an : $j < k$ asking verdict the
content the order trace's the makes what is which nothing. predicted has
 $\square$ verdict. this of

saying worth is it and ,0 not is that paper this in first the is bound The
 π system a way: either same the reads 5.11 Corollary changes. that what
the under $q_g q_r 2^{-n} + \varepsilon$ within unpredictability has $\{\mathcal{F}_{\text{Rand}}\}$ of place in put
adding. simply error emulation the and bound ideal the reading, search
realization a – free longer no is side ideal the that is changes it What
dif- of are terms two the and nontrivial. already was that bound a inherits
than better does protocol no and own, sampling's the is one kinds: ferent
cheap as is itself Transfer costs. realization the what is other the while it.
Re- so otherwise, or responsive call. no place cores 's $\mathcal{F}_{\text{Rand}}$ gets: it as
as exactly untouched. is 4.29 Theorem and arises never caveat 's9.4 mark
(16.2) (Section $\mathcal{G}_{\text{PKI}}$ for

Storage

Functionality 14.1

be them lets and values produces one where companion: ' $\mathcal{F}_{\text{Rand}}$ is $\mathcal{F}_{\text{Store}}$ party A are. they until them holds and values takes other the forgotten. is what reports leak the it: erases and back, it gets index, an at value a puts keeping protocol a by means 1 Chapter what are they Together held. still persis- all, $\mathcal{F}_{\text{Rand}}$ through randomness all – tape machine a on state no .Leak own its through holds it what exposing each, $\mathcal{F}_{\text{Store}}$ through tence

parameters, **N** and **P**.pid parameter: the save exactly, ' $\mathcal{F}_{\text{Rand}}$ **fields. The** is state its ' $\mathcal{F}_{\text{Rand}}$ Like none. reading store a ,par := $\perp$ and ,**U** := $\emptyset$ put, another what get may party one – serves it parties the across shared what and tape, per-party a than rather store a it makes what is which point. the being sharing by means 1 Chapter

the and conditions, three of one in is index An **two. not states. Three** there: put ever was nothing if $\square$ is It purpose. on apart them keeps code was it if $\perp$ is it and erased: not and put was something if value a holds it empty either in index an accepts **Put** operation The erased. then and put get a so , $\square$ only refuses **Get** reusable: is index erased an so condition. distinction The refused. than rather $\perp$ answered is index erased an on can caller a answers: two those between difference the in place its earns which gone, is and here was something from here ever was nothing tell rather it performed that protocol the to observable erasure makes what is alike both refuse would difference the hid that store A hole. silent a than happened. had erasure an whether asked be not could and

$\mathcal{F}_{\text{Store}}$ Functionality		
$\text{par} := \perp, \text{.U} := \emptyset, \text{.N}, \text{.P}, \text{.pid}$		
$\text{:8 require } \mathbb{L}[i] \neq \square$ $\text{:9 return } \mathbb{L}[i]$	$\text{id}' \text{ from } \text{id.Get}(i)$ $\text{/ never written refuses}$ $\text{/ } \perp \text{ if erased}$	:Initialize()
$\text{:10 return } \mathbb{L}$	$\text{id}' \text{ from } \text{id.Leak}()$	$\text{id}' \text{ from } \text{id.Put}(i, x)$
		$\text{:3 require } \mathbb{L}[i] \in \{\perp, \square\}$ / an occupied index refuses $\text{:4 } \mathbb{L}[i] \leftarrow x$:5 return ok
		$\text{id}' \text{ from } \text{id.Erase}(i)$
		$\text{:6 } \mathbb{L}[i] \leftarrow \perp$:7 return ok

rather **require** as written are refusals The code. the on remarks Two framework's the in said thing same the is which rej of returns as than 3 lines so rej with **require** refused a answers 3.6 Section words: own paper this of core no and owes. store a refusals two the exactly give 8 and that index an erasing nothing: refuses Erase And it. writing by rej returns erasing protocol a because ok answers that no-op a is nothing holds there. anything was there whether know to have not should defensively

Properties 14.2

.FinKeep for. is store a what of halves two the are they and properties. Two **.FinGone** back. got value the is erased not and put value a :persistence reading, search the take Both back. got not is erased value a :erasure trace same the read they – game one of finalizations two are both and test the is which alone. verdict the in differ and relays three same the of shell. a sharing for gives 17.2 Section

$\mathcal{F}_{\text{Store}}.\mathbf{N} := \{\mathcal{G}\mathbf{m}.\text{pid}\}$ with $\gamma := \{\mathcal{G}\mathbf{m}\} \cup \{\mathcal{F}_{\text{Store}}\}$ over played are They reason: same the for and before as tightness, $\mathcal{F}_{\text{Store}}.\mathbf{P} \subseteq \mathcal{G}\mathbf{m}.\mathbf{P}$ and pinned The for. account cannot verdict the value a is see not does game the put a compare verdicts both reason: 's13.2 Section for root. the refuse relays the in making own game's the of rej a so one. put a against value got a either. win could and value a as read be would trace

follow finalizations the $\mathcal{F}_{\text{Store}}$ over shell Game

$\mathcal{U} := \{(\mathcal{F}_{\text{Store}}, \text{serves}_{\text{gm}})\}$ $\mathcal{N} := \mathbb{Z}$ $\mathcal{P} := \mathcal{F}_{\text{Store}} \cdot \mathcal{P} \cup \{\mathbb{Z}\}$ $\text{pid} := (\text{gm.F}, 0, 0)$
 $\text{par} := \perp$

<pre> :13 require $\neg \text{done}$:14 require $\text{id.P} \neq \mathbb{Z}$:15 $y \leftarrow \text{id}_{\mathcal{F}_{\text{Store}}}.\text{FullGet}(i)$ as id :16 $\text{tr} \leftarrow \text{tr} \cdot (\text{get}, \text{id.P}, i, y)$:17 return y id' from <u>id.Get(i)</u> :18 return $\perp$ </pre>	<pre> :1 done $\leftarrow$ false :2 $\text{tr} \leftarrow ()$ id' from <u>id.Put(i, x)</u> :3 require $\neg \text{done}$:4 require $\text{id.P} \neq \mathbb{Z}$:5 $a \leftarrow \text{id}_{\mathcal{F}_{\text{Store}}}.\text{FullPut}(i, x)$ as id :6 $\text{tr} \leftarrow \text{tr} \cdot (\text{put}, \text{id.P}, i, x, a)$:7 return a id' from <u>id.Erase(i)</u> :8 require $\neg \text{done}$:9 require $\text{id.P} \neq \mathbb{Z}$:10 $\text{id}_{\mathcal{F}_{\text{Store}}}.\text{FullErase}(i)$ as id :11 $\text{tr} \leftarrow \text{tr} \cdot (\text{erase}, \text{id.P}, i)$:12 return ok </pre>
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shared is $\mathcal{F}_{\text{Store}}$ over shell the $\mathcal{G}_m$ of finalizations The

<pre> :19 require $\neg \text{done}$:20 require $\text{id.P} = \mathbb{Z}$:21 $\text{done} \leftarrow \text{true}; \text{parse } \text{tr} \text{ as } (e_j)_{j=1}^m$:22 $w \leftarrow 1$ if $\exists j < k \exists i, x, y : e_j = (\text{put}, \cdot, i, x, \text{ok}) \wedge e_k = (\text{get}, \cdot, i, y)$:23 $\wedge y \neq x \wedge \neg \exists l \in (j, k) : e_l = (\text{erase}, \cdot, i)$:24 else $w \leftarrow 0$:25 return w </pre>	<pre> (persistence) id' from <u>id.FinKeep()</u> </pre>
<pre> :26 require $\neg \text{done}$:27 require $\text{id.P} = \mathbb{Z}$:28 $\text{done} \leftarrow \text{true}; \text{parse } \text{tr} \text{ as } (e_j)_{j=1}^m$:29 $w \leftarrow 1$ if $\exists j < k < l \exists i, x : e_j = (\text{put}, \cdot, i, x, \text{ok})$:30 $\wedge e_k = (\text{erase}, \cdot, i) \wedge e_l = (\text{get}, \cdot, i, x)$:31 $\wedge \neg \exists l' \in (k, l) : e_{l'} = (\text{put}, \cdot, i, \cdot, \text{ok})$:32 else $w \leftarrow 0$:33 return w </pre>	<pre> (erasure) id' from <u>id.FinGone()</u> </pre>

some if 1 returns FinKeep finalization The words. in verdicts. The index that at put successful last the value the than other answered get if 1 returns FinGone finalization The intervening. erasure no there. left have to supposed was erasure an value the exactly answered get some or- trace's the read Both between. in it restored having put no removed. is store a reason: same the for and does steadiness's 17.2 Section as der.

between and next, the and call one between happens what about claim a ordering. an is

گزاره 14.1 $\mathcal{F}_{\text{Store}}$ both has $\mathcal{F}_{\text{Store}}$ (properties) of either FinFor and FinKeep adversary the of $\mathcal{X}$ occupant every and $\mathcal{Z} \in \mathbb{Z}_{\gamma, \gamma}$ every for, FinGone against 0 within property each has $\{\mathcal{F}_{\text{Store}}\}$ So $\text{Pr}[\text{Win}_{\text{Fin}}] = 0$ slot, occupants. all of class the $\mathcal{X}$ for, 5.8 Definition of sense the in $(\mathbb{Z}_{\gamma, \gamma}, \mathcal{X})$

اثبات. One invariant both carries $\mathbb{L}[i]$ inside a call tight- from comes It names. call that value the to and records. trace the own game's the from guard), the passes game the but caller (no ness from and runs, core any before party corrupt a at intercepting wrapper and call relay's a between runs nothing so call, no placing cores 's $\mathcal{F}_{\text{Store}}$ then are properties Both order. execution in append entries and return its track and conjuncts, verdict's the assume move: same the by it off read Persis- named. entries the between $\mathbb{L}[i]$ written have could lines which can put later no so index, occupied an refusing put the on turns tence $\square$ nor $\perp$ neither containing $\{0, 1\}^*$ on turns erasure unrecorded: intervene it. before put value the be cannot erase an after returns get a value the so trace the call a inside only changes $\mathbb{L}[i]$ invariant: one on rests Everything $\mathcal{G}_m$ but caller no tightness By names. call that value the to and records. shell's the through only it reaches $\mathcal{G}_m$ and, $\mathcal{F}_{\text{Store}}$ at 2 line 'sGuard passes wrapper own game's the party corrupt a at (15 and 11, 5 (lines relays three 's $\mathcal{F}_{\text{Store}}$. (5.4 Remark all at runs γ of core no and earlier step a intercepts call relay's a between runs γ of machine other no so call, no place cores below numbers Line order. execution in append entries and return, its and

otherwise. said unless cores own 's $\mathcal{F}_{\text{Store}}$ name

$e_k = (\text{get}, \cdot, i, y)$ and $e_j = (\text{put}, \cdot, i, x, \text{ok})$ suppose persistence: For 'sPut so .ok answered put The between. i at erase no and $j < k$ with 6 line 'sErase and line that Only $\mathbb{L}[i] \leftarrow x$ set 4 line its and passed 3 line lies i at entry erase no recorded: is each invariant the by and $\mathbb{L}[i]$ write either, between lie cannot ok answering i at put further a and between. So on. j from occupied being $\mathbb{L}[i]$ and index occupied an refusing 3 line hence :x returns 9 line its and passes, 8 line 'sGet ,k at still $\mathbb{L}[i] = x \neq \square$. 0 gives 22 line at verdict the and $y = x$

$e_l =$ and $e_k = (\text{erase}, \cdot, i)$, $e_j = (\text{put}, \cdot, i, x, \text{ok})$ suppose erasure: For .l and k between i at put successful no and $j < k < l$ with $(\text{get}, \cdot, i, x)$ that line only the invariant the by and ,k at $\mathbb{L}[i] \leftarrow \perp$ set 6 line 'sErase answering i at put recorded a inside 4 line 'sPut is value a restore could

9 line 'sGet and $\perp$ at $\perp[i] = \perp$ So excludes. hypothesis the which ok argument. 'sPut a from 4 line by written was $x : x \neq \perp$ Now $\perp$ returns x answer not did get the Hence $\square$ nor $\perp$ neither contains $\{0, 1\}^*$ and $\square$.0 gives 29 line at verdict the and e_1 to contrary

beyond discharge to nothing with 5.11 Corollary by transfer Both the so all. at call no place cores 's $\mathcal{F}_{\text{Store}}$ conditions: standing four its complete- dummy and way cheap the met is hypothesis responsive-call functionalities three The $\mathcal{G}_{\text{PKI}}$ and $\mathcal{F}_{\text{Rand}}$ for as exactly untouched. is ness pat- the and nothing. pay that three the are nothing adversary the ask that the is clock the charges 17.2 Section what – coincidence a not is tern bill. no has notification no with functionality a and notification.

Signatures Digital

Functionality 15.1

own Canetti's of tradition the in signatures, digital idealizes below $\mathcal{F}_{\text{Sig}}$ not and parameter a is id process Its [6] functionality signature ideal subscript the what is that – fixed is component name The constant, a and session the but – 1 Chapter of convention naming the by records, several and session per copy a hold may system one so not, are instance Def- one and handle, to exists composition situation the session, a within The pinned, were id process whole the if outright forbid would 1.2 inition any at read be it lets 7 Chapter so component, neither reads below code global any which, 7.2 Definition of condition the meeting $\mathbf{N}$ an for – other abbreviates $\mathcal{A}(x)$ that 3.4 Section from Recall does, one local purely or ideal the in served; being party the of behalf on adversary the to call a here them of one every and calls, those answers simulator the execution use, before 12 Chapter of sense the in sanitized is

throughout, same the is style The **functionalities. ideal write we How** pin not does specification the and produced be to has value a Wherever fly, the on choice the sanitize and it choose simulator the let we down, it goal security the as freedom much as have to meant then is simulator The goal the and – theorem a not intention, design a – more no and permits code, the in buried than rather predicates the in visible stays correct- protect two first the and times, three happens this $\mathcal{F}_{\text{Sig}}$ In adversary, the from key verification the takes **Gen** operation The ness, issued already key every from distinct key, actual an it makes Clean_{vk} and generation so it, under valid recorded already signature any of free and oper- The pre-forged, arrive nor collide neither keys and succeeds always not signature actual an it makes Clean_σ and signature, the takes **Sign** ation been has key a whenever succeeds signing so invalid, recorded already

asks and that, for asks Locality verifies, produces it what and generated, elsewhere, token the passing without answer simulator the that addition in The code, the of property a than rather class simulator the on condition a outside verdict a instead: unforgeability for sanitized is **Verify** operation key a under triple fresh a on 1 of verdict any does so and, 0 becomes $\{0, 1\}$ unforgeable, strongly signatures makes which generated, party honest an rather inline written is sanitization the so unique, is value good the There so authenticated, assumed not are keys Verification, **San** through than own, its of vk a takes Verify

ad- the hands functionality, ideal an of requires 3.2 Section which, Leak recorded signatures the with together key verification party's the versary fixes Initialize operation The state, the of else nothing and it, under valid **Ver** and party served per key one holds **VK** types: their and tables two the ev- starting both signature, and message key, of triple each on verdict the recorded a from decided" yet "not separates that value the, $\perp$ at erywhere infinite: $\perp$ nor $\perp$ neither contain and infinite are Σ and $\mathcal{K}$ Both, 1 or 0 two the of clean and values, good of out run never sanitizers the that so them, refuses already membership that so markers

to restricted be must class simulator the wanted, is locality Where placing without call simulator each answering those ones: responsive the are calls simulator and once at returns token the so own, their of call any enforces 9 Chapter discipline the — calls coroutine than rather subroutine the nothing, costs this signatures stateless non-interactive, For code, in already, responsive being simulators usual

Sec- re-entrance, against guards **Verify** and **Gen** in line further One operations both and re-entered, be instance suspended a lets 3.6 tion writing and entry table a testing between call adversary an at suspend recorded value a and resumption, on entry the re-tests therefore each it: two issue could calls nested two re-test the Without wins, meanwhile con- very the — triple one on verdicts different return or party, one to keys record, to exists **Ver** sistency

$\mathcal{F}_{\text{Sig}}$ Functionality	
$\text{par} := \perp$ $\mathcal{U} := \{(\mathcal{A}, \text{serves})\}$ $\mathcal{N}$ $\mathcal{P}$ pid	
id' from <u>$\text{id.Verify}(\text{vk}, \text{msg}, \sigma)$</u> :20 if $\text{Ver}[\text{vk}, \text{msg}, \sigma] \neq \square$ then :21 $\text{return } \text{Ver}[\text{vk}, \text{msg}, \sigma]$:22 $b \leftarrow \mathcal{A}(\text{id.Verify}, \text{vk}, \text{msg}, \sigma)$:23 if $\text{Ver}[\text{vk}, \text{msg}, \sigma] \neq \square$ then :24 $\text{return } \text{Ver}[\text{vk}, \text{msg}, \sigma]$ / recorded while suspended :25 if $b \notin \{0, 1\}$ then :26 $b \leftarrow 0$:27 if $\exists P' \notin \mathcal{C} : \text{vk} = \text{vk}[P']$ then :28 $b \leftarrow 0$ / no forgery under an honest key :29 $\text{Ver}[\text{vk}, \text{msg}, \sigma] \leftarrow b$:30 $\text{return } b$ id' from <u>$\text{id.Leak}()$</u> :31 if $\text{vk}[\text{id.P}] = \square$ then :32 $\text{return } (\perp, \emptyset)$:33 $\text{return } (\text{vk}[\text{id.P}],$ $\{(\text{msg}, \sigma) : \text{Ver}[\text{vk}[\text{id.P}], \text{msg}, \sigma] = 1\})$ <u>$:\text{Clean}_{\text{vk}}(\text{vk}, \text{vk}, \text{Ver})$</u> :34 $\text{return } \text{vk} \in \mathcal{K} \wedge \neg \exists P : \text{vk}[P] = \text{vk}$:35 $\wedge \neg \exists \text{msg}, \sigma : \text{Ver}[\text{vk}, \text{msg}, \sigma] = 1$ <u>$:\text{Clean}_{\sigma}(\sigma; \text{msg}, \text{vk}, \text{Ver})$</u> :36 $\text{return } \sigma \in \Sigma \wedge \text{Ver}[\text{vk}, \text{msg}, \sigma] \neq 0$	<u>$:\text{Initialize}()$</u> :1 $\text{vk} : \mathcal{F}_{\text{Sig}}.\mathcal{P} \rightarrow \mathcal{K} \cup \{\square\}$:2 $\text{vk}[*] \leftarrow \square$:3 $\text{Ver} : \mathcal{K} \times \mathcal{M} \times \Sigma \rightarrow \{0, 1\} \cup \{\square\}$:4 $\text{Ver}[*, *, *] \leftarrow \square$ id' from <u>$\text{id.Gen}()$</u> :5 if $\text{vk}[\text{id.P}] \neq \square$ then :6 $\text{return } \text{vk}[\text{id.P}]$:7 $\text{vk} \leftarrow \mathcal{A}(\text{id.Gen})$:8 if $\text{vk}[\text{id.P}] \neq \square$ then :9 $\text{return } \text{vk}[\text{id.P}]$ / recorded while suspended :10 $\text{vk} \leftarrow \text{San}[\text{Clean}_{\text{vk}}](\text{vk}; \text{vk}, \text{Ver})$:11 $\text{vk}[\text{id.P}] \leftarrow \text{vk}$:12 $\text{return } \text{vk}$ id' from <u>$\text{id.Sign}(\text{msg})$</u> :13 $\text{vk} \leftarrow \text{vk}[\text{id.P}]$:14 if $\text{vk} = \square$ then :15 $\text{return } \perp$:16 $\sigma \leftarrow \mathcal{A}(\text{id.Sign}, \text{msg})$:17 $\sigma \leftarrow \text{San}[\text{Clean}_{\sigma}](\sigma; \text{msg}, \text{vk}, \text{Ver})$:18 $\text{Ver}[\text{vk}, \text{msg}, \sigma] \leftarrow 1$:19 $\text{return } \sigma$

Properties 15.2

one give We instance. an worth is shape a and shape. a is 5.1 Definition the what that saying ,FinCor properties: two carrying $\mathcal{F}_{\text{Sig}}$ over $\mathcal{G}_{\text{m}}$ game else nothing that saying ,FinUF and verifies, produces interface signing which games. two than rather shell one of finalizations two are They does. read they — for ask two these what and allows now shape the what is The alone. verdict the in differ and oracles, same the of trace. same the and call. to which choosing by on judged is it which chooses environment other. the against game the closes one calling

5.8 Definition of reading search the take Both

$$\text{Adv}_{\mathcal{G}_{\text{m}}, \mathcal{F}_{\text{Sig}}, \mathcal{Z}, \mathcal{X}}^{\text{FinCor}} := \Pr[\text{Win}_{\text{FinCor}}],$$

$$\text{Adv}_{\mathcal{G}_{\text{m}}, \mathcal{F}_{\text{Sig}}, \mathcal{Z}, \mathcal{X}}^{\text{FinUF}} := \Pr[\text{Win}_{\text{FinUF}}],$$

better no merely not and impossible be to meant is either winning because is time the half succeed forgeries whose scheme signature a coin: a than

shared is $\mathcal{F}_{\text{Sig}}$ over shell the $\mathcal{G}_{\text{m}}$ of finalizations The

```

                                (correctness)  id' from  id.FinCor()

:16 require  $\neg \text{done}$ 
:17 require  $\text{id}.P = Z$ 
:18  $\text{done} \leftarrow \text{true}$ 
:19  $\mathbf{C} \leftarrow (\mathbf{C}, Z).\text{FullStatus}()$  as id
:20  $w \leftarrow 1$  if  $\exists P, Q, vk, \text{msg}, \sigma : P \notin \mathbf{C} \wedge Q \notin \mathbf{C} \wedge \sigma \in \Sigma$ 
:21    $\wedge (\text{gen}, P, vk) \in \text{tr} \wedge (\text{sign}, P, \text{msg}, \sigma) \in \text{tr} \wedge (\text{ver}, Q, vk, \text{msg}, \sigma, 0) \in \text{tr}$ ;
:22 else  $w \leftarrow 0$ 
:23 return  $w$ 

                                (unforgeability)  id' from  id.FinUF()

:24 require  $\neg \text{done}$ 
:25 require  $\text{id}.P = Z$ 
:26  $\text{done} \leftarrow \text{true}$ 
:27  $\mathbf{C} \leftarrow (\mathbf{C}, Z).\text{FullStatus}()$  as id
:28  $w \leftarrow 1$  if  $\exists P, Q, vk, \text{msg}, \sigma : P \notin \mathbf{C} \wedge Q \notin \mathbf{C}$ 
:29    $\wedge (\text{gen}, P, vk) \in \text{tr} \wedge (\text{ver}, Q, vk, \text{msg}, \sigma, 1) \in \text{tr} \wedge (\text{sign}, P, \text{msg}, \sigma) \notin \text{tr}$ ;
:30 else  $w \leftarrow 0$ 
:31 return  $w$ 

```

oc- means $\in$ sequence. a is trace The code. the about things Four
 and tr read verdicts The order. the reads verdict no and it. in currence
 serves $\mathcal{G}_{\text{m}}$ in exemption root the why is which nothing. call and register the
 rather end the at $P \notin \mathbf{C}$ ask Both .Z serve not need $\mathcal{F}_{\text{Sig}}$ and suffices
 that makes (3.5 (Section monotonicity and signing. of time the at than
 honest was closes game the when honest party a reading: stronger the
 no where $\perp$ answers Sign because $\sigma \in \Sigma$ asks Correctness throughout.
 func- the against hold to signature no is $\perp$ and generated. been has key
 the not triple. the excludes 28 line :strong is unforgeability And tionality.
 forgery a as counts message signed a on signature second a so message.
 give. to written was sanitizer 's15 Chapter notion the is which –

گزاره 15.1 $\mathcal{F}_{\text{Sig}}$ both has $\mathcal{F}_{\text{Sig}}$.properties) both has $\mathcal{F}_{\text{Sig}}$.FinUF and FinCor of either Fin For
 slot. adversary the of $\mathcal{X}$ occupant every and $\mathcal{Z} \in \mathcal{Z}_{\gamma, \gamma}$ every for
 $(\mathcal{Z}_{\gamma, \gamma}, \mathcal{X})$ against 0 within property each has $\{\mathcal{F}_{\text{Sig}}\}$ So . $\Pr[\text{Win}_{\text{Fin}}] = 0$
 occupants. all of class the $\mathcal{X}$ for .5.8 Definition of sense the in

اثبات. One preliminary and two. The verdicts. both that is preliminary
 a at bookkeeping: not is it and honest. be must verdict a by named ties
 so adversary's. the is answer the and run not does core the party corrupt
 forgery. a on 1 a or signature good a on 0 a report could Verify mediated a

Cor- wrong. nothing done having $\mathcal{F}_{\text{Sig}}$ with fire would finalization either and Verify and $\text{ver}[\text{vk}, \text{msg}, \sigma] \leftarrow 1$ recording Sign from follows then rectness whether on splits Unforgeability line. first its at entry recorded a returning under 1 a computed have cannot it one: computed or entry an read Verify records Sign only and recorded. was entry the so key. party's honest an honest them makes Monotonicity close. the at honest parties Take one. firing 4 line mediated. is them at places game the call no so throughout. ver- Both $\{0, 1\}^*$ in lying name own game's the and party corrupt a at only both: of it need both and Q verifier the of and P signer the of this ask dicts adversary's. the is answer the and run not does core the party corrupt a at signature. good a on 0 a — likes it whatever reports Verify mediated a so having $\mathcal{F}_{\text{Sig}}$ with 1 return would finalization either and — forgery a on 1 a or wrong. nothing done

$\sigma \in \Sigma$ with $(\text{sign}, P, \text{msg}, \sigma) \in \text{tr}$ suppose correctness. For set then and Clean_σ against σ sanitized it so ran. Sign of core The never and once fixed Gen which $\text{vk} = \text{vk}[P]$ for $\text{ver}[\text{vk}, \text{msg}, \sigma] \leftarrow 1$ at it returns and set entry that finds Verify($\text{vk}, \text{msg}, \sigma$) later Any changes. Q every for $(\text{ver}, Q, \text{vk}, \text{msg}, \sigma, 0) \notin \text{tr}$ thus :1 returns it so line. first its .0 gives 20 line and

$\text{vk} =$ with $(\text{ver}, Q, \text{vk}, \text{msg}, \sigma, 1) \in \text{tr}$ suppose unforgeability. For computed it or entry recorded a read Verify Either . P honest an for $\text{vk}[P]$ sets ver writing before test last the :1 a computed have cannot It one. So witnesses. P which $P' \notin \mathbf{C}$ some for $\text{vk} = \text{vk}[P']$ whenever $b \leftarrow 0$ whose party the at — 1 a records Sign only and recorded. was entry the Hence twice. key no issues Clean_{vk} since P being party that vk is key $\square$.0 gives 28 line and $(\text{sign}, P, \text{msg}, \sigma) \in \text{tr}$

ideal the says merely 0 is it that part: interesting the not is bound The functional- ideal an as properties. these have to written was functionality the and it. with does then 5.11 Corollary what is content The be. should ity $\{\mathcal{F}_{\text{Sig}}\}$ of place in put π system a useful: that makes what is reading search adversary every against ε most at probability with finalization either wins is probability that reading search the under and dummy. the only not and reading decision the Had itself. property the bounds ε so advantage. the which $2 \Pr[\text{Win}_{\text{Fin}}] - 1$ on 2ε read would conclusion same the taken been suc- forger a — all at constraint no is unwinnable be to meant game a for ap- 5 Chapter of rule calibration the is That it. meets time the half ceeding each displays: both carries 5.10 Proposition why and bites. it where plied

inherit two these and fixes, reading own its scale the at inherits finalization
 .ε at

by discharged is none and corollary the with come conditions Four
 well-formed: be must π by $\{\mathcal{F}_{\text{Sig}}\}$ of replacement The alone. emulation
 pin to members 's π asks which system, game a be again must $\{\mathcal{G}_m\} \cup \pi$
 simu- the :5.1 Definition of part is tightness since did, $\mathcal{F}_{\text{Sig}}$ where $\mathcal{G}_m$.pid
 environments the and refusal-oblivious: be must adversary the and lator
 argued separately never thus is realization A -identity.Z no silence must
 emulation one the from both inherits it — unforgeable or correct be to
 above finalizations two the and checks, four those with together proof,
 of. is inheritance the what are

5.11 Corollary bites. it where is this and it. with travels caveat One
 $\mathcal{F}_{\text{Sig}}$ and call. responsive no place system game the of cores the that asks
 Should applies. corollary the so „ $\mathcal{A}(\cdot)$ through slot the reaches written as
 which — Verify and Sign .Gen in $\mathcal{A}^1$ write and advice 's9 Chapter adopt one
 reason the for instead, directly argued be must transfer the — locality buys
 priced stand, things as are, inheritance and Locality gives. 9.4 Remark
 other. each against

Infrastructure Public-Key

Functionality 16.1

Verify — out hands it key the authenticate not does it that plainly said $\mathcal{F}_{\text{Sig}}$ keys verification because caller. any to unconnected own. its of vk a takes be- $\mathcal{G}_{\text{PKI}}$ functionality The .(15.1 (Section authenticated assumed not are cer- own Canetti's of shape the in gap. the closes that setup the is low the to key a binding directory shared single. a :[6] functionality tification because trusted be can Verify to handed vk a that so it. claims that party believed. simply was it because than rather here from came it only and key. one register may party Each said. quickly is does it What one conjures which .**Gen** unlike — input as key the takes **Register** one: first the keeps and — factory a not and directory a is this slot. the from two no that so file on already key every against sanitized given. is it key anyone answers **Retrieve** operation The one. same the hold ever parties has one no if nothing or name. they party the for file on is whatever with registered.

id process The reason. same the for and 's. $\mathcal{G}_{\text{Clock}}$ are They **fields. The** copy: one at system the caps 1.2 Definition so .(GPKI, 0, 0) constant. a is key. which owns who of notion second a be would directory second a global($\mathcal{G}_{\text{PKI}}$) makes $\mathbf{N} := \mathbf{Std} \cup \mathbf{A} \cup \mathbf{Z}$ out. rule to exists PKI a what exactly the universe the again is $\mathbf{P}$ and clock's. the as disjunct same the by hold in state. is registration — it used has who than rather covers directory The . $\mathbf{U} := \emptyset$ clock's: the than simpler is field One membership. not .vk with arriving key the all. at call no place Retrieve and Register operations on depends $\mathcal{G}_{\text{PKI}}$ nothing is there so slot. the from than rather request the has. already functionality every machinery ambient the beyond

ex- two the of each from line one takes Register **guards. borrowed Two** for acting Z or A named caller a 's: $\mathcal{G}_{\text{Clock}}$ is first The it. before amplex

nothing, changes and key own party's that answered is party honest an honest an on tick to allowed than rather time the answered is it as exactly goes what decide not file, on is what ask may outsider an – behalf party's the – proceeds call the so fire, not does check the party corrupt a At there. and one, ticks it as freely as behalf party's corrupt a on registers adversary letting mediation show, will directory the what is submits it key whatever sends corrupt 's $\mathcal{F}_{AC}$ for does it as unmediated through call this exactly later wins, succeed to call first the 's: $\mathcal{F}_{Sig}$ is second The .(19.1 (Section as exactly it, overwriting than rather file on already key the returning ones from differs here Sanitizing exists, one once key a re-issues never Gen a fix each $Clean_\sigma$ and $Clean_{vk}$ repairs: it what in predicates signature both the is which supplied, caller the one fixes $Clean_{reg}$ proposed: slot the value already presumably is key own party's honest an – position exposed more one the and in, wrote adversary the whatever is one's corrupt a but good, the key a claiming party second a is through let cannot directory a thing reason same the for here exists always value good A holds, already first those exactly are excluded keys the and infinite is $\mathcal{K} : Clean_{vk}$ for does it .(12 Chapter of (footnote many finitely recorded, have can run bounded a

$\mathcal{G}_{PKI}$ Functionality	
$par := \perp$	$\mathcal{U} := \emptyset$
$\mathcal{N} := Std \cup A \cup Z$	$\mathcal{P} := (GPKI, 0, 0)$
$id' \text{ from } id_Retrieve(P)$	$:Initialize()$
:10 return $vk[P]$	:1 $vk : \mathcal{G}_{PKI}.P \rightarrow \mathcal{K} \cup \{\square\}$
$id' \text{ from } id_Leak()$	:2 $vk[*] \leftarrow \square$
:11 return vk	$id' \text{ from } id_Register(vk)$
$:Clean_{reg}(vk; vk)$	:3 if $id'.F \in \{A, Z\} \wedge id.P \notin \mathcal{C}$ then
:12 return $vk \in \mathcal{K} \wedge \neg \exists P : vk[P] = vk$	:4 return $vk[id.P]$ / a peek, not a write
	:5 if $vk[id.P] \neq \square$ then
	:6 return $vk[id.P]$
	:7 $vk \leftarrow San[Clean_{reg}](vk; vk)$
	:8 $vk[id.P] \leftarrow vk$
	:9 return vk

party, the against recorded is Registration code. the on remarks Three the tied already has **Guard** are: ticks reason same the for caller, the not reg- P when token the holds machine whichever so, 2 line by id.P to call none and, Deregister no is There entry, own 'sP on lands key the isters, the is bound once key a but revocation, support can PKI real a added: is for left addition genuine a is revocation and is, there directory simplest the with answers Leak And over, papered gap a than rather elsewhere at entry one out gives already Retrieve than more no is which table, whole

this only not interfaces. its of one every at public is directory the — time a times. enough asked ,Retrieve nothing adversary the tells Leak so one. not. would

No either. than less needs $\mathcal{G}_{PKI}$ it. before examples both against Set key the — Gen in as nothing. of out choose to simulator's the is value than rather input caller's a repairs here sanitizing so — call the with arrives no And it. before **San** every from differs this place one the output. slot's a about responsive be to notification no is there so all. at core the leaves call as transfer below properties two the settle: to **9** Chapter for price no and can. property a as cheaply

Properties 16.2

local a — promise of kinds two having for properties two carried $\mathcal{G}_{Clock}$ moves it that one. global a and coded. as moves counter the that one. same the splits $\mathcal{G}_{PKI}$ directory The agreed. has honest everyone when only regis- a says fixity. ,**FinFix** game. one of finalizations two into and way. entry. own party's one about promise. local the — changes never key tered global the — key same the hold ever parties two no says binding. ,**FinBind** of reading search the take Both once. at $\mathcal{G}_{PKI}$. **P** of all over quantified one. **.5.8** Definition

$\mathcal{G}_{PKI}$ with , $\gamma := \{\mathcal{G}_m\} \cup \{\mathcal{G}_{PKI}\}$ system game the over played are Both Re- — pinned ,**N** := $\{\mathcal{G}_m.pid\}$ field: one save **16.1** Section in as taken iden- the for and ,(**17.2** (Section clock the for as exactly forcing. 's**6.5** mark adversary an against property a is $\mathcal{F}$ global a of property a reason: tical there key every about claims both are binding and fixity and it. shares that is. there all is trace game's the where only certifiable is. session one for asks **3** line so global. longer no is directory the Pinned. passes $\mathcal{G}_m$ but caller No it. meets 0 session own experiment's the and adversary's the not and hosts. it machine a not environment. the not **:2** line ever $\mathcal{G}_{PKI}$ key every So apply. even could **4** line before refused -identities. A whose game a — recorded is and below relay the through enters holds an Inside it. know to way other no has key" a share parties "two is verdict $.(\mathcal{G}_{PKI}.pid, id.P)$ for $id_{\mathcal{G}_{PKI}}$ write we id at interface

follow finalizations the $\mathcal{G}_{PKI}$ over shell Game

par := $\perp$ $\mathcal{U} := \{(\mathcal{G}_{PKI}, \text{serve}_{\mathcal{G}_m})\}$ $\mathcal{N} := \mathbb{Z}$ $\mathcal{P} := \mathcal{G}_{PKI}.P \cup \{Z\}$ pid := $(\mathcal{G}_m.F, 0, 0)$

<pre> id' from <u>id.Retrieve(P)</u> :9 require $\neg$done :10 require id.P $\neq$ Z :11 vk' $\leftarrow$ id$_{\mathcal{G}_{PKI}}$.FullRetrieve(P) as id :12 return vk' </pre>	<pre> :Initialize() :1 done $\leftarrow$ false :2 tr $\leftarrow$ () // the trace, a sequence id' from <u>id.Register(vk)</u> :3 require $\neg$done :4 require id.P $\neq$ Z :5 vk' $\leftarrow$ id$_{\mathcal{G}_{PKI}}$.FullRegister(vk) as id :6 tr $\leftarrow$ tr $\cdot$ (reg, id.P, vk') :7 return vk' id' from <u>id.Leak()</u> :8 return $\perp$ </pre>
--	---

shared is $\mathcal{G}_{PKI}$ over shell the $\mathcal{G}_m$ of finalizations The

```

(fixity) id' from id.FinFix()
:13 require  $\neg$ done
:14 require id.P = Z
:15 done  $\leftarrow$  true
:16 C  $\leftarrow$  (C, Z).FullStatus() as id
:17 w  $\leftarrow$  1 if  $\exists P, vk_1, vk_2 : P \notin C \wedge vk_1 \neq vk_2$ 
:18     $\wedge$  (reg, P, vk1)  $\in$  tr  $\wedge$  (reg, P, vk2)  $\in$  tr
:19 else w  $\leftarrow$  0
:20 return w

(binding) id' from id.FinBind()
:21 require  $\neg$ done
:22 require id.P = Z
:23 done  $\leftarrow$  true
:24 C  $\leftarrow$  (C, Z).FullStatus() as id
:25 w  $\leftarrow$  1 if  $\exists P, Q, vk : P \notin C \wedge Q \notin C \wedge P \neq Q$ 
:26     $\wedge$  (reg, P, vk)  $\in$  tr  $\wedge$  (reg, Q, vk)  $\in$  tr
:27 else w  $\leftarrow$  0
:28 return w

```

and relayed is Retrieve operation The code. the about things Two look not could that game a trace: its reads verdict neither though logged inter- the leaving and all. at $\mathcal{G}_{PKI}$ exercising be not would up key anyone's games for Read relayed shell 's $\mathcal{G}_{Clock}$ as exactly nothing, costs whole face and vk₁ need not does verdict 's FinFix And mutators. the only read that means idempotence trace: its of reading steadiness's unlike ordered, vk₂ the write, the be can them produced that calls two the of one most at a identically, game the loses order either and it. of read later a only other

first. came one whichever key changed a being key changed

گزاره 16.1 $\mathcal{G}_{PKI}$ has (fixity) every For $\mathcal{Z} \in \mathbb{Z}_{\gamma, \gamma}$ every and $\mathcal{X}$ occupant every within fixity has $\{\mathcal{G}_{PKI}\}$ So $\Pr[\text{Win}_{\text{FinFix}}] = 0$ slot, adversary the of all of class the $\mathcal{X}$ for **5.8** Definition of sense the in $(\mathbb{Z}_{\gamma, \gamma}, \mathcal{X})$ against occupants.

اثبات. Suppose $(\text{reg}, P, vk_1), (\text{reg}, P, vk_2) \in \text{tr}$ with $P \notin \mathbf{C}$. By tightness P is at core own's $\mathcal{G}_{PKI}$. Register running each **5** line at logged were both intercepts never **4** line so, Z or A not's. $\mathcal{G}_m$ is name own relay's the P once and $vk[P] = \square$ while only reached **8** line is write only core's The it. call later every – good for value sanitized some to $vk[P]$ fixes it reached unchanged. one. same the value. **6** line returns and $vk[P] \neq \square$ finds P at wrote first ran whichever agree: vk_2 and vk_1 producing calls two the So gives **17** line at verdict The back. it read only other the and value. the $\square$.

گزاره 16.2 $\mathcal{G}_{PKI}$ has (binding) every For $\mathcal{Z} \in \mathbb{Z}_{\gamma, \gamma}$ every and $\mathcal{X}$ occupant every within binding has $\{\mathcal{G}_{PKI}\}$ So $\Pr[\text{Win}_{\text{FinBind}}] = 0$ slot, adversary the of all of class the $\mathcal{X}$ for **5.8** Definition of sense the in $(\mathbb{Z}_{\gamma, \gamma}, \mathcal{X})$ against 0 occupants.

اثبات. Suppose $(\text{reg}, P, vk), (\text{reg}, Q, vk) \in \text{tr}$ and $P, Q \notin \mathbf{C}$. By **16.1** Proposition moment the from $vk = vk[Q]$ and $vk = vk[P]$ argument's **16.1** Proposition 'sP case the second, came write 'sQ Say onward. written first was each $vk \leftarrow \text{San}[\text{Clean}_{\text{reg}}](\cdot; vk)$ ran **8** line 'sRegister moment that at symmetric: output its that is **(12** Chapter guarantee 'sSan and set, already $vk[P]$ with freshness and $\mathcal{K}$ in membership – input the whatever $\text{Clean}_{\text{reg}}$ satisfies cannot write 'sQ So them. among 'sP file. on already key every against **25** line at verdict The $vk = vk[P] = vk[Q]$ contradicting, $vk[P]$ equal $\square$ gives **0**.

hypo- The argue. to extra nothing with **5.11** Corollary by transfer Both is – call responsive no place cores system's game the that – asks it esis or responsive kind, any of call no place cores 's $\mathcal{G}_{PKI}$ way: cheap the met (Sec- completeness dummy and arises never caveat 's**9.4** Remark so not. at locality bought notification whose $\mathcal{G}_{\text{Clock}}$ Unlike untouched. is **(4.4** tion priced calls slot whose $\mathcal{F}_{AC}$ and $\mathcal{F}_{\text{Net}}$ unlike and expense, inheritance's $\{\mathcal{G}_{PKI}\}$ of place in put π system a nothing: pays $\mathcal{G}_{PKI}$ transfer, own their

's15.2 Section of conditions four the directly, binding and fixity inherits that directory A discharge. to left ones only the being paragraph closing spend. to nothing has nothing adversary the asks

Clock Global

Functionality 17.1

con- it about said was what of most and randomized, and local is $\mathcal{F}_{\text{Sig}}$ counts. both on opposite its is example second The simulator. the cerned and Tackmann Maurer, Katz, by introduced was functionality clock A Maurer, Badertscher, by recast :[11] UC in synchrony model to Zikas it ,[3] counter monotone a around setup shared a as Zikas and Tschudi composable the one the functionality, global canonical the become has be- $\mathcal{G}_{\text{Clock}}$ clock The by, time keep successors its and Bitcoin of analyses paper. this of language the in functionality that is low only and 0 at starts `now` counter The said. quickly is does it What enter Parties it. read may serves clock the Anyone forward. steps ever the and ,**Deregister** and Register through structure round the leave and **Update** to: it asked has party registered honest every when steps counter registered honest the once and party, calling the against tick a records them. clears and counter the advances 17 line on test the in, all are ticks the and done. is it in party honest last the when exactly ends round A code the how – early it close nor round the stall neither can adversary adversary the to reported is tick recorded Each below. is each refuses the with and (9 (Chapter responsively ,20 line on returns, Update before at settled is costs it what and buys notification the what discarded, answer stands it as counter the answers operation Every .17.2 Section of end the ticks the and time new the learns round a closes that tick the so return, on again. reads stands it where know must that party a old: the it before a was 's $\mathcal{F}_{\text{Sig}}$ where $(\text{GClock}, 0, 0)$ constant, a is id process The **fields. The** which system. any in copy one most at puts then 1.2 Definition parameter. would clock second a and shared. instance. one means global point: the is first its by hold global($\mathcal{G}_{\text{Clock}}$) makes $\mathbf{N} := \mathbf{Std} \cup \mathbf{A} \cup \mathbf{Z}$ time. second a be

session every and check session the waives **Guard** of 3 line so disjunct, component session the separation very the – instance one the reaches time keeps that protocol A here. made not deliberately make. to exists 1.3 Definition of disjunct global the and ,U its in ($\mathcal{G}_{\text{Clock}}$, serves) declares copy. shared the at entry the meet session every of members lets what is notification the places. code the call one the for $\mathbf{U} := \{(\mathcal{A}, \text{serves})\}$ And the than rather universe the as but parameter a stays **P** And .20 line of in is who while for, exists interface an parties the fixes it membership: through left and entered state, is moment given a at structure round the table, more one is membership dynamic – [3] in as Deregister and Register field. of kind new a not

operation, an of not functionality, the of field a is **N ticking. and Reading** :**Read** for one right the is above set the and core, every governs set one so and both, lets [3] – time the ask may environment the and adversary the global a makes what of much is clock shared the reading environment the free adversary an where mutators, three the for wide too is It global. setup it drop or finished, not had party the round a close could party a for act to of line first the as sits refinement The altogether. structure round the from answered is party honest an for acting Z or A named caller a core: each the – passes it party corrupt a at while nothing, changes and time the ,[3] in as owns, it parties the for ticks and deregisters registers, adversary corrupt a reading never 17 line matters, it where inert are ticks those and is party honest no while naming: deserves consequence One flag, party's a register may adversary the so vacuous, is 17 line of test the registered is time registration, honest first the until – freely it tick and party corrupt .[3] in as too there spend, to adversary's the

is it and party", registered honest "every reads 17 line of quantifier The stan- a party corrupt a At itself. enforces $\mathcal{G}_{\text{Clock}}$ it of little how seeing worth inter- full the of 4 line by adversary the to handed is Update caller's dard party same the at call own adversary's the and runs, core the before face adversary's the are ticks party's corrupt a way either :13 line by admitted is conven- the by register the reads which – $\forall$ the and withhold, or give to therefore parties Corrupt regardless, them ignores – 3.1 Section of tion both party honest an At one. close help nor open round a hold neither ticks observers' the turns 13 line and fire, not does **Mediate** shut: routes tick- after corrupted party a piece: last the adds Monotonicity reads, into on, then from quantifier the outside is it but behind, flag stale a leaves ing it. clears advance next the and

$\mathcal{G}_{\text{Clock}}$ Functionality	
$\text{par} := \perp$ $\mathcal{U} := \{(\mathcal{A}, \text{serves})\}$ $\mathcal{N} := \text{Std} \cup \mathcal{A} \cup \mathcal{Z}$ $\mathcal{P}$ $\text{pid} := (\text{GClock}, 0, 0)$	
id' from <u>$\text{id.Update}()$</u> :12 if $\text{id}'.F \in \{\mathcal{A}, \mathcal{Z}\} \wedge \text{id}.P \notin \mathcal{C}$ then :13 return now / reading, not ticking :14 if $\text{reg}[\text{id}.P] = 0$ then :15 return now / not in the round structure :16 $\text{tick}[\text{id}.P] \leftarrow 1$:17 if $\forall P \in \mathcal{G}_{\text{Clock}}.\mathcal{P} \setminus \mathcal{C} : \text{reg}[P] = 1 \Rightarrow$:18 $\text{tick}[P] = 1$ then :18 $\text{now} \leftarrow \text{now} + 1$:19 $\text{tick}[*] \leftarrow 0$ / the round is over :20 $\mathcal{A}'(\text{id.Update}, \text{now})$ / notify: the answer is :21 discarded :21 return now id' from <u>$\text{id.Read}()$</u> :22 return now id' from <u>$\text{id.Leak}()$</u> :23 return $(\text{now}, \text{reg}, \text{tick})$	:Initialize() :1 $\text{now} \leftarrow 0$:2 $\text{reg}, \text{tick} : \mathcal{G}_{\text{Clock}}.\mathcal{P} \rightarrow \{0, 1\}$:3 $\text{reg}[*] \leftarrow 0; \text{tick}[*] \leftarrow 0$ id' from <u>$\text{id.Register}()$</u> :4 if $\text{id}'.F \in \{\mathcal{A}, \mathcal{Z}\} \wedge \text{id}.P \notin \mathcal{C}$ then :5 return now :6 $\text{reg}[\text{id}.P] \leftarrow 1; \text{tick}[\text{id}.P] \leftarrow 0$ / a fresh :7 member owes a tick :7 return now id' from <u>$\text{id.Deregister}()$</u> :8 if $\text{id}'.F \in \{\mathcal{A}, \mathcal{Z}\} \wedge \text{id}.P \notin \mathcal{C}$ then :9 return now :10 $\text{reg}[\text{id}.P] \leftarrow 0$:11 return now

recorded are ticks and Registrations code. the on remarks Three register may P for acting machine any caller: the not party, the against in- [3] .2 line on party the to call the tied already having Guard ,P tick or before separately tick functionality and party registered every has stead identity by indexed tables is wanted, if discipline, that closes: round the 'sUpdate is notification The changes. else nothing and party, than rather and too, departures and registrations about adversary the tells [3] alone: And mechanical. is mutators two other the to pattern 's20 line extending is counter the secrets: no keeping clock a state, whole the returns Leak and round the in is who only say tables the and anyway, read to anyone's it. with done is who

still is needed example signature the what of most, $\mathcal{F}_{\text{Sig}}$ against Set to simulator's the is value No code. the in legible absence each absent, to nothing is there so – discarded is answer notification's the – choose its re-tested and $\mathcal{A}(\cdot)$ through slot the reached $\mathcal{F}_{\text{Sig}}$ where And sanitize. re- is call one clock's the hung, it while happen could what against tables and .20 line precedes Update of mutation every construction: by sponsive re-entered ever is instance no so leaving, from token the keeps Respond oc- every against enforced – re-test to nothing is there and mid-operation class. simulator a advise only could 15 Chapter where slot. the of cupant plays 17.2 Section bites: it where recorded is costs notification the What there. it prices and clock the over properties two

Properties 17.2

second, a worth is clock the once: 5.1 Definition instanced 15.2 Section is order whose paper this in thing one the is time because least not properties: two carrying $\mathcal{G}_{\text{Clock}}$ over $\mathcal{G}_m$ game one give We content. the promises code the as moves counter the that saying steadiness. **FinStead** saying unanimity. **FinUna** and — pushed when only and one. by forward. — Both pushed. has party registered honest every when only moves it that be to meant being either winning 5.8 Definition of reading search the take hard. than rather impossible

the with $\gamma := \{\mathcal{G}_m\} \cup \{\mathcal{G}_{\text{Clock}}\}$ system game the over played are They The pinned. $\mathbf{N} := \{\mathcal{G}_m.\text{pid}\}$ field: one save 17 Chapter in as taken clock asks requires. 5.1 Definition which $\{\mathcal{G}_m\}$ at Tightness forced. is change is **Std** bare a and ids. process whole admit to test under members the operational in why says 5.7 Definition refuses: 6.5 Remark what exactly any- as it touching functionality. global a shares environment an — terms adversary an against property a is $\mathcal{F}$ global a of property a and may. one the but nuisance a not is sharing properties two these For it. shares that all about assertions are unanimity and steadiness claim: the of negation all is trace its where only them certify can game a and are. there ticks the are certified properties the and privately. played is clock the So is. there 15.2 Section of situation the — change not does sharing which code. its of under $\mathcal{F}_{\text{Sig}}$ the and protocol its names ships that $\mathcal{F}_{\text{Sig}}$ the where exactly. game. the names test

one for asks **Guard** of 3 line so global. longer no is clock the Pinned. carries id process clock's the and 0 session in sits experiment the session: at 2 line passes now $\mathcal{G}_m$ but caller no And passes. check the so $s = 0$ lie claims whose hosts. it machine a or environment the not clock: the refused are -identitiesA whose adversary. the not and $\mathbf{N}$ pinned a outside changes counter the So time. the them answer even could 13 line before Register the inside only membership the and below 15 line inside only game a — recorded is change every and it. beside relays Deregister and way other no has pushed" one no when moved time "the is verdict whose and $(\mathcal{G}_{\text{Clock}}, \text{pid}, \text{id}, P)$ for $\text{id}_{\mathcal{G}_{\text{Clock}}}$ write we id at interface an Inside know. to gave. clock the τ answer the carries entry trace every

root. the refuse relays the :15.2 Section in analogue no has line One were and serve not does $\mathcal{F}_{\text{Sig}}$ party a addressed root the at relays the There well-shaped over existential verdicts. the which .1 line at **rej** answered game's the of **rej** a so entry. every reads Steadiness read. never tuples.

nothing relay, the at refused trace; the into allowed be cannot making own gave. actually clock the answer an is τx in entry every and appended. is honest an answering π a rightly: and bites. this replacement a Against that clock a spot, the on steadiness loses number a but anything read moves. it however broken being time the say cannot prop- single a that for asks shell shared a thing first the is line That like existential. is verdict its — it need not does Unanimity not. would erty steady- is refusal the so — did those as rej a past read would and 's: $\mathcal{F}_{\text{Sig}}$ finalizations both because it carries shell the and alone, demand ness's serves. it verdicts the of strictest the to built is shell A shell. the off hang trace a on played is unanimity others: the by paid is that of cost the and thing one the is it because naming, worth cost a is It needs. it than cleaner by weakened be would property a Where away. take can shell a sharing second a is answer the over-served, merely than rather demand another's finalization. second a not and game

follow finalizations the $\mathcal{G}_{\text{Clock}}$ over shell Game

$\text{par} := \text{.U} := \{\{\mathcal{G}_{\text{Clock}}, \text{serves}_{\mathcal{G}_{\text{m}}}\}\} \quad \text{.N} := \mathbb{Z} \quad \text{.P} := \mathcal{G}_{\text{Clock}} \cdot \text{PU}\{\mathbb{Z}\} \quad \text{.pid} := (\mathcal{G}_{\text{m}}.F, 0, 0)$

$\perp$

```

                                id' from id.Read()                                :Initialize()
:18 require  $\neg \text{done}$ 
:19 require  $\text{id.P} \neq \mathbb{Z}$ 
:20  $\tau \leftarrow \text{id}_{\mathcal{G}_{\text{Clock}}}.\text{FullRead}()$  as id
:21  $\tau x \leftarrow \tau x \cdot (\text{read}, \text{id.P}, \tau)$ 
:22 return  $\tau$ 

                                id' from id.Leak()
:23 return  $\perp$ 

                                id' from id.Register()
:3 require  $\neg \text{done}$ 
:4 require  $\text{id.P} \neq \mathbb{Z}$  // the root only finalizes
:5  $\tau \leftarrow \text{id}_{\mathcal{G}_{\text{Clock}}}.\text{FullRegister}()$  as id
:6  $\tau x \leftarrow \tau x \cdot (\text{reg}, \text{id.P}, \tau)$ 
:7 return  $\tau$ 

                                id' from id.Deregister()
:8 require  $\neg \text{done}$ 
:9 require  $\text{id.P} \neq \mathbb{Z}$ 
:10  $\tau \leftarrow \text{id}_{\mathcal{G}_{\text{Clock}}}.\text{FullDeregister}()$  as id
:11  $\tau x \leftarrow \tau x \cdot (\text{dereg}, \text{id.P}, \tau)$ 
:12 return  $\tau$ 

                                id' from id.Tick()
:13 require  $\neg \text{done}$ 
:14 require  $\text{id.P} \neq \mathbb{Z}$ 
:15  $\tau \leftarrow \text{id}_{\mathcal{G}_{\text{Clock}}}.\text{FullUpdate}()$  as id
:16  $\tau x \leftarrow \tau x \cdot (\text{tick}, \text{id.P}, \tau)$ 
:17 return  $\tau$ 

```

shared is $\mathcal{G}_{\text{Clock}}$ over shell the $\mathcal{G}_{\text{m}}$ of finalizations The

(steadiness) id' from [id.FinStead\(\)](#)

```

:24 require  $\neg \text{done}$ 
:25 require  $\text{id}.P = Z$ 
:26  $\text{done} \leftarrow \text{true}$ 
:27 parse  $\text{tr}$  as  $((\text{op}_j, P_j, \tau_j))_{j=1}^n$ :  $\tau_0 := 0$ 
:28  $w \leftarrow 1$  if  $\exists j \in \{1, \dots, n\}$ :  $\tau_j \notin \{\tau_{j-1}, \tau_{j-1} + 1\}$ 
:29  $\vee (\tau_j = \tau_{j-1} + 1 \wedge \text{op}_j \neq \text{tick})$ :
:30 else  $w \leftarrow 0$ 
:31 return  $w$ 

```

(unanimity) id' from [id.FinUna\(\)](#)

```

:32 require  $\neg \text{done}$ 
:33 require  $\text{id}.P = Z$ 
:34  $\text{done} \leftarrow \text{true}$ 
:35  $\mathbf{C} \leftarrow (C, Z).\text{FullStatus}()$  as  $\text{id}$ 
:36  $w \leftarrow 1$  if  $\exists t \geq 0 \exists P \in \mathcal{G}_{\text{Clock}}.P \setminus \mathbf{C}$ :
:37  $(\exists \tau_r \leq t : (\text{reg}, P, \tau_r) \in \text{tr}) \wedge (\text{dereg}, P, \cdot) \notin \text{tr}$ 
:38  $\wedge (\exists \tau > t : (\cdot, \cdot, \tau) \in \text{tr}) \wedge (\forall \tau' \in \{t, t+1\} : (\text{tick}, P, \tau') \notin \text{tr})$ :
:39 else  $w \leftarrow 0$ 
:40 return  $w$ 

```

an- an on 1 returns FinStead finalization The words. in verdicts. The an or it. above one than more answer an predecessor. its below swer must departures and registrations reads. tick: a at but anywhere increase is $\tau_{j-1} + 1$ answered tick A it. left answer last the where time the find 'sInitialize is $\tau_0 := 0$ and – round the closed that tick the is it – allowed already is 1 answered read first a so observation. first the as taken word re- FinUna finalization The pushed. anyone before moved that counter a the at honest – party some although t some passed counter the if 1 turns t answered tick no has – released never and t round by registered close. covering $t + 1$ answer the is. that t round during placed tick no $:t + 1$ or the not trace. the read conjuncts registration The it. closed that one the did. itself game the what only name too they so tables. clock's was trace the There instructive. both .15.2 Section with contrasts Two else. little reads FinStead here read: verdict no order whose sequence a had verdicts both there And besides. much not and order an being time any- of nothing asks FinStead named: they party every of $P \notin \mathbf{C}$ ask to be- intercepts wrapper own game's the party corrupt a at because one. an is tr in entry every so .(5.4 (Remark records and runs core the fore is honesty name must that verdict one The already. answer core's honest it names it and call. not did that parties concerns claim whose 's.FinUna

stronger the that making again (3.5 (Section monotonicity – close the at not; other the register, the reading one – so differ two the That reading. for: is shell shared a what is – membership other the order. reading one them made that verdicts, the not common, in have they oracles the is it game. one

نکته 17.1 (Agreement across read steadiness is (property The .parties) hold parties honest :agreement is for wanted usually is clock shared a corol- a but game third a not is it Here time. the of view common a the over quantifies 28 Line design. by and above, verdict the of lary en- tick consecutive two between party: by party not interleaved, trace two so asked, whoever predecessor, its equal must answer every tries the answered are ticks two same the between reading parties honest own party's each – party per Stated won. is game the or value same agree- and weaker, strictly be would verdict the – steady subsequence 5 with P answer then could replacement a property: separate a ment What property. neither fail and activations adjacent in 3 with Q and to- single a model: execution the is claims both carry verdict one lets and exist not do reads" "simultaneous so trace, the orders totally ken Two entries. adjacent of consistency the to reduces view common a a only, parties honest among is Agreement naming. worth are limits says it and :(5.4 (Remark adversary's the being answers party's corrupt while stale grow may view whose ask, not does that party a of nothing the of disagreement not party, the of staleness – ahead runs clock the never that caller a promise could functionality a nothing and clock, syn- loosely the , Δ some within agreeing views – variant drift A calls. ,28 line of weakening genuine a be would – [11] of clocks chronized third a than rather finalization third a is it shape present the under and the in differs and relays same the of trace same the reads it game: shell. this off it hanging for test the exactly is which alone, verdict

گزاره 17.2 (both has $\mathcal{G}_{\text{Clock}}$.properties) and FinStead of either Fin For .properties) slot, adversary the of $\mathcal{X}$ occupant every and $\mathcal{Z} \in \mathbb{Z}_{\gamma, \gamma}$ every for .FinUna $(\mathbb{Z}_{\gamma, \gamma}, \mathbb{X})$ against 0 within property each has $\{\mathcal{G}_{\text{Clock}}\}$ So . $\Pr[\text{Win}_{\text{Fin}}] = 0$ occupants. all of class the $\mathbb{X}$ for ,5.8 Definition of sense the in

اثبات. One carries invariant the properties: both changes counter the only membership and one. exactly by and records trace the tick a side

first the is it Establishing departures. and registrations recorded inside caller no so tightness. – things three on turns it and work. paragraph's only them reaching game the cores: clock's the reaches game the but handing γ of core no and mediated: is answer no so parties. honest at cores clock's the call single the one. run could that anything to token the each order. execution in append therefore Entries responsive. being place com- by follows then Steadiness returned. call its as counter the carrying Unanimity runs. nothing them between since entries. consecutive paring de- and verdict the of conjuncts four the assumes it half: longer the is corruption of monotonicity using invariant. the from contradiction a rives invariant: one on rests Everything throughout. honest party the keep to exactly by and records. trace the tick a inside only changes counter the departures. and registrations recorded inside only membership the one: at counter the – 17 Chapter of cores the in only written are tables Both `reg` passes. 17 line of test the when one by stepping .Update of line one passing calls on only run cores those and – Deregister and Register in only calls `gm` and .2 line passes `gm` but caller no pinned **N** with :Guard wrapper own its one corrupt a at – parties honest at relays. the through mediated is call no so – ran γ of core no and earlier step a intercepted γ of core no Moreover recorded. answer the is answer core's each and clock's the call one the one: run could that anything to token the hands Respond – responsive is which .20 line of notification the is place cores slot. the occupies whatever place. would answerer its call every refuses relays' the and – case any in it precedes Update of mutation every and γ of machine other no with returning chain that clock. the on is call one each order. execution in append therefore Entries between. in run having returned. call its as counter the carrying

.0 = τ_0 'sInitialize is counter the entry first the before steadiness: For entry at counter the so runs. γ of core no entries consecutive Between own 'sj entry inside taken steps the by $j - 1$ entry at that from differs j tick. a for one or none departure. a or registration a read. a for none call: tick. a at only increment the with throughout. $\tau_j \in \{\tau_{j-1}, \tau_{j-1} + 1\}$ Hence .0 gives 28 line and

some of hold 36 line of conjuncts four the suppose unanimity: For mono- by throughout. honest – close the at $P \in \mathcal{G}_{\text{Clock}} \cdot \mathbf{P} \setminus \mathbf{C}$ some and t the only `reg[P] ← 1` set $\tau_r \leq t$ answered registration recorded A tonicity. Deregister no placed game the .`reg` write Deregister and Register of cores honest an reach cannot observers the and – entry `dereg` no is there – P at $\tau_r > t$ answers entry Some on. moment that from `reg[P] = 1` so party.

stepped it so took, counter the values are answers the invariant the by and ev- of flags the read 17 line its tick; recorded some inside $t + 1$ to t from them, among P – then registered and moment that at honest party ery flag That $\text{tick}[P] = 1$ so $-\tau_r \leq t$ and growing ever only register the the of step every at cleared and P at Update of core the by only written is stood counter the while set was it so re-registration, every at and counter itself was it if $t + 1$ or t answered tick that P at game the of tick a by t at $(\text{tick}, P, \tau') \in \tau_r$ So recorded. was it and round, the closed that one the 36 line and conjunct, fourth the contradicting $\tau' \in \{t, t + 1\}$ some for $\square$.0 gives

was specification the that only says 0 being bound the $\mathcal{F}_{\text{Sig}}$ with As no- the here and transfer, the is content the properties: its have to written cores the that asks 5.11 Corollary price. its collects 20 line of tification the so one, is line that and call, responsive no place system game the of for directly, argued be must $\{\mathcal{G}_{\text{Clock}}\}$ of place in put π system a to transfer Theo- guards hypothesis same the and – gives 9.4 Remark reason the com- dummy forgoes clock notifying the over built system a so ,4.29 rem and 20 line delete – notification the without clock A it, with pleteness them inherits and proof same the by properties both has – entry **U** the discharge, to nothing with corollary the through notification A whether, not price, which only is note, choice, The to token the hand would took, $\mathcal{F}_{\text{Sig}}$ route the $\mathcal{A}(\cdot)$ ordinary the through would answers answering; before ticks nested drive to free adversary an ideal the of fail would steadiness and order, trace of out append then because only $\mathcal{G}_{\text{Clock}}$ of properties are above verdicts two the – itself clock chose $\mathcal{F}_{\text{Sig}}$ where So returns, Update before on acted be cannot 20 line locality choose must clock notifying a locality, with paid and inheritance tells that one the is neither pays that clock the inheritance; with pay and nothing, adversary the

Network - Delayed Δ

Functionality 18.1

Communicate- network. a is reader first the and read, be to exists clock The measured delays form, diffusion the]'s:11] is delay bounded a with tion analyses blockchain composable the network the is clock, shared a on language the in functionality that is below $\mathcal{F}_{\text{Net}}$ network The .[3] over run it. be than rather clock the use to example first the and – paper this of is $\mathcal{F}_{\text{Net}}$ code: no owed and $\mathcal{F}_{\text{Diffuse}}$ channel diffusion a named 1 Chapter attached. time with like looks channel a such what

mes- the timestamps **Send** operation The said. quickly is does it What inter- clock's the $\text{id}_{\mathcal{G}_{\text{Clock}}} := (\mathcal{G}_{\text{Clock}}, \text{pid}, \text{id.P})$ through – clock the by sage tells it. stores – 17.2 Section of notation the served, being party the at face The id. message a answers and – public is network the – adversary the ev- answers then early, reveal to what adversary the asks **Fetch** operation The old. Δ everything with together party fetching the to released erything and Δ within when chooses adversary the point: the is power of division with- is do cannot it what own: its of traffic add may and first, whom to last 'sFetch in sits bound that and time, its past message honest an hold is τ time at sent message A it. reaches slot the of answer no where line. first. favoured adversary the whoever on, $\tau + \Delta$ time from fetch every in

are **N** and **P** .pid clock's. the not pattern, 's $\mathcal{F}_{\text{Sig}}$ follow They **fields. The** naming **N** its session, per instance one local, is network a parameters: in- every and – 1 Chapter of pattern local the in serves it protocol the com- " τ " "time makes what is which clock, shared one the reads stance en- first the $:\mathbf{U} := \{(\mathcal{G}_{\text{Clock}}, \text{serves}), (\mathcal{A}, \text{serves})\}$ sessions. across parable met. is and does network the party every serve to clock the requires try global the by copy shared the at in, sits network the session whatever only well-formed is $\mathcal{F}_{\text{Net}}$ containing system a – 1.3 Definition of disjoint

be- calls slot two the covers second the and — alongside clock the with
 is $n \cdot s\mathcal{F}_{\text{Rand}}$ as reads, actually code of line a parameter a ,par := Δ low.
 .(13 (Chapter

mes- a deliver adversary the lets [3] **called. not answered. is Scheduling**
 lit- Transcribed accord. own its of network the on acting by early sage
A containing **N** local a so calls, adversary the interface an is that erally
 shape one the other, no in and 0 session in slot the from reachable —
 asked is slot the So carry. cannot relocation and excludes 7.2 Definition
 (S, I) pair a back takes and time the adversary the hands Fetch instead:
 in- sanitized — traffic fresh of injections ids. stored the among releases —
 are calls slot Both .15 Chapter in as canonical, being repair good the line.
 it before mutates Send operation The discipline. clock's the responsive.
 responsive- and answer. the reading mutations its cannot, Fetch notifies:
 straight comes token the immaterial: difference the makes what is ness
 re-tests re-entrance $s\mathcal{F}_{\text{Sig}}$ and call. either inside runs else nothing back.
 too. here unnecessary stay

are Fetch and Send party's corrupt A **code. of line a without Corruption.**
 buffer the so run. cores the before interface full the of 4 line by intercepted
 — mediated never are reads clock cores' the and only, sends honest holds
 "sends", party corrupt a What word. own clock's the always is timestamp a
 of injections the instead: side receiving the from supplies adversary the
 the to released ∞ timestamp and $\perp$ sender with buffer the enter 16 line
 guarantee. 's20 line into ripening no so and age. no — alone party fetching
 traffic. own adversary's the conscripts never and honest the for is which

$\mathcal{F}_{\text{Net}}$ Functionality	
$\text{par} := \Delta \quad .\mathbf{U} := \{(\mathcal{G}_{\text{Clock}}, \text{serve}), (\mathcal{A}, \text{serve})\} \quad .\mathbf{N} \quad .\mathbf{P} \quad .\text{pid}$	
$\text{id}' \text{ from } \text{id.Fetch()}$:11 $\tau \leftarrow \text{id}_{\mathcal{G}_{\text{Clock}}}.\text{FullRead}() \text{ as id}$:12 $(S, I) \leftarrow \mathcal{A}'(\text{id.Fetch}, \tau) \quad / \text{ a non-pair}$ reads as $(\emptyset, \emptyset)$:13 $S \leftarrow S \cap \{m : (m, \cdot, \cdot) \in \text{buf}\}$:14 $I \leftarrow I \cap \mathcal{M} \quad / \text{ sanitized inline}$:15 $\text{rel}[m, \text{id.P}] \leftarrow 1 \text{ for each } m \in S$:16 for each $\text{msg} \in I$: :17 $\text{buf} \leftarrow \text{buf} \cup \{(\text{ctr}, \perp, \text{msg}, \infty)\}$:18 $\text{rel}[\text{ctr}, \text{id.P}] \leftarrow 1$; $\text{ctr} \leftarrow \text{ctr} + 1$:19 return $\{(m, \text{msg}) : (m, P', \text{msg}, \tau_s) \in \text{buf}$:20 $\wedge (\text{rel}[m, \text{id.P}] = 1 \vee \tau_s + \Delta \leq \tau)\}$	: Initialize() :1 $\text{ctr} \leftarrow 0$:2 $\text{buf} \leftarrow \emptyset \quad / \text{ entries } (m, P, \text{msg}, \tau)$:3 $\text{rel} : \mathbb{N} \times \mathcal{F}_{\text{Net}}.\mathbf{P} \rightarrow \{0, 1\}$:4 $\text{rel}[*] \leftarrow 0 \quad / \text{ what is released to whom}$ $\text{id}' \text{ from } \text{id.Send}(\text{msg})$:5 $\tau \leftarrow \text{id}_{\mathcal{G}_{\text{Clock}}}.\text{FullRead}() \text{ as id}$:6 $m \leftarrow \text{ctr}$; $\text{ctr} \leftarrow \text{ctr} + 1$:7 $\text{buf} \leftarrow \text{buf} \cup \{(m, \text{id.P}, \text{msg}, \tau)\}$:8 $\mathcal{A}'(\text{id.Send}, m, \text{msg}) \quad / \text{ the network is public}$:9 return m $\text{id}' \text{ from } \text{id.Leak}()$:10 return (buf, rel)

net- the sender: no carry pairs Fetched code. the on remarks Three what are injections the and]'s.3] diffusion. unauthenticated an is work sent. one no messages contain may fetch a – means unauthenticated one is diffusion authenticated An write. to adversary's the being wire the drop keeps. already buffer the sender the return – away line one and field dif- a and step further a is addressing with Authentication injections. the fetch one what sticky: is rel table The it. gives 19 Chapter object: ferent grows only network the of view party's a so hide. may fetch later no reveals Δ past silence adversary's the so deadline. the or rel reads 20 line and – state. whole the returns Leak And punished. than rather ignored simply is adversary the to told were contents its secrets: no keeping network the doing. own slot's the is table release the and arrived. they as 8 line at

Properties 18.2

monotone Δ by delivery unformulated: properties three left 18.1 Section all: at game no needs last The sending. before delivery no and views. buf with intersecting 20 line name. can S nothing is buf in yet not id an release ever can slot the of occupant no so runs. else anything before unformulated. leave still we views Monotone sent. been yet not has what formalize we What timing. to orthogonal and alone rel of property a being by ripened message a :liveness. **FinLive** name the under first. the is here property a formalize we And fetch. honest later no from missing is Δ an message a :authentication. **FinAuth** have. to claimed never was $\mathcal{F}_{\text{Net}}$

reading search the take Both someone. by sent was fetches party honest
5.8 Definition of

shape that where is this and game. one of finalizations two are They
 unau- an being and, outright first the has $\mathcal{F}_{\text{Net}}$ network The most. the earns
 injec- the on remark own's **18.1 Section** — design by diffusion thenticated
 oracles. of set one shell. one second: the lacks provably — **16** line of tions
 the and unwinnable is one which of it off read verdicts two and trace. one
 be would that games separate two Under .1 probability with won is other
 twice. measured object one is it here objects: two about statements two
 en- an flag: single's **5.3 Remark** is apart measurements the keeps What
 no so ,FinLive against game the closed has FinAuth calls that vironment
 conjunction. a is pair the about nothing and both. on judged is run
 with $\gamma := \{\mathcal{G}_m\} \cup \{\mathcal{F}_{\text{Net}}, \mathcal{G}_{\text{Clock}}\}$ system game the over played are Both
 tight- 's **15.2 Section** — $\mathcal{F}_{\text{Net}}.\mathbf{P} \subseteq \mathcal{G}_m.\mathbf{P}$ and pinned $\mathcal{F}_{\text{Net}}.\mathbf{N} := \{\mathcal{G}_m.\text{pid}\}$
 follow already fields own's $\mathcal{F}_{\text{Net}}$ since whole. transplanted argument ness
 be- box interface The .(**18.1** (Section clock's the than rather pattern's $\mathcal{F}_{\text{Sig}}$
 the not : $\mathcal{F}_{\text{Net}}$ at **Guard** passes $\mathcal{G}_m$ but caller No full. in $\mathcal{G}_m.\mathbf{U}$ states low
 in- invented an at hosted machine a not adversary. the not environment.
 stores. ever core's $\mathcal{F}_{\text{Net}}$ message every So name. game's the of stance
 be- relays the through passes .Fetch a to returns ever it answer every and
 someone" by "sent names verdict whose game a — recorded is and low
 and clock the on dependence own's $\mathcal{F}_{\text{Net}}$ it. know to way other no has
 $\mathcal{F}_{\text{Net}}$ for was it as exactly met is and this of any by unaffected is slot the
1.3 Definition of disjunct global the by copy shared the at clock the itself.
 for $\text{id}_{\mathcal{G}_{\text{Clock}}}$ and $(\mathcal{F}_{\text{Net}}.\text{pid}, \text{id}.\mathbf{P})$ for $\text{id}_{\mathcal{F}_{\text{Net}}}$ write we id at interface an Inside
17.2 Section of notation the second the ,($\mathcal{G}_{\text{Clock}}.\text{pid}, \text{id}.\mathbf{P}$)

follow finalizations the : $\mathcal{G}_{\text{Clock}}$ and $\mathcal{F}_{\text{Net}}$ over shell Game

$\mathbf{U} := \mathbf{N} := \mathbf{Z} .\mathbf{P} := \mathcal{F}_{\text{Net}}.\mathbf{P} \cup \{\mathbf{Z}\} .\text{pid} := (\mathcal{G}_m.F, 0, 0)$
 $\text{par} := \perp \quad \{(\mathcal{F}_{\text{Net}}, \text{serve}_{\mathcal{G}_m}), (\mathcal{G}_{\text{Clock}}, \text{serve})\}$

<pre> :8 require $\neg$done :9 $\tau \leftarrow \text{id}_{\mathcal{G}_{\text{Clock}}}.\text{FullRead}()$ as id :10 $M \leftarrow \text{id}_{\mathcal{F}_{\text{Net}}}.\text{FullFetch}()$ as id :11 $\text{tr} \leftarrow \text{tr} \cdot (\text{fetch}, \text{id}.\mathbf{P}, \tau, M)$:12 return M :13 return $\perp$ </pre>	<pre> id' from <u>id.Fetch()</u> :1 done $\leftarrow$ false :2 $\text{tr} \leftarrow ()$ / the trace. a sequence id' from <u>id.Send(msg)</u> :3 require $\neg$done :4 $\tau \leftarrow \text{id}_{\mathcal{G}_{\text{Clock}}}.\text{FullRead}()$ as id :5 $m \leftarrow \text{id}_{\mathcal{F}_{\text{Net}}}.\text{FullSend}(msg)$ as id :6 $\text{tr} \leftarrow \text{tr} \cdot (\text{send}, \text{id}.\mathbf{P}, m, \tau, msg)$:7 return m id' from <u>id.Leak()</u> </pre>
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shared is $\mathcal{G}_{\text{Clock}}$ and $\mathcal{F}_{\text{Net}}$ over shell the $\mathcal{G}_m$ of finalizations The

(liveness) id' from $\text{id.FinLive}()$

```
:14 require  $\neg \text{done}$ 
:15 require  $\text{id.P} = Z$ 
:16  $\text{done} \leftarrow \text{true}$ 
:17  $\mathbf{C} \leftarrow (\mathbf{C}, Z).\text{FullStatus}()$  as  $\text{id}$ 
:18  $w \leftarrow 1$  if  $\exists P, Q, m, \tau_s, \text{msg}, \tau, M : P \notin \mathbf{C} \wedge Q \notin \mathbf{C}$ 
:19  $\wedge (\text{send}, P, m, \tau_s, \text{msg}) \in \text{tr} \wedge (\text{fetch}, Q, \tau, M) \in \text{tr}$ 
:20  $\wedge \tau \geq \tau_s + \Delta \wedge (m, \text{msg}) \notin M$ 
:21 else  $w \leftarrow 0$ 
:22 return  $w$ 
```

(authentication) id' from $\text{id.FinAuth}()$

```
:23 require  $\neg \text{done}$ 
:24 require  $\text{id.P} = Z$ 
:25  $\text{done} \leftarrow \text{true}$ 
:26  $\mathbf{C} \leftarrow (\mathbf{C}, Z).\text{FullStatus}()$  as  $\text{id}$ 
:27  $w \leftarrow 1$  if  $\exists Q, \tau, M, m, \text{msg} :$ 
:28  $Q \notin \mathbf{C} \wedge (\text{fetch}, Q, \tau, M) \in \text{tr} \wedge (m, \text{msg}) \in M$ 
:29  $\wedge \neg \exists P, \tau_s : (\text{send}, P, m, \tau_s, \text{msg}) \in \text{tr}$ 
:30 else  $w \leftarrow 0$ 
:31 return  $w$ 
```

'sFetch or 'sSend reads relay Neither code. the about things Three read both so – read to none is there – timestamp a for value return own readings two the $\mathcal{F}_{\text{Net}}$ to call the above line one themselves. clock the guard own 's $\mathcal{F}_{\text{Net}}$ not them. between runs γ of core no because coincide relay the honest. id.P finds and $\mathbf{C}$ reads only which check. corruption and .(17.1 (Section call no places which itself. Read not and all. at run having follow fields 's $\mathcal{F}_{\text{Net}}$'s:17.2 Section unlike $\text{id.P} = Z$ excludes relay Neither worse the any reads verdict neither and clock's. the not and pattern 's $\mathcal{F}_{\text{Sig}}$ demand the spared is shell the so – too fetching or sending root the for they shell the get finalizations both and clock's. the of made steadiness of nothing asks verdict authentication's And alone. for asked have would is pair fetched the – about ask to none returns Fetch because sender. a exactly is whom. by sent. ever was m some whether and alone. (m, msg) whole the what is which recover. otherwise can $\mathcal{F}_{\text{Net}}$ of caller no fact the testing. is game

occupant every and $\mathcal{Z} \in \mathcal{Z}_{\gamma, \gamma}$ every For .liveness) has $\mathcal{F}_{\text{Net}}$ 18.1 گزاره
liveness has $\{\mathcal{F}_{\text{Net}}, \mathcal{G}_{\text{Clock}}\}$ So . $\Pr[\text{Win}_{\text{FinLive}}] = 0$ slot. adversary the of $\mathcal{X}$
class the $\mathbb{X}$ for .5.8 Definition of sense the in $(\mathcal{Z}_{\gamma, \gamma}, \mathbb{X})$ against 0 within

occupants. all of

اثبات. Suppose $(\text{send}, P, m, \tau_s, \text{msg}) \in \text{tr}$. (Sec- one internal 's $\mathcal{F}_{\text{Net}}$.Send and read clock own Its relay. Send the core the so given, just argument the by value. same the answer (18.1 tion
 buf in $(m, P, \text{msg}, \tau_s)$ exactly stores
 's $\mathcal{F}_{\text{Net}}$.Fetch. $\tau \geq \tau_s + \Delta$ with $(\text{fetch}, Q, \tau, M) \in \text{tr}$ also suppose Now
 and $(m', P', \text{msg}', \tau_{s'}) \in \text{buf}$ with (m', msg') pair every returns core
 clock the of reading own its τ . (20 (line $\text{rel}[m', Q] = 1 \vee \tau_{s'} + \Delta \leq \tau$
 $(m, P, \text{msg}, \tau_s)$ is entry buffer the : $m' = m$ Taking relay's. the again, and.
 what- met is disjunct second the so hypothesis, by holds $\tau_s + \Delta \leq \tau$ and
 settles which, 12 line at answered slot the whatever and is $\text{rel}[m, Q]$ ever
 gives 18 line and $(m, \text{msg}) \in M$ Hence disjunct. neither and I and S only
 $\square$.0

گزاره 18.2 $\mathcal{Z}^* \in \mathbb{Z}_{\gamma, \gamma}$ are There .authentication) have not does $\mathcal{F}_{\text{Net}}$
 $\varepsilon < \text{no} : \Pr[\text{Win}_{\text{FinAuth}}] = 1$ with slot adversary the of $\mathcal{X}^*$ occupant an
 containing class any against authentication 's $\{\mathcal{F}_{\text{Net}}, \mathcal{G}_{\text{Clock}}\}$ bounds 1
 them.

اثبات. The machines. two in counterexample A. answers $\mathcal{X}^*$ occupant
 ex- is which – call no places and injection one with query fetch 's $\mathcal{F}_{\text{Net}}$
 the to relay to is answer to way only whose dummy, the not is it why actly
 point whole the is difference That refuses. Respond call a environment.
 the than rather slot the of occupant every over properties these stating of
 and satisfied, out come would FinAuth dummy, the at read alone: dummy
 on finalizes and once fetches then $\mathcal{Z}^*$ environment The vacuously. only
 the reaches sanitizing, survives injection The else. nothing and FinAuth
 .1 advantage with fires verdict the so ran, ever relay Send no and trace.
 $\text{id}.A(\text{id}''.\text{Fetch}, \tau)$ query the On call. no place but fields 's $\mathcal{A}$ have $\mathcal{X}^*$ Let
 for $(\emptyset, \{\text{msg}^*\})$ returns it – 12 line of shape the – $\text{id}''.\text{F} = \mathcal{F}_{\text{Net}}.\text{pid}$ with
 is it call. no Placing . $\perp$ returns it else anything on : $\text{msg}^* \in \mathcal{M}$ fixed one
 dummy, the unlike calls. slot 's $\mathcal{F}_{\text{Net}}$ of either at Respond by refused never
 does Respond call a .(5 (line $\mathcal{Z}$ to relay to is answer to way only whose
 exactly and , $\mathcal{D}$ from differs $\mathcal{X}^*$ where exactly is This .(9.4 (Remark refuse
 of occupant every over stated are 18.1 and 17.2 ,15.1 Propositions why
 .5.8 Definition of case dummy's the – alone dummy the not and slot the
 and satisfied, as FinAuth read would subscript. the from dropped $\mathcal{X}$ with
 all. at inject to unable being $\mathcal{D}$ vacuously, only

claim then once. $\text{Fetch}()$ call, $Q \in \mathcal{F}_{\text{Net}}$. $\mathbf{P}$ for $(\mathcal{G}_m.\text{pid}, Q)$ claim $\mathcal{Z}^*$ Let the is which other. no and FinAuth and $-\text{FinAuth}()$ call and $(\mathcal{G}_m.\text{pid}, Z)$ beside one the not and property this on judged be to takes it what of whole 's $\mathcal{G}_m$ of identities only claims and nothing. hosts one. no corrupts It it.

$\mathcal{Z}^* \in \mathbb{Z}_{\gamma, \gamma}$ so name. own

of loop the so. $(I \leftarrow \text{INM})$ sanitizing survives $I = \{\text{msg}^*\}$ fetch. that At $\text{rel}[m, Q] \leftarrow 1$ sets and buf to $(m, \perp, \text{msg}^*, \infty)$ fresh some adds 16 line No $(m, \text{msg}^*) \in M$ with $(\text{fetch}, Q, \tau, M) \in \text{tr}$ and returned. is (m, msg^*) indeed $-(\text{send}, \cdot, m, \cdot, \text{msg}^*)$ entry no carries tr so ran. ever relay Send $Q \notin \mathbf{C}$ So injection. the to fresh being m all. at m naming entry send no .1 gives 27 line and throughout.

$\mathcal{Z}^*$ environment The .18.1 Proposition touches run that about Nothing $\text{Win}_{\text{FinLive}}$ and answered never was FinLive so. FinAuth at game the closed same the and .5.3 Remark by disjoint are events two the occur: not did always it one beside keeps always it property a host therefore can shell $\square$ other. each about anything saying two the without loses

of cores the if only 5.11 Corollary by transfers liveness. $\mathcal{G}_{\text{Clock}}$ with As 8 lines do. 's $\mathcal{F}_{\text{Net}}$ of both and call. responsive no place system game the transfer the have then must $\{\mathcal{F}_{\text{Net}}, \mathcal{G}_{\text{Clock}}\}$ of place in put π system a :12 and clock's the as exactly gives. 9.4 Remark reason the for directly. argued no did note. Tightness. thing. same the 17.2 Section cost notification own unlike system: game a as γ licensing beyond above proof either in work a out rule to needs here verdict neither unforgeability. and correctness what off read to only sees. never $\mathcal{G}_m$ route some by buf entering message a survive would propositions both so $-\text{placed itself } \mathcal{G}_m$ what to happens it. for asks above nothing though wanted. one were dependence. weaker Proposi- transfer: to nothing having nothing. transfers Authentication exactly specification. the about fact a but inherit to bound a not is 18.2 tion than rather now proved $-\text{prose in asserted already 18.1 Section fact the its ignores that one than stranger no adversary an by and said. merely does What it. on improve to asked is } \{\mathcal{F}_{\text{Net}}, \mathcal{G}_{\text{Clock}}\}$ of replacement No mail. traffic sender's corrupt a : $\mathcal{F}_{\text{AC}}$'s 19 Chapter is property analogous the have through than rather identity. own its under itself. Send through enters there adver- the is "Authentication. 19 (Chapter answer slot unaccountable an verdict a for sender a carries pair fetched the so interface)". own sary's addressed. fetched. $-\text{condition win transplanted the there and name. to met. not is } -\text{sent never and}$

Authenticated –Delayed Δ Channel

Functionality 19.1

one the is object complementary The signs. nobody and diffuses $\mathcal{F}_{\text{Net}}$ arrives none and attached sender its with arrives message every where and Δ same the given .[7] of channel authenticated the – nobody from two the and functionality. that is below $\mathcal{F}_{\text{AC}}$ channel The clock. same the the than more deal good a in differ they side: by side reading worth are lives. actually authentication where is difference the and field. sender as recipient a takes Send operation The said. quickly is does it What fresh a under it files clock. the by pair the stamps message. a as well this and secrecy. not is authentication – adversary the tells and index. ad- everything gathers Fetch operation The all. at nothing hides channel adversary the asks age. of come has that party fetching the to dressed the empties and union. the answers early. release to rest the of which sender. record: whole the is gets receiver the What answered. it entries ad- the – ' $\mathcal{F}_{\text{Net}}$ is power of division The message. time. send recipient. bound: the is so and – order what in and Δ within when chooses versary slot the whatever names. it party the of fetch next the in is old Δ entry an answers.

pa- are **N** and **P** ,pid reasons. ' $\mathcal{F}_{\text{Net}}$ for ' $\mathcal{F}_{\text{Net}}$ are They **fields. The** served protocol the naming **N** with session. per instance one rameters. is field **U** The .par := Δ and .1 Chapter of pattern local the in the by copy shared the at met entry clock its . $\{(S_{\text{Clock}}, \text{serves}), (\mathcal{A}, \text{serves})\}$ and needed are serves of conjuncts Both .1.3 Definition of disjunct global chan- the party every serve to clock the asks first the formality: a is neither be not would name own channel's the second the without and does. nel

.4 line on read every refuse would **Guard** so clock, the at admitted

be- 16 line needs $\mathcal{F}_{\text{Net}}$ **interface. own adversary's the is Authentication**
 under: enter to name no traffic adversarial gives channel diffusion a cause
 it on write slot the letting by modelled is forgery so anonymous, is wire the
 At it. delivers already interface full the and name, a is there Here directly.
 A admit 4 and 3 lines — core the reaches adversary the sender corrupt a
 test- both calls, its on fires hook mediation neither and party corrupt a at
 sender's the under Send through enters traffic corrupt so — id'. $F \neq A$ ing
 $\mathcal{F}_{\text{AC}}$ channel The anyone's, like clock the by timestamped identity, own
 an of field sender the and none, needs and injections no has therefore
 the interface which about fact a but wire the about report a not is entry
 is forgery is: channel authenticated an what is That through, came call
 from, forge to nowhere is there because impossible
 adversary the message A ripens, traffic adversarial that is price The
 12 line so timestamp, real a with list the in is party corrupt a from sends
 $\mathcal{F}_{\text{Net}}$ network The it, retract can answer later no and Δ by it deliver will
 the for is guarantee the so ∞ at injections filing deliberately, this refuses
 con- a is difference The traffic, own slot's the conscripts never and honest
 message a it: against made choice a than rather addressing of sequence
 is sender the message a is recipient named a and sender named a with
 accepted it deadline the to adversary the holding and for, record the on
 field sender the keeps that reading the is party corrupt a as spoke it when
 says, it what meaning

dis- it answering so recipient, one names entry Each **empties. Fetching**
 them between slot the and 12 line what clears 16 line and it, charges
 owed is message diffused a — this do cannot $\mathcal{F}_{\text{Net}}$ network The selected.
 and instead table `rel` sticky the carries it why is which — everyone to
 differ- are guarantees two The answered, already has it what re-answers
 only view party's a that is 's $\mathcal{F}_{\text{Net}}$ other: the implies neither and kind in ent
 over protocol A once, exactly delivered is message a that 's $\mathcal{F}_{\text{AC}}$ grows.
 fetched, it what lose not must $\mathcal{F}_{\text{AC}}$ over one and deduplicate must $\mathcal{F}_{\text{Net}}$
 absence, other's the is obligation each and

release to indices of set a is answer slot's The **release. the Sanitizing**
 are that entries naming set a is be can it wrong thing only the and early,
 procedure 's12 Chapter all, at entries not or — party's fetching the not
 it through answer the runs 14 line so fault, of kind this exactly repairs

under

$$\text{Clean}_{\text{addr}}(S; \text{list}, P) : \Longleftrightarrow S \subseteq \{i : \text{list}[i] = (\cdot, P, \cdot, \cdot)\},$$

things Two context. read-only the as party fetching the and list the with right the **San** makes what are they because saying, worth are that follow so exists, always value good A assertion. an than rather here instrument satisfies $\emptyset$ case: this in prove to nothing has 12 Chapter of footnote the lex- the also is it set empty the being and holds, list the whatever $\text{Clean}_{\text{addr}}$ release early no to repaired is answer failing a so such, first icographically had, have not could it nothing adversary the costs repair that And all, at may released subset good particular some wanting adversary an since 's**San** in assertion An untouched, it returned be and subset that submit of denial a slot the hand and answers good on same the read would place dies fetch party's honest the and index stray one – ones bad on service out, rule to exists 12 Chapter failure the is which –

load- is it here and discipline, clock's The **responsive. are calls slot Both** time the reads Fetch operation The out, spelling worth way a in bearing responsive not call that Were .13 line on slot the asks and 10 line on clock the drive could and between in token the hold would adversary the used 12 line time the by stale be then would τ answering: before forward deadline its past message a and $\tau_i + \Delta \leq \tau$ clear would entries fewer it, and read the makes what is Responsiveness withheld, lawfully be would and notifies it before mutates Send operation The moment, one test the nothing reason: 's $\mathcal{F}_{\text{Net}}$ for too, responsive is but argument, such no needs call, either inside run may else

$\mathcal{F}_{\text{AC}}$ Functionality	
$\text{par} := \Delta$	$\text{.U} := \{(\mathcal{G}_{\text{Clock}}, \text{serves}), (\mathcal{A}, \text{serves})\}$
.N	.P
.pid	
id' from <u>id.Fetch()</u> :10 $\tau \leftarrow \text{id}_{\mathcal{G}_{\text{Clock}}}.\text{FullRead}()$ as id :11 $R \leftarrow \{m : \text{list}[m] = (\cdot, \text{id.P}, \tau_m, \cdot)\}$:12 $\wedge \tau_m + \Delta \leq \tau$ / come of age :13 $S \leftarrow \mathcal{A}^i(\text{id.Fetch}, \tau)$ / released early :14 $S \leftarrow \text{San}[\text{Clean}_{\text{addr}}](S; \text{list}, \text{id.P})$:15 $M \leftarrow \{\text{list}[m] : m \in R \cup S\}$:16 $\text{list}[m] \leftarrow \perp$ for each $m \in R \cup S$:17 return M	:Initialize() :1 $\text{ctr} \leftarrow 0$:2 $\text{list} : \mathbb{N} \rightarrow (\mathcal{F}_{\text{AC}}.\mathbf{P}^2 \times \mathbb{N} \times \mathbb{M}) \cup \{\perp\}$:3 $\text{list}[*] \leftarrow \perp$ / entries (P, Q, τ , msg) id' from <u>$\text{id.Send}(Q, \text{msg})$</u> :4 $\tau \leftarrow \text{id}_{\mathcal{G}_{\text{Clock}}}.\text{FullRead}()$ as id :5 $\text{ctr} \leftarrow \text{ctr} + 1$:6 $\text{list}[\text{ctr}] \leftarrow (\text{id.P}, Q, \tau, \text{msg})$:7 $\mathcal{A}^i(\text{id.Send}, \text{ctr}, Q, \tau, \text{msg})$:8 return ok id' from <u>$\text{id.Leak}()$</u> :9 return list

P against checked not is recipient The code. the on remarks Three and filed is parties channel's the outside addressed message a :Send on fetch to declines who party a silence same the is which fetched, never entries about statement a is below guarantee the and produce, would receiver a so message, the with travels time send The fetches, someone re- fetch 's $\mathcal{F}_{\text{Net}}$ — itself deadline the check and receives it what date can auditable locally is Δ 's $\mathcal{F}_{\text{AC}}$ that is difference the and thing, such no turns the returns **Leak** And functionality, the about fact a only is 's $\mathcal{F}_{\text{Net}}$ where the to told were contents its secrets: no keeping channel the entire, list is drained been since has what and arrived, they as 7 line at adversary leave, watched itself slot the what

Properties 19.2

$\mathcal{F}_{\text{Net}}$ one the and Δ by delivery claims: two named above paragraph The A names, it party the by sent, was fetched message every — make cannot in yet not index an game: no needs again sending, before delivery no third, nothing and $\emptyset$ to S repairs **San** so, $\text{Clean}_{\text{addr}}$ fails drained, already or, list for point same the made 18.2 (Section either early released is real yet not the formalize below FinAuth and FinLive finalizations The buffer). 's $\mathcal{F}_{\text{Net}}$ the — over starting than rather pair 's18.2 Section transplanting two, other channel diffusion a between change not does finalize log, relay, of shell same the being these names, the do neither and one, addressed an and reading, search the take Both channel, different a of asked properties two the and time, this both, keep: can $\mathcal{F}_{\text{AC}}$ two the of which is changes What for name is transplant the That design, whole the of point the is second channels two the between difference the put to way sharpest the is name differently, answered, $\mathcal{F}_{\text{AC}}$ of and $\mathcal{F}_{\text{Net}}$ of asked game, one — with $\gamma := \{\mathcal{G}_{\text{m}}\} \cup \{\mathcal{F}_{\text{AC}}, \mathcal{G}_{\text{Clock}}\}$ system game the over played are Both ar- tightness same the — $\mathcal{F}_{\text{AC}}.\mathbf{P} \subseteq \mathcal{G}_{\text{m}}.\mathbf{P}$ and pinned $\mathcal{F}_{\text{AC}}.\mathbf{N} := \{\mathcal{G}_{\text{m}}.\text{pid}\}$ already fields 's $\mathcal{F}_{\text{AC}}$ reason, same the for and, 18.2 Section as gument **Guard** passes $\mathcal{G}_{\text{m}}$ but caller No, (19.1 (Section pattern 's $\mathcal{F}_{\text{Net}}$ following Fetch record every and, list in places ever Send entry every so, $\mathcal{F}_{\text{AC}}$ at inter- The recorded, is and below relays the through passes returns, ever for $\text{id}_{\mathcal{F}_{\text{AC}}}$ write we id at interface an Inside full, in $\mathcal{G}_{\text{m}}.\mathbf{U}$ states box face of notation the second the, $(\mathcal{G}_{\text{Clock}}.\text{pid}, \text{id}, \text{P})$ for $\text{id}_{\mathcal{G}_{\text{Clock}}}$ and $(\mathcal{F}_{\text{AC}}.\text{pid}, \text{id}, \text{P})$.17.2 Section

follow finalizations the $\mathcal{G}_{\text{Clock}}$ and $\mathcal{F}_{\text{AC}}$ over shell Game

$U := .N := Z .P := \mathcal{F}_{\text{AC}}.P \cup \{Z\}$,pid := ($\mathcal{G}_{\text{m}}.F, 0, 0$)
 $\text{par} := \perp \quad \{(\mathcal{F}_{\text{AC}}, \text{serve}_{\mathcal{G}_{\text{m}}}), (\mathcal{G}_{\text{Clock}}, \text{serve})\}$

<pre> :8 require $\neg$done :9 $\tau \leftarrow \text{id}_{\mathcal{G}_{\text{Clock}}}.\text{FullRead}()$ as id :10 $M \leftarrow \text{id}_{\mathcal{F}_{\text{AC}}}.\text{FullFetch}()$ as id :11 $\text{tr} \leftarrow \text{tr} \cdot (\text{fetch}, \text{id}.P, \tau, M)$:12 return M :13 return $\perp$ </pre>	<pre> :1 done $\leftarrow$ false :2 $\text{tr} \leftarrow ()$ // the trace. a sequence :3 require $\neg$done :4 $\tau \leftarrow \text{id}_{\mathcal{G}_{\text{Clock}}}.\text{FullRead}()$ as id :5 $r \leftarrow \text{id}_{\mathcal{F}_{\text{AC}}}.\text{FullSend}(Q, \text{msg})$ as id :6 $\text{tr} \leftarrow \text{tr} \cdot (\text{send}, \text{id}.P, Q, \tau, \text{msg})$:7 return τ </pre>
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shared is $\mathcal{G}_{\text{Clock}}$ and $\mathcal{F}_{\text{AC}}$ over shell the $\mathcal{G}_{\text{m}}$ of finalizations The

```

:14 require  $\neg$ done
:15 require id.P = Z
:16 done  $\leftarrow$  true
:17  $C \leftarrow (C, Z).\text{FullStatus}()$  as id
:18  $w \leftarrow 1$  if  $\exists P, Q, \tau_s, \text{msg}, \tau, M : P \notin C \wedge Q \notin C$ 
:19  $\wedge (\text{send}, P, Q, \tau_s, \text{msg}) \in \text{tr} \wedge (\text{fetch}, Q, \tau, M) \in \text{tr}$ 
:20  $\wedge \tau \geq \tau_s + \Delta \wedge \neg \exists \tau', M' :$ 
:21  $(\text{fetch}, Q, \tau', M') \in \text{tr} \wedge (P, Q, \tau_s, \text{msg}) \in M' :$ 
:22 else  $w \leftarrow 0$ 
:23 return w

(authentication) id' from id.FinAuth()

:24 require  $\neg$ done
:25 require id.P = Z
:26 done  $\leftarrow$  true
:27  $C \leftarrow (C, Z).\text{FullStatus}()$  as id
:28  $w \leftarrow 1$  if  $\exists Q, \tau, M, P, \tau_s, \text{msg} :$ 
:29  $Q \notin C \wedge (\text{fetch}, Q, \tau, M) \in \text{tr} \wedge (P, Q, \tau_s, \text{msg}) \in M$ 
:30  $\wedge (\text{send}, P, Q, \tau_s, \text{msg}) \notin \text{tr} :$ 
:31 else  $w \leftarrow 0$ 
:32 return w

```

the read relays Both familiar. them of two code. the about things Three readings two the 's:18.2 Section as reason same the for themselves clock call no placing Read them. between runs γ of core no because coincide nothing having .ok return. own 's $\mathcal{F}_{\text{AC}}$.Send keeps relay 'sSend own. its of and back pass to id message a had which 's. $\mathcal{F}_{\text{Net}}$ unlike – report to else over search no needs verdict authentication's And out. hands never $\mathcal{F}_{\text{AC}}$ names already record fetched a namesake's: its unlike times. or senders

was tuple exact the that check direct a is condition win the so , τ_s and P the — it produced have might what over one existential an not sent, never the at does it what exactly verdict, the in doing, is carries $\mathcal{F}_{AC}$ address interface.

occupant every and $\mathcal{Z} \in \mathbb{Z}_{\gamma, \gamma}$ every For .liveness) has $\mathcal{F}_{AC}$ 19.1 گزاره
liveness has $\{\mathcal{F}_{AC}, \mathcal{G}_{Clock}\}$ So . $\Pr[\text{Win}_{FinLive}] = 0$ slot, adversary the of $\mathbb{X}$
class the $\mathbb{X}$ for ,5.8 Definition of sense the in $(\mathbb{Z}_{\gamma, \gamma}, \mathbb{X})$ against 0 within
occupants. all of

of activation some at written , $(\text{send}, P, Q, \tau_s, \text{msg}) \in \text{tr}$ Suppose اثبات.
one internal ' $s\mathcal{F}_{AC}$.Send and read clock own Its .(5 (line relay Send the
between runs γ of core no :18.2 Section of argument the by agree, (4 (line
relay the since honest id.P finding check corruption and guard ' $s\mathcal{F}_{AC}$ them,
fresh some wrote core the So call. no placing itself Read and all, at ran
moment. that at $\text{list}[m] = (P, Q, \tau_s, \text{msg})$ with m
 $\text{list}[m]$ If . $\tau \geq \tau_s + \Delta$ with $(\text{fetch}, Q, \tau, M) \in \text{tr}$ too Suppose
by $m \in R$ then runs, core fetch's that when $(P, Q, \tau_s, \text{msg})$ holds still
its τ for hypothesis, by $\tau_s + \Delta \leq \tau$ and ,Q names entry the — 12 line
— relay's the again, argument same the by and. clock the of reading own
whatever and 13 line at answers slot the whatever $(P, Q, \tau_s, \text{msg}) \in M$ so
been already had $\text{list}[m]$ instead If .R touching neither it. with does San
the — Q by fetch earlier some of 16 line at only happened that cleared.
in placed already indices on only and — entry an clears ever that line one
.M own fetch's earlier that in present already hence ,RUS own fetch's that
verdict the and 's.Q of fetch some in appears $(P, Q, \tau_s, \text{msg})$ way Either
 $\square$.0 gives 18 line at

oc- every and $\mathcal{Z} \in \mathbb{Z}_{\gamma, \gamma}$ every For .authentication) has $\mathcal{F}_{AC}$ 19.2 گزاره
has $\{\mathcal{F}_{AC}, \mathcal{G}_{Clock}\}$ So . $\Pr[\text{Win}_{FinAuth}] = 0$ slot, adversary the of $\mathbb{X}$ cupant
,5.8 Definition of sense the in $(\mathbb{Z}_{\gamma, \gamma}, \mathbb{X})$ against 0 within authentication
 $\mathcal{F}_{Net}$ proved 18.2 Section property the — occupants all of class the $\mathbb{X}$ for
instead. here held lacks.

$(P, Q, \tau_s, \text{msg}) \in$ and $Q \notin \mathbf{C}$ with $(\text{fetch}, Q, \tau, M) \in \text{tr}$ Suppose اثبات.
index both and , $M = \{\text{list}[m] : m \in R \cup S\}$ code, ' $s\mathcal{F}_{AC}$.Fetch By .M
match pattern own its by R stands: already it as list on only draw sets
with i indices only admits Clean_{addr} since sanitizing, after S and ,(12 (line
values among picks (12 (Chapter 3 line and already $\text{list}[i] = (\cdot, Q, \cdot, \cdot)$

not does list index an manufactures neither – predicate that meeting some for $\text{list}[m] = (P, Q, \tau_s, \text{msg})$ if only $(P, Q, \tau_s, \text{msg}) \in M$ So hold. point. some at m

is $\perp$ than other value a to $\text{list}[m]$ writes ever that line only The that – pinned $\mathcal{F}_{AC}.\mathbf{N} = \{\mathcal{G}m.\text{pid}\}$ – tightness by and core. own 'sSend the not environment. the not places: relay 's $\mathcal{G}m$ calls on only runs core game's the of instance invented an at hosted machine a not adversary. activa- some at happened $(P, Q, \tau_s, \text{msg})$ producing write the So name. pre- the of argument coinciding-read the by and relay. Send the of tion Hence $.(send, P, Q, \tau_s, \text{msg})$ exactly as there logged was it proof vious $\square$.0 gives 28 line at verdict the and $.(send, P, Q, \tau_s, \text{msg}) \in \text{tr}$

only 5.11 Corollary by transfer properties both $\mathcal{F}_{Net}$ and $\mathcal{G}_{Clock}$ with As there. as here. and – call responsive no place cores system's game the if 'sSend covers responsive" are calls slot "Both 's19.1 Section does: neither $\{\mathcal{F}_{AC}, \mathcal{G}_{Clock}\}$ of place in put π system A alike. query 'sFetch and notification 9.4 Remark reason the for directly. argued transfers both have then must thing. same the 18.2 Section cost notification own 's $\mathcal{F}_{Net}$ as exactly gives. re- negative a as only standing left is here nothing network. the Unlike from transplanted 18.2 Section condition win the is 19.2 Proposition sult. laid proofs. two the and – instead here closed close. not could and $\mathcal{F}_{Net}$ tightness whether fact: one in difference whole the locate side. by side it $\mathcal{F}_{AC}$ For it. produced that Send a to back record fetched a trace can cannot. it $\mathcal{F}_{Net}$ for entry: an gain to way other no having list can. always trans- game one chapters. Two in. way other the exactly being 16 line them. between difference entire the is address the and twice. planted comparison the makes what is names the keeps transplant the That resem- that properties two not is FinAuth finalization The all. at sayable – sent never addressed. fetched. – condition win one but other each ble Under $\mathcal{F}_{AC}$ for 0 and $\mathcal{F}_{Net}$ for 1 is answer the and channels. two of asked game. own its carried property each where replaced. 5 Chapter shape the would sameness the and $\mathcal{G}m'_{Auth}$ and $\mathcal{G}m_{Auth}$ been have would two the notation. the is it here remark: a been have

Authenticated the Realizing Channel

proto- a it: earns chapter This postulated. was 19 Chapter of channel The
 unauthen- the directory, public-key a — subroutines ideal three over ρ col
 that — 15 Chapter of signatures the and ,18 Chapter of network ticated
 ev- ,0 within exactly, so does and channel, authenticated an UC-realizes
 information-theoretic. reduction whole the and ideal being subroutine ery
 the exercises that one the and paper, the in construction largest the is It
 sanitization, discipline, responsive the composition, once: at it of most
 do- each appear, all 11 Chapter of lemma fundamental the and tightness.
 could. other no job a ing

from key signing a draws time first the for sending party A **does. ρ What**
 the reads it Q to msg send To $\mathcal{F}_{PKI}$ directory the in it registers and $\mathcal{F}_{Sig}$
 signs ,s number sequence per-party a increments , τ timestamp a for clock
 over signature its with tuple the diffuses and ,($\mathbb{I}, Q, \tau, s, msg$) tuple the
 item diffused a keeps and delivers. $\mathcal{F}_{Net}$ what collects it fetch, To $\mathcal{F}_{Net}$
 un- verifies that signature a carries fetcher, the to addressed is it if only
 seen not pair (sender, s) a has and key, directory sender's claimed the der
 carries Diffusion $(\mathbb{I}, Q, \tau, msg)$ as up hands it survives What before.
 and second the supplies signature the authenticity: no and addresses no
 a trusts receiver a that so first, the tuple signed the inside address the
 sig- own sender's the under is name that because exactly name sender's
 nature.

and obstruction, the is sender corrupt A **$\mathcal{F}_{AC}$ not , $\mathcal{F}_{AC}^\circ$ is target the Why**
 ad- the network diffusion a Over incidental. than rather instructive is it
 — fetch a during particular in likes, it whenever wire the on writes versary
 the of window responsive the inside chosen are (16 (line injections 's $\mathcal{F}_{Net}$

the key a under signed injected, so message A them. carries that fetch and receiver the by accepted is party, corrupt a for registered adversary only delivers which, $\mathcal{F}_{AC}$ so it, placed ever Send prior no but delivered: call can simulator no and it, produce cannot stored. Send own its what Worse. gap. the repair to window responsive fetch's the inside $\mathcal{F}_{AC}$. Send val- signature the — choose to own its is timestamp sender's corrupt the time true the stamps $\mathcal{F}_{AC}$ where — carries payload the τ whatever idates au- an what are both ρ in defect a is gap Neither witnessed. it call a of 20 Chapter and party, corrupt a gives unavoidably diffusion thenticated does subroutines these over protocol no that argument the with closes traffic sender's corrupt a with $\mathcal{F}_{AC}$: $\mathcal{F}_{AC}^\circ$ is object realized the So better. exactly is which timestamped. freely and unstored fetch. the at admitted honest- The less. no and channel authenticated]'s7] of level guarantee the party's honest an that authenticity, and, Δ by delivery — guarantees sender survive — names it party honest the by sent genuinely was record fetched objects two the one no corrupts that adversary an against And whole. a for $\Pr[\text{Bad}]$ at difference the prices lemma fundamental the coincide: exactly, itself, $\mathcal{F}_{AC}$ realizes ρ so raise. can sender corrupt a only that bad .(20.2 (Corollary corrupt is nobody whenever

re- the are signatures The **shelf. the off quite not are subroutines Two** and — **Verify** and **Sign**, Gen in $\mathcal{A}^1$ — 9 Chapter of $\mathcal{F}_{Sig}^1$ variant sponive respon- is call slot one its atomic: is fetch 's $\mathcal{F}_{AC}$ token. the is reason the answer. its and call the between execution the drives occupant no so sive. would verification a $\mathcal{A}(\cdot)$ ordinary the through $\mathcal{F}_{Sig}$ reach to fetch 'sp Were re- before elsewhere act it let and mid-fetch token the adversary the hand simulator no and show cannot fetch atomic 's $\mathcal{F}_{AC}^\circ$ interleaving an turning, 's $\mathcal{F}_{Net}$ escape: the removes $\mathcal{F}_{Sig}^1$ variant responsive The reproduce. could is fetch every and send every so responsive, already are calls 's $\mathcal{F}_{AC}^\circ$ and of $\mathcal{G}_{PKI}$ global the not local, is directory the And sides. both on atomic real the in ρ . Send honest an by written is directory shared a :16 Chapter where, **Retrieve** through environment the by back straight read and world peek the only holds simulator the and it writes nothing world ideal the in commit- the difference. unsimulatable an — parties honest at grants $\mathcal{G}_{PKI}$, $\mathcal{F}_{PKI}$ local A coordinates. paper's this in setup global a of problem ment out kept is Z it: closes hybrid, any like session the to private identities its a being environment the by retrieval direct a reason. same the for **N** its of serve. to machine no has world ideal the call

key registered a sanitizes $\mathcal{F}_{PKI}$ **purpose. on bindings forgets directory The**

fresh- cross-party 's $\mathcal{G}_{PKI}$ dropping slot, per – more nothing and key a be to
ad- the weapon: a be would freshness The. (Clean_{reg}, 16 (Chapter ness
 $\mathcal{F}_{Sig}$. Gen value the being it key, signing party's honest an chooses versary
the have and slot corrupt a at key that pre-register could it so returns.
direc- leaving one, different a to repaired registration own party's honest
rejected message every party's the and disagreeing key signing and tory
Per-slot unconditionally, promises $\mathcal{F}_{AC}^{\circ}$ liveness the of denial a – ever for
sender's the binding: on rested never attribution because enough is fixity
and up, name that looks receiver the tuple, signed the inside travels name
different a naming tuple a on key party's honest an under valid signature a
costs and here nothing buys Binding suppresses. $\mathcal{F}_{Sig}$ forgery a is sender
goes, it so liveness.

$\mathcal{F}_{PKI}$ directory Local	
par := $\perp$	.U := $\emptyset$
	.N .P ,pid
id' from <u>id.Retrieve(P)</u>	:Initialize()
:10 return vk[P]	:1 vk : $\mathcal{F}_{PKI}.P \rightarrow \mathcal{K} \cup \{\square\}$
id' from <u>id.Leak()</u>	:2 vk[*] $\leftarrow \square$
:11 return vk	id' from <u>id.Register(vk)</u>
:Clean _{key} (vk)	:3 if id'.F $\in \{A, Z\} \wedge$ id.P $\notin \mathbf{C}$ then
:12 return vk $\in \mathcal{K}$	:4 return vk[id.P] / a peek, not a write
	:5 if vk[id.P] $\neq \square$ then
	:6 return vk[id.P]
	:7 vk $\leftarrow$ San[Clean _{key}](vk)
	:8 vk[id.P] $\leftarrow$ vk
	:9 return vk

(S, I) pair a back takes Fetch changed: operation one with $\mathcal{F}_{AC}$ is $\mathcal{F}_{AC}^{\circ}$
of set a I before, as ids stored among releases early the S slot, the from
injection each pins Sanitizing unstored, delivers slot the records injected
ever only can injection an so party, fetching the to and sender corrupt a to
.Send own 's $\mathcal{F}_{AC}$ it: for asked that fetch the into word party's corrupt a put
unchanged, are Initialize and Leak

replaced, $\mathcal{F}_{AC}$ of fetch the $:\mathcal{F}_{AC}^\circ$ Functionality(19.1) (Section $\mathcal{F}_{AC}$ as fields

id' from id.Fetch()

```

:1  $\tau \leftarrow \text{id}_{\mathcal{G}_{\text{Clock}}}.\text{FullRead}()$  as id
:2  $\mathbf{C} \leftarrow (\mathbf{C}, \text{id.P}).\text{FullStatus}()$  as id
:3  $\mathbf{R} \leftarrow \{m : \text{list}[m] = (\cdot, \text{id.P}, \tau_m, \cdot) \wedge \tau_m + \Delta \leq \tau\}$ 
:4  $(S, I) \leftarrow \mathcal{A}^I(\text{id.Fetch}, \tau)$  / releases and injections
:5  $S \leftarrow \text{San}[\text{Clean}_{\text{addr}}](S; \text{list}, \text{id.P})$ 
:6  $I \leftarrow \text{San}[\text{Clean}_{\text{inj}}](I; \mathbf{C}, \text{id.P})$ 
:7  $M \leftarrow \{\text{list}[m] : m \in R \cup S\} \cup I$ 
:8  $\text{list}[m] \leftarrow \perp$  for each  $m \in R \cup S$ 
:9 return M

:Cleaninj(I; C, Pf)

:1 return  $I \subseteq \{(P, Q, \tau, \text{msg}) : P \in \mathbf{C} \wedge Q = P_f\}$ 
```

the at interfaces the through subroutines four its reads protocol The
U Its .18 Chapter of $\text{id}_{\mathcal{F}} := (\mathcal{F}.\text{pid}, \text{id.P})$ notation the served, being party
 payloads own its stamps ρ since them, among clock the four, all names
 returns. never $\mathcal{F}_{\text{Net}}$ which stamp, 's $\mathcal{F}_{\text{Net}}$ recover not does and

 $\{\mathcal{F}_{PKI}, \mathcal{F}_{\text{Net}}, \mathcal{F}_{\text{Sig}}^I\}$ over ρ Protocol

$\mathcal{U} := \{(\mathcal{F}_{PKI}, \text{serves}), (\mathcal{F}_{\text{Net}}, \text{serves}), (\mathcal{F}_{\text{Sig}}^I, \text{serves}), (\mathcal{G}_{\text{Clock}}, \text{serves})\}$ $\mathcal{N}$ $\mathcal{P}$ pid
 par := $\perp$

id' from id.Fetch() :Initialize()

```

:15  $\mathbf{N} \leftarrow \text{id}_{\mathcal{F}_{\text{Net}}}.\text{FullFetch}()$  as id
:16  $M \leftarrow \emptyset$ 
:17 for  $(m, y) \in \mathbf{N}$  do
:18   parse y as  $(x, \sigma)$ .  $x = (P, Q, \tau, s, \text{msg})$  /
      else skip
:19   if  $Q \neq \text{id.P}$  then skip
:20    $\text{vk} \leftarrow \text{id}_{\mathcal{F}_{PKI}}.\text{FullRetrieve}(P)$  as id
:21   if  $\text{vk} = \square$  then skip
:22    $b \leftarrow \text{id}_{\mathcal{F}_{\text{Sig}}}.\text{FullVerify}(\text{vk}, x, \sigma)$ 
      asid
:23   if  $b \neq 1$  then skip
:24   if  $(P, s) \in \text{seen}[\text{id.P}]$  then skip
:25    $\text{seen}[\text{id.P}] \leftarrow \text{seen}[\text{id.P}] \cup \{(P, s)\}$ 
:26    $M \leftarrow M \cup \{(P, Q, \tau, \text{msg})\}$ 
:27 end for
:28 return M

:1 gen :  $\mathbf{P} \rightarrow \{0, 1\}$ ; gen[*]  $\leftarrow 0$ 
:2 seq :  $\mathbf{P} \rightarrow \mathbb{N}$ ; seq[*]  $\leftarrow 0$ 
:3 seen :  $\mathbf{P} \rightarrow \mathcal{P}(\mathbf{P} \times \mathbb{N})$ 
:4 seen[*]  $\leftarrow \emptyset$ 

      id' from id.Send(Q, msg)
:5 if  $\text{gen}[\text{id.P}] = 0$  then
:6    $\text{vk} \leftarrow \text{id}_{\mathcal{F}_{\text{Sig}}}.\text{FullGen}()$  as id
:7    $\text{id}_{\mathcal{F}_{PKI}}.\text{FullRegister}(\text{vk})$  as id
:8    $\text{gen}[\text{id.P}] \leftarrow 1$ 
:9  $\tau \leftarrow \text{id}_{\mathcal{G}_{\text{Clock}}}.\text{FullRead}()$  as id
:10  $\text{seq}[\text{id.P}] \leftarrow \text{seq}[\text{id.P}] + 1$ 
:11  $x \leftarrow (\text{id.P}, Q, \tau, \text{seq}[\text{id.P}], \text{msg})$ 
:12  $\sigma \leftarrow \text{id}_{\mathcal{F}_{\text{Sig}}}.\text{FullSign}(x)$  as id
:13  $\text{id}_{\mathcal{F}_{\text{Net}}}.\text{FullSend}((x, \sigma))$  as id
:14 return ok
```

world real the In slot. adversary the of $\mathcal{A}$ occupant an Fix **simulator. The**
 must $\mathcal{S}_{\mathcal{A}}$ and gone, are they world ideal the in hybrids: three the faces $\mathcal{A}$
 .4.26 (Definition absorption an is It itself. inside from faces their present
 each copy internal one with $\mathcal{A}$ bundles $\mathcal{S}_{\mathcal{A}}$ side): adversary the on read
 execution real the as exactly them against $\mathcal{A}$ runs, $\mathcal{F}_{\text{Sig}}^I$ and $\mathcal{F}_{\text{Net}}, \mathcal{F}_{PKI}$ of

no- Send honest an On it. tells $\mathcal{F}_{AC}^\circ$ what from them drives and would. $\mathbb{I}$ for seq internal the advances it $:(\mathbb{I}, Q, \tau, \text{msg})$ it hands $\mathcal{F}_{AC}^\circ$ tification internal the into steps own 'sp replays and $\mathcal{x} = (\mathbb{I}, Q, \tau, s, \text{msg})$ forms answers. $\mathcal{A}$ hosted the query $\mathcal{A}^!$ whose $\mathcal{F}_{Sig}^!$.Sign internal an – hybrids hosted the so – delivers it notification whose $\mathcal{F}_{Net}$.Send internal an and makes $\mathcal{F}_{AC}^\circ$ When place. would world real the queries the precisely sees $\mathcal{A}$'s $\mathcal{F}_{Net}$ internal the over filter 'sp.Fetch runs $\mathcal{S}_{\mathcal{A}}$ query. fetch responsive its hand would ρ records which exactly learns $\mathcal{A}$ hosted the against delivery their by entries honest early-released or ripe the $:(S, I)$ answers and up. key a under verifying those – records corrupt-sender the and S in ids $\mathcal{F}_{AC}^\circ$ through. straight passes it Corruption I in – slot corrupt a at registered does. 's3.2 Remark as bundle the surviving Corrupt

passes above paragraph the state of pieces two explicit: Made chosen. than rather it above just code the by forced both silently. over with slot the notifies (19.1 Section $\mathcal{F}_{AC}$ from (inherited Send own 's $\mathcal{F}_{AC}^\circ$ the alone. $(\mathbb{I}, Q, \tau, \text{msg})$ not – index list own its m . (Send, $m, Q, \tau, \text{msg})$ The .(9.2 (Definition at arrives query the identity the off read $\mathbb{I}$ sender it later: entry the name to below. fid index. that keep must $\mathcal{S}_{\mathcal{A}}$ simulator And itself. $\mathcal{F}_{AC}^\circ.\text{list}$ to access no having time. fetch at it recover cannot reads test freshness its seen the and seq .5 line at gen – tables own 'sp world ideal the of part no being ρ host. not does $\mathcal{S}_{\mathcal{A}}$ tables are – 24 line at shad - needs filter its reproducing so here. internal subroutines its only and reading by than rather construction by 'sp to equal kept own. its of ows them.

whether decide both and open. leaves prose the questions routing Two $\mathcal{A}$ hosted the call a of becomes what is first The all. at works simulator the neither and party corrupt a at $\mathcal{A}$ admits 3 Line identity. own 'sp on places reaches call a such world real the in so calls. its on fires hook mediation mediated is places then core that call subroutine every and – core 'sp so corrupt). party the and ρ being caller the .4 (line $\mathcal{A}$ to back straight instead would it outward Relayed buffer. 's $\mathcal{F}_{Net}$ reaches ever it of nothing Δ by delivered is and ripens that record a files Send whose $\mathcal{F}_{AC}^\circ$ reach real-world no with delivery a – (19.1 (Section answers slot the whatever rout- explicit 's4.13 Definition relayed: not are calls these So counterpart. against core 'sp runs below handler last the where inside. them keeps ing declared is set outward The really. does Mediate as exactly $\mathcal{A}$ hosted the accordingly.

handlers Both reaches. answer responsive a far how is second The call every refuses (9.1 (Definition Respond and call. responsive a answer

placed not are machines hosted two between calls :places responder the nothing but reachable. are copies internal the so , (4.13 (Definition all at reached. be otherwise would it outside things Two is. bundle the outside each. avoids code 's $\mathcal{S}_{\mathcal{A}}$ and register. corruption the reading by opens interface full copy's hosted A computed is answer responsive a — outright refuses 9.2 Definition which copies the drive handlers the So knows. already responder the what from nothing and above. FullGen than rather Gen written cores. their through before mediated is fetch the or send the at corrupt party a it: by lost is and behalf its on issued ever is query slot no so runs. core own 's $\mathcal{F}_{AC}^{\circ}$ adds interface full the party honest an at and one. serves ever handler no opens $\mathcal{F}_{Net}$ And fires. hook neither and passes. gate the — silencer its only hosted. than rather shared is which , $\mathcal{G}_{Clock}$ reading by Fetch and Send both carries. already query or notification the τ the with answered is read That it send the on one: convenient a merely not and value right the is which the on and share. reads own 's $\mathcal{F}_{Net}$ and 'sp shows proof the stamp the is past advanced have can call responsive no which read. 's $\mathcal{F}_{AC}^{\circ}$ is it fetch .(9.3 (Proposition

$\mathcal{S}_A$ Simulator

4.26 (Definition $\mathcal{F}_{\text{Sig}}^1, \mathcal{F}_{\text{Net}}, \mathcal{F}_{\text{PKI}}$ of each copy internal one and $\mathcal{A}$ hosts $!D_s(\mathcal{A})$ occupies serves. still execution ideal the identities the set outward side): adversary the on read
(4.13 (Definition hand by routes handler last the which own. 'sp less

Q at (τ) query Fetch responsive 's $\mathcal{F}_{AC}^*$ on :Initialize()

```

:14 N  $\leftarrow$  internal id $_{\mathcal{F}_{\text{Net}}}$ .Fetch() / clock
    answered  $\tau$ :  $\mathcal{A}$  answers the release query
:15 S, I  $\leftarrow \emptyset, \emptyset$ 
:16 for (m, y)  $\in$  N do
:17   parse y as (x,  $\sigma$ ): x = ( $\mathbb{Q}$ , Q',  $\tau_x$ , s, msg)
    / else skip
:18   if Q'  $\neq$  Q then skip
:19   vk  $\leftarrow$  internal id $_{\mathcal{F}_{\text{PKI}}}$ .Retrieve( $\mathbb{Q}$ )
:20   if vk =  $\square$  then skip
:21   b  $\leftarrow$  internal id $_{\mathcal{F}_{\text{Sig}}}$ .Verify(vk, x,  $\sigma$ )
:22   if b  $\neq$  1 then skip
:23   if ( $\mathbb{Q}$ , s)  $\in$  seen[Q] then skip
:24   seen[Q]  $\leftarrow$  seen[Q]  $\cup$  {( $\mathbb{Q}$ , s)}
:25   if fid[ $\mathbb{Q}$ , s]  $\neq$   $\square$ 
:26     then S  $\leftarrow$  S  $\cup$  {fid[ $\mathbb{Q}$ , s]}
:27     else I  $\leftarrow$  I  $\cup$  {( $\mathbb{Q}$ , Q',  $\tau_x$ , msg)}
:28 end for
:29 return (S, I)

```

$\mathbb{Q}$ corrupt a at Fetch or Send 's $\mathcal{A}$ on

```

:30 run p's own core for that operation (the protocol box above) on gen, seq, seen, with every subroutine call it places — to  $\mathcal{F}_{\text{Sig}}^1, \mathcal{F}_{\text{PKI}}, \mathcal{F}_{\text{Net}}$  or  $\mathcal{G}_{\text{Clock}}$  — handed to the hosted  $\mathcal{A}$  rather than placed, as the full interface's line 4 hands it really: return the core's own value
:31 never relayed outward: no  $\mathcal{F}_{AC}^*$ .list entry, and none drained

```

Corrupt and Routing.

```

:32 any other call between the hosted  $\mathcal{A}$  and an internal hybrid: served inside, the bundle's ordinary routing (Definition 4.13.) (A Corrupt leaves under the hosted  $\mathcal{A}$ 's own claim, the bundle occupying  $ID_s(\mathcal{A})$ , so Corrupt's own gate id'.F = A (its line 3) is met without rewriting

```

$\mathbb{Q}$ at (m, Q, τ , msg) note Send 's $\mathcal{F}_{AC}^*$ on

```

:1 gen : P  $\rightarrow$  {0, 1}: gen[*]  $\leftarrow$  0
:2 seq : P  $\rightarrow$  N: seq[*]  $\leftarrow$  0
:3 seen : P  $\rightarrow$   $\mathcal{P}(P \times N)$ : seen[*]  $\leftarrow$   $\emptyset$ 
:4 fid : P  $\times$  N  $\rightarrow$  N  $\cup$  { $\square$ }: fid[*]  $\leftarrow$   $\square$  / own record  $\mapsto$   $\mathcal{F}_{AC}^*$ 's index

```

$\mathbb{Q}$ at (m, Q, τ , msg) note Send 's $\mathcal{F}_{AC}^*$ on

```

:5 if gen[ $\mathbb{Q}$ ] = 0 then
:6   vk  $\leftarrow$  internal id $_{\mathcal{F}_{\text{Sig}}}$ .Gen()
:7   internal id $_{\mathcal{F}_{\text{PKI}}}$ .Register(vk)
:8   gen[ $\mathbb{Q}$ ]  $\leftarrow$  1
:9   seq[ $\mathbb{Q}$ ]  $\leftarrow$  seq[ $\mathbb{Q}$ ] + 1: s  $\leftarrow$  seq[ $\mathbb{Q}$ ]
:10 x  $\leftarrow$  ( $\mathbb{Q}$ , Q,  $\tau$ , s, msg)
:11  $\sigma$   $\leftarrow$  internal id $_{\mathcal{F}_{\text{Sig}}}$ .Sign(x) / hosted  $\mathcal{A}$  answers the query
:12 internal id $_{\mathcal{F}_{\text{Net}}}$ .Send((x,  $\sigma$ )) / its clock read answered  $\tau$ 
:13 fid[ $\mathbb{Q}$ , s]  $\leftarrow$  m

```

read to easy but proof the by used all code. 's $\mathcal{S}_A$ 'about things Three entry list undrained "an exactly is 26 line at fid[$\mathbb{Q}$, s] $\neq$ $\square$ First. past. record a recognizes $\mathcal{S}_A$ words: own proof's the in record" the matches is which it. produced that Send honest the relayed once it when precisely all. at fid into gets entry an way only the in- the of reading a not and flag own 'sp of shadow a is gen Second. Register 's $\mathcal{F}_{\text{PKI}}$ — agree two the party honest an at though $\sqrt{VK}$'s $\mathcal{F}_{\text{PKI}}$ ternal

ρ only so ,4 line its at peek a into there call -caller'sZ or -A an turns is which one. corrupt a at company part They entry. that writes ever before away mediated is Register 'sp there taken: not is shortcut the why regardless. $\text{gen}[\mathbb{Q}] \leftarrow 1$ sets ρ while unset $\text{vk}[\mathbb{Q}]$ leaving runs. core 's $\mathcal{F}_{\text{PKI}}$ re- would vk testing simulator A answer. the reading not 5 line 'sp.Send hosted the show and send later every party's the on Register and Gen run sees. never $\mathcal{A}$ real the queries of pair a $\mathcal{A}$ in- ones ripe ,S into record honest matching every puts 26 line Third, own. its on R through ones ripe the taken have would $\mathcal{F}_{\text{AC}}^\circ$ where cluded. entry undrained any admits $\text{Clean}_{\text{addr}}$ nothing: costs and deliberate is This their drains and 7 line its at S with R unions $\mathcal{F}_{\text{AC}}^\circ$.Fetch and ,Q to addressed it Naming once. it drains and delivers twice entry ripe a naming so union. with query the inside own its of read clock no having do. can $\mathcal{S}_{\mathcal{A}}$ all also is early. from ripe tell to which

$\pi := \text{Let } .\text{channel}) \text{ authenticated the realizes } \rho)$ **قضيه 20.1**
 well- is $\pi \cup \{\mathcal{G}_{\text{Clock}}\}$ that so matched parameters with $\{\rho, \mathcal{F}_{\text{PKI}}, \mathcal{F}_{\text{Net}}, \mathcal{F}_{\text{Sig}}^1\}$
 UC- π Then $\mathcal{F}_{\text{Net}}.\text{par} = \mathcal{F}_{\text{AC}}^\circ.\text{par} = \Delta$ and $\rho.\mathbf{N} = \mathcal{F}_{\text{AC}}^\circ.\mathbf{N}$ formed.
 slot adversary the of $\mathcal{A}$ occupant every for :0 within $\{\mathcal{F}_{\text{AC}}^\circ\}$ emulates
 $\mathcal{Z} \in \mathbb{Z}_{\pi, \{\mathcal{F}_{\text{AC}}^\circ\}}$ every for gives. above $\mathcal{S}_{\mathcal{A}}$ simulator the

$$\text{Exec}(\pi, \mathcal{Z}, \mathcal{A}, \mathcal{C}) \equiv \text{Exec}(\{\mathcal{F}_{\text{AC}}^\circ\}, \mathcal{Z}, \mathcal{S}_{\mathcal{A}}, \mathcal{C}),$$

the not adversary. every over is quantifier The distributions. as equal
 Theo- so responsive. is world either in call slot every alone: dummy
 against exhibited is simulator the and this reach not does 4.29 rem
 directly. $\mathcal{A}$ each

parts. three in invariant an carrying activations. on induction One اثبات.
 routing. the so ,4.18 Lemma in as machine to machine pairs coupling The
 state the is new is what lemma's: that are arguments refusal and claims
 the of whole the is it on invariant the and cut. $\mathcal{F}_{\text{AC}}^\circ/\rho$ the across carried
 hold hybrids internal the parts: three the states invariant The argument.
 $\mathcal{F}_{\text{AC}}^\circ.\text{list}$ and holds. ρ what hold shadows 's $\mathcal{S}_{\mathcal{A}}$ hold. ones real the what
 Send it. preserve cases Three sends. honest undrained the exactly holds
 replaying that and time. same the at send a stamp worlds two the shows
 carries Fetch equal. them keeps copies internal the into steps own 'sp
 an whether on but status current sender's the on not splits it weight: the
 unmatched the that shows and record. the matches entry list undrained

and here, buys unforgeability what is which – sender corrupt a forces case activation one the covers traffic Corrupt-party trips. never Clean_{inj} why what is it withholding that shows and withholds, set outward declared the paired tapes, equal on executions two the Couple needs, invariant the to each $\mathcal{G}_{\text{Clock}}$ and $\mathcal{C}$, $\mathcal{Z}$:4.18 Lemma in as machine paired to machine the and $\mathcal{A}$ real the to $\mathcal{S}_{\mathcal{A}}$ of hybrids internal the and $\mathcal{A}$ hosted the itself. Everything $\mathcal{F}_{\text{AC}}^{\circ}$ to hold hybrids the what with together ρ and hybrids, real the :4.17 Definition of shape the in activations, on induction one is below routing its and verbatim, 's4.18 Lemma are paragraphs refusal and claims that calls the and bundle, a is $\mathcal{S}_{\mathcal{A}}$ – one but call every of holds paragraph those save relays, inward and outward ordinary the are boundary its cross set outward declared the which identities, own 'sp on places $\mathcal{A}$ hosted the carried state the only so – inside serves handler last the and withholds the of whole the is it on invariant the and new, is cut $\mathcal{F}_{\text{AC}}^{\circ}/\rho$ the across argument.

the (i) configurations: of pair corresponding every At invariant. The hybrids, real the as tables same the hold $\mathcal{S}_{\mathcal{A}}$ of $\mathcal{F}_{\text{Sig}}^1$ and $\mathcal{F}_{\text{Net}}$, $\mathcal{F}_{\text{PKI}}$ internal or- same the in calls same the by driven and alike initialised been having holds $\mathcal{F}_{\text{AC}}^{\circ}.\text{list}$ (iii) and 's: ρ equal seen and seq, gen shadow 's $\mathcal{S}_{\mathcal{A}}$ (ii) der: party's a through made ρ sends the of (P, Q, τ , msg) records the exactly at $\rho.\text{Send}$ such per entry one – honest was party that while Send own others, no and drained, since has fetch no that – read it timestamp the being corruption entry, the disturb not does sender the of corruption Later paired because hold (ii) and (i) placed, already record the and monotone an at driving: same the under tapes equal on code same the run copies their drive handlers 's $\mathcal{S}_{\mathcal{A}}$ and hybrids real the drives core 'sp party honest mediate sides both one corrupt a at and steps, same the through copies third the as runs, core callee's any before away calls subroutine same the preserve, cases three the what is (iii) checks, below case $\mathcal{F}_{\text{AC}}^{\circ}.\text{Send}$ side ideal the On $\mathbb{I}$ honest an at Send(Q, msg) calls $\mathcal{Z}$ Send, respon- $\mathcal{S}_{\mathcal{A}}$ notifies and list to ($\mathbb{I}$, Q, τ , msg) appends clock, the reads $\mathcal{F}_{\text{Net}}.\text{Send}$ internal its and clock, the reads $\rho.\text{Send}$ side real the On sively, Sec- of argument the by coincide, reads clock two the again: it reads $\mathcal{G}_{\text{Clock}}.\text{Read}$ them, between runs execution either of core no – 18.2 tion notified, $\mathcal{S}_{\mathcal{A}}$ sides, both on τ stamped is send the so – call no placing that copies into runs, ρ steps Send/Sign/Register/Gen the exactly replays with answered read clock own 's $\mathcal{F}_{\text{Net}}$ internal the – 'sp to equal keeps (i) respon- the since stamp, same that is which carried, notification the τ the answers $\mathcal{A}$ hosted the and – it make to own its of call no place may der

one ok return Both tapes. equal over does $\mathcal{A}$ real the as query $\mathcal{A}^1$ each holds. (iii) send: one the matches entry list new deliv- worlds two the on turns Everything. Q at $\text{Fetch}()$ calls $\mathcal{Z}$ Fetch . $\text{Clean}_{\text{inj}}$ within it name to able being $\mathcal{S}_{\mathcal{A}}$ on and $\mathcal{M}$ set same the ering record delivered a internally: mirrors $\mathcal{S}_{\mathcal{A}}$ which filter. 'sp. Fetch Run is $\mathcal{x} = (P, Q, \tau, s, \text{msg})$. $(\mathcal{x}, \sigma)$ tuple diffused whose one is (P, Q, τ, msg) fresh is and $\text{VK}_{\mathcal{F}_{\text{PKI}}}[P]$ under verifying signature a carries. Q to addressed undrained an whether on but status current 'sP on not Split. $\text{seen}[Q]$ in case injected the and case honest the – record the matches entry list two. these exactly being honest. was P while P for made ρ send a is it (iii) by matches: one If id its $\mathcal{S}$ through and ripe. if R through – store from it delivers $\mathcal{F}_{\text{AC}}^\circ$ and match to early it releases $\mathcal{S}_{\mathcal{A}}$ if Q to addressed as $\text{Clean}_{\text{addr}}$ by admitted the now. by corrupt is sender the not or Whether release. early 's $\mathcal{F}_{\text{Net}}$ delivered is none so injected. never and list from delivered is record twice.

un- where is this and fetch. the at corrupt is P then matches: none If honest be would it fetch the at honest P Were place. its earns forgeability – key $\mathcal{F}_{\text{Sig}}$ own 'sP be would $\text{VK}_{\mathcal{F}_{\text{PKI}}}[P]$ so monotonicity. by throughout. under verifying signature a and – returned Gen what exactly registers ρ (Sec- test honest-key 'sVerify forgery: a never is key party's honest an $b = 1$ so key. a such under triple fresh any on $b \leftarrow 0$ sets (15.1 tion by which one. records Sign honest an only and recorded. triple the forces the $\{P \in \mathbf{C}\}$ So case. the against – entry list matching a leave would (iii) signature a or injection $\mathcal{F}_{\text{Net}}$ an produced. have to adversary's the is record $\text{Clean}_{\text{inj}}$ which $\mathcal{I}$ in (P, Q, τ, msg) places $\mathcal{S}_{\mathcal{A}}$ and party. corrupt a as made it delivers $\mathcal{F}_{\text{AC}}^\circ$ channel The fetcher. the Q and corrupt being P admits. record no buys: unforgeability what is This up. it hands ρ as unstored. the so name. honest an wear ever can it behind send genuine a lacking never $\text{Clean}_{\text{inj}}$ and senders. corrupt for asked ever only is channel injection trips.

a matches: draining and (ii). by sides both on matches Freshness re- seen marked and ideally list from drained is delivered record stored and really seen by filtered is tuple same the of re-delivery 's $\mathcal{F}_{\text{Net}}$ so ally. the So time. second a case neither into falling ideally. entry list no finds configurations: corresponding to step and $\mathcal{M}$ same the return es Fetch two lockstep. in drained entries stored its preserved. is (iii) own 'sp on $\mathcal{F}$ corrupt a at Fetch or Send calls $\mathcal{A}$ traffic. Corrupt-party the case the and withholds. set outward the activation one the – identity

being ¶ ,3 line at it admits interface full The reach. not do above two core 'sp side real the on so -caller. A an on fires hook neither and corrupt. a at name 'sp carries places then core that call subroutine Every runs. the before A to each hands interface same that of 4 line so party, corrupt 's $\mathcal{F}_{PKI}$ buffer. 's $\mathcal{F}_{Net}$ reaches activation the of nothing runs: core callee's 's $\mathcal{S}_A$ supply. to 'sA is timestamp very its and tables. 's $\mathcal{F}_{Sig}$ or directory so shadows. same the on A hosted the against core same that runs 30 line untouched. hybrids the leave and alike seen and seq, gen move sides both the because ideally – list touches side neither and (ii); and (i) preserving is (iii) so – touch to $\mathcal{F}_{AC}^\circ$ no is there because really, $\mathcal{S}_A$ leaves never call check. to nothing with preserved

why saying worth is it and needs. (iii) what is calls these Withholding reach would call the outward Relayed routing. the to it leaving than rather in- $\mathcal{F}_{AC}^\circ$ test ripeness the time: true the at record a files which, $\mathcal{F}_{AC}^\circ$.Send whatever Δ within Q to it deliver then would 12 line 's $\mathcal{F}_{AC}$.Fetch at herits A unless all at nothing delivers world real the where answered. slot the was it moment the (iii) break would entry The account. own its on injects com- to 's $\mathcal{F}_{AC}$ being R – it retract could 's $\mathcal{S}_A$ of answer later no and filed. sender corrupt a to charges 19.1 Section that ripening the is This pute. speak parties corrupt whose protocol a and, $\mathcal{F}_{AC}$.Send through speaking it. pay not does instead $\mathcal{F}_{Net}$ through

and $\mathcal{C}$, $\mathcal{G}_{Clock}$, $\mathcal{Z}$ among call a cut: the touches activation other No on identically claimed and routed is identities 'sp of none targets that A Cor- governs. invariant the state no reads and ,4.18 Lemma by sides both executions The output. one 's $\mathcal{Z}$ return configurations halting responding an but bound a not – 0 is advantage the and distributions. as equal are forgery no and fixed are tapes the once deterministic step every identity. □ refuse. would sanitizer the injection an force to through slipping ever

was weaker nothing why on clear being worth is it and .0 is bound The so ideal. is subroutine Every needed. was stronger nothing and available identity an is reduction the for: charge to gap computational no is there does note. Transfer, is. theorem's composition the as distributions. of and call. responsive no placing cores wants 5.11 Corollary here: run not $\mathcal{F}_{AC}^\circ$ of property a so responsively, all clock the and $\mathcal{F}_{Net}$, $\mathcal{F}_{Sig}^!$ reach 'sp as exactly, π over re-proved be must it – free for π to descend not does the What clock. notifying the over built anything for warned 17.2 Section bound a not realization: a for thing stronger the is instead gives theorem adversary. every against equality an but dummy the against

In- .itself) $\mathcal{F}_{AC}$ realizes ρ classes. non-corrupting (Against 20.2 نتیجه
 $\mathcal{F}_{AC}^b$ let and $I \neq \emptyset$ when 7 line at $bad \leftarrow \text{true}$ set to $\mathcal{F}_{AC}^\circ$ strument
 (S, I) taking 's. $\mathcal{F}_{AC}$ fetch its – I ignores that $\mathcal{F}_{AC}$ pair-shaped the be
 (Defini- bad until identical are $\mathcal{F}_{AC}^b$ and $\mathcal{F}_{AC}^\circ$ Now alone. S using and
 occu- for So difference. bad-guarded one the injection the. (11.1 tion
 $I = \emptyset$ forces Clean_{inj} nobody. corrupt that environments and pants
 indistinguish- are two the 11.2 Lemma by whence $\Pr[\text{Bad}] = 0$ and
 non- the against 0 within $\{\mathcal{F}_{AC}\}$ hence $\{\mathcal{F}_{AC}^b\}$ UC-emulates π and able
 less: no 's. $\mathcal{F}_{AC}^\circ$ is guarantee the and party a Corrupt classes. corrupting
 -Send its transfers. (19.2 (Section FinAuth of half honest-sender the
 FinAuth full the while records. honest only reading argument tracing
 reason. same the for and $\mathcal{F}_{Net}$ of does it as exactly $\mathcal{F}_{AC}^\circ$ of fails

$\{\mathcal{F}_{PKI}, \mathcal{F}_{Net}, \mathcal{F}_{Sig}^1\}$ over protocol No .forced) is weakening (The 20.3 نکته
 target. true the but convenience a not is $\mathcal{F}_{AC}^\circ$ so shipped. as $\mathcal{F}_{AC}$ realizes
 responsive fetch's a inside diffuses. party corrupt A did. one Suppose
 exactly grants $\mathcal{F}_{Net}$ – accepts receiver its item well-formed a window.
 carrying – (18.1 (Section write to adversary's the being wire the this.
 ac- the side ideal the On choosing. adversary's the of τ^* timestamp a
 $\mathcal{F}_{AC}$. Send called have must simulator the so atomic. is fetch cepting
 the call that stamps $\mathcal{F}_{AC}$ but entry; matching a place to earlier strictly
 de- the off τ^* reads receiver the and τ^* not made. was it time true
 its has and time one at sends that environment An record. livered
 The certainty. with distinguishes another claim confederate corrupt
 since it. removes check receiver-side no and give to 's $\mathcal{F}_{Net}$ is freedom
 weak- only carries: payload the time whatever validates signature the
 gap. the closes – $\mathcal{F}_{AC}^\circ$ is which – it admit to object ideal the ening
 's $\mathcal{F}_{Net}$ forces that impossibility the of analogue diffusion the is This
 guarantee. delivery its outside sit to injections own

run- of point the and construction. single longest paper's the is This
 load-bearing. up shows framework the of layer every that is full in it ning
 responsive the hybrids: its and ρ between cut the supplies Composition
 can schedules the so atomic fetches both makes 9 Chapter of discipline
 senders: corrupt to slot corrupt a confines sanitization all: at coupled be
 its and descend. to needs $\mathcal{F}_{AC}^\circ$ of property a what be would tightness
 directly proved is realization the why is cores responsive under absence

what is 11 Chapter of lemma fundamental the and inherited: than rather realization exact the into object weaker the of realization exact the turns authenticated The play. in is corruption no wherever one stronger the of theorem. a now is ,19 Chapter in postulated channel.

Dive Deeper A : $\mathcal{F}_{\text{Sig}}$

of properties two proved and functionality signature a gave 15 Chapter a does What open. leaves that questions three the asks chapter This. properties those inherit to it on built protocol a for satisfy to have scheme own framework's this in drawn like. look protocol the does what and — they says 3.2 Section where held state its and randomness its with terms. char- to enough properties two the are hold: converse the Does be? must realizes them satisfying interfaces its with anything that so $\mathcal{F}_{\text{Sig}}$ acterize it of realization a does or shape. its of object strongest the $\mathcal{F}_{\text{Sig}}$ is And it? not? does it something satisfy no. bound: the as security own scheme's a with yes are: answers The which stronger. strictly be can realization a — no and so: instructively and general. in functionalities ideal about knowing worth is

games their and schemes Signature 21.1

a over $\text{DS} = (\text{Gen}, \text{Sign}, \text{Ver})$ algorithms of triple a is scheme signature A space coin a and Σ space signature a $\mathcal{M}$ space message a $\mathcal{K}$ space key explicitly: coins their taking Sign and Gen with $\{0, 1\}^n$

$$\begin{aligned} (\text{vk}, \text{sk}) &\leftarrow \text{Gen}(r), & \sigma &\leftarrow \text{Sign}(\text{sk}, \text{msg}; r), \\ b &\leftarrow \text{Ver}(\text{vk}, \text{msg}, \sigma), \end{aligned}$$

tape implicit an than rather arguments are coins The deterministic. Ver ran- whose scheme a and $\mathcal{F}_{\text{Rand}}$ through them draws below π_{Sig} because function- randomness a to wired be not could it inside hidden is domness all. at ality

property-based ordinary the in games are notions security three Its framework. no — alone DS against $\mathcal{B}$ adversary an by played each sense. un- anyway. down write would cryptographer a notions the execution. no One .[10] them among chief attack chosen-message under forgeability

pa- the idiom same the in three all by shared oracle, signing one shell.
(Def- finalizations several across shell a share systems game own per's
calling by halts then likes, it as Sign queries ,vk receives $\mathcal{B}$: (5.1 inition
wins. it whether decides which below, procedures three the of one exactly

shell the :DS scheme signature a for Game

```

:1 (vk, sk) ← DS.Gen(r)                                     :Initialize()
:2 return vk                                                  / r uniform

:3 σ ← DS.Sign(sk, msg; r)                                    :Sign(msg)
:4 tr ← tr · (msg, σ)                                         / r uniform
:5 return σ

```

shared is above shell the notion: per one game. the of Finalizations

```

:6 σ ← Sign(msg)                                              (correctness) Cor(msg)
:7 w ← 1 if DS.Ver(vk, msg, σ) = 1: else w ← 0                / a fresh call. on the game's own oracle
:8 return w

:9 w ← 1 if DS.Ver(vk, msg, σ) = 1 ∧ (msg, σ) ∉ tr: else w ← 0 (unforgeability) Uf(msg, σ)
:10 return w

:11 w ← 1 if msg ≠ msg' ∧ DS.Ver(vk, msg, σ) = DS.Ver(vk, msg', σ) = 1: else (resistance) (collision) Col(msg, msg', σ)
    w ← 0
:12 return w

```

and Uf, Cor that probabilities the for $\epsilon_{\text{col}}(\mathcal{B})$ and $\epsilon_{\text{uf}}(\mathcal{B})$, $\epsilon_{\text{cor}}(\mathcal{B})$ Write
have schemes most which, $\epsilon_{\text{cor}} = 0$ is correctness Perfect. 1 return Col
ex- 9 line :[1] strong is Unforgeability needs. follows what of none and
match- alone, message the than rather returned oracle the pair the cludes
standard not is resistance Collision. 15.2 Section in reading's FinUF ing
place. its earns it where is 21.4 Section two: other the by implied not is and

π_{Sig} protocol The 21.2

every own: its of nothing holds and interfaces three 's $\mathcal{F}_{\text{Sig}}$ carries π_{Sig}
invocation an outlives that value every and $\mathcal{F}_{\text{Rand}}$ through drawn is coin
the and shape leakage-well-formed 's 3.2 Section is which, $\mathcal{F}_{\text{Store}}$ in sits
Leak- decoration. not is That it. to written paper this in protocol first

Leak functionality's a but (4 (line corrupt being party the on gated is age would parties across shared $\mathcal{F}_{\text{Rand}}$ single a so entire, state its over hands corrupted. was party one any moment the coins party's every surrender The it. survive could unforgeability no — key signing every them with and $\mathcal{F}_{\text{Rand}}^{\text{P}}$ party, per each of instance one addresses therefore π_{Sig} protocol guard's the and allows already 5.5 Remark which, $\mathbf{P} = \{\text{P}\}$ with $\mathcal{F}_{\text{Store}}^{\text{P}}$ and each erases it And reaches, already number instance the to blindness put was Erase 's $\mathcal{F}_{\text{Rand}}$ what is which used, been has it moment the coin corrupt a so leak, cannot erased been has what :13.1 (Section for there own its even not — else nothing and key signing own its surrenders party coins. past

$\{\mathcal{F}_{\text{Rand}}^{\text{P}}, \mathcal{F}_{\text{Store}}^{\text{P}}\}_{\text{P} \in \mathbf{P}}$ over π_{Sig} Protocol	
$\text{par} := \text{DS} \quad .\mathbf{U} := \{(\mathcal{F}_{\text{Rand}}, \text{serves}), (\mathcal{F}_{\text{Store}}, \text{serves})\} \quad .\mathbf{N} \quad .\mathbf{P} \quad .\text{pid}$	
$\text{id}' \text{ from } \text{id}.\text{Sign}(\text{msg})$	$\text{id}' \text{ from } \text{id}.\text{Gen}()$
:11 $s_k \leftarrow \text{id}_{\mathcal{F}_{\text{Store}}}.\text{FullGet}(s_k) \text{ as id}$	:1 $vk \leftarrow \text{id}_{\mathcal{F}_{\text{Store}}}.\text{FullGet}(vk) \text{ as id}$
:12 if $s_k = \text{rej}$ then	:2 if $vk \neq \text{rej}$ then
:13 return $\perp$ / no key yet	:3 return vk / one key per party
:14 $(i, r) \leftarrow \text{id}_{\mathcal{F}_{\text{Rand}}}.\text{FullRnd}() \text{ as id}$	:4 $(i, r) \leftarrow \text{id}_{\mathcal{F}_{\text{Rand}}}.\text{FullRnd}() \text{ as id}$
:15 $\sigma \leftarrow \text{DS}.\text{Sign}(s_k, \text{msg}; r)$	:5 $(vk, s_k) \leftarrow \text{DS}.\text{Gen}(r)$
:16 $\text{id}_{\mathcal{F}_{\text{Rand}}}.\text{FullErase}(i) \text{ as id}$ / erased as soon as used	:6 $\text{id}_{\mathcal{F}_{\text{Rand}}}.\text{FullErase}(i) \text{ as id}$ / the coins cannot leak
:17 return σ	:7 $\text{id}_{\mathcal{F}_{\text{Store}}}.\text{FullPut}(vk, vk) \text{ as id}$
$\text{id}' \text{ from } \text{id}.\text{Leak}()$	:8 $\text{id}_{\mathcal{F}_{\text{Store}}}.\text{FullPut}(s_k, s_k) \text{ as id}$
:18 return $\perp$ / it holds nothing to leak	:9 return vk
	$\text{id}' \text{ from } \text{id}.\text{Verify}(vk, \text{msg}, \sigma)$
	:10 return $\text{DS}.\text{Ver}(vk, \text{msg}, \sigma)$

the with answers Rnd 's $\mathcal{F}_{\text{Rand}}$ because pairs as back come draws The the name π_{Sig} lets what is which, (7 (line string the beside wrote it index index the to indifferent is and strings the reads FinUnp erases: then it entry them. beside travelling

state no has it because $\perp$ returns **Leak** own 's π_{Sig} remarks. Three there, it reads adversary party's corrupt a and 's $\mathcal{F}_{\text{Store}}^{\text{P}}$ is key the give: to where is leak the — asks leakage-well-formedness what precisely is which all, at call no places **Verify** operation The else, nowhere and is, state the nor draws neither and local is verification a so deterministic, being **Ver** be cannot **Verify** whose here protocol one the is π_{Sig} why is that stores: fetched is key the And movement, token's the by 's $\mathcal{F}_{\text{Sig}}$ from distinguished key "one makes what is which, π_{Sig} in cached than rather store the from rather (**FinKeep**, 14.2 (Section fixity per-slot 's $\mathcal{F}_{\text{Store}}$ about fact a party" per itself, about makes π_{Sig} promise a than

گزاره 21.1 $\gamma := \{\mathcal{Gm}\} \cup \pi_{\text{Sig}} \cup \text{Let}.$ properties) two the inherits π_{Sig} with 5.1 Definition of sense the in system game a be $\{\mathcal{F}_{\text{Rand}}^P, \mathcal{F}_{\text{Store}}^P\}_P$ q most at placing $\mathcal{Z} \in \mathbb{Z}_{\gamma, \gamma}$ every For 15.2 Section of shell the $\mathcal{Gm}$ there slot adversary the of $\mathcal{X}$ occupant every and relays the on calls once $\mathcal{X}$ and $\mathcal{Z}$ running each. DS against $\mathcal{B}_{\text{uf}}$ and $\mathcal{B}_{\text{cor}}$ adversaries are that such overhead, $O(q)$ with

$$\Pr[\text{Win}_{\text{FinCor}}] \leq q \cdot \varepsilon_{\text{cor}}(\mathcal{B}_{\text{cor}}),$$

$$\Pr[\text{Win}_{\text{FinUF}}] \leq q \cdot \varepsilon_{\text{uf}}(\mathcal{B}_{\text{uf}}) + q^2 2^{-n}.$$

اثبات. One preliminary paragraph and two. The reductions. preliminary The col- en- trace every makes tightness give: already subroutines the what lects 14.1 Proposition party, honest an at gave core own 's π_{Sig} answer an try good, for party per issued is key one so faithful and fixed store the makes unread- erased, being and, uniform coins the makes 13.1 Proposition and same the by game own scheme's the to reduces then property Each able. failure a as names it triple the identify fires, verdict the assume move: at the of which advance in guess reduction the have and outright, DS of from, comes q factor the where is which — one failing the is calls q most ,gen every so , π_{Sig} reaches $\mathcal{Gm}$ but caller no tightness By bounds, both in honest an at gave core own 's π_{Sig} answer an is τ_r of entry ver and sign noth- records and intercepts wrapper game's the one corrupt a at — party the faithful: and fixed is store the 14.1 Proposition By (5.4 Remark ing 1 line on guard 'sGen and gets, later it what are puts party a s_k and v_k coins the 13.1 Proposition By good, for party per key one issues therefore are ,Sign in counterpart its and 6 line at erased being and, uniform are alone, P serves $\mathcal{F}_{\text{Rand}}^P$ and them of empty is leak 's $\mathcal{F}_{\text{Rand}}^P$ nobody: by read them, reaches party another of corruption no so a , (gen, P, v_k) a ,Q ,P honest fires: 20 line suppose correctness, For verification The , $(\text{ver}, Q, v_k, \text{msg}, \sigma, 0)$ a and , $\sigma \in \Sigma$ with $(\text{sign}, P, \text{msg}, \sigma)$ DS.Sign($s_k, \text{msg}; \tau$) is signature the and ,10 line by DS.Ver(v_k, msg, σ) is the by uniform coins on ,5 line by v_k very that with paired s_k the under The outright, DS of failure correctness a is triple the So above, paragraph at the of which advance in guesses execution, the runs $\mathcal{B}_{\text{cor}}$ adversary wins it message: its outputs and one, failing the is calls signing q most ,q factor the is which right, is guess the and does game the whenever

.(gen, P, vk) ,Q ,P honest fires: 28 line suppose unforgeability, For is vk Either cases. Two .(sign, P, msg, σ) no and (ver, Q, vk, msg, σ , 1) re- (msg, σ) with $\text{Ver}(\text{vk}, \text{msg}, \sigma) = 1$ then and ,P to issued Gen key the planting by harvests $\mathcal{B}_{\text{uf}}$ which forgery. strong a – P at Sign no by turned parties generating q most at the of one guessed a at key challenge its party other every oracle. its from calls signing party's that answering and 's.P equal to happens that key party's honest another is vk or itself: runs it collided. coins -bitn uniform on draws $\text{Gen}(\tau)$ independent two then and era- the where Note . $q^2 2^{-n}$ most at is generations q most at over which of coins the $\mathcal{X}$ hand would party any corrupting it. without spent: is sure does it oracle an with answer not could $\mathcal{B}_{\text{uf}}$ and .Sign and Gen own 'sP $\square$ for. coins the have not

information- nor 0 neither is that paper the in bound first the is This $\mathcal{F}_{\text{Sig}}$ functionality The changed. has what saying worth is it and theoretic. has π_{Sig} :(15.1 (Proposition to written was it because properties its has a of price ordinary the are -factorsq two the and does. DS because them legitimate exchange the makes what is 5.11 Corollary reduction. guessing of realization any to descends $\mathcal{F}_{\text{Sig}}$ of property a – direction other the in re- the from approached statement same the is proposition this and – it by $\mathcal{F}_{\text{Sig}}$ from obtained not is π_{Sig} because directly proved side. alization's scheme. a from built but emulation

reach cannot properties what and converse, The 21.3

converse The properties. the to scheme a from runs 21.1 Proposition charac- properties two the that further: much and way other the run would satisfying and interfaces its carrying functionality any that so , $\mathcal{F}_{\text{Sig}}$ terize of ask we what are properties – conjecture natural a is It it. realizes them the as good as be to ought them satisfying object an so object. ideal an false. is it and – of them asked we one

a is There .($\mathcal{F}_{\text{Sig}}$ characterize not do properties two (The 21.2 گزاره Both having interfaces. and fields 's $\mathcal{F}_{\text{Sig}}$ with $\mathcal{F}'$ functionality standard occu- every and environment every against 0 within FinUF and FinCor any within $\{\mathcal{F}_{\text{Sig}}\}$ UC-emulate not does $\{\mathcal{F}'\}$ that such slot. the of pant $\epsilon < \frac{1}{2}$.

party's honest no when .Verify in change: one with $\mathcal{F}_{\text{Sig}}$ be $\mathcal{F}'$ Let .اثبات. – $P' \notin \mathbf{C}$ every for $\text{vk} \neq \text{vk}[P']$ when is. that – vk queried the equals key

query later a that so nowhere, recorded bit, uniform fresh a is answer the is $\mathcal{F}'$ triple other every On independently, answered is triple same the on included, memoization exactly, $\mathcal{F}_{\text{Sig}}$

existential is 28 and 20 lines of Each .0 within properties both has $\mathcal{F}'$ honest no is vk whose triple a so , $P \notin \mathbf{C}$ with entry $(\text{gen}, P, \text{vk})$ a over $\mathcal{F}_{\text{Sig}}$ as answers $\mathcal{F}'$ triples other all on and neither, witnesses key party's .15.1 Proposition by 0 are verdicts whose does.

generated vk^* with σ , msg, vk^* any pick $\mathcal{Z}$ Let emulate, not does It honest an at twice $\text{Verify}(\text{vk}^*, \text{msg}, \sigma)$ call and one, no corrupt nobody, by in- are answers the $\mathcal{F}'$ Against differ, answers two the if 1 returning party, the $\mathcal{F}_{\text{Sig}}$ Against $\frac{1}{2}$ probability with 1 returns it so bits, uniform dependent answers the so it, returns second the and $\text{Ver}[\text{vk}^*, \text{msg}, \sigma]$ fixes call first does, simulator the whatever — 0 probability with 1 returns it and agree to able slot's the of value no and code own 's $\mathcal{F}_{\text{Sig}}$ being memoization the $\square$ $\frac{1}{2}$ is advantage The it, unfix

pins Emulation counterexample, the than useful more is diagnosis The things two pin properties two the behaviour, observable whole object's an un- forgeries that and verify, signatures produced honestly that — it about gap the and reach, their of out is else Everything not, do keys honest der honest, P with $(\text{gen}, P, \text{vk})$ over existential are verdicts both shape: a has and generated, party honest no keys about said is whatever nothing so in- verdict either all being answer one consistency, about said is nothing two but unforgeability of notion stronger a not is missing is What spectrs, once obvious makes code 's $\mathcal{F}_{\text{Sig}}$ ones the are they and properties, further party, per key one issues Gen its and memoizes, Verify its for: looked

15.2 Section of shell the of finalizations further Two

```

                                (consistency)  id' from  id.FinCons()
:1 require  $\neg \text{done}$ :
:2 require  $\text{id.P} = \text{Z}$ :  $\text{done} \leftarrow \text{true}$ 
:3  $w \leftarrow 1$  if  $\exists Q, Q', \text{vk}, \text{msg}, \sigma : (\text{ver}, Q, \text{vk}, \text{msg}, \sigma, 0) \in \text{tr}$ 
:4    $\wedge (\text{ver}, Q', \text{vk}, \text{msg}, \sigma, 1) \in \text{tr}$ : else  $w \leftarrow 0$ 
:5 return  $w$ 

                                party) per key (one  id' from  id.FinOne()
:6 require  $\neg \text{done}$ :
:7 require  $\text{id.P} = \text{Z}$ :  $\text{done} \leftarrow \text{true}$ 
:8  $\mathbf{C} \leftarrow (\mathbf{C}, \mathbf{Z}).\text{FullStatus}()$  as  $\text{id}$ 
:9  $w \leftarrow 1$  if  $\exists P \notin \mathbf{C}, \text{vk} \neq \text{vk}' : (\text{gen}, P, \text{vk}) \in \text{tr} \wedge (\text{gen}, P, \text{vk}') \in \text{tr}$ :
:10 else  $w \leftarrow 0$ 
:11 return  $w$ 

```

FinCons — $\mathcal{F}_{\text{Sig}}$ by 0 at attained are both reading, search the take Both because **FinOne** thereafter, read and triple per once written is Ver because first-write- same the by each thereafter, read and once written is $\text{vk}[P]$ are both and — directory the for makes 16.1 Proposition argument wins respec- FinKeep 's $\mathcal{F}_{\text{Store}}$ and determinism 's Ver by π_{Sig} by 0 at attained party no names **FinCons** party: corrupt a of anything asks Neither tively. out answers corrupt kept already having wrapper own game's the all. at .tr of

stan- a be $\mathcal{F}'$ Let .properties) four the from converse. (The 21.3 قضيه $\mathcal{F}'$ returns Sign whose interfaces, and fields 's $\mathcal{F}_{\text{Sig}}$ with functionality dard let and party, calling the at generated been has key no when exactly $\perp$ and $\mathbb{Z}_{\gamma, \gamma}$ against 0 within **FinOne** and **FinCons**. **FinUF**, **FinCor** have $\mathcal{F}'$ with 15.2 Section of system game the γ for slot, the of occupant every every for :0 within $\{\mathcal{F}_{\text{Sig}}\}$ UC-emulates $\{\mathcal{F}'\}$ Then . $\mathcal{F}_{\text{Sig}}$ of place in $\mathcal{F}'$ with $\mathcal{S}_{\mathcal{A}}$ simulator a is there $\mathcal{A}$ occupant

$$\text{Exec}(\{\mathcal{F}'\}, \mathcal{Z}, \mathcal{A}, \mathcal{C}) \equiv \text{Exec}(\{\mathcal{F}_{\text{Sig}}\}, \mathcal{Z}, \mathcal{S}_{\mathcal{A}}, \mathcal{C})$$

$\mathcal{Z} \in \mathbb{Z}_{\{\mathcal{F}'\}, \{\mathcal{F}_{\text{Sig}}\}}$ every for

copy internal an with bundled $\mathcal{A}$ — one obvious the is simulator The اثبات. same the putting by queries slot three 's $\mathcal{F}_{\text{Sig}}$ of each answering, $\mathcal{F}'$ of sides two the that is argument the of whole the so — copy that to call an- slot's the is answer 's $\mathcal{F}_{\text{Sig}}$ Since value. returned a on disagree never override no showing to reduces that overrides, code own its unless swer

sanitizer, 'sGen memoization, 'sGen four: exactly are there and fires, ever
 ex- and turn in taken is Each test. honest-key 'sVerify and sanitizer, 'sSign
 cases sanitizer the .0 at properties hypothesised four the of one by cluded
 where is That make. would sanitizer the repair which by further splitting
 them. exactly needs theorem the why is it and spent, are properties four all
 the on it runs and $\mathcal{F}'$ of copy internal an with $\mathcal{A}$ bundles $\mathcal{S}_{\mathcal{A}}$ simulator The
 three the exactly for slot the asks $\mathcal{F}_{\text{Sig}}$ functionality The reports. $\mathcal{F}_{\text{Sig}}$ calls
 at bit the .Sign at signature the .Gen at key the — pin not does it values
 $\mathcal{F}'$ internal its to call same the putting by each answers $\mathcal{S}_{\mathcal{A}}$ and — Verify
 hosted the letting back, comes what forwarding and identity same the at
 Leakage slot. a to put would code own 's $\mathcal{F}'$ whatever answer, and see, $\mathcal{A}$
 note likewise: copy internal the from serves it instructions corruption and
 environment the all, at 's $\mathcal{F}_{\text{Sig}}$ resemble not need **Leak** 's $\mathcal{F}'$ why is this that
 occupies. $\mathcal{S}_{\mathcal{A}}$ which slot, the through only it reaching

Every- .11.2 Lemma in as tapes equal on executions two the Couple
 $\mathcal{F}'$ internal the what returns $\mathcal{F}_{\text{Sig}}$ call, every at claim: one to reduces thing
 sees. environment the value the on agree sides two the that so returned,
 it overrides. code own its unless answer slot's the is answer 's $\mathcal{F}_{\text{Sig}}$ Since
 four. exactly are there and fires, ever override no that enough is

$\mathcal{F}_{\text{Sig}}$ set already is $\text{vk}[\text{id.P}]$ If sanitizer, 'sGen and memoization 'sGen
 that at key same the returns $\mathcal{F}'$ internal the says 0 at FinOne it: returns
 offered the replaces **San**[Clean_{vk}] Otherwise agree. two the so too. party
 al- signature a has or issued, already one duplicates or key, a not is it if vk
 non- a with Gen answered $\mathcal{F}'$ then key: a Not it. under valid recorded ready
 (ver, ., vk, ., ., 0) with $\sigma \in \Sigma$ carry would party that at Sign later a and key,
 environ- the once 0 at FinCor by excluded — environment the to available
 then parties honest two Duplicate: may. it which verifies, and signs ment
 ex- .FinUF wins other the at verifying and one at signing and key, a share
 issued, is vk before stands $\text{ver}[\text{vk}, \text{msg}, \sigma] = 1$ some Pre-signed: cluded.
 key a under signed party honest an so .Sign at only 1 a records $\mathcal{F}_{\text{Sig}}$ and
 the which and, $\perp$ returning refuses. Sign 's $\mathcal{F}_{\text{Sig}}$ which — generated yet not
 idle. is sanitizer the So too. refuses Sign 's $\mathcal{F}'$ on hypothesis

sig- a not is it if σ replaces **San**[Clean _{σ}] sanitizer The sanitizer, 'sSign
 $\sigma \in \Sigma$ 'sFinCor signature: a Not already. $\text{ver}[\text{vk}, \text{msg}, \sigma] = 0$ if or nature
 envi- the excluded, be case this lets that reading the exactly is conjunct
 :0 Already .0 a forbidding FinCor and value returned the verifying ronment
 $\mathcal{F}'$ internal the — now 1 verifies and earlier 0 verified triple same the then
 excluded .FinCons is which — accept must it signature a produced having
 idle. is too it So .0 at

re- first The suppression. honest-key its and memoization 'sVerify the says 0 at FinCons and exists. one where verdict recorded the turns is this agree: they so way, same the triple same the answers $\mathcal{F}'$ internal is it and exploited. 21.2 Proposition of counterexample the override the triple fresh a on $b \leftarrow 0$ sets second The closes. FinCons what exactly answers $\mathcal{F}'$ internal the says 0 at FinUF and key. party's honest an under agree. both So condition. win its being 1 a – too there 0 's $\mathcal{F}'$ internal the is sees environment the value every fires. override No in as exactly halt. same the to step in proceed runs coupled the and own. advantage the and equal are distributions The case. no-flag 's11.2 Lemma $\square$.0 is

prop- Two point. the is hold to took it what and holds. converse the So not are added two the and are. four characterization: a not were erties table a – commits code 's $\mathcal{F}_{\text{Sig}}$ places two the but signing about facts deep properties functionality's A over. twice thereafter. read and once written read- and commitment. such every cover they when exactly it characterize missing the find to way better a is commitments its for specification a ing true. be to like would one else what asking than property

lacks $\mathcal{F}_{\text{Sig}}$ that π_{Sig} of property A 21.4

is It shape. its of object strongest the is $\mathcal{F}_{\text{Sig}}$ whether is question last The costs. sanitization what is it oversight: an not is gap the and not.

resistance collision finalization: further A

```

resistance) (collision id' from id.FinColl())
:1 require  $\neg \text{done}$ ;
:2 require id.P = Z; done  $\leftarrow$  true
:3 C  $\leftarrow$  (C, Z).FullStatus() as id
:4 w  $\leftarrow$  1 if  $\exists P \notin \mathbf{C}, Q, Q' \notin \mathbf{C}, \text{vk}, \text{msg} \neq \text{msg}', \sigma$  :
:5   (gen, P, vk)  $\in \text{tr} \wedge (\text{ver}, Q, \text{vk}, \text{msg}, \sigma, 1) \in \text{tr}$ 
:6    $\wedge (\text{ver}, Q', \text{vk}, \text{msg}', \sigma, 1) \in \text{tr}$ ; else w  $\leftarrow$  0
:7 return w

```

than rather hard be to meant being winning reading, search the takes It guess. a than better no

گزاره 21.4 $(\text{not does } \mathcal{F}_{\text{Sig}} \text{ it. has } \pi_{\text{Sig}})$ $q \cdot \varepsilon_{\text{col}}(\mathcal{B}_{\text{col}})$ within FinColl has π_{Sig} .not) does $\mathcal{F}_{\text{Sig}}$ it. has π_{Sig}) 21.1 Proposition in as built DS against $\mathcal{B}_{\text{col}}$ adversary an for

an and $\mathcal{Z}^*$ are there $:\varepsilon < 1$ any within it have not does $\mathcal{F}_{\text{Sig}}$ tionality
 $\text{Pr}[\text{Win}_{\text{FinColl}}] = 1$ with $\mathcal{X}^*$ occupant

honest an under verifying σ one exhibits trace winning a $:\pi_{\text{Sig}}$ For اثبات.
 .10 line by DS.Ver is verification and messages. distinct two on vk party's
 The coins. uniform on drawn key a under DS for collision a is pair the so
 q most at the of one guessed a at key challenge its plants $\mathcal{B}_{\text{col}}$ adversary
 runs and oracle its from signing party's that answers parties. generating
 signature. the and messages two the outputs and itself. rest the
 Verify every key. fresh a with query Gen every answer $\mathcal{X}^*$ let $:\mathcal{F}_{\text{Sig}}$ For
 cor- $\mathcal{Z}^*$ Let $.\sigma^* \in \Sigma$ fixed one with queries Sign both and $.\emptyset$ with query
 $\text{msg} \neq \text{msg}'$ for $\text{Sign}(\text{msg}')$ and $\text{Sign}(\text{msg})$ then $\text{Gen}()$ call nobody. rupt
 sig- first The finalize. then triples. both on Verify then party. honest an at
 that and signature a be σ^* that only asking Clean_σ accepted. is nature
 $.\text{Ver}[\text{vk}, \text{msg}, \sigma^*] \leftarrow 1$ unset: being it holds. which $.\text{Ver}[\text{vk}, \text{msg}, \sigma^*] \neq 0$
 message. other the at reason same the for accepted is second The
 Both $.\text{Ver}[\text{vk}, \text{msg}', \sigma^*] \leftarrow 1$ and well. as unset being $\text{Ver}[\text{vk}, \text{msg}', \sigma^*]$
 1 gives 4 line at verdict the and $.\text{1}$ recorded the return then verifications
 □ certainty. with

patched. be can it rather. or — patched be to defect a not is attack The
 other no for valid recorded be σ that conjunct the Clean_σ to adding by
 Sanitiza- cost. would that what in is interest the and $.\text{vk}$ under message
 slot the on depend not do guarantees functionality's a that so exists tion
 a is predicate Clean a of conjunct each $:(12$ (Chapter helpfully answering
 longer no may simulator the value a also is each and bought. guarantee
 unforge- and correctness buys written as $\mathcal{F}_{\text{Sig}}$ functionality The choose.
 read- defensible a is which resistance. collision buy to declines and ability
 made consequence. the and — for is functionality signature a what of ing
 stronger strictly be may realization a that is $.\text{21.4}$ Proposition by precise
 realizes. it object ideal the than

An intuition. usual the inverts it because with. sitting worth is That
 re- with behaviour. possible best the as read often is functionality ideal
 $\mathcal{F}_{\text{Sig}}$ property a satisfies π_{Sig} here below: from it approximating alizations
 Corol- — downward it transfers emulation of amount no and fails. provably
 never realization. the to object ideal the from properties carries 5.11 lary
 may protocol a what is fixes functionality ideal an What way. other the
 realization a anything achieve: to happens it what not for. on relied be

goes that argument every to invisible is and real, is that beyond achieves
object. ideal the through

Issues Pending

it. raises body the places the from gathered open, leaves paper this What
and Presentation cover. not does and does review its what with together
listed. not are typesetting

hypothe- 4.29 Theorem **calls. responsive at stops completeness Dummy**
relay dummy's The call. responsive no place cores whose systems sises
adversary real a where $\perp$ answers and call such first the at refused is
com- Recovering fails. regrouping the so substance. with answer would
do we and side. environment's the on notion matching a needs pleteness
one. give not

in- results property-transfer three All **theory. responsive no have Games**
reaches $\mathcal{F}$ whose system game A .4.29 Theorem from condition that herit
dummy-only its have must and three all outside sits $\mathcal{A}^1$ through slot the
functionalities own paper's this of Four directly. established hypothesis
by transfers their argue and reason this exactly for hypothesis the decline
hand.

simulator the fixes 4.11 Definition **converse. no has order quantifier The**
sim- the which in reading weaker the implies and environment, the before
the Whether prove. not do we converse The both. on depend may ulator
open. is separate notions two

compar- blocked a keeps Tightness **characterized. not are pairs Blocked**
.I of member no constrains it but members, private the for faithful ison
a by reachable is name used bare a admitting member interface an and
the where $\mathbf{Z} \cup \mathbf{A}$ only admitting or those. Pinning instance. fake hosted
general no give We it. close would directly, them reaches environment
characterization. no and condition sufficient

clo- name whole the withholds Admissibility **cost. would spawning What**
simulator a Letting needs. strictly comparison a than more is which sure.

only leave would occupy not does φ identities at functionalities spawn
 relax- that What pairing. usual the for empty – withhold to $\text{IDs}(\varphi) \setminus \text{IDs}(\pi)$
 out. worked not is composition in costs ation

assem- is entry trace Every **consistency. temporal not is atomicity Trace**
 an Whether checked. was paper the in append every and atomically, bled
 dif- a is time in agree it alongside fetched set the and timestamp entry's
 The atomicity. by than rather responsiveness by secured property, ferent
 one in them separates nowhere paper the and conflate to easy are two
 place.

in examined were units audit twenty-one of Six **covers. review the What**
 the than rather pass single a in and criticality, by leaves-first chosen full.
 clustered found defect Every for. asks protocol the ones independent two
 may it whether and outward, places bundle a what – mechanism one on
 were all and – call responsive a answering while members hosted its use
 the for: look not did pass one what in therefore is exposure The fixed.
 should pass further a where is channel authenticated the of realization
 defects. contained actually that unit the being go.

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